

On the Zeros of Riemann's Zeta-function

by

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To the memory of

Alfonso Torio Cruz

my grandfather and first teacher

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Biographical Sketch

The author was born in Metro Manila, Philippines. He attended the Ateneo de Manila University, Philippines, and graduated with a Bachelor of Science degree in Mathematics in 2007 and a Master of Science degree in Mathematics in 2010. He began doctoral studies in Mathematics at the University of Rochester in 2011. In 2013, he received a Master of Arts degree in Mathematics from the University of Rochester. He pursued his research in analytic number theory under the supervision of Professor Steven M. Gonek.

The following research publications were a result of work conducted during doctoral study, and form part of this thesis:

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Abstract

This thesis has three parts. In the first part, we combine the mollifier method with a zero detection method of Atkinson to prove in a new way that a positive proportion of the nontrivial zeros of the Riemann zeta-function $\zeta(s)$ are on the critical line. One of the main ingredients of the proof is an estimate for a mollified fourth moment of $\zeta(\frac{1}{2} + it)$. We deduce this estimate from the twisted fourth moment formula that has been recently developed by Hughes and Young.

The second part of this thesis is concerned with bounding the number $N(\sigma, T)$ of zeros of $\zeta(s)$ that have real parts $> \sigma$ and imaginary parts between 0 and T . We prove a claim of Conrey that improves previous bounds for $N(\sigma, T)$ due to Selberg and Jutila. We do this by constructing a new mollifier that allows us to apply Conrey's technique of using Kloosterman sum estimates to deduce an asymptotic formula for the mollified second moment of $\zeta(\sigma + it)$ when σ is $> \frac{1}{2}$ and close to $\frac{1}{2}$.

In the third part of the thesis, we give a conditional proof of the equivalence of certain asymptotic formulas for (a) averages over intervals for the 2-point form factor $F(\alpha, T)$ for the zeros of $\zeta(s)$, (b) the mean square of the logarithmic derivative of $\zeta(s)$, (c) a variance for the number of primes in short intervals, and (d) the number of pairs of zeros of $\zeta(s)$ with small gaps. The main result is a generalization of the fusion of a theorem of Goldston and a theorem of Goldston, Gonek, and Montgomery. We apply our result to deduce several consequences of the Alternative Hypothesis.

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1 Introduction and results

The function $\zeta(s)$ now known as the Riemann zeta-function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

when the complex variable s has real part > 1 . In his epoch-making memoir of 1859, Riemann [55] outlined how one can study the prime numbers through this function. We can see the connection between $\zeta(s)$ and the primes via the Euler product formula

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1},$$

where the product on the right-hand side is over the set of all primes p . In fact, Riemann showed that $\zeta(s)$ has an analytic continuation to the entire complex plane, and revealed the link between the zeros of $\zeta(s)$ and the primes. This link can be seen more clearly from the *explicit formula* (see, for example, §17 of Davenport [21])

$$\sum_{n \leq x} \Lambda(n) - \frac{1}{2} \Lambda(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}), \quad x > 1.$$

Here, $\Lambda(n) = \log p$ if $n > 1$ and n is an integer power of a prime p , and $\Lambda(n) = 0$ otherwise. The sum on the right-hand side is over all the non-real zeros ρ of $\zeta(s)$. The left-hand side of the explicit formula is an expression involving the primes,

while the right-hand side involves the zeros of $\zeta(s)$. This enables us to deduce properties of the primes from properties of the zeros, and vice versa. Because of this connection, a huge amount of work over the past century and a half has been put towards locating the zeros.

Today we know a good deal of information about the zeros of $\zeta(s)$. Riemann [55] observed that $\zeta(s)$ has zeros at $s = -2, -4, -6, \dots$, called the *trivial zeros*, and no other real zeros. He also observed that the non-real zeros, also called the *nontrivial zeros*, are all contained in the *critical strip*

$$\{s \in \mathbb{C} : 0 \leq \operatorname{Re}(s) \leq 1\}.$$

In this strip, the zeros are symmetric about the line $\operatorname{Re}(s) = \frac{1}{2}$. If $N(T)$ is the number of nontrivial zeros, counting multiplicity, that have imaginary parts between 0 and T , then (see, for example, §15 of Davenport [21])

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T) \quad (1.0.1)$$

as $T \rightarrow \infty$. This formula was proved by von Mangoldt first in 1895 with a slightly weaker bound for the error term, and then fully in 1905. Hadamard and de la Vallée Poussin [64, §3.1] independently proved in 1896 that $\zeta(s)$ has no zeros with real part equal to 1. Their discovery led them to the first proofs of the prime number theorem, which states that if $\pi(x)$ is the number of primes $\leq x$, then

$$\pi(x) \sim \frac{x}{\log x} \quad (1.0.2)$$

as $x \rightarrow \infty$. This had been conjectured a century earlier by Gauss and Legendre, and its proof highlights the striking power of $\zeta(s)$. The story, however, is far from over. It turns out that the degree of precision of (1.0.2) depends on how near the zeros are to the *critical line* $\operatorname{Re}(s) = \frac{1}{2}$. In fact, Riemann conjectured in his original memoir that all the nontrivial zeros lie on the critical line. This statement is the famous *Riemann Hypothesis*, and to date is still an open problem.

The past decades have seen a lot of progress towards solving this and many other problems about $\zeta(s)$, with many statements having been proved, and even more having been conjectured. We will see indications of these throughout this thesis.

In this thesis, we consider three problems about the distribution of zeros of $\zeta(s)$. The first problem is that of showing that there are many zeros on the critical line. In Chapter 2, we devise a new method of proving that the critical line contains a positive proportion of the nontrivial zeros of $\zeta(s)$. The second problem is to bound the number of zeros that are off the critical line. Our current state of knowledge is yet unable to rule out the existence of such zeros, but there are bounds for the number of those inside certain regions of the critical strip. We improve one such bound in Chapter 3. Finally, the third problem is to determine implications of the vertical distribution of the zeros along the critical line. In Chapter 4, we consider hypothetical statements about this distribution, and ascertain consequences for the prime numbers and other properties of $\zeta(s)$.

Throughout this thesis, we use the following standard notations in analytic number theory. The equation

$$f(T) = g(T) + O(h(T)) \tag{1.0.3}$$

means that there exists some constant A , called the *implied constant*, such that

$$|f(T) - g(T)| \leq Ah(T) \tag{1.0.4}$$

for all values of the variable T in a certain range. This range will be clear from the context, and usually contains all $T \geq 0$ or all large enough T . The expression

$$f(T) = g(T) + o(h(T))$$

means that

$$\lim_{T \rightarrow \infty} \frac{f(T) - g(T)}{h(T)} = 0.$$

We write

$$f(T) \sim g(T)$$

to mean

$$f(T) = (1 + o(1))g(T),$$

which is the same as $\lim_{T \rightarrow \infty} f(T)/g(T) = 1$ when $g(T)$ is nonzero. As an alternative to (1.0.3), we may write

$$f(T) - g(T) \ll h(T).$$

We also write $a \gg b$ to mean $b \ll a$. The expression

$$f(T) \asymp g(T)$$

means that both $f(T) \ll g(T)$ and $f(T) \gg g(T)$ hold. We sometimes use subscripts to indicate the dependence of the estimates on certain parameters. For example,

$$f(T) - g(T) \ll_b h(T)$$

means that the implied constant A in (1.0.4) is a function of a parameter b .

1.1 Zeros on the critical line

Recall that $N(T)$ is the number of zeros of $\zeta(s)$ in the critical strip that have imaginary parts between 0 and T . Let $N_0(T)$ denote the number of these zeros that are on the critical line $\operatorname{Re}(s) = 1/2$. In Chapter 2 we give a new proof of the following.

Theorem 1.1.1. *The quantity*

$$\kappa = \liminf_{T \rightarrow \infty} \frac{N_0(T)}{N(T)}$$

is positive.

We may interpret this theorem by saying that a positive proportion of the nontrivial zeros of $\zeta(s)$ are on the critical line. This theorem was first proved

by Selberg [58] in 1942. The main new idea in Selberg's work is to replace $\zeta(s)$ by $\zeta(s)M(s)$ in a method of Hardy and Littlewood [34] for detecting zeros of $\zeta(s)$. Here, $M(s)$ is a certain function called a *mollifier*, and its purpose is to diminish losses incurred in estimations. The idea is that inequalities such as the Cauchy-Schwarz inequality become sharper when used with $\zeta(s)M(s)$ instead of the function $\zeta(s)$ without a mollifier. Selberg observed that introducing a mollifier in Hardy and Littlewood's method improves their lower bound $N_0(T) \gg T$ to $N_0(T) \gg T \log T$. This implies that $\kappa > 0$ because of the asymptotic formula (1.0.1). The lower bound for κ that is implicit in Selberg's proof is very small [64, §10.9].

Since Selberg's introduction of the mollifier method, there has been a lot of progress towards improving the lower bound for κ . All recent advances in the subject are based on a second approach to the problem, developed by Levinson [50]. The key component in Levinson's technique is a method devised by Siegel [60] to detect zeros of $\zeta(s)$. Siegel's method alone leads only to $N_0(T) \gg T$, which is the same thing that Hardy and Littlewood proved. Levinson combined this method with the mollifier method, and the result is the second proof of Selberg's theorem with the stronger bound $\kappa \geq .3474$. This approach has been improved further by Conrey [15], who arrived at $\kappa \geq .4088$ by using deep techniques involving Kloosterman sums. Conrey's innovation is now the principal tool for pushing the lower bound of κ higher, and more recent developments involve using stronger mollifiers with Conrey's results. One improvement is due to Bui, Conrey, and Young [12], who showed that $\kappa \geq .4105$. Another is due to Feng [27], who has claimed to have proved $\kappa \geq .4128$. However, Bui, Conrey, and Young [12] have observed that there might be an error in his work. More recently, Robles, Roy, and Zaharescu [56] have proved that $\kappa \geq .410725$. A further step forward is due to Bui [11], who used Conrey's ideas together with a mollified third moment and a mollified fourth moment of $\zeta(s)$ to prove $\kappa \geq .410918$.

In Chapter 2, we develop a third approach to the problem due to Atkinson [2]. The basic idea in Atkinson's technique is that one can estimate the number of zeros of a real-valued function $f(t)$ by counting the number of times it changes sign. This makes it similar to the method of Hardy and Littlewood, but Atkinson counts the sign changes in a different way. The approach needs two main ingredients. The first is an asymptotic formula for the mean-value

$$\int_T^{2T} e^{ia} \zeta\left(\frac{1}{2} + it'\right) \zeta\left(\frac{1}{2} - it\right) dt, \quad (1.1.1)$$

where a is a fixed real number and t' is defined in a way that makes the integrand real-valued for all t . The second ingredient is an estimate for the fourth moment

$$\int_T^{2T} |\zeta\left(\frac{1}{2} + it\right)|^4 dt. \quad (1.1.2)$$

Atkinson proves an asymptotic formula for (1.1.1) himself in [2], while he applies the theorem of Ingham [42] to estimate the fourth moment. He uses these two components together with the Cauchy-Schwarz inequality to show that the integrand in (1.1.1) is often negative. This allowed him to estimate the number of sign changes of the Hardy Z -function and deduce the lower bound $N_0(T) \gg T(\log T)^{-1}$. We give a more detailed description of Atkinson's method in Section 2.1.

The main idea in our proof of Theorem 1.1.1 is to combine the mollifier method with Atkinson's zero detection technique. We do this by inserting a mollifier into each of the mean-values (1.1.1) and (1.1.2). Using the resulting expressions in Atkinson's argument makes the Cauchy-Schwarz inequality sharper and thus reduces losses that occur from its use. The end result is an improvement of Atkinson's lower bound for $N_0(T)$ by two factors of $\log T$. To reach this conclusion, we need to evaluate both of the mollified mean-values. The first main result of Chapter 2 is an asymptotic formula for a smoothed version of (1.1.1) with a mollifier. We prove this using a method developed by Young [66] for estimating

a smoothed second moment of $\zeta(s)$ times a mollifier. The result is Theorem 1.1.2 below.

The second main result of Chapter 2 is an asymptotic formula for a mollified fourth moment of $\zeta(s)$. This is considerably more difficult to evaluate than the second moment. The mollified fourth moment, or more generally the “twisted” fourth moment, has received considerable attention in the past few decades. One thing that makes the subject difficult is that any progress in it seems to depend on the theory of Kloosterman sums. Iwaniec [46] was the first to give a nontrivial upper bound for the twisted fourth moment

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 \left| \sum_{n \leq N} a_n n^{it} \right|^2 dt$$

with an arbitrary Dirichlet polynomial $\sum_{n \leq N} a_n n^{it}$. His bound was improved by Deshouillers and Iwaniec [22; 23], and later by Watt [65]. Their bounds hold when N is small compared to T , and are of size $T^{1+\varepsilon}$. The first asymptotic formula for a twisted fourth moment was proved by Jose Gaggero Jara [28] in a Ph.D. thesis. The formula turns out to be quite complicated, and Gaggero did not use it to study the case when the Dirichlet polynomial is a mollifier. More recently, Hughes and Young [40] have found a different way of proving an asymptotic formula. Their method is based on the delta method of Duke, Friedlander, and Iwaniec [24], which uses Weil’s bound for Kloosterman sums. Still another way of evaluating the twisted fourth moment has been developed by Motohashi [53]. His method uses the spectral theory of Kloosterman sums, and allows for a closer study of the error terms in the asymptotic formula.

Among the results we have described, the one we use to prove our formula for the mollified fourth moment is the theorem of Hughes and Young [40]. Their result reveals a permutational structure that is present in the main terms of the asymptotic formula for the twisted fourth moment. This allows us to use a lemma developed by Conrey, Farmer, Keating, Rubinstein, and Snaith [17] precisely for

evaluating functions with this structure. This lemma is Lemma 2.5.1 in [17], and the special case we use is Lemma 2.5.1 below. To the author's knowledge, the mollified fourth moment that we consider in this thesis has not been evaluated before. However, Conrey and Snaith [19] have formulated specific conjectures for it by using the Ratios Conjectures. Their hypotheses also apply to the fourth moment with a mollifier that is different from the one in this thesis. Moreover, Bui [11] has proved a formula for this other fourth moment. To do this, he also used a form of the theorem of Hughes and Young. Our result confirms a special case of one of the two conjectures of Conrey and Snaith, while Bui's verifies a special case of the other conjecture. Our asymptotic formula for the mollified fourth moment is Theorem 1.1.3 below.

To state our results, we need to make a few definitions. Let

$$Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$$

be the Hardy Z -function, where $\vartheta(t)$ is the Riemann-Siegel theta function defined by $e^{-2i\vartheta(t)} = \chi\left(\frac{1}{2} + it\right)$, where

$$\chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi}{2}s\right) \Gamma(1-s).$$

Stirling's formula implies that $\vartheta(t)$ is strictly increasing for large t (see (2.2.8) below). Thus, if a is a fixed real number, then we can define a unique t' for each large t by

$$\vartheta(t') - \vartheta(t) = a. \tag{1.1.3}$$

Let $M(t)$ be the mollifier defined by

$$M(t) = \sum_{n \leq y} \frac{\mu(n)}{n^{1/2+it}} \left(\frac{\log(y/n)}{\log y} \right)^2, \tag{1.1.4}$$

where $y = T^\theta$ and $\mu(n)$ is the Möbius function. For a function $w(t)$, let

$$W = \int_{-\infty}^{\infty} w(t) dt. \tag{1.1.5}$$

All instances of $w(t)$ in this section and Chapter 2 depend on T ; thus W and other integrals involving $w(t)$ also depend on T . We use ε to denote an arbitrarily small positive constant that is not necessarily the same for each instance.

Our first main result is the evaluation of a mollified version of Atkinson's mean-value (1.1.1). We prove it by using the method devised by Young [66].

Theorem 1.1.2. *Let $a \neq 0$ and $\theta > 0$ be fixed real numbers. Suppose $w(t)$ is a smooth, nonnegative function that is supported on $[T/2, 4T]$ and satisfies $w^{(j)}(t) \ll_j T_0^{-j}$ for all $j = 0, 1, 2, \dots$, where $T^{1/2+\varepsilon} \ll T_0 \ll T$. If $\ell \geq 0$ is a fixed integer, then*

$$\begin{aligned} \int_{-\infty}^{\infty} Z(t)Z(t')|M(t)|^2 w(t) dt &= W \cos a + \frac{4W}{3\theta} \left(\frac{\sin a}{a} \right) \\ &+ O\left(\frac{W}{\log T}\right) + O\left(y^{\ell+1} \frac{T^{\frac{3}{2}+\frac{\ell}{2}+\varepsilon}}{T_0^\ell}\right) + O(y^{1/2} T^{3/4} \log^3 T). \end{aligned} \quad (1.1.6)$$

Corollary 1.1.1. *Let $a \neq 0$ and $0 < \theta < \frac{1}{2}$ be real numbers, and let $b \leq 1$ satisfy $\theta < 2b - \frac{3}{2}$. Then there exists a smooth function $w(t)$ that equals 1 on $[T, T+T^b]$ such that*

$$\int_{-\infty}^{\infty} Z(t)Z(t')|M(t)|^2 w(t) dt = \left(\cos a + \frac{4 \sin a}{3a\theta} + o(1) \right) T^b \quad (1.1.7)$$

as $T \rightarrow \infty$.

We will not need the variants of the above results with $a = 0$. Those versions can be proved by taking the limiting case $\alpha = \beta = 0$ of a result of Young [66, Lemma 3].

Our second main result is the following asymptotic formula for a mollified fourth moment. We prove this by applying the recent theorem of Hughes and Young [40]. In fact, we generalize slightly and consider the fourth moment with a small “shift” ρ . We need to introduce this shift to be able to use our result to prove Theorem 1.1.1.

Theorem 1.1.3. *Let $0 < \theta < \frac{1}{22}$ and $a \in \mathbb{R}$ be fixed. Suppose $w(t)$ is a smooth, nonnegative function that is supported on $[T/2, 4T]$ and satisfies $w^{(j)}(t) \ll_j T_0^{-j}$ for all $j = 0, 1, 2, \dots$, where $T^{1/2+\varepsilon} \ll T_0 \ll T$. If $\rho = 2a/\log(T/2\pi)$, then*

$$\begin{aligned} \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i\rho + it)|^4 |M(t)|^4 w(t) dt &= C(a, \theta) W \\ &+ O\left(\frac{W}{\log T}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned} \quad (1.1.8)$$

where $C(a, \theta)$ is a computable constant that depends only on a and θ .

Corollary 1.1.2. *Let $a \in \mathbb{R}$, $0 < \theta < \frac{1}{22}$, and $\rho = 2a/\log(T/2\pi)$, and let $b \leq 1$ satisfy $\theta < \frac{13}{22}b - \frac{6}{11}$. Then there exists a smooth function $w(t)$ that equals 1 on $[T, T + T^b]$ such that*

$$\int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i\rho + it)|^4 |M(t)|^4 w(t) dt = C(a, \theta) T^b + o(T^b)$$

as $T \rightarrow \infty$, where $C(a, \theta)$ is a computable constant that depends only on a and θ .

A few remarks about Theorem 1.1.3 are in order. First, note that we have left $C(a, \theta)$ unspecified. In principle, we can compute the exact value of $C(a, \theta)$. However, doing so by hand turns out to be a formidable task, and so we resort to numerical calculations via computer software. We calculate $C(a, \theta)$ by writing it in terms of the polynomial $\mathcal{R}(t; y, \rho)$ in (2.15.2) below. Hence we only need to find the coefficients of $\mathcal{R}(t; y, \rho)$, and we can use any computer software that can compute coefficients of polynomials. The resulting value we obtain for $C(a, \theta)$ is in Section 2.19. The equation (2.15.2) results from simplifying certain path integrals by writing their integrands as power series. Alternatively, we can evaluate these path integrals by using the method of Conrey and Snaith [19, §6] of using matrix determinants. This will lessen the amount of work needed to compute $C(a, \theta)$. However, it does not seem to do so to the extent that we are able to calculate by hand. Thus we adhere to our approach of using power series because it seems to be the most straightforward.

Aside from the main term $C(a, \theta)$, one may also take notice of the form (1.1.4) of the mollifier $M(t)$. In their proofs of Theorem 1.1.1, both Selberg [58] and Levinson [50] use the “weight” factor

$$\varpi = \frac{\log(y/n)}{\log y}$$

in their mollifiers. On the other hand, we use the weight ϖ^2 , as one can see in the definition (1.1.4) of $M(t)$. Conrey [15] uses more general weights of the form $P(\varpi)$, where $P(x)$ is a polynomial with $P(0) = 0$. This weight is also used in the above mentioned works of Bui [11], Bui, Conrey, and Young [12], and Feng [27]. We use the weight ϖ^2 so that we can deduce a useful formula, namely (1.1.8), for the mollified fourth moment in a way that is not too cumbersome. At one extreme, using the simpler weight ϖ leads to a different asymptotic formula that has a main term of size $W \log T$. Such a result cannot be applied to prove Theorem 1.1.1, because our proof needs a main term of size W . At the opposite extreme, working with a general weight $P(\varpi)$ makes the computations more complicated. Since our goal is merely to show that $\kappa > 0$, we do not need the most general version of Theorem 1.1.3, and we can focus on the simplest case with weight ϖ^2 . Our method also works for the general case with weight $P(\varpi)$, where $P(x)$ is a polynomial satisfying $P(0) = P'(0) = 0$. It is left to the reader to make the slight changes necessary to prove the general case.

Theorem 1.1.3 does not overlap with the result of Bui [11]. Indeed, Bui uses a different form of a mollifier, namely

$$M_2(t) = \sum_{n \leq y} \frac{\mu_2(n)}{n^{1/2+it}} P\left(\frac{\log(y/n)}{\log y}\right),$$

where $\mu_2(n)$ is the n th coefficient of the Dirichlet series for $\zeta^{-2}(s)$. To mollify the fourth moment of $\zeta(s)$ with this mollifier is to multiply $|\zeta(s)|^4$ by $|M_2(t)|^2$ in the integral. In our case, we multiply $|\zeta(s)|^4$ by $|M(t)|^4$.

Bettin, Bui, Li, and Radziwiłł [6] have recently improved the result of Hughes and Young by increasing the range of h and k on which an asymptotic formula

for the twisted fourth moment holds. The theorem of Hughes and Young is valid for $hk \leq T^{2/11-\varepsilon}$. Bettin et al. have proved an asymptotic formula that is true for $h \leq T^{1/4-\varepsilon}$ and $k \leq T^{1/4-\varepsilon}$. Moreover, they show that the formula still holds even when one averages over such h and k . Using their result in our proof of Theorem 1.1.3, we would be able to deduce a version of Theorem 1.1.3 that is true for θ in the extended range $0 < \theta < 1/8$, at least for certain forms of $w(t)$.

Having made our remarks about Theorem 1.1.3, we now return to the main problem of this section. The lower bound for κ that we arrive at is very small; we only reach

$$\kappa > 0.0001049.$$

Moreover, we are only able to increase this by a small amount when we assume the (still unproven) assertion that it is possible to take any positive θ in the hypotheses of Theorems 1.1.2 and 1.1.3. Indeed, the largest lower bound for κ that we obtain through numerical computations is ≈ 0.0086729 when $\theta \approx 0.9967280$. This seems to indicate that Atkinson's method of detecting zeros has an inherent weakness that makes it difficult for us to apply it effectively. We deduced these estimates by using the weight ϖ^2 . It is possible that these bounds can be improved by employing a more general weight $P(\varpi)$ and choosing an optimal polynomial $P(x)$. However, we do not expect any improvements to be significant.

One possible way of improving this approach is to modify it so that the lower bound for κ tends to 1 as $\theta \rightarrow \infty$. This phenomenon was observed to hold for Levinson's method by Farmer [25]. As we have found in the above paragraph, this does not hold for the current version of our approach. Sandro Bettin has suggested to the author that perhaps this might not be the case if one instead uses the mollifier

$$M_{1/2}(t) = \sum_{n \leq y} \frac{\lambda(n)}{n^{1/2+it}} \left(\frac{\log(y/n)}{\log y} \right)^2, \quad (1.1.9)$$

where $\lambda(n)$ is the n th coefficient of the Dirichlet series for $\zeta^{-1/2}(s)$. His suggestion is to use the factor $|M_{1/2}(t)M_{1/2}(t')|^2$ instead of $|M(t)|^2$ on the left-hand side of (1.1.6). Applying Cauchy-Schwarz as in (2.1.1) below would then result to inequalities involving $|Z(t)|^4|M_{1/2}(t)|^8$ and $|Z(t')|^4|M_{1/2}(t')|^8$ instead of $|Z(t)M(t)|^4$ and $|Z(t')M(t')|^4$. The idea is that $M(t)$ is an effective mollifier for $Z(t)$, but a poor one for $Z(t')$. Hence, using the mollifier (1.1.9) instead of (1.1.4) would eliminate losses caused by the incompatibility of $M(t)$ and $Z(t')$. However, evaluating the mollified mean-values of $\zeta(s)$ in this case is much more complicated. Thus this is left for future work. We do not make any claims that this suggestion will lead to any desirable results. In any case, it would be interesting to see how Atkinson's method can be further improved in this or in other ways.

This section and Chapter 2, with slight changes, have been published in [5] (<http://dx.doi.org/10.1016/j.jnt.2016.01.026>).

1.2 Zero-density estimates

Let $N(\sigma, T)$ denote the number of zeros $\beta + i\gamma$ of $\zeta(s)$ that satisfy $\beta > \sigma$ and $0 < \gamma \leq T$. Upper bounds for $N(\sigma, T)$ have a large number of applications to problems in number theory. Because of this, a great deal of effort has been put toward the estimation of $N(\sigma, T)$. Bounds for $N(\sigma, T)$ are usually called *zero-density estimates*. The earliest zero-density estimate is due to Bohr and Landau, who showed in 1914 that [64, §9.15]

$$N(\sigma, T) \ll T$$

as $T \rightarrow \infty$ for fixed $\sigma > \frac{1}{2}$. This was later improved by Carlson, who proved [64, §9.17]

$$N(\sigma, T) \ll T^{4\sigma(1-\sigma)+\varepsilon}$$

as $T \rightarrow \infty$ for fixed $\sigma > \frac{1}{2}$. Hoheisel [38] used a refinement of Carlson's theorem together with a zero-free region due to Littlewood to show that there exists an absolute constant $\vartheta < 1$ such that

$$p_{n+1} - p_n \ll p_n^\vartheta$$

for all n , where $p_1 < p_2 < \dots$ are the prime numbers. The value of ϑ that Hoheisel arrived at was $\vartheta = \frac{32999}{33000}$, which is very close to 1. Later, Ingham [43] proved that if $\zeta(\frac{1}{2} + it) \ll t^b$ then

$$N(\sigma, T) \ll T^{2(1+2b)(1-\sigma)} (\log T)^5$$

uniformly for $\frac{1}{2} \leq \sigma \leq 1$. Ingham then used this to improve Hoheisel's theorem by showing that it holds true for

$$\vartheta = \frac{1 + 4b}{2 + 4b} + \varepsilon.$$

This increased the range of admissible values of ϑ to $\frac{577}{925} < \vartheta < 1$, by the known values for b then. Note that if the Lindelöf Hypothesis $\zeta(\frac{1}{2} + it) \ll t^\varepsilon$ is true, then Ingham's bound for $N(\sigma, T)$ implies

$$N(\sigma, T) \ll T^{2(1-\sigma)+\varepsilon} \tag{1.2.1}$$

uniformly for $\frac{1}{2} \leq \sigma \leq 1$. In this case, Hoheisel's theorem holds with $\vartheta = \frac{1}{2} + \varepsilon$, which is almost as strong as the bound

$$p_{n+1} - p_n \ll p_n^{1/2} \log p_n$$

that would hold if the Riemann Hypothesis is true [20]. This is only one of many examples for which a strong upper bound for $N(\sigma, T)$ can give results of nearly the same quality as those implied by the Riemann Hypothesis.

The bound (1.2.1) is called the *density hypothesis*, and proving it is currently an open problem. However, much progress has been made toward it. Ingham [44] proved that

$$N(\sigma, T) \ll T^{\frac{3(1-\sigma)}{2-\sigma}} (\log T)^5 \tag{1.2.2}$$

uniformly for $\frac{1}{2} \leq \sigma \leq 1$. Using a method due to Halász, Montgomery [51] showed that

$$N(\sigma, T) \ll T^{\frac{2(1-\sigma)}{\sigma}} (\log T)^{14}$$

uniformly for $\frac{4}{5} \leq \sigma \leq 1$. The Halász-Montgomery method has been further developed by Huxley [41], who proved

$$N(\sigma, T) \ll T^{\frac{3(1-\sigma)}{3\sigma-1}} (\log T)^{44}$$

uniformly for $\frac{3}{4} \leq \sigma \leq 1$. Note that the density hypothesis (1.2.1) for the range $\frac{5}{6} \leq \sigma \leq 1$ follows from Huxley's theorem. Building on these developments, Jutila [48] has extended the range on which (1.2.1) is known to hold to $\frac{11}{14} \leq \sigma \leq 1$. A considerable number of other bounds for various ranges have also been proved by Heath-Brown [36] and Ivić [45, Chapter 11]. More recently, Bourgain [10] has proved that (1.2.1) holds for $\sigma > \frac{25}{32}$, thus improving Jutila's result.

A large number of advances have been made toward proving the density hypothesis (1.2.1) for $\sigma > \frac{1}{2}$ in ranges that are far from $\frac{1}{2}$. However, progress for σ close to $\frac{1}{2}$ has been slower. One reason for this is that the Halász-Montgomery method only seems to work well when σ is far from $\frac{1}{2}$. The standard approach to bounding $N(\sigma, T)$ when σ is close to $\frac{1}{2}$ is to apply Littlewood's lemma (see, for example, (9.9.1) of Titchmarsh [64]). Using this approach, Selberg [59] proved that

$$N(\sigma, T) \ll T^{1+\theta(1-2\sigma)} \log T \tag{1.2.3}$$

uniformly for $\frac{1}{2} \leq \sigma \leq 1$, with $\theta = \frac{1}{8}$. This supersedes (1.2.2) when

$$\frac{1}{2} \leq \sigma \leq \frac{1}{2} + \frac{48 \log \log T}{13 \log T}.$$

Bounds of the form (1.2.3) are useful for studying $\arg \zeta(\frac{1}{2} + it)$. In fact, this is what Selberg does in [59]. Later, Jutila [49] improved Selberg's theorem by showing that (1.2.3) holds for $0 < \theta < \frac{1}{2}$. Note that the density hypothesis (1.2.1) is equivalent to the statement that (1.2.3) holds for all $0 < \theta < 1$.

Selberg proves (1.2.3) by bounding a mollified second moment

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \quad (1.2.4)$$

for $\frac{1}{2} < \sigma \leq 1$, where $M(s)$ is a certain function called a mollifier, as introduced in Section 1.1 above. Selberg used a mollifier of the form

$$M(s) = \sum_{n \leq T^\theta} \frac{\lambda_n}{n^s}, \quad (1.2.5)$$

for some specific coefficients λ_n . Mollified moments such as (1.2.4) have a wide number of applications to the study of $\zeta(s)$ and other L -functions. In using a mollifier such as (1.2.5), it is most often the case that one can deduce better estimates by increasing the value of θ . For instance, the upper bound (1.2.3) decreases as θ increases. However, the evaluation of the mollified moment becomes more delicate as the value of θ increases. To address this issue, Conrey [15] has devised a technique that gives nontrivial estimates for values of θ larger than what previous methods can handle. The key point of his technique is to use Kloosterman sum estimates to approximate the sums arising from the analysis of the mollified moment. As we have seen in Section 1.1, his technique allowed him to prove that more than 40% of the zeros of $\zeta(s)$ lie on the critical line [15].

In Chapter 3, we apply Conrey's technique to the estimation of $N(\sigma, T)$ and improve the bound (1.2.3). Our main result is the following.

Theorem 1.2.1. *If $0 < \theta < \frac{4}{7}$, then*

$$N(\sigma, T) \ll_{\theta} T^{1+\theta(1-2\sigma)} \log T$$

uniformly for $\sigma \geq \frac{1}{2}$ and large T .

As mentioned above, this bound has been proved by Selberg for $\theta = \frac{1}{8}$ and by Jutila for $0 < \theta < \frac{1}{2}$. Theorem 1.2.1 is not entirely new; in fact, it has been announced by Conrey [14] in 1989. However, as far as the author knows, a proof has never been published.

The main new contribution in our proof of Theorem 1.2.1 is the use of a new mollifier $M(s)$ in (1.2.4) that is different from Selberg's, Jutila's, or Conrey's. This mollifier is

$$M(s) = \sum_{n \leq y^{1+\varepsilon}} \frac{\mu(n)}{n^s} \left(\frac{W(y/n)}{W(y)} \right), \quad (1.2.6)$$

where $y = T^\theta$ and the function $W(\xi)$ is defined by

$$W(\xi) = W(\xi; \sigma) = \frac{1}{2\pi i} \int_{(r)} \Gamma(s+1) \frac{\zeta(2\sigma+s)}{\zeta(2\sigma)} \zeta(1+s) \xi^s ds \quad (1.2.7)$$

for $\sigma > \frac{1}{2}$, $\xi > 0$, and $r > 0$. The use of (1.2.6) gives us relative ease in evaluating the main terms in our asymptotic formula for (1.2.4), and at the same time allows us to use the full power of Conrey's technique in estimating the error terms. The function $W(\xi)$ is reminiscent of the functions $\omega_j(\xi)$ of Soundararajan [61]. By choosing the mollifier $M(s)$ in (1.2.4) to be (1.2.6), we prove the following asymptotic formula, from which Theorem 1.2.1 will follow by standard arguments.

Theorem 1.2.2. *Let $M(s)$ be defined by (1.2.6) with $y = T^\theta$. If $0 < \theta < \frac{4}{7}$, then there exists an $\epsilon_0 > 0$ such that*

$$\frac{1}{T} \int_T^{2T} |\zeta(\sigma + it) M(\sigma + it)|^2 dt = 1 + O(y^{1-2\sigma})$$

uniformly for $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$.

In fact, we only need the upper bound for (1.2.4) implied by Theorem 1.2.2 in order to prove Theorem 1.2.1.

1.3 The Alternative Hypothesis

The results described in Sections 1.1 and 1.2 are unconditional. That is, their truth does not rely on any unproved hypothesis. On the other hand, in this section and Chapter 4, we assume that the Riemann Hypothesis (RH) is true.

Let $\frac{1}{2} + i\gamma$ denote a nontrivial zero of $\zeta(s)$. In the early 1970's Montgomery [52] introduced a new method of studying the distribution of zeros of $\zeta(s)$. Assuming RH, he defined the function

$$F(\alpha) = F(\alpha, T) = \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{0 < \gamma, \gamma' \leq T} T^{i\alpha(\gamma - \gamma')} w(\gamma - \gamma'),$$

where the sum is over all pairs γ, γ' of ordinates of zeros counted according to multiplicity. Here α is real, $T \geq 2$, and $w(u) = 4/(4 + u^2)$. He then observed that F is a real valued even function of α , and proved that

$$F(\alpha, T) = (1 + o(1))T^{-2\alpha} \log T + \alpha + o(1) \quad (1.3.1)$$

uniformly for $0 \leq \alpha \leq 1 - \varepsilon$ for any fixed $\varepsilon > 0$. From this he deduced that if RH is true, then at least two-thirds of the zeros of $\zeta(s)$ are simple. He also showed that if $0 < \gamma_1 \leq \gamma_2 \leq \gamma_3 \cdots$ is a list of the ordinates of all the zeros above the real line, counted according to their multiplicities, then the differences

$$\frac{\gamma_{n+1} \log \gamma_{n+1}}{2\pi} - \frac{\gamma_n \log \gamma_n}{2\pi} \quad (1.3.2)$$

are less than 0.68 infinitely often. Later, Mueller and Heath-Brown noted that F is nonnegative (see [13]), and Goldston and Montgomery [33, Lemma 8] showed that (1.3.1) holds uniformly for $0 \leq \alpha \leq 1$.

The usefulness of $F(\alpha)$ for deducing information on the distribution of the zeros is limited by the fact that the asymptotic behavior of $F(\alpha)$ is known only for $|\alpha| \leq 1$. To get around this difficulty, Montgomery [52] used a quantitative form of the Hardy-Littlewood twin prime hypothesis to conjecture that for any fixed $M \geq 1$,

$$F(\alpha, T) = 1 + o(1) \quad (1.3.3)$$

as $T \rightarrow \infty$, uniformly for $1 \leq \alpha \leq M$. This led him to make the following conjecture.

Conjecture (The Pair Correlation Conjecture). *For any fixed $\beta > 0$, we have*

$$N(T, \beta) \stackrel{\text{def}}{=} \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\substack{0 < \gamma, \gamma' \leq T \\ 0 < \gamma - \gamma' \leq \frac{2\pi\beta}{\log T}}} 1 \sim \int_0^\beta 1 - \left(\frac{\sin \pi u}{\pi u} \right)^2 du$$

as $T \rightarrow \infty$.

The integrand above is called the *pair correlation function* of the zeros. It happens that the eigenvalues of large random Hermitian matrices (the Gaussian Unitary Ensemble or GUE), which are used in the study of particle physics, have exactly the same pair correlation function. Thus the pair correlation conjecture provided a surprising connection between number theory and random matrix theory, two fields that seemed unrelated at the time. Later, Odlyzko [54] found strong numerical evidence supporting the pair correlation conjecture. Bolanz [9], building on Montgomery's ideas, proved that (1.3.3) holds for $1 \leq M < \frac{3}{2}$ provided that a strong form of the Hardy-Littlewood twin prime conjecture is true. He later (unpublished) extended the range to $1 \leq M < 2$ under an additional assumption. Hejhal [37] proved results similar to Montgomery's for the triple correlation function of the zeros of $\zeta(s)$, and Rudnick and Sarnak [57] proved results for the n -correlation function, providing even more theoretical evidence that the zeros are distributed like the eigenvalues of matrices from the Gaussian Unitary Ensemble. Bogomolny and Keating [7; 8] used a prime-twin type conjecture to heuristically extend the range of the n -correlation result of Rudnick and Sarnak.

In spite of the overwhelming numerical and theoretical evidence, the pair correlation conjecture has yet to be proved. Thus, it is of interest to determine consequences of other conjectures about the spacings of the zeros. One well known alternative to the pair correlation conjecture is the *Alternative Hypothesis*. There are various formulations of it, but they all essentially say that almost all the differences (1.3.2) are close to half-integers. In a sense, the Alternative Hypothesis is antithetical to the pair correlation conjecture in that the latter says the zeros

are randomly distributed, whereas the former says they are quite regular. Besides supporting the pair correlation conjecture, a disproof of the Alternative Hypothesis would be useful in showing that Landau-Siegel zeros do not exist. Indeed, Conrey and Iwaniec [18] have obtained a relation between the spacings (1.3.2) and the size of $L(1, \chi)$ for real primitive Dirichlet characters χ . A corollary of their Theorem 1.2 is that if the number of ordinates $\gamma_n \leq T$ for which the spacings (1.3.2) are less than 0.49, say, is $\gg T(\log T)^{4/5}$ (as $T \rightarrow \infty$), then

$$L(1, \chi) \gg (\log q)^{-90}.$$

The implied constant in their result is effectively computable.

Conrey [16] formulates the Alternative Hypothesis as the existence of a function $h(T)$ that goes to 0 as $T \rightarrow \infty$ such that if $\gamma_n \geq T_0$, then the difference (1.3.2) is within $h(T_0)$ of a half-integer. A more precise formulation is implicit in a lecture of Heath-Brown [35]. He observed that if $L(1, \chi) \ll q^{-1/4-\varepsilon}$, then there is a sequence of points t_n that are close to the zeros of $\zeta(s)L(s, \chi)$ such that if t_m and t_n are about the size of T , with T much larger than q , then there is an integer k with

$$(t_m - t_n) \log T = \pi k + O(|t_m - t_n| \{\log q + \log^{2/3} T\}).$$

We will base our formulation of the Alternative Hypothesis on this. We let

$$\tilde{\gamma} = \frac{\gamma}{2\pi} \log \gamma$$

denote the normalized ordinate of a zero, so that the average spacing between consecutive $\tilde{\gamma}_n$'s equals 1. Our version of the hypothesis is the following.

Hypothesis (The Alternative Hypothesis). *For each n there is an integer k_n with*

$$\tilde{\gamma}_{n+1} - \tilde{\gamma}_n = \frac{1}{2}k_n + O(|\gamma_{n+1} - \gamma_n| \psi(\gamma_n)), \quad (1.3.4)$$

where $\psi(\gamma)$ is a function such that $\psi(\gamma) \rightarrow \infty$ and $\psi(\gamma) = o(\log \gamma)$ as $\gamma \rightarrow \infty$.

Our aim in Chapter 4 is to deduce consequences of the Alternative Hypothesis. To do this, we look at known consequences of the pair correlation conjecture and determine their corresponding forms under the Alternative Hypothesis.

Goldston [30] showed that the pair correlation conjecture is actually equivalent to a weaker variant of (1.3.3), namely

$$\int_b^{b+\delta} F(\alpha, T) d\alpha \sim \delta \quad (1.3.5)$$

as $T \rightarrow \infty$ for all fixed $b \geq 1$ and $\delta > 0$. He also showed that the pair correlation conjecture is equivalent to

$$J(\beta, T) \stackrel{\text{def}}{=} \int_1^{T^\beta} \left(\psi\left(x + \frac{x}{T}\right) - \psi(x) - \frac{x}{T} \right)^2 \frac{dx}{x^2} \sim \left(\beta - \frac{1}{2} \right) \frac{\log^2 T}{T}$$

as $T \rightarrow \infty$, for all fixed $\beta \geq 1$. Here, $\psi(x) = \sum_{n \leq x} \Lambda(n)$, where Λ is the von Mangoldt function defined by $\Lambda(p^m) = \log p$ for prime powers $p^m > 1$ and $\Lambda(n) = 0$ for all other n . Later, Goldston, Gonek, and Montgomery [32] showed that (1.3.5) is equivalent to

$$I(b, T) \stackrel{\text{def}}{=} \int_1^T \left| \frac{\zeta'}{\zeta} \left(\frac{1}{2} + \frac{b}{\log T} + it \right) \right|^2 dt \sim \left(\frac{1 - e^{-2b}}{4b^2} \right) T \log^2 T$$

as $T \rightarrow \infty$, for all fixed $b > 0$. Thus, collecting these results together, we have a four-way equivalence between the asymptotic formulas for $F(\alpha, T)$, $I(b, T)$, $J(\beta, T)$, and $N(T, \beta)$.

Theorem 1.3.1 (Goldston, Gonek, Montgomery). *Assume RH. The following statements are equivalent.*

$$(A1) \quad \int_1^b F(\alpha, T) d\alpha \sim b - 1 \quad \text{as } T \rightarrow \infty, \text{ for all fixed } b \geq 1.$$

$$(B1) \quad I(b, T) \sim \left(\frac{1 - e^{-2b}}{4b^2} \right) T \log^2 T \quad \text{as } T \rightarrow \infty, \text{ for all fixed } b > 0.$$

$$(C1) \quad J(\beta, T) \sim \left(\beta - \frac{1}{2}\right) \frac{\log^2 T}{T} \quad \text{as } T \rightarrow \infty, \text{ for all fixed } \beta \geq 1.$$

$$(D1) \quad N(T, \beta) \sim \int_0^\beta 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2 du \quad \text{as } T \rightarrow \infty, \text{ for all fixed } \beta > 0.$$

Note that by (1.3.1) we can evaluate the integral of $F(\alpha, T)$ on any subinterval of $[0, 1]$ and write

$$\int_0^b F(\alpha, T) d\alpha \sim \frac{1}{2} + \frac{b^2}{2}$$

as $T \rightarrow \infty$, for all fixed $0 < b \leq 1$. This gives a version of the statement (A1) for $0 < b \leq 1$ that is true in any case (on RH). Also, a result of Gallagher and Mueller [29] states that

$$J(\beta, T) \sim \frac{\beta^2 \log^2 T}{2T} \quad \text{as } T \rightarrow \infty, \text{ for all fixed } 0 < \beta \leq 1.$$

This is a version of (C1) for $0 < \beta \leq 1$ that holds true in any case (unconditionally).

In Chapter 4, we prove a generalization of Theorem 1.3.1 (Theorem 1.3.2). Afterwards, we apply our result to obtain an analogue of Theorem 1.3.1 for the Alternative Hypothesis (Theorem 1.3.3). To state our main result, we set

$$N^*(T) \stackrel{\text{def}}{=} \left(\frac{T}{2\pi} \log T\right)^{-1} \sum_{0 < \gamma \leq T} m_\gamma,$$

where m_γ denotes the multiplicity of the zero $\frac{1}{2} + i\gamma$ and the sum again counts ordinates according to multiplicity. Also, for a general measure μ , we shall mean by $\int_a^b f(\alpha) d\mu(\alpha)$ the integral of f over the set $[a, b)$.

Theorem 1.3.2. *Assume RH. Let μ be a positive Borel measure on $[0, \infty)$ for which the function $\alpha \mapsto \min\{1, \alpha^{-2}\}$ is integrable over $[0, \infty)$. The following statements are equivalent*

$$(A) \quad \mu[0, b) + o(1) \leq \int_0^b F(\alpha, T) d\alpha \leq \mu[0, b] + o(1)$$

as $T \rightarrow \infty$, for all fixed $b > 0$.

$$(B) \quad I(b, T) \sim \left(\int_0^\infty e^{-2b\alpha} d\mu(\alpha) - \frac{1}{2} \right) T \log^2 T$$

as $T \rightarrow \infty$, for all fixed $b > 0$.

$$(C) \quad \mu[0, \beta] - \frac{1}{2} + o(1) \leq J(\beta, T) \frac{T}{\log^2 T} \leq \mu[0, \beta] - \frac{1}{2} + o(1)$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$.

$$(D) \quad \frac{\beta}{2} N^*(T) + \int_0^\beta N(T, u) du \sim \int_0^\infty \left(\frac{\sin \pi \beta \alpha}{\pi \alpha} \right)^2 d\mu(\alpha)$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$.

The measure that makes the statement (A) consistent with (1.3.1) and (1.3.5) is the measure with $\mu(0) = 1/2$, $d\mu(\alpha) = \alpha d\alpha$ for $0 < \alpha \leq 1$, and $d\mu(\alpha) = d\alpha$ for $\alpha > 1$. Thus we see that Theorem 1.3.1 corresponds to taking this measure in Theorem 1.3.2. Keeping in mind the remarks below Theorem 1.3.1, we see from a straightforward calculation that (A), (B), and (C) with this measure are equivalent to (A1), (B1), and (C1), respectively. However, to prove that (D) with this choice of μ is equivalent to (D1) is not as straightforward. We shall not carry this out here because the argument is similar to the one we shall use in Section 4.3 to prove Theorem 1.3.3.

Let $g(\alpha) = |\alpha|$ for $|\alpha| \leq 1$ and be extended to $\mathbb{R}$ by periodicity. In Section 4.4 we shall prove that the Alternative Hypothesis and the assumption that $N^*(T) \sim 1$ imply that

$$\frac{1}{2} + \sum_{n < b/2} 1 + \int_0^b g + o(1) \leq \int_0^b F(\alpha, T) d\alpha \leq \frac{1}{2} + \sum_{n \leq b/2} 1 + \int_0^b g + o(1)$$

for any fixed $b > 0$ as $T \rightarrow \infty$. Thus, the analogue of Theorem 1.3.1 for the Alternative Hypothesis is

Theorem 1.3.3. *Assume RH. The following statements are equivalent.*

$$(A2) \quad \frac{1}{2} + \sum_{n < b/2} 1 + \int_0^b g + o(1) \leq \int_0^b F(\alpha, T) d\alpha \leq \frac{1}{2} + \sum_{n \leq b/2} 1 + \int_0^b g + o(1)$$

as $T \rightarrow \infty$, for all fixed $b > 0$.

$$(B2) \quad I(b, T) \sim \left(\frac{e^{2b} - 1}{4b^2(e^{2b} + 1)} + \frac{1}{e^{4b} - 1} \right) T \log^2 T \quad \text{as } T \rightarrow \infty, \text{ for all fixed } b > 0.$$

$$(C2) \quad \sum_{n < \beta/2} 1 + \int_0^\beta g + o(1) \leq J(\beta, T) \frac{T}{\log^2 T} \leq \sum_{n \leq \beta/2} 1 + \int_0^\beta g + o(1)$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$.

$$(D2) \quad N(T, \beta) \sim \begin{cases} m - \frac{1}{2\pi^2} \sum_{n=0}^{m-1} \frac{4}{(2n+1)^2} & \text{if } m < \beta < m + \frac{1}{2}, \\ m + \frac{1}{2} - \frac{1}{2\pi^2} \sum_{n=0}^m \frac{4}{(2n+1)^2} & \text{if } m + \frac{1}{2} < \beta < m + 1, \end{cases}$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$ not a half-integer. Here $m = [\beta]$, the greatest integer $\leq \beta$.

By Proposition 4.4.1 in Section 4.4, if the Alternative Hypothesis is true and $N^*(T) \sim 1$ holds, then the statement (A2) holds. An immediate consequence of this and Theorem 1.3.3 is

Corollary 1.3.1. *Assume RH, the Alternative Hypothesis, and that $N^*(T) \sim 1$ as $T \rightarrow \infty$. Then each of the statements (A2), (B2), (C2), and (D2) is true.*

If we assume the Alternative Hypothesis and that all zeros are simple, then we can use Corollary 1.3.1 to estimate the proportion of zeros γ_n for which $\tilde{\gamma}_{n+1} - \tilde{\gamma}_n$ is near a fixed half-integer $k/2$. Let $B_{k/2}(T)$ be the set of zeros $\gamma_n \leq T$ such that $k/2$ is closest among all half-integers to $\tilde{\gamma}_{n+1} - \tilde{\gamma}_n$. We also define

$$p_{k/2} = p_{k/2}(T) \stackrel{\text{def}}{=} \left(\frac{T}{2\pi} \log T \right)^{-1} |B_{k/2}(T)|.$$

Theorem 1.3.4. *Assume RH, the Alternative Hypothesis, and that all zeros of $\zeta(s)$ are simple. Then as $T \rightarrow \infty$ we have*

$$p_0 = o(1),$$

$$p_{1/2} = \frac{1}{2} - \frac{2}{\pi^2} + o(1), \tag{1.3.6}$$

$$\frac{4}{\pi^2} + o(1) \leq p_1 \leq \frac{1}{2} + o(1), \tag{1.3.7}$$

$$\frac{6}{\pi^2} - \frac{1}{2} + o(1) \leq p_{3/2} \leq \frac{1}{2} - \frac{2}{\pi^2} + o(1), \tag{1.3.8}$$

and

$$p_{k/2} \leq \frac{1}{k} \left(3 - \frac{24}{\pi^2} \right) + o(1) \quad \text{for } k \geq 4. \tag{1.3.9}$$

Our estimates for p_0 , $p_{1/2}$, and p_1 agree with those in Section 2 of Farmer, Gonek, and Lee [26]. However, one should note that the formulation of the Alternative Hypothesis in [26] is stronger than ours.

This section and Chapter 4, with slight changes, have been published in [4] (<http://dx.doi.org/10.1016/j.jnt.2016.05.007>).

2 Zeros of the Riemann zeta-function on the critical line

In this chapter, we prove the results described in Section 1.1. All instances of the function $w(t)$ in this chapter depend on T . Thus the number W defined by (1.1.5) and other integrals involving $w(t)$, particularly those in Theorems 1.1.2 and 1.1.3, also depend on T . We follow standard practice in analytic number theory and use ε to denote an arbitrarily small positive constant that is not necessarily the same for each instance. We consider the variable $T > 1$ as a large parameter going to infinity.

2.1 Atkinson's method and our starting point

In this section, we describe the key idea in the method of Atkinson [2] for deducing a lower bound for the number of zeros of $\zeta(s)$ on the critical line. We present it in a form that enables us to use Theorems 1.1.2 and 1.1.3 to prove Theorem 1.1.1.

Since $|Z(t)| = |\zeta(\frac{1}{2} + it)|$, each real zero of $Z(t)$ corresponds to a unique zero of $\zeta(s)$ on the critical line. Thus we can prove Theorem 1.1.1 by showing that $Z(t)$ has many zeros as a function of the real variable t . To do this, we show that the

sign of $Z(t)$ changes often. We define $\mathcal{S}$ to be the set of $t \in [T/2, 4T]$ for which $Z(t)Z(t') < 0$. Atkinson's approach is to prove a lower bound for the measure of $\mathcal{S}$. Using such an estimate, we can deduce a lower bound for the number of critical zeros of $\zeta(s)$ by an elementary argument (see Section 2.18).

To find a lower bound for the measure of $\mathcal{S}$, let $w(t)$ be a smooth nonnegative function supported on $[T/2, 4T]$. Apply the Cauchy-Schwarz inequality twice (or Hölder's inequality once) to write

$$\begin{aligned} & \left(\int_{\mathcal{S}} |Z(t)Z(t')| |M(t)|^2 w(t) dt \right)^2 \\ & \leq \mathcal{N} \left(\int_{-\infty}^{\infty} |Z(t)|^4 |M(t)|^4 w(t) dt \right)^{1/2} \left(\int_{-\infty}^{\infty} |Z(t')|^4 |M(t)|^4 w(t) dt \right)^{1/2}, \end{aligned} \quad (2.1.1)$$

where

$$\mathcal{N} = \int_{-\infty}^{\infty} \mathbf{1}_{\mathcal{S}}(t) w(t) dt$$

and $\mathbf{1}_{\mathcal{S}}(t)$ is the characteristic function of $\mathcal{S}$. We can estimate the measure of $\mathcal{S}$ from the size of $\mathcal{N}$ (see Section 2.18), and so it suffices to bound $\mathcal{N}$ from below. We do this by dividing both sides of (2.1.1) by the two factors on its right-hand side that are different from $\mathcal{N}$. Hence, to prove a lower bound for $\mathcal{N}$, we need a lower bound for the mean-value on the left-hand side of (2.1.1) and an upper bound for each of the fourth power moments on the right-hand side.

We estimate the left-hand side of (2.1.1) as follows. Since $Z(t)Z(t') < 0$ for $t \in \mathcal{S}$, it is true that

$$|Z(t)Z(t')| = -Z(t)Z(t')$$

for such t . Moreover, if $t \in [T/2, 4T]$ is not in $\mathcal{S}$, then $Z(t)Z(t') \geq 0$. It follows from these observations that

$$\begin{aligned} \int_{\mathcal{S}} |Z(t)Z(t')| |M(t)|^2 w(t) dt &= - \int_{\mathcal{S}} Z(t)Z(t') |M(t)|^2 w(t) dt \\ &\geq - \int_{-\infty}^{\infty} Z(t)Z(t') |M(t)|^2 w(t) dt. \end{aligned} \quad (2.1.2)$$

Thus we can use Theorem 1.1.2 to bound the left-hand side of (2.1.1) from below. For the lower bound to be nontrivial, we will need to choose a value of a that makes the right-hand side of (1.1.6) negative.

We can use Theorem 1.1.3 to deduce upper bounds for the fourth power moments on the right-hand side of (2.1.1). The special case of Theorem 1.1.3 with $a = 0$ immediately gives a formula for the fourth moment that does not involve t' . For the other one, we approximate it by the left-hand side of (1.1.8). We fill-in the details and finish this argument in Section 2.18.

The structure of the rest of the chapter is as follows. In Section 2.2, we state basic lemmas and develop the machinery we need in our proof of Theorem 1.1.2. In Section 2.3, we prove an asymptotic formula for a smoothed version of the mean-value (1.1.1) with a factor $(h/k)^{-it}$. We then use the formula to prove Theorem 1.1.2 in Section 2.4. The starting point of our proof of Theorem 1.1.3 is in Section 2.5. There, we apply Theorem 1.1 of Hughes and Young [40] and Lemma 2.5.1 of Conrey et al. [17] to write the mollified fourth moment as an iterated integral. Sections 2.6 through 2.15 are devoted to simplifying this iterated integral. We complete the proof of Theorem 1.1.3 in Section 2.15. In Section 2.16, we construct a specific smooth function $w(t)$ and prove Corollaries 1.1.1 and 1.1.2. In Section 2.17, we apply Theorem 1.1.3 to evaluate the integral with $Z(t')$ on the right-hand side of (2.1.1). We use our results from Sections 2.16 and 2.17 to finish our proof of Theorem 1.1.1 in Section 2.18. Finally, we describe the results of our numerical computations in Section 2.19.

2.2 Lemmas

Our first lemma is a generalization of the well-known Euler product formula. Special cases of this have been implicitly used in various places in the literature. We will need only the cases $k = 2$ and $k = 4$ in this thesis.

Lemma 2.2.1. *Let $f(n_1, \dots, n_k)$ be a complex-valued function such that*

$$f(h_1 j_1, \dots, h_k j_k) = f(h_1, \dots, h_k) f(j_1, \dots, j_k)$$

whenever the products $h_1 \cdots h_k$ and $j_1 \cdots j_k$ are relatively prime. If

$$\sum_{1 \leq n_1, \dots, n_k < \infty} |f(n_1, \dots, n_k)| < \infty \quad (2.2.1)$$

or

$$\prod_p \left(\sum_{0 \leq \ell_1, \dots, \ell_k < \infty} |f(p^{\ell_1}, \dots, p^{\ell_k})| \right) < \infty, \quad (2.2.2)$$

then

$$\sum_{1 \leq n_1, \dots, n_k < \infty} f(n_1, \dots, n_k) = \prod_p \left(\sum_{0 \leq \ell_1, \dots, \ell_k < \infty} f(p^{\ell_1}, \dots, p^{\ell_k}) \right). \quad (2.2.3)$$

Proof. Assume first that (2.2.1) holds. Let

$$g(n) = \sum_{n_1 \cdots n_k = n} f(n_1, \dots, n_k),$$

where the sum is over all k -tuples $(n_1, \dots, n_k)$ of positive integers that have product equal to n . It follows from our hypothesis for $f(n_1, \dots, n_k)$ that $g(n)$ is a multiplicative function, i.e. $g(mn) = g(m)g(n)$ for relatively prime m and n . This implies the Euler product formula (see, for example, Theorem 11.6 of [1])

$$\sum_{n=1}^{\infty} g(n) = \prod_p \left(\sum_{\ell=0}^{\infty} g(p^{\ell}) \right), \quad (2.2.4)$$

because the sum on the left-hand side is absolutely convergent by (2.2.1). By inserting the definition of $g(n)$ in (2.2.4) and rearranging the terms, we arrive at (2.2.3).

Now suppose that (2.2.2) holds. Let

$$g_0(n) = \sum_{n_1 \cdots n_k = n} |f(n_1, \dots, n_k)|.$$

Our hypothesis for $f(n_1, \dots, n_k)$ again implies that $g_0(n)$ is a multiplicative function. Therefore, for any integer $N \geq 2$, it is true that

$$\sum_{n=1}^N g_0(n) \leq \prod_{p \leq N} \left(\sum_{\ell=0}^N g_0(p^\ell) \right) \leq \prod_p \left(\sum_{0 \leq \ell_1, \dots, \ell_k < \infty} |f(p^{\ell_1}, \dots, p^{\ell_k})| \right).$$

Thus the sum $\sum_{n=1}^{\infty} g_0(n)$ converges absolutely by (2.2.2). Hence (2.2.1) holds, and so (2.2.3) is true by the above paragraph. $\square$

The following are basic estimates for $\zeta(s)$ from Chapter 3 of Titchmarsh [64].

Lemma 2.2.2. *Let $z = u + iv$. There is an absolute constant $c_0 > 0$ such that if $-\infty < v < \infty$ and*

$$-\frac{c_0}{\log(|v| + 2)} \leq u \leq 1,$$

then $\zeta(1 + z) \neq 0$ and the following bounds hold:

$$\zeta(1 + z) \ll \max \left\{ \frac{1}{|z|}, \log(|v| + 2) \right\},$$

$$\frac{1}{\zeta(1 + z)} \ll \log(|v| + 2),$$

and

$$\frac{\zeta'(1 + z)}{\zeta(1 + z)} \ll \log(|v| + 2).$$

In the rest of this section, we develop the basic machinery used in our proof of Theorem 1.1.2. The implied constants in our estimates never depend on t , but may depend on other parameters, such as a , that we take to be fixed. The next lemma can be easily proved by induction.

Lemma 2.2.3. *If f and g are differentiable functions, then for $n \geq 2$,*

$$\frac{d^n}{dt^n}(f \circ g)(t) = (f' \circ g)(t)g^{(n)}(t) + (f^{(n)} \circ g)(t)(g'(t))^n + \Sigma,$$

where Σ is a sum of terms of the form

$$(f^{(m_1+m_2+\dots+m_{n-1})} \circ g)(t)(g'(t))^{m_1}(g''(t))^{m_2} \dots (g^{(n-1)}(t))^{m_{n-1}}$$

with $\sum_{j=1}^{n-1} jm_j = n$, $m_j \geq 0$ for all j , and $m_j \geq 1$ for some $j \geq 2$. The sum Σ is empty when $n = 2$.

Lemma 2.2.4. *Let a be a fixed real number. If t' is defined by (1.1.3) for large t , then the following formulas hold for large enough t :*

$$t' - t = \frac{2a}{\log \frac{t}{2\pi}} + O\left(\frac{1}{t \log^3 t}\right), \quad (2.2.5)$$

$$\frac{dt'}{dt} = 1 + O\left(\frac{1}{t \log^2 t}\right), \quad (2.2.6)$$

and

$$\frac{d^n}{dt^n} t' \ll_n \frac{1}{t^n \log t}, \quad n \geq 2. \quad (2.2.7)$$

Proof. Equations (2.2.5) and (2.2.6) are proved in Atkinson [2], but we include their proofs here for completeness. We will use the formulas

$$\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O\left(\frac{1}{t^2}\right), \quad (2.2.8)$$

and

$$\vartheta^{(n)}(t) = \frac{(-1)^n (n-2)!}{2t^{n-1}} + O\left(\frac{1}{t^n}\right), \quad n \geq 2, \quad (2.2.9)$$

which both easily follow from Stirling's formula.

To prove (2.2.5), observe that the mean value theorem of differential calculus gives

$$t' - t = \frac{\vartheta(t') - \vartheta(t)}{\vartheta'(x_1)}$$

for some x_1 between t and t' . Hence $t' - t \ll (\log t)^{-1}$ by (1.1.3) and (2.2.8). This and the second-order Taylor expansion of ϑ at t imply that

$$\vartheta(t') - \vartheta(t) = (t' - t)\vartheta'(t) + O\left(\frac{\vartheta''(x_2)}{\log^2 t}\right)$$

for some x_2 between t and t' . We arrive at (2.2.5) upon dividing both sides by $\vartheta'(t)$ and using (1.1.3), (2.2.8), and (2.2.9) with $n = 2$.

To show (2.2.6), differentiate both sides of (1.1.3) to deduce that

$$\vartheta'(t') \frac{dt'}{dt} - \vartheta'(t) = 0.$$

From this and (2.2.8), it follows that

$$\frac{dt'}{dt} = \frac{\vartheta'(t)}{\vartheta'(t')} = \frac{\log \frac{t}{2\pi} + O\left(\frac{1}{t^2}\right)}{\log \frac{t'}{2\pi} + O\left(\frac{1}{t'^2}\right)}.$$

We can approximate $\log t'$ by $\log t$ using (2.2.5), and the result is (2.2.6).

We next prove (2.2.7). Suppose, by way of induction, that $(d^j/dt^j)t' \ll (t^j \log t)^{-1}$ for $2 \leq j \leq n-1$. We differentiate both sides of (1.1.3) n times and use Lemma 2.2.3 with $f(t) = \vartheta(t)$ and $g(t) = t'$ to write

$$\vartheta'(t') \left(\frac{d^n}{dt^n} t' \right) + \vartheta^{(n)}(t') \left(\frac{dt'}{dt} \right)^n + \Sigma - \vartheta^{(n)}(t) = 0.$$

We apply (2.2.8), (2.2.9), (2.2.6), and our inductive assumption to estimate all the terms except for the first term on the left-hand side. After some rearrangement, the result is

$$\frac{d^n}{dt^n} t' = \frac{1}{\vartheta'(t')} \left(\frac{c_n}{t^{n-1}} - \frac{c_n}{(t')^{n-1}} + O\left(\frac{1}{t^n}\right) \right),$$

where $c_n = (-1)^n(n-2)!/2$. It follows from (2.2.8) and (2.2.5) that the right-hand side is $\ll (t^n \log t)^{-1}$. This proves (2.2.7). $\square$

Let $H(t, s)$ be defined by

$$H(t, s) = \frac{\Gamma\left(\frac{\frac{1}{2}+it'+s}{2}\right) \Gamma\left(\frac{\frac{1}{2}-it+s}{2}\right)}{\Gamma\left(\frac{\frac{1}{2}+it'}{2}\right) \Gamma\left(\frac{\frac{1}{2}-it}{2}\right)}. \quad (2.2.10)$$

Lemma 2.2.5. *If $\operatorname{Re}(s)$ is fixed and $> -\frac{1}{2}$, then for large enough t ,*

$$H(t, s) \ll t^{1/2} e^{\frac{\pi}{2}t} \quad (2.2.11)$$

uniformly for $-\infty < \operatorname{Im}(s) < \infty$ and

$$H(t, s) = \left(\frac{t}{2}\right)^s \left(1 + O\left(\frac{1+|s|^2}{t}\right)\right) \quad (2.2.12)$$

uniformly for $|\operatorname{Im}(s)|^2 \ll t$.

Proof. We first prove (2.2.12). If t is large and $|\operatorname{Im}(s)|^2 \ll t$, then it follows from Stirling's formula that

$$\Gamma\left(\frac{\frac{1}{2} + it + s}{2}\right) = \sqrt{2\pi} \left(\frac{t}{2}\right)^{z - \frac{1}{2}} \exp\left(i\frac{\pi z}{2} - i\frac{\pi}{4} - i\frac{t}{2}\right) \left(1 + O\left(\frac{1 + |s|^2}{t}\right)\right), \quad (2.2.13)$$

where $z = \frac{1}{2}(\frac{1}{2} + it + s)$. We use this and its complex conjugate to estimate each gamma function in the definition (2.2.10) of $H(t, s)$. The result is

$$H(t, s) = \left(\frac{t'}{2}\right)^{s/2} \left(\frac{t}{2}\right)^{s/2} \left(1 + O\left(\frac{1 + |s|^2}{t}\right)\right).$$

The estimate (2.2.12) now follows from this and (2.2.5).

To prove (2.2.11), apply the trivial bound $\Gamma \ll 1$ to estimate the numerator of $H(t, s)$ and (2.2.13) to estimate the denominator. $\square$

Lemma 2.2.6. *If $\operatorname{Re}(s) > -\frac{1}{2}$ and $n \geq 1$ are fixed, then for large enough t ,*

$$\frac{d^n}{dt^n} \log H(t, s) \ll \log |s| \quad (2.2.14)$$

uniformly for $t \leq 2|\operatorname{Im}(s)|$ and

$$\frac{d^n}{dt^n} \log H(t, s) \ll \frac{1 + |s|}{t^n} \quad (2.2.15)$$

uniformly for $t \geq 2|\operatorname{Im}(s)|$.

Proof. We prove (2.2.15) first. Observe that

$$\frac{d^n}{dt^n} \log H(t, s) = \frac{d^n}{dt^n} \Upsilon(t') + \frac{d^n}{dt^n} \Upsilon(-t), \quad (2.2.16)$$

where $\Upsilon(t)$ is defined by

$$\Upsilon(t) = \Upsilon(t, s) = \log \Gamma\left(\frac{\frac{1}{2} + it + s}{2}\right) - \log \Gamma\left(\frac{\frac{1}{2} + it}{2}\right).$$

If $t \geq 2|\operatorname{Im}(s)|$ and $m \geq 1$, then we see from Stirling's formula that

$$\begin{aligned} \frac{d^m}{dt^m} \Upsilon(t) &= \left(\frac{\Gamma'}{\Gamma}\right)^{(m-1)} \left(\frac{\frac{1}{2} + it + s}{2}\right) \left(\frac{i}{2}\right)^m - \left(\frac{\Gamma'}{\Gamma}\right)^{(m-1)} \left(\frac{\frac{1}{2} + it}{2}\right) \left(\frac{i}{2}\right)^m \\ &\ll \frac{1 + |s|}{t^m}. \end{aligned} \quad (2.2.17)$$

To estimate the derivatives of $\Upsilon(t')$, we apply Lemma 2.2.3 with $f(t) = \Upsilon(t)$ and $g(t) = t'$. We use (2.2.17) to estimate the derivatives of $f(t)$ and (2.2.6) or (2.2.7) to bound the derivatives of $g(t)$. The result is

$$\frac{d^n}{dt^n} \Upsilon(t') \ll \frac{1 + |s|}{t^n}$$

for $t \geq 2|\operatorname{Im}(s)|$ and $n \geq 1$. We now insert this and the bound (2.2.17), which also holds for $\Upsilon(-t)$, into (2.2.16) to deduce (2.2.15).

The proof of (2.2.14) is the same, except that we use the bound

$$\frac{d^m}{dt^m} \Upsilon(t) \ll \begin{cases} \log |s| & \text{if } m = 1 \\ 1 & \text{if } m \geq 2 \end{cases}$$

instead of (2.2.17). This bound follows from Stirling's formula when $t \leq 2|\operatorname{Im}(s)|$. $\square$

Lemma 2.2.7. *If $\operatorname{Re}(s) > -\frac{1}{2}$ and $n \geq 0$ are fixed, then for large enough t ,*

$$\frac{d^n}{dt^n} H(t, s) \ll t^{1/2} e^{\frac{\pi}{2}t} \log^n(|s| + 2) \quad (2.2.18)$$

uniformly for $-\infty < \operatorname{Im}(s) < \infty$ and

$$\frac{d^n}{dt^n} H(t, s) \ll t^{\operatorname{Re}(s)-n} (1 + |s|)^n \quad (2.2.19)$$

uniformly for $|\operatorname{Im}(s)|^2 \ll t$.

Proof. By Lemma 2.2.3 with $f(t) = e^t$ and $g(t) = \log H(t, s)$, we see that $(d^n/dt^n)H(t, s)$ equals a sum of terms of the form

$$H(t, s) \left(\frac{d}{dt} \log H(t, s) \right)^{m_1} \cdots \left(\frac{d^n}{dt^n} \log H(t, s) \right)^{m_n} \quad (2.2.20)$$

with $m_1 + 2m_2 + 3m_3 + \cdots + nm_n = n$. To prove (2.2.18), observe that (2.2.14) and (2.2.15) together give

$$\frac{d^j}{dt^j} \log H(t, s) \ll_j \log(|s| + 2)$$

for $-\infty < \text{Im}(s) < \infty$ and $j \geq 1$. We use this and (2.2.11) to bound each term of the form (2.2.20) and thus deduce (2.2.18).

Now we show (2.2.19). If v and t are positive real numbers with $v^2 \leq Bt$ and $t \geq 4B$, then $v \geq 2B$ implies $2v \leq v^2/B \leq t$, while $v \leq 2B$ implies $2v \leq 4B \leq t$. Hence $|\text{Im}(s)|^2 \ll t$ implies that $2|\text{Im}(s)| \leq t$ and so (2.2.15) holds for large enough t . Using (2.2.15) and (2.2.12) to bound each term of the form (2.2.20), we arrive at (2.2.19). $\square$

Let $K(t, s)$ be defined by

$$K(t, s) = \pi^{it'-it} \frac{\Gamma\left(\frac{\frac{1}{2}-it'+s}{2}\right) \Gamma\left(\frac{\frac{1}{2}+it+s}{2}\right)}{\Gamma\left(\frac{\frac{1}{2}+it'}{2}\right) \Gamma\left(\frac{\frac{1}{2}-it}{2}\right)}. \quad (2.2.21)$$

Since $\Gamma(\bar{z}) = \overline{\Gamma(z)}$, we can write

$$K(t, s) = \pi^{it'-it} X(t) \overline{H(t, \bar{s})}, \quad (2.2.22)$$

where

$$X(t) = \frac{\Gamma\left(\frac{\frac{1}{2}-it'}{2}\right) \Gamma\left(\frac{\frac{1}{2}+it}{2}\right)}{\Gamma\left(\frac{\frac{1}{2}+it'}{2}\right) \Gamma\left(\frac{\frac{1}{2}-it}{2}\right)}. \quad (2.2.23)$$

The following lemma states that Lemma 2.2.7 holds verbatim when $H(t, s)$ is replaced by $K(t, s)$.

Lemma 2.2.8. *If $\text{Re}(s) > -\frac{1}{2}$ and $n \geq 0$ are fixed, then for large enough t ,*

$$\frac{d^n}{dt^n} K(t, s) \ll t^{1/2} e^{\frac{\pi}{2}t} \log^n(|s| + 2)$$

uniformly for $-\infty < \text{Im}(s) < \infty$ and

$$\frac{d^n}{dt^n} K(t, s) \ll t^{\text{Re}(s)-n} (1 + |s|)^n$$

uniformly for $|\text{Im}(s)|^2 \ll t$.

Proof. In view of (2.2.22) and Lemma 2.2.7, we only need to bound the derivatives of $\pi^{it'-it}$ and $X(t)$. First observe that from (2.2.6), (2.2.7), and Lemma 2.2.3 with $f(t) = e^t$ and $g(t) = i(t' - t) \log \pi$, it follows that

$$\frac{d^j}{dt^j} \pi^{it'-it} \ll_j \frac{1}{t^j} \quad (2.2.24)$$

for $j \geq 0$.

The next task is to estimate the derivatives of $X(t)$. We first consider $\log X(t)$. Write

$$\frac{d^j}{dt^j} \log X(t) = \frac{d^j}{dt^j} Y(t) - \frac{d^j}{dt^j} \overline{Y(t)}, \quad (2.2.25)$$

where

$$Y(t) = \log \Gamma\left(\frac{\frac{1}{2} + it}{2}\right) - \log \Gamma\left(\frac{\frac{1}{2} + it'}{2}\right).$$

It follows from Lemma 2.2.3 with $f(t) = \log \Gamma(t)$ and $g(t) = \frac{1}{2}(\frac{1}{2} + it')$ that

$$\begin{aligned} \frac{d^j}{dt^j} Y(t) &= \left(\frac{\Gamma'}{\Gamma}\right)^{(j-1)} \left(\frac{\frac{1}{2} + it}{2}\right) \left(\frac{i}{2}\right)^j - \left(\frac{\Gamma'}{\Gamma}\right)^{(j-1)} \left(\frac{\frac{1}{2} + it'}{2}\right) \left(\frac{i}{2}\right)^j \left(\frac{dt'}{dt}\right)^j \\ &\quad - \frac{\Gamma'}{\Gamma} \left(\frac{\frac{1}{2} + it'}{2}\right) \left(\frac{i}{2}\right) \left(\frac{d^j}{dt^j} t'\right) - \Sigma \end{aligned}$$

for $j \geq 1$. The last two terms on the right-hand side do not appear when $j = 1$. Use (2.2.6), (2.2.7), and Stirling's formula to bound the last two terms on the right-hand side and deduce that

$$\begin{aligned} \frac{d^j}{dt^j} Y(t) &= \left(\frac{\Gamma'}{\Gamma}\right)^{(j-1)} \left(\frac{\frac{1}{2} + it}{2}\right) \left(\frac{i}{2}\right)^j \\ &\quad - \left(\frac{\Gamma'}{\Gamma}\right)^{(j-1)} \left(\frac{\frac{1}{2} + it'}{2}\right) \left(\frac{i}{2}\right)^j \left(\frac{dt'}{dt}\right)^j + O\left(\frac{1}{t^j}\right). \end{aligned}$$

We estimate the first two terms on the right-hand side via Stirling's formula and (2.2.6). The result is

$$\frac{d^j}{dt^j} Y(t) = \begin{cases} \frac{i \log t}{2} - \frac{i \log t'}{2} + O\left(\frac{1}{t}\right) & \text{if } j = 1, \\ \frac{b_j}{(\frac{1}{2} + it)^{j-1}} - \frac{b_j}{(\frac{1}{2} + it')^{j-1}} + O\left(\frac{1}{t^j}\right) & \text{if } j \geq 2, \end{cases}$$

where $b_j = (-i)^j(j-2)!/2$. From this and (2.2.5), we arrive at $(d^j/dt^j)Y(t) \ll t^{-j}$ for $j \geq 1$. This bound and the equation (2.2.25) imply that

$$\frac{d^j}{dt^j} \log X(t) \ll_j \frac{1}{t^j} \quad (2.2.26)$$

for $j \geq 1$.

Now we can estimate the derivatives of $X(t)$. It follows from Lemma 2.2.3 with $f(t) = e^t$ and $g(t) = \log X(t)$ that $(d^j/dt^j)X(t)$ equals a sum of terms of the form

$$X(t) \left(\frac{d}{dt} \log X(t) \right)^{m_1} \cdots \left(\frac{d^j}{dt^j} \log X(t) \right)^{m_j} \quad (2.2.27)$$

with $m_1 + 2m_2 + 3m_3 + \cdots + jm_j = j$. We apply the estimate (2.2.13) to each gamma function in the definition (2.2.23) of $X(t)$ to deduce that

$$X(t) = \left(\frac{t'}{2} \right)^{-it'} \left(\frac{t}{2} \right)^{it} e^{i(t'-t)} \left(1 + O\left(\frac{1}{t} \right) \right). \quad (2.2.28)$$

Hence $X(t) \ll 1$. Use this and (2.2.26) to see that each term of the form (2.2.27) is $\ll t^{-j}$. Therefore

$$\frac{d^j}{dt^j} X(t) \ll_j \frac{1}{t^j} \quad (2.2.29)$$

for $j \geq 0$.

Lemma 2.2.8 now follows from (2.2.22), the product rule, (2.2.24), (2.2.29), and the bounds (2.2.18) and (2.2.19). $\square$

Let $q(t, s)$ be defined by

$$q(t, s) = \frac{(t' - t)^2 + 4s^2}{(t' - t)^2}. \quad (2.2.30)$$

Using Lemmas 2.2.7 and 2.2.8, we can estimate the derivatives of $H(t, s)q(t, s)$ and $K(t, s)q(t, s)$, as follows.

Lemma 2.2.9. *If $\operatorname{Re}(s) > -\frac{1}{2}$ and $n \geq 0$ are fixed, then for large enough t ,*

$$\frac{d^n}{dt^n} H(t, s)q(t, s) \ll t^{1/2} (\log t)^{n+2} e^{\frac{\pi}{2}t} (1 + |s|)^2 \log^n(|s| + 2) \quad (2.2.31)$$

uniformly for $-\infty < \operatorname{Im}(s) < \infty$ and

$$\frac{d^n}{dt^n} H(t, s) q(t, s) \ll t^{\operatorname{Re}(s)-n} (\log t)^{n+2} (1 + |s|)^{n+2} \quad (2.2.32)$$

uniformly for $|\operatorname{Im}(s)|^2 \ll t$. The same bounds hold when the expression $H(t, s)$ is replaced by $K(t, s)$.

Proof. We use Lemma 2.2.3 with $f(t) = (t^2 + 4s^2)/t^2$ and $g(t) = t' - t$ and the estimates (2.2.5), (2.2.6), and (2.2.7) to deduce that

$$\frac{d^m}{dt^m} q(t, s) \ll_m |s|^2 \frac{(\log t)^{m+2}}{t^m}$$

for $m \geq 1$. For $m = 0$, the bound is $q(t, s) \ll 1 + |s|^2 \log^2 t$. The lemma now follows from these, the product rule, and Lemmas 2.2.7 and 2.2.8. $\square$

For a function $F(s)$, we use $\int_{(c)} F(s) ds$ to denote the path integral of $F(s)$ along the line from $c - i\infty$ to $c + i\infty$. Let $V_1(t, x)$ and $V_2(t, x)$ be defined by

$$V_1(t, x) = \frac{1}{2\pi i} \int_{(1)} H(t, s) q(t, s) \frac{\exp(s^2)}{x^s} \frac{ds}{s} \quad (2.2.33)$$

and

$$V_2(t, x) = \frac{1}{2\pi i} \int_{(1)} K(t, s) q(t, s) \frac{\exp(s^2)}{x^s} \frac{ds}{s}, \quad (2.2.34)$$

where $H(t, s)$ and $K(t, s)$ are defined by (2.2.10) and (2.2.21), respectively.

Lemma 2.2.10. *Let $n \geq 0$ and $B > 0$ be fixed. For $j = 1$ or 2 , if t is large enough then*

$$\frac{d^n}{dt^n} V_j(t, x) \ll \frac{t^B}{x^B} \left(\frac{(\log t)^{n+2}}{t^n} \right)$$

uniformly for $x > 0$.

Proof. We first treat the case with $j = 1$. By Cauchy's theorem and the rapid decay of $\exp(s^2)$ as $|\operatorname{Im}(s)| \rightarrow \infty$, we can move the line of integration in the

definition (2.2.33) of $V_1(t, x)$ to $\operatorname{Re}(s) = B$. Then we differentiate under the integral sign to deduce that

$$\frac{d^n}{dt^n} V_1(t, x) = \frac{1}{2\pi i} \int_{(B)} \frac{d^n}{dt^n} H(t, s) q(t, s) \frac{\exp(s^2)}{x^s} \frac{ds}{s}.$$

Write this integral as $I_1 + I_2$, where I_1 is the part with $|\operatorname{Im}(s)|^2 \leq 100t$ and I_2 is the rest. To bound I_1 , use (2.2.32) to deduce that

$$I_1 \ll_{n,B} \frac{t^B}{x^B} \left(\frac{(\log t)^{n+2}}{t^n} \right).$$

On the other hand, we see from (2.2.31) that

$$I_2 \ll_{n,B} x^{-B} e^{-t}.$$

The conclusion of Lemma 2.2.10 for $j = 1$ now follows from these estimates for I_1 and I_2 . We can reach the same conclusion for $j = 2$ in the exact same way because of the last sentence of Lemma 2.2.9. $\square$

2.3 The twisted Atkinson mean value formula

In this section, we prove the following.

Lemma 2.3.1. *Let $w(t)$ be a smooth function with support in $[T/2, 4T]$ such that $w^{(j)}(t) \ll_j T_0^{-j}$ for $j \geq 0$, where $T^{1/2+\varepsilon} \ll T_0 \ll T$. If h and k are relatively prime positive integers, then*

$$\begin{aligned} & \int_{-\infty}^{\infty} \left(\frac{h}{k} \right)^{-it} \zeta\left(\frac{1}{2} + it'\right) \zeta\left(\frac{1}{2} - it\right) w(t) dt \\ &= \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} k^{-it'+it} \zeta(1 + it' - it) w(t) dt + \frac{e^{-i2a}}{\sqrt{hk}} \int_{-\infty}^{\infty} h^{it'-it} \zeta(1 - it' + it) w(t) dt \\ & \quad + O\left(\frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} (hk)^{\ell/2} \right) + O((hk)^{-1/4} T^{3/4} \log^2 T) \end{aligned}$$

for any fixed integer $\ell \geq 0$.

Our starting point in proving Lemma 2.3.1 is the following approximate functional equation. Its proof closely resembles the proof of Proposition 2.1 of Hughes and Young [40] (see also Theorem 5.3 of [47] and Lemma 4 of [66]).

Proposition 2.3.1. *If t is large enough, then*

$$\zeta(\tfrac{1}{2} + it')\zeta(\tfrac{1}{2} - it) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{V_1(t, \pi mn)}{m^{\frac{1}{2}+it'} n^{\frac{1}{2}-it}} + \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{V_2(t, \pi mn)}{m^{\frac{1}{2}+it} n^{\frac{1}{2}-it'}} + O(e^{-t^2/4}),$$

where V_1 and V_2 are defined by (2.2.33) and (2.2.34), respectively.

Proof. Define $\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, so that the functional equation for $\zeta(s)$ implies $\Lambda(s) = \Lambda(1-s)$. Let I be the integral

$$I = \frac{1}{2\pi i} \int_{(1)} \Lambda(\tfrac{1}{2} + it' + s) \Lambda(\tfrac{1}{2} - it + s) q(t, s) \exp(s^2) \frac{ds}{s}. \quad (2.3.1)$$

The poles of the integrand between the lines $\operatorname{Re}(s) = 1$ and $\operatorname{Re}(s) = -1$ are at $s = 0$, $s = \pm \frac{1}{2} - it'$, and $s = \pm \frac{1}{2} + it$. The residues of the poles different from $s = 0$ are of negligible size due to the rapid decay of $\exp(s^2)$ as $|\operatorname{Im}(s)| \rightarrow \infty$. More precisely, we can bound them by $O(e^{-t^2/2})$. Thus, the residue theorem gives

$$I = \Lambda(\tfrac{1}{2} + it') \Lambda(\tfrac{1}{2} - it) + J + O(e^{-t^2/2}), \quad (2.3.2)$$

where J is defined in the same way as I , except with $\operatorname{Re}(s) = -1$ as its line of integration. We can easily deduce from the functional equation $\Lambda(s) = \Lambda(1-s)$ and a change of variable that

$$J = -\frac{1}{2\pi i} \int_{(1)} \Lambda(\tfrac{1}{2} - it' + s) \Lambda(\tfrac{1}{2} + it + s) q(t, s) \exp(s^2) \frac{ds}{s}. \quad (2.3.3)$$

Now recall the definitions (2.2.10) of $H(t, s)$ and (2.2.21) of $K(t, s)$. Insert (2.3.1) and (2.3.3) into (2.3.2) and rearrange the result to deduce that

$$\begin{aligned} & \zeta(\tfrac{1}{2} + it') \zeta(\tfrac{1}{2} - it) \\ &= \frac{1}{2\pi i} \int_{(1)} \pi^{-s} H(t, s) \zeta(\tfrac{1}{2} + it' + s) \zeta(\tfrac{1}{2} - it + s) q(t, s) \exp(s^2) \frac{ds}{s} \\ &+ \frac{1}{2\pi i} \int_{(1)} \pi^{-s} K(t, s) \zeta(\tfrac{1}{2} - it' + s) \zeta(\tfrac{1}{2} + it + s) q(t, s) \exp(s^2) \frac{ds}{s} + O(e^{-t^2/4}). \end{aligned}$$

We expand each zeta-function into its absolutely convergent Dirichlet series, interchange the order of summation, and then use the definitions (2.2.33) and (2.2.34) to arrive at the conclusion of Proposition 2.3.1. $\square$

We now begin our proof of Lemma 2.3.1. For brevity, we define

$$I(h, k) = \int_{-\infty}^{\infty} \left(\frac{h}{k} \right)^{-it} \zeta\left(\frac{1}{2} + it'\right) \zeta\left(\frac{1}{2} - it\right) w(t) dt$$

We apply Proposition 2.3.1 to deduce that

$$\begin{aligned} I(h, k) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} m^{-it'+it} V_1(t, \pi mn) w(t) dt \\ &\quad + \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} n^{it'-it} V_2(t, \pi mn) w(t) dt + O(e^{-T}), \end{aligned} \quad (2.3.4)$$

where we have interchanged the order of summation on the right-hand side.

We will first estimate the contribution of the “off-diagonal” terms, which have $hm \neq kn$. Before we can do so, we need to bound the derivative

$$\frac{d^\ell}{dt^\ell} m^{-it'+it} V_1(t, \pi mn) w(t).$$

We restrict our attention to those t with $T/2 \leq t \leq 4T$ since we are assuming that the support of $w(t)$ is contained in $[T/2, 4T]$. Recall that we are also assuming that

$$w^{(j)}(t) \ll_j \frac{1}{T_0^j}, \quad (2.3.5)$$

where $T^{\frac{1}{2}+\varepsilon} \ll T_0 \ll T$. From (2.2.6), (2.2.7), and Lemma 2.2.3 with $f(t) = e^t$ and $g(t) = -i(t' - t) \log m$, it follows that

$$\frac{d^j}{dt^j} m^{-it'+it} \ll_j \frac{\log^j(m+2)}{T^j}$$

for $j \geq 0$. This implies

$$\frac{d^j}{dt^j} m^{-it'+it} \ll_j \frac{\log^j(m+2)}{T_0^j} \quad (2.3.6)$$

since $T_0 \ll T$. Moreover, from Lemma 2.2.10 and the fact that $T_0 \ll T$, it follows that

$$\frac{d^j}{dt^j} V_1(t, \pi mn) \ll_{j,B} \frac{T^B}{(mn)^B} \left(\frac{(\log T)^{j+2}}{T_0^j} \right)$$

for any integer $j \geq 0$ and real number $B > 0$. We apply this, (2.3.5), and (2.3.6) together with the product rule to deduce that

$$\frac{d^\ell}{dt^\ell} m^{-it'+it} V_1(t, \pi mn) w(t) \ll_{\ell,B} \frac{T^B}{T_0^\ell (mn)^B} (\log T)^{\ell+2} \log^\ell(m+2). \quad (2.3.7)$$

for $\ell \geq 0$ and $B > 0$. A similar argument leads to

$$\frac{d^\ell}{dt^\ell} n^{it'-it} V_2(t, \pi mn) w(t) \ll_{\ell,B} \frac{T^B}{T_0^\ell (mn)^B} (\log T)^{\ell+2} \log^\ell(n+2). \quad (2.3.8)$$

for any integer $\ell \geq 0$ and real number $B > 0$.

Using these, we can now estimate the contribution of the off-diagonal terms on the right-hand side of (2.3.4). By integrating by parts ℓ times and applying (2.3.7), we see that

$$\begin{aligned} & \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} m^{-it'+it} V_1(t, \pi mn) w(t) dt \\ & \ll_{\ell,B} \frac{T}{|\log(hm/kn)|^\ell} \left\{ \frac{T^B}{T_0^\ell (mn)^B} (\log T)^{\ell+2} \log^\ell(m+2) \right\} \end{aligned}$$

for any integer $\ell \geq 0$ and real number $B > 0$. We multiply both sides by $(mn)^{-1/2}$ and use the fact that $|\log(hm/kn)| \gg 1/\sqrt{hkmn}$ to arrive at

$$\begin{aligned} & \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} m^{-it'+it} V_1(t, \pi mn) w(t) dt \\ & \ll_{\ell,B} (hk)^{\ell/2} \left\{ \frac{T^{B+1}}{T_0^\ell (mn)^{B-\frac{\ell}{2}+\frac{1}{2}}} (\log T)^{\ell+2} \log^\ell(m+2) \right\}. \end{aligned} \quad (2.3.9)$$

We choose $B = \frac{\ell}{2} + \frac{1}{2} + \varepsilon$ so that the sum of the right-hand side over all $m, n \geq 1$ converges. Summing (2.3.9) over all $m, n \geq 1$ with $hm \neq kn$, we now deduce that

$$\sum_{hm \neq kn} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} m^{-it'+it} V_1(t, \pi mn) w(t) dt \ll_{\ell,\varepsilon} (hk)^{\ell/2} \left(\frac{T^{\frac{3}{2}+\frac{\ell}{2}+\varepsilon}}{T_0^\ell} \right).$$

A similar argument using (2.3.8) leads to

$$\sum_{hm \neq kn} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} \left(\frac{hm}{kn} \right)^{-it} n^{it'-it} V_2(t, \pi mn) w(t) dt \ll_{\ell, \varepsilon} (hk)^{\ell/2} \left(\frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} \right).$$

From these estimates and (2.3.4), we conclude that

$$\begin{aligned} I(h, k) &= \sum_{hm=kn} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} m^{-it'+it} V_1(t, \pi mn) w(t) dt \\ &\quad + \sum_{hm=kn} \frac{1}{\sqrt{mn}} \int_{-\infty}^{\infty} n^{it'-it} V_2(t, \pi mn) w(t) dt + O\left(\frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} (hk)^{\ell/2} \right) \end{aligned} \quad (2.3.10)$$

for fixed $\ell \geq 0$ and $\varepsilon > 0$.

Having bounded the contribution of the off-diagonal terms, our next task is to estimate the sums on the right-hand side of (2.3.10). Since h and k are relatively prime, the relation $hm = kn$ holds if and only if $m = k\nu$ and $n = h\nu$ for some integer ν . Thus we can write (2.3.10) as

$$\begin{aligned} I(h, k) &= \frac{1}{\sqrt{hk}} \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_{-\infty}^{\infty} (k\nu)^{-it'+it} V_1(t, \pi hk\nu^2) w(t) dt \\ &\quad + \frac{1}{\sqrt{hk}} \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_{-\infty}^{\infty} (h\nu)^{it'-it} V_2(t, \pi hk\nu^2) w(t) dt + O\left(\frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} (hk)^{\ell/2} \right). \end{aligned} \quad (2.3.11)$$

Let Σ_1 and Σ_2 denote the first and second terms, respectively, on the right-hand side of (2.3.11).

We first estimate Σ_1 . Replacing $V_1(t, \pi hk\nu^2)$ by its definition (2.2.33) and interchanging the order of summation, we see that

$$\Sigma_1 = \frac{(hk)^{-1/2}}{2\pi i} \int_{-\infty}^{\infty} k^{-it'+it} \int_{(1)} H(t, s) q(t, s) \frac{\exp(s^2)}{(\pi hk)^s} \zeta(1 + it' - it + 2s) \frac{ds}{s} w(t) dt.$$

The interchange of order of summation is justified by absolute convergence. Observe that the pole of the zeta-function in the above integral is canceled by a zero of $q(t, s)$. Thus $s = 0$ is the only pole of the integrand that lies in the region

bounded by the lines $\operatorname{Re}(s) = -\frac{1}{4}$ and $\operatorname{Re}(s) = 1$. Since $H(t, 0)q(t, 0) = 1$ by (2.2.10) and (2.2.30), we therefore deduce from the residue theorem that

$$\begin{aligned} \Sigma_1 &= \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} k^{-it'+it} \zeta(1+it'-it) w(t) dt \\ &+ \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{k^{-it'+it}}{\sqrt{hk}} \int_{(-\frac{1}{4})} H(t, s) q(t, s) \frac{\exp(s^2)}{(\pi hk)^s} \zeta(1+it'-it+2s) \frac{ds}{s} w(t) dt. \end{aligned} \quad (2.3.12)$$

To complete our evaluation of Σ_1 , we need to estimate the last term in (2.3.12). Write the s -integral in (2.3.12) as

$$\frac{1}{2\pi i} \int_{(-\frac{1}{4})} H(t, s) q(t, s) \frac{\exp(s^2)}{(\pi hk)^s} \zeta(1+it'-it+2s) \frac{ds}{s} = I_1 + I_2,$$

where I_1 is the part with $|\operatorname{Im}(s)|^2 \leq 100t$ and I_2 is the rest. Note that it follows from (2.2.5) and a well-known bound for $\zeta(s)$ (see (2.12.2) of Titchmarsh [64]) that

$$\zeta(1+it'-it+2s) \ll 1 + |\operatorname{Im}(s)| \quad (2.3.13)$$

uniformly for $\operatorname{Re}(s) = -\frac{1}{4}$. To bound I_1 , use (2.3.13) and (2.2.32) with $n = 0$ to deduce that

$$I_1 \ll \frac{\log^2 t}{t^{1/4}} (hk)^{1/4}.$$

On the other hand, we see from (2.3.13) and (2.2.31) with $n = 0$ that

$$I_2 \ll (hk)^{1/4} e^{-t}.$$

From these estimates for I_1 and I_2 , it follows that

$$\frac{1}{2\pi i} \int_{(-\frac{1}{4})} H(t, s) q(t, s) \frac{\exp(s^2)}{(\pi hk)^s} \zeta(1+it'-it+2s) \frac{ds}{s} \ll \frac{\log^2 t}{t^{1/4}} (hk)^{1/4}.$$

Insert this into (2.3.12) and use our assumptions that $w(t) \ll 1$ and $w(t) = 0$ for $t \notin [T/2, 4T]$ to conclude that

$$\Sigma_1 = \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} k^{-it'+it} \zeta(1+it'-it) w(t) dt + O((hk)^{-1/4} T^{3/4} \log^2 T). \quad (2.3.14)$$

We next estimate the second term Σ_2 on the right-hand side of (2.3.11). We repeat the arguments leading up to (2.3.12) to deduce from the definition (2.2.34) of $V_2(t, \pi h k \nu^2)$ and the residue theorem that

$$\begin{aligned} \Sigma_2 &= \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} h^{it'-it} K(t, 0) \zeta(1 - it' + it) w(t) dt \\ &\quad + \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{h^{it'-it}}{\sqrt{hk}} \int_{(-\frac{1}{4})} K(t, s) q(t, s) \frac{\exp(s^2)}{(\pi h k)^s} \zeta(1 - it' + it + 2s) \frac{ds}{s} w(t) dt. \end{aligned} \quad (2.3.15)$$

By the last sentence in Lemma 2.2.9, we can carry out the same reasoning leading up to (2.3.14) to deduce from (2.3.15) that

$$\Sigma_2 = \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} h^{it'-it} K(t, 0) \zeta(1 - it' + it) w(t) dt + O((hk)^{-1/4} T^{3/4} \log^2 T). \quad (2.3.16)$$

To evaluate $K(t, 0)$, we apply (2.2.22) and the fact that $H(t, 0) = 1$ to write $K(t, 0) = \pi^{it'-it} X(t)$. Hence, by (2.2.28) and (2.2.5), it follows that

$$K(t, 0) = e^{-i2a} \left(1 + O\left(\frac{1}{t}\right) \right).$$

We insert this into (2.3.16) and arrive at

$$\Sigma_2 = \frac{e^{-i2a}}{\sqrt{hk}} \int_{-\infty}^{\infty} h^{it'-it} \zeta(1 - it' + it) w(t) dt + O((hk)^{-1/4} T^{3/4} \log^2 T), \quad (2.3.17)$$

where we have used (2.2.5) and the Laurent series expansion of $\zeta(s)$ near $s = 1$ to bound the contribution of $O(1/t)$.

Lemma 2.3.1 now follows from (2.3.11), (2.3.14), and (2.3.17).

2.4 Proof of Theorem 1.1.2

We now begin our proof of Theorem 1.1.2. For brevity, we define

$$\mathcal{I} = \int_{-\infty}^{\infty} Z(t') Z(t) |M(t)|^2 w(t) dt.$$

Since $Z(t)$ is real for real t , it follows that

$$Z(t) = e^{i\vartheta(t)}\zeta(\tfrac{1}{2} + it) = e^{-i\vartheta(t)}\zeta(\tfrac{1}{2} - it).$$

Use this and the definitions (1.1.3) and (1.1.4) to deduce that

$$\begin{aligned} \mathcal{I} &= e^{ia} \int_{-\infty}^{\infty} \zeta(\tfrac{1}{2} + it')\zeta(\tfrac{1}{2} - it)|M(t)|^2 w(t) dt \\ &= e^{ia} \sum_{m,n \leq y} \frac{\mu(m)\mu(n)}{\sqrt{mn}} \left(\frac{\log(y/m)}{\log y} \right)^2 \left(\frac{\log(y/n)}{\log y} \right)^2 \\ &\quad \times \int_{-\infty}^{\infty} \left(\frac{m'}{n'} \right)^{-it} \zeta(\tfrac{1}{2} + it')\zeta(\tfrac{1}{2} - it)w(t) dt, \end{aligned} \quad (2.4.1)$$

where $m' = m/(m, n)$ and $n' = n/(m, n)$. We apply Lemma 2.3.1 to the inner integral. To estimate the contribution of the second error term in the conclusion of Lemma 2.3.1, we let $\varphi(d)$ be the Euler totient function and write

$$\begin{aligned} \sum_{m,n \leq y} \frac{(m'n')^{-1/4}}{\sqrt{mn}} &= \sum_{m,n \leq y} \frac{(m, n)^{1/2}}{(mn)^{3/4}} \leq \sum_{m,n \leq y} \frac{(m, n)}{(mn)^{3/4}} \\ &= \sum_{m,n \leq y} \frac{1}{(mn)^{3/4}} \sum_{d|(m,n)} \varphi(d) = \sum_{d \leq y} \frac{\varphi(d)}{d^{3/2}} \left(\sum_{j \leq y/d} \frac{1}{j^{3/4}} \right)^2 \\ &\ll y^{1/2} \log y. \end{aligned}$$

Therefore the result of applying Lemma 2.3.1 to the inner integral in (2.4.1) is

$$\begin{aligned} \mathcal{I} &= \frac{e^{ia}}{\log^4 y} \int_{-\infty}^{\infty} \zeta(1 + it' - it)w(t)\mathbf{S}(t' - t) dt \\ &\quad + \frac{e^{-ia}}{\log^4 y} \int_{-\infty}^{\infty} \zeta(1 - it' + it)w(t)\mathbf{S}(-t' + t) dt \\ &\quad + O\left(y^{\ell+1} \frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} \right) + O(y^{1/2} T^{3/4} \log^3 T), \end{aligned} \quad (2.4.2)$$

where

$$\mathbf{S}(\varrho) = \sum_{m,n \leq y} \frac{\mu(m)\mu(n)}{\sqrt{mn}} \left(\frac{\log(y/m)}{\log y} \right)^2 \left(\frac{\log(y/n)}{\log y} \right)^2 \frac{(n')^{-i\varrho}}{\sqrt{m'n'}}.$$

Since $m' = m/(m, n)$ and $n' = n/(m, n)$, we can write this as

$$\mathbf{S}(\varrho) = \sum_{m, n \leq y} \frac{\mu(m)\mu(n)}{mn^{1+i\varrho}} (m, n)^{1+i\varrho} \left(\log(y/m) \right)^2 \left(\log(y/n) \right)^2. \quad (2.4.3)$$

Our next task is to evaluate $\mathbf{S}(\varrho)$. To do this, we apply Perron's formula

$$\frac{2}{2\pi i} \int_{(\eta)} \frac{x^s}{s^3} ds = \begin{cases} \log^2 x & \text{if } x \geq 1 \\ 0 & \text{if } 0 < x \leq 1, \end{cases} \quad (2.4.4)$$

which holds for any $\eta > 0$, to the right-hand side of (2.4.3). The result is

$$\mathbf{S}(\varrho) = \frac{4}{(2\pi i)^2} \int_{(\eta)} \int_{(\eta)} \frac{y^{s_1+s_2}}{s_1^3 s_2^3} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\mu(m)\mu(n)}{m^{1+s_1} n^{1+s_2+i\varrho}} (m, n)^{1+i\varrho} ds_2 ds_1, \quad (2.4.5)$$

where the interchange of the order of summation is valid by absolute convergence.

To facilitate later discussions, we choose the lines of integration in (2.4.5) to have

$\eta = 1/L$, where $L = \log T$. Let

$$g(m, n) = g(m, n; s_1, s_2, \varrho) = \frac{\mu(m)\mu(n)}{m^{1+s_1} n^{1+s_2+i\varrho}} (m, n)^{1+i\varrho}$$

be the summand of the inner sum in (2.4.5). This function satisfies the multiplicative property

$$g(h_1 j_1, h_2 j_2) = g(h_1, h_2) g(j_1, j_2)$$

for $(h_1 h_2, j_1 j_2) = 1$. Thus, it follows from Lemma 2.2.1 with $k = 2$ that

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} g(m, n) = \prod_p \left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(p^m, p^n) \right), \quad (2.4.6)$$

so long as either side is absolutely convergent. We will see in what follows that the right-hand side is absolutely convergent when ϱ is real and both s_1 and s_2

have positive real parts. Indeed, observe that $g(p^m, p^n) = 0$ when either $m \geq 2$ or $n \geq 2$. Thus if ϱ is real and s_1 and s_2 have positive real parts, then

$$\begin{aligned} \prod_p \left(\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(p^m, p^n) \right) &= \prod_p \left(g(1, 1) + g(p, 1) + g(1, p) + g(p, p) \right) \\ &= \prod_p \left(1 - \frac{1}{p^{1+s_1}} - \frac{1}{p^{1+s_2+i\varrho}} + \frac{1}{p^{1+s_1+s_2}} \right) \quad (2.4.7) \\ &= \frac{\zeta(1+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2+i\varrho)} G(s_1, s_2, \varrho), \end{aligned}$$

where $G(s_1, s_2, \varrho)$ is an Euler product such that

$$G(s_1, s_2, z) = \prod_p \left(1 + O\left(\frac{1}{p^{3/2}}\right) \right) \quad (2.4.8)$$

for $\operatorname{Re}(s_1) \geq -\frac{1}{8}$, $\operatorname{Re}(s_2) \geq -\frac{1}{8}$, and $|z| < \frac{1}{8}$. In view of (2.4.6) and (2.4.7), we now deduce from (2.4.5) that

$$\mathbf{S}(\varrho) = \frac{4}{(2\pi i)^2} \int_{(\eta)} \int_{(\eta)} \frac{y^{s_1+s_2}}{s_1^3 s_2^3} \frac{\zeta(1+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2+i\varrho)} G(s_1, s_2, \varrho) ds_2 ds_1 \quad (2.4.9)$$

for real ϱ , where we have chosen $\eta = 1/L$. For brevity, we define

$$\mathcal{G}(s_1, s_2, \varrho; y) = \frac{y^{s_1+s_2}}{s_1^3 s_2^3} \frac{\zeta(1+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2+i\varrho)} G(s_1, s_2, \varrho). \quad (2.4.10)$$

Hence we can write (2.4.9) as

$$\mathbf{S}(\varrho) = \frac{4}{(2\pi i)^2} \int_{(\eta)} \int_{(\eta)} \mathcal{G}(s_1, s_2, \varrho; y) ds_2 ds_1 \quad (2.4.11)$$

where ϱ is real and $\eta = 1/L$.

To complete our evaluation of $\mathbf{S}(\varrho)$, the idea is to move the lines of integration in (2.4.11) to the left and use the residue theorem. In the rest of the chapter, σ_j denotes the real part of s_j , and t_j the imaginary part. We assume that ϱ is a real number such that $\varrho \ll 1/L$. Also, we use A to denote some positive absolute constant that is not necessarily the same for each instance. The more trivial estimations will be left to the reader.

Let E_1 be the line segment from $\eta - iL^5$ to $\eta + iL^5$, and let E_2 be the line segment from $\eta - i2L^5$ to $\eta + i2L^5$. The exponent 5 of L^5 is arbitrary, and the discussions that follow remain valid if we replace 5 by any larger real number. Since $\eta = 1/L$ and $y = T^\theta$, it follows that $y^{s_j} \ll 1$ for $\sigma_j = \eta$. Using this, Lemma 2.2.2, and (2.4.8), we can show that the part of the integral in (2.4.11) that has large $|t_1|$ or $|t_2|$ is small. In other words,

$$\mathbf{S}(\varrho) = \frac{4}{(2\pi i)^2} \int_{E_1} \int_{E_2} \mathcal{G}(s_1, s_2, \varrho; y) ds_2 ds_1 + O(L^{-A}). \quad (2.4.12)$$

Next, we let D_2 be the line segment from $-c(\log L)^{-1} - i2L^5$ to $-c(\log L)^{-1} + i2L^5$, where $c > 0$ is an absolute constant that is so small that we can apply Lemma 2.2.2 to bound the zeta-functions in (2.4.10) when $-c(\log L)^{-1} \leq \sigma_2 \leq 1$ and $s_1 \in E_1$. One can take $c = c_0/6$, say. We “move” E_2 to D_2 . That is, we integrate $\mathcal{G}$ with respect to s_2 along the rectangle with right side E_2 and left side D_2 and deduce from the residue theorem that

$$\frac{1}{2\pi i} \int_{E_2} \mathcal{G} ds_2 = \operatorname{Res}_{s_2=0} \mathcal{G} + \operatorname{Res}_{s_2=-s_1} \mathcal{G} + \frac{1}{2\pi i} \int_{D_2} \mathcal{G} ds_2 + \frac{1}{2\pi i} \int_{\mathcal{H}} \mathcal{G} ds_2, \quad (2.4.13)$$

where $\mathcal{H}$ is the union of the horizontal sides of the rectangle. Here and in the sequel, $\operatorname{Res}_{s=\omega} f$ denotes the residue at $s = \omega$ of a function $f(s)$. We multiply both sides of (2.4.13) by $(2\pi i)^{-1}$ and integrate with respect to s_1 along E_1 to arrive at

$$\begin{aligned} \frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} \mathcal{G} ds_2 ds_1 &= \frac{1}{2\pi i} \int_{E_1} \left\{ \operatorname{Res}_{s_2=0} \mathcal{G} + \operatorname{Res}_{s_2=-s_1} \mathcal{G} \right\} ds_1 \\ &\quad + \frac{1}{(2\pi i)^2} \int_{E_1} \int_{D_2} \mathcal{G} ds_2 ds_1 + \frac{1}{(2\pi i)^2} \int_{E_1} \int_{\mathcal{H}} \mathcal{G} ds_2 ds_1. \end{aligned} \quad (2.4.14)$$

Using Lemma 2.2.2, (2.4.8), and the fact that $|s_2| \geq |t_2| = 2L^5$ for $s_2 \in \mathcal{H}$, we can easily prove that

$$\frac{1}{(2\pi i)^2} \int_{E_1} \int_{\mathcal{H}} \mathcal{G} ds_2 ds_1 \ll L^{-A}. \quad (2.4.15)$$

Similarly, applying Lemma 2.2.2, (2.4.8), and the bound $y^{s_2} \ll y^{-c/\log L}$ for $s_2 \in D_2$, we can show that

$$\frac{1}{(2\pi i)^2} \int_{E_1} \int_{D_2} \mathcal{G} ds_2 ds_1 \ll L^A y^{-c/\log L} \ll L^{-A}. \quad (2.4.16)$$

It now follows from (2.4.12), (2.4.14), (2.4.15), and (2.4.16) that

$$\mathbf{S}(\varrho) = \frac{4}{2\pi i} \int_{E_1} \left\{ \operatorname{Res}_{s_2=0} \mathcal{G} + \operatorname{Res}_{s_2=-s_1} \mathcal{G} \right\} ds_1 + O(L^{-A}). \quad (2.4.17)$$

Now we estimate the integral on the right-hand side of (2.4.17). We first bound the integral of the residue of $\mathcal{G}$ at $s_2 = -s_1$. From the definition (2.4.10) of $\mathcal{G}$ and the fact that $\operatorname{Res}_{s=1} \zeta = 1$, it follows that

$$\operatorname{Res}_{s_2=-s_1} \mathcal{G} = -\frac{G(s_1, -s_1, \varrho)}{s_1^6 \zeta(1+s_1) \zeta(1-s_1+i\varrho)}. \quad (2.4.18)$$

Let Γ_1 be the path defined by the equation

$$\Gamma_1(t_1) = \frac{c}{\log(|t_1|+3)} + it_1, \quad -L^5 \leq t_1 \leq L^5.$$

We move E_1 to Γ_1 , i.e. we integrate $\operatorname{Res}_{s_2=-s_1} \mathcal{G}$ along the closed curve obtained by connecting the endpoints of E_1 and Γ_1 with horizontal line segments. By Cauchy's theorem, the result is

$$\int_{E_1} \operatorname{Res}_{s_2=-s_1} \mathcal{G} ds_1 = \int_{\Gamma_1} \operatorname{Res}_{s_2=-s_1} \mathcal{G} ds_1 + O(L^{-A}), \quad (2.4.19)$$

where we have bounded the integrals along the horizontal line segments using (2.4.18), (2.4.8), Lemma 2.2.2, and the fact that $|s_1|^6 \geq |t_1|^6 = L^{30}$ for s_1 on those line segments. We can also bound the integral along Γ_1 via (2.4.18), (2.4.8), and Lemma 2.2.2 to deduce that

$$\int_{\Gamma_1} \operatorname{Res}_{s_2=-s_1} \mathcal{G} ds_1 \ll 1.$$

From this and (2.4.19), it follows that

$$\int_{E_1} \operatorname{Res}_{s_2=-s_1} \mathcal{G} ds_1 \ll 1. \quad (2.4.20)$$

We next estimate the integral of $\text{Res}_{s_2=0}\mathcal{G}$ on the right-hand side of (2.4.17). By the residue theorem and the definition (2.4.10) of $\mathcal{G}$, we can write

$$\text{Res}_{s_2=0}\mathcal{G} = \frac{1}{2\pi i} \oint_{\mathcal{C}_2} \frac{y^{s_1+s_2}}{s_1^3 s_2^3} \frac{\zeta(1+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2+i\varrho)} G(s_1, s_2, \varrho) ds_2, \quad (2.4.21)$$

where $\mathcal{C}_2$ is the circle $|s_2| = 1/2L$, taken once in the positive direction. Let D_1 be the line segment from $-c(\log L)^{-1} - iL^5$ to $-c(\log L)^{-1} + iL^5$. We move E_1 to D_1 . That is, we integrate $\text{Res}_{s_2=0}\mathcal{G}$ along the rectangle determined by E_1 and D_1 and use the residue theorem. We bound the integral along the horizontal line segments using (2.4.21), (2.4.8), Lemma 2.2.2, and the fact that $|s_1| \geq |t_1| = L^5$, while we estimate the integral along D_1 via (2.4.21), (2.4.8), Lemma 2.2.2, and the bound $y^{s_1} \ll y^{-c/\log L} \ll L^{-A}$. The result is

$$\frac{1}{2\pi i} \int_{E_1} \text{Res}_{s_2=0}\mathcal{G} ds_1 = \text{Res}_{s_1=0} \text{Res}_{s_2=0}\mathcal{G} + O(L^{-A}).$$

From this, (2.4.20), and (2.4.17), it follows that

$$\mathbf{S}(\varrho) = 4 \text{Res}_{s_1=0} \text{Res}_{s_2=0}\mathcal{G} + O(1). \quad (2.4.22)$$

It is left to evaluate the main term on the right-hand side. By (2.4.21) and the residue theorem, we can write

$$\text{Res}_{s_1=0} \text{Res}_{s_2=0}\mathcal{G} = \frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \frac{y^{s_1+s_2}}{s_1^3 s_2^3} \frac{\zeta(1+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2+i\varrho)} G(s_1, s_2, \varrho) ds_2 ds_1, \quad (2.4.23)$$

where $\mathcal{C}_1$ is the circle $|s_1| = 1/L$, taken once in the positive direction. If $f(z)$ is analytic and bounded for $|z| < \frac{1}{8}$, then Cauchy's integral formula gives

$$f(s) = f(0) + \frac{s}{2\pi i} \oint_{|z|=\frac{3}{32}} \frac{f(z)}{z(z-s)} dz = f(0) + O(|s|) \quad (2.4.24)$$

uniformly for $|s| < 1/16$. By (2.4.8) and uniform convergence over compact sets, we see that $G(s_1, s_2, z)$ is analytic in each of its variables for $|s_1|, |s_2|, |z| < \frac{1}{8}$. Thus we can apply (2.4.24) repeatedly to $G(s_1, s_2, \varrho)$ to deduce for $s_1, s_2, \varrho \ll 1/L$ that

$$G(s_1, s_2, \varrho) = G(0, s_2, \varrho) + O(L^{-1}) = G(0, 0, \varrho) + O(L^{-1}) = G(0, 0, 0) + O(L^{-1}).$$

A straightforward computation using the definition (2.4.7) of $G(s_1, s_2, \varrho)$ shows us that $G(0, 0, 0) = 1$. Thus

$$G(s_1, s_2, \varrho) = 1 + O(L^{-1})$$

for $s_1, s_2, \varrho \ll 1/L$. Also, for the same s_1, s_2 , and ϱ , we have the Laurent series expansions

$$\begin{aligned} \zeta(1 + s_1 + s_2) &= \frac{1}{s_1 + s_2} + O(1), \\ \frac{1}{\zeta(1 + s_1)} &= s_1 + O(|s_1|^2) = s_1 + O(L^{-2}), \end{aligned}$$

and

$$\frac{1}{\zeta(1 + s_2 + i\varrho)} = s_2 + i\varrho + O(|s_2 + i\varrho|^2) = s_2 + i\varrho + O(L^{-2}).$$

We insert all these into (2.4.23) and arrive at

$$\operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} \mathcal{G} = \frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \frac{y^{s_1+s_2}}{s_1^2 s_2^3} \left(\frac{s_2 + i\varrho}{s_1 + s_2} \right) ds_2 ds_1 + O(L^2) \quad (2.4.25)$$

for real ϱ such that $\varrho \ll 1/L$.

Next, we use the fact that

$$\frac{1}{s_1 + s_2} = \frac{1}{s_1(1 + s_2/s_1)} = \frac{1}{s_1} - \frac{s_2}{s_1^2} + \frac{s_2^2}{s_1^3} + \dots$$

to deduce from (2.4.25) that

$$\begin{aligned} &\operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} \mathcal{G} \\ &= \frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} y^{s_1+s_2} \left(\frac{1}{s_1^3 s_2^2} - \frac{1}{s_1^4 s_2} + \frac{i\varrho}{s_1^3 s_2^3} - \frac{i\varrho}{s_1^4 s_2^2} + \frac{i\varrho}{s_1^5 s_2} + \dots \right) ds_2 ds_1 \\ &\quad + O(L^2) \end{aligned} \quad (2.4.26)$$

for $\varrho \ll 1/L$. For integers j and k , it follows from the residue theorem that

$$\frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \frac{y^{s_1+s_2}}{s_1^j s_2^k} ds_2 ds_1 = \begin{cases} \frac{(\log y)^{j+k-2}}{(j-1)!(k-1)!} & \text{if } j, k \geq 1, \\ 0 & \text{else.} \end{cases}$$

From this and (2.4.26), we arrive at

$$\begin{aligned} \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} \mathcal{G} &= \frac{(\log y)^3}{2} - \frac{(\log y)^3}{6} + \frac{i\varrho(\log y)^4}{4} - \frac{i\varrho(\log y)^4}{6} + \frac{i\varrho(\log y)^4}{24} + O(L^2) \\ &= \frac{\log^3 y}{3} + \frac{i\varrho \log^4 y}{8} + O(L^2) \end{aligned}$$

for real ϱ with $\varrho \ll 1/L$. We insert this into (2.4.22) to conclude that

$$\mathbf{S}(\varrho) = \frac{4 \log^3 y}{3} + \frac{i\varrho \log^4 y}{2} + O(L^2) \quad (2.4.27)$$

for real ϱ with $\varrho \ll 1/L$. This completes our evaluation of $\mathbf{S}(\varrho)$.

We now finish our proof of Theorem 1.1.2. By (2.2.5), if $T/2 \leq t \leq 4T$ and $\varrho = t' - t$ or $\varrho = -t' + t$, then $\varrho \ll 1/L$ and so (2.4.27) holds for such ϱ . It follows from this and (2.4.2) that

$$\begin{aligned} \mathcal{I} &= \frac{e^{ia}}{\log^4 y} \int_{-\infty}^{\infty} \zeta(1 + it' - it) w(t) \left(\frac{4 \log^3 y}{3} + \frac{i(t' - t) \log^4 y}{2} + O(L^2) \right) dt \\ &\quad + \frac{e^{-ia}}{\log^4 y} \int_{-\infty}^{\infty} \zeta(1 - it' + it) w(t) \left(\frac{4 \log^3 y}{3} - \frac{i(t' - t) \log^4 y}{2} + O(L^2) \right) dt \\ &\quad + O \left(y^{\ell+1} \frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} \right) + O(y^{1/2} T^{3/4} \log^3 T). \end{aligned}$$

For brevity, write this as

$$\mathcal{I} = \mathcal{I}_1 + \mathcal{I}_2 + O \left(y^{\ell+1} \frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell} \right) + O(y^{1/2} T^{3/4} \log^3 T). \quad (2.4.28)$$

By (2.2.5) and the Laurent series expansion of $\zeta(s)$ near $s = 1$, it is true that

$$\zeta(1 + it' - it) = \frac{\log \frac{t}{2\pi}}{i2a} + O(1)$$

and

$$i(t' - t)\zeta(1 + it' - it) = 1 + O(L^{-1}).$$

We insert these into the definition of $\mathcal{I}_1$ to deduce that

$$\mathcal{I}_1 = \frac{2e^{ia}}{i3a \log y} \int_{-\infty}^{\infty} w(t) \log \frac{t}{2\pi} dt + \frac{e^{ia}}{2} W + O\left(\frac{W}{L}\right), \quad (2.4.29)$$

where W is defined by (1.1.5). Since the support of $w(t)$ is contained in $[T/2, 4T]$, we can write

$$\int_{-\infty}^{\infty} w(t) \log \frac{t}{2\pi} dt = W \log T + O(W).$$

From this and the definition $y = T^\theta$, we see that (2.4.29) simplifies to

$$\mathcal{I}_1 = \frac{2e^{ia}}{i3a\theta} W + \frac{e^{ia}}{2} W + O\left(\frac{W}{L}\right). \quad (2.4.30)$$

A similar argument leads to

$$\mathcal{I}_2 = -\frac{2e^{-ia}}{i3a\theta} W + \frac{e^{-ia}}{2} W + O\left(\frac{W}{L}\right).$$

From this, (2.4.30), and (2.4.28), we conclude that

$$\begin{aligned} \mathcal{I} &= \frac{4W}{3} \left(\frac{\sin a}{a\theta} \right) + W \cos a + O\left(\frac{W}{L}\right) \\ &\quad + O\left(y^{\ell+1} \frac{T^{\frac{3}{2} + \frac{\ell}{2} + \varepsilon}}{T_0^\ell}\right) + O(y^{1/2} T^{3/4} \log^3 T). \end{aligned}$$

This completes the proof of Theorem 1.1.2.

2.5 Application of the theorem of Hughes and Young

Our main tool in proving Theorem 1.1.3 is the formula for the twisted fourth moment of $\zeta(s)$ that is due to Hughes and Young [40, Theorem 1.1]. In this section, we apply their theorem to write the mollified fourth moment in terms of an integral that can be evaluated using residue theory. We assume that $w(t)$ is a function satisfying the properties in the hypothesis of Theorem 1.1.3.

We first express the formula of Hughes and Young in a form that is easier to handle. To do this, we make a few definitions. Let Ξ be the set of permu-

tations σ of $\{1, 2, 3, 4\}$ such that $\sigma(1) < \sigma(2)$ and $\sigma(3) < \sigma(4)$. For a function $K(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ of four variables, define

$$(\Psi K)(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \sum_{\sigma \in \Xi} K(\alpha_{\sigma(1)}, \alpha_{\sigma(2)}, \alpha_{\sigma(3)}, \alpha_{\sigma(4)}).$$

Let $\sigma_{\alpha, \beta}(\ell) = \sum_{mn=\ell} m^{-\alpha} n^{-\beta}$, and for a positive integer h , let h_p be the exponent of p in the prime factorization $h = \prod_p p^{h_p}$. As in Hughes and Young [40], we define

$$A_{\alpha, \beta, \gamma, \delta}(0) = \frac{\zeta(1 + \alpha + \gamma) \zeta(1 + \alpha + \delta) \zeta(1 + \beta + \gamma) \zeta(1 + \beta + \delta)}{\zeta(2 + \alpha + \beta + \gamma + \delta)}, \quad (2.5.1)$$

$$\begin{aligned} B_{\alpha, \beta, \gamma, \delta, h, k}(0) &= \prod_{p|h} \left(\frac{\sum_{j=0}^{\infty} \sigma_{\alpha, \beta}(p^j) \sigma_{\gamma, \delta}(p^{j+h_p}) p^{-j}}{\sum_{j=0}^{\infty} \sigma_{\alpha, \beta}(p^j) \sigma_{\gamma, \delta}(p^j) p^{-j}} \right) \\ &\quad \times \prod_{p|k} \left(\frac{\sum_{j=0}^{\infty} \sigma_{\alpha, \beta}(p^{j+k_p}) \sigma_{\gamma, \delta}(p^j) p^{-j}}{\sum_{j=0}^{\infty} \sigma_{\alpha, \beta}(p^j) \sigma_{\gamma, \delta}(p^j) p^{-j}} \right), \end{aligned} \quad (2.5.2)$$

and

$$Z_{\alpha, \beta, \gamma, \delta, h, k}(0) = A_{\alpha, \beta, \gamma, \delta}(0) B_{\alpha, \beta, \gamma, \delta, h, k}(0). \quad (2.5.3)$$

Since the value of $Z_{\alpha, \beta, \gamma, \delta, h, k}(0)$ does not change when α and β are interchanged or when γ and δ are interchanged, we can write the conclusion of Theorem 1.1 of Hughes and Young [40] in the form (note the minus signs before γ and δ)

$$\begin{aligned} &\int_{-\infty}^{\infty} \left(\frac{h}{k} \right)^{-it} \zeta\left(\frac{1}{2} + \alpha + it\right) \zeta\left(\frac{1}{2} + \beta + it\right) \zeta\left(\frac{1}{2} - \gamma - it\right) \zeta\left(\frac{1}{2} - \delta - it\right) w(t) dt \\ &= \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} w(t) \left(\frac{t}{2\pi} \right)^{-\frac{\alpha + \beta - \gamma - \delta}{2}} (\Psi K_0)(\alpha, \beta, \gamma, \delta) dt \\ &\quad + O(T^{3/4+\varepsilon} (hk)^{7/8} (T/T_0)^{9/4}), \end{aligned} \quad (2.5.4)$$

where

$$K_0(\alpha, \beta, \gamma, \delta) = \left(\frac{t}{2\pi} \right)^{\frac{\alpha + \beta - \gamma - \delta}{2}} Z_{\alpha, \beta, -\gamma, -\delta, h, k}(0).$$

As stated in the hypothesis of Theorem 1.1 of Hughes and Young [40], the formula (2.5.4) holds when $\alpha, \beta, \gamma, \delta$ are complex numbers that are $\ll 1/L$ (recall that we use $L = \log T$) and h and k are relatively prime positive integers such that $hk \leq T^{2/11-\varepsilon}$.

To be able to use (2.5.4) in our proof of Theorem 1.1.3, we replace each of the variables $\alpha, \beta, \gamma, \delta$ by itself plus $i\rho$, where

$$\rho = \frac{2a}{\log \frac{T}{2\pi}} \quad (2.5.5)$$

and a is a fixed real number. The result is

$$\begin{aligned} & \int_{-\infty}^{\infty} \left(\frac{h}{k} \right)^{-it} \zeta\left(\frac{1}{2} + \alpha + i\rho + it\right) \zeta\left(\frac{1}{2} + \beta + i\rho + it\right) \\ & \quad \times \zeta\left(\frac{1}{2} - \gamma - i\rho - it\right) \zeta\left(\frac{1}{2} - \delta - i\rho - it\right) w(t) dt \\ & = \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} w(t) \left(\frac{t}{2\pi} \right)^{-\frac{\alpha+\beta-\gamma-\delta}{2}} (\Psi K_{\rho})(\alpha, \beta, \gamma, \delta) dt \\ & \quad + O(T^{3/4+\varepsilon} (hk)^{7/8} (T/T_0)^{9/4}), \end{aligned} \quad (2.5.6)$$

where $K_{\rho}(\alpha, \beta, \gamma, \delta)$ is defined by

$$K_{\rho}(\alpha, \beta, \gamma, \delta) = K_{\rho}(\alpha, \beta, \gamma, \delta; h, k) = \left(\frac{t}{2\pi} \right)^{\frac{\alpha+\beta-\gamma-\delta}{2}} Z_{\alpha+i\rho, \beta+i\rho, -\gamma-i\rho, -\delta-i\rho, h, k}(0). \quad (2.5.7)$$

We next express the factor $(\Psi K_{\rho})(\alpha, \beta, \gamma, \delta)$ in (2.5.6) as an integral. To do this, we use the following lemma, which is a special case of Lemma 2.5.1 of Conrey et al. [17].

Lemma 2.5.1. *Let f be analytic in a neighborhood of $s = 0$ except for a simple pole of residue 1 at $s = 0$. Suppose that $F(\alpha, \beta, \gamma, \delta)$ is a function that is analytic in each variable in a neighborhood of $(0, 0, 0, 0)$ such that $F(\alpha, \beta, \gamma, \delta) = F(\beta, \alpha, \gamma, \delta)$ and $F(\alpha, \beta, \gamma, \delta) = F(\alpha, \beta, \delta, \gamma)$. Let*

$$K(\alpha, \beta, \gamma, \delta) = F(\alpha, \beta, \gamma, \delta) f(\alpha - \gamma) f(\alpha - \delta) f(\beta - \gamma) f(\beta - \delta).$$

If $\alpha \neq \gamma, \delta$ and $\beta \neq \gamma, \delta$, then

$$\begin{aligned} & (\Psi K)(\alpha, \beta, \gamma, \delta) \\ &= \frac{1}{4(2\pi i)^4} \oint \oint \oint \oint \frac{K(z_1, z_2, z_3, z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell)^2}{\prod_{j=1}^4 (z_j - \alpha)(z_j - \beta)(z_j - \gamma)(z_j - \delta)} dz_4 dz_3 dz_2 dz_1, \end{aligned}$$

where the path of integration is the same for all four integrals and encircles each of the four points $\alpha, \beta, \gamma, \delta$ exactly once in the positive direction. In particular, ΨK is analytic at $(0, 0, 0, 0)$.

Now let $\omega \neq 0$ and set $\alpha = \beta = \omega$ and $\gamma = \delta = 2\omega$ in (2.5.6). We apply Lemma 2.5.1 to the right-hand side of the resulting equation, and then take the limit of both sides as $\omega \rightarrow 0$. The result we arrive at is stated in the following proposition.

Proposition 2.5.1. *Let $w(t)$ be a function satisfying the conditions in the hypothesis of Theorem 1.1.3. Suppose that h and k are relatively prime positive integers with $hk \leq T^{2/11-\varepsilon}$. If ρ is defined by (2.5.5), then*

$$\begin{aligned} & \int_{-\infty}^{\infty} \left(\frac{h}{k}\right)^{-it} \zeta\left(\frac{1}{2} + i\rho + it\right) \zeta\left(\frac{1}{2} + i\rho + it\right) \zeta\left(\frac{1}{2} - i\rho - it\right) \zeta\left(\frac{1}{2} - i\rho - it\right) w(t) dt \\ &= \frac{1}{\sqrt{hk}} \int_{-\infty}^{\infty} \frac{w(t)}{4(2\pi i)^4} \oint \oint \oint \oint K_{\rho}(z_1, z_2, z_3, z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell)^2 \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \\ & \quad + O(T^{3/4+\varepsilon} (hk)^{7/8} (T/T_0)^{9/4}), \end{aligned} \tag{2.5.8}$$

where the path of integration of each line integral encircles 0 exactly once in the positive direction.

In view of Proposition 2.5.1, we can now write the mollified fourth moment in terms of path integrals. For brevity, we define

$$\mathcal{J} = \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + i\rho + it)|^4 |M(t)|^4 w(t) dt,$$

where $0 < \theta < 1/22$ in the definition (1.1.4) of $M(t)$. We start by inserting the definition of $M(t)$ into the above equation. Since $\theta < 1/22$, we can apply Proposition 2.5.1 to the resulting expression and deduce that

$$\begin{aligned} \mathcal{J} = & \sum_{n_1, n_2, n_3, n_4 \leq y} \left\{ \prod_{j=1}^4 \frac{\mu(n_j)}{n_j^{1/2}} \left(\frac{\log(y/n_j)}{\log y} \right)^2 \right\} \frac{1}{\sqrt{(n_1 n_2)' (n_3 n_4)'}} \\ & \times \int_{-\infty}^{\infty} \frac{w(t)}{4(2\pi i)^4} \oint \oint \oint \oint K_{\rho}(z_1, z_2, z_3, z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_{\ell})^2 \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \\ & + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned} \quad (2.5.9)$$

where $(n_1 n_2)' = n_1 n_2 / (n_1 n_2, n_3 n_4)$ and $(n_3 n_4)' = n_3 n_4 / (n_1 n_2, n_3 n_4)$. Note that we have trivially bounded the contribution of the error term in (2.5.8). Using the definition (2.5.7) of $K_{\rho}(z_1, z_2, z_3, z_4)$ with $h = (n_1 n_2)'$ and $k = (n_3 n_4)'$ and the definition (2.5.3) of Z , we rearrange the factors in (2.5.9) and write it as

$$\begin{aligned} \mathcal{J} = & \int_{-\infty}^{\infty} \frac{w(t)}{4(2\pi i)^4} \oint \oint \oint \oint \left(\frac{t}{2\pi} \right)^{\frac{z_1 + z_2 - z_3 - z_4}{2}} A_{z_1, z_2, -z_3, -z_4}(0) \prod_{1 \leq j < \ell \leq 4} (z_j - z_{\ell})^2 \\ & \times \left(\frac{16}{(\log y)^8} \right) Q(z_1 + i\rho, z_2 + i\rho, -z_3 - i\rho, -z_4 - i\rho) \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \\ & + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned} \quad (2.5.10)$$

where $Q(\alpha, \beta, \gamma, \delta) = Q(\alpha, \beta, \gamma, \delta; y)$ is defined by

$$\begin{aligned} & Q(\alpha, \beta, \gamma, \delta) \\ = & \sum_{n_1, n_2, n_3, n_4 \leq y} \left\{ \prod_{j=1}^4 \frac{\mu(n_j)}{n_j} \left(\frac{\log^2(y/n_j)}{2} \right) \right\} (n_1 n_2, n_3 n_4) B_{\alpha, \beta, \gamma, \delta, (n_1 n_2)', (n_3 n_4)'}(0). \end{aligned} \quad (2.5.11)$$

To facilitate later discussions, we choose the path of integration of each line integral in (2.5.10) to be a circle with center 0 and radius $1/L$.

2.6 Evaluation of Q

The purpose of this and the next few sections is to evaluate the sum $Q(\alpha, \beta, \gamma, \delta)$ defined by (2.5.11). We may assume throughout that $\alpha, \beta, \gamma, \delta \ll 1/L$ since we will use our formula for $Q(\alpha, \beta, \gamma, \delta)$ only to estimate the right-hand side of (2.5.10).

We apply Perron's formula (2.4.4) to deduce from (2.5.11) that

$$Q(\alpha, \beta, \gamma, \delta) = \frac{1}{(2\pi i)^4} \int_{(\eta_1)} \int_{(\eta_2)} \int_{(\eta_3)} \int_{(\eta_4)} \frac{y^{s_1+s_2+s_3+s_4}}{s_1^3 s_2^3 s_3^3 s_4^3} \\ \times \sum_{1 \leq n_1, n_2, n_3, n_4 < \infty} \left\{ \prod_{j=1}^4 \frac{\mu(n_j)}{n_j^{1+s_j}} \right\} (n_1 n_2, n_3 n_4) B_{\alpha, \beta, \gamma, \delta, (n_1 n_2)', (n_3 n_4)'}(0) \prod_{j=1}^4 ds_j, \quad (2.6.1)$$

where the interchange of order of summation is valid by absolute convergence. For ease in later discussions, we choose the lines of integration in (2.6.1) to have $\eta_j = c_j/L$ for $j = 1, 2, 3, 4$, where c_1, c_2, c_3 , and c_4 are positive constants that satisfy the inequalities

$$|\alpha|, |\beta|, |\gamma|, |\delta| < \frac{c_1}{L} < \frac{c_2}{L} < \frac{c_3}{L} < \frac{c_4}{L}. \quad (2.6.2)$$

We next write the inner sum in (2.6.1) as an Euler product. Let

$$f(n_1, n_2, n_3, n_4) = \left\{ \prod_{j=1}^4 \frac{\mu(n_j)}{n_j^{1+s_j}} \right\} (n_1 n_2, n_3 n_4) B_{\alpha, \beta, \gamma, \delta, (n_1 n_2)', (n_3 n_4)'}(0). \quad (2.6.3)$$

It follows from the definition (2.5.2) and from the multiplicativity of the gcd function that f is multiplicative, i.e.

$$f(h_1 j_1, h_2 j_2, h_3 j_3, h_4 j_4) = f(h_1, h_2, h_3, h_4) \cdot f(j_1, j_2, j_3, j_4)$$

whenever $h_1 h_2 h_3 h_4$ and $j_1 j_2 j_3 j_4$ are relatively prime. Hence, it follows from Lemma 2.2.1 with $k = 4$ that

$$\sum_{1 \leq n_1, n_2, n_3, n_4 < \infty} f(n_1, n_2, n_3, n_4) = \prod_p \left(\sum_{0 \leq \ell_1, \ell_2, \ell_3, \ell_4 < \infty} f(p^{\ell_1}, p^{\ell_2}, p^{\ell_3}, p^{\ell_4}) \right), \quad (2.6.4)$$

so long as either side is absolutely convergent. We will see in what follows that the right-hand side is absolutely convergent when α, β, γ , and δ are small and the real part σ_j of s_j is larger than each of $|\alpha|, |\beta|, |\gamma|, |\delta|$ for $j = 1, 2, 3, 4$. Observe that $f(n_1, n_2, n_3, n_4)$ is zero when at least one of n_1, n_2, n_3, n_4 is divisible by the square of a prime. Thus, to evaluate the right-hand side of (2.6.4), we only need to calculate $f(p^{\ell_1}, p^{\ell_2}, p^{\ell_3}, p^{\ell_4})$ for $0 \leq \ell_1, \ell_2, \ell_3, \ell_4 \leq 1$. To do this, we use the following lemma, which is a special case of Lemma 6.9 of Hughes and Young [40]. Recall the definition (2.5.2) of $B_{\alpha, \beta, \gamma, \delta, h, k}(0)$.

Lemma 2.6.1. *For small enough $\alpha, \beta, \gamma, \delta$, it is true that*

$$B_{\alpha, \beta, \gamma, \delta, h, k}(0) = B_{\alpha, \beta, \gamma, \delta, h} B_{\gamma, \delta, \alpha, \beta, k},$$

where $B_{\alpha, \beta, \gamma, \delta, h}$ is defined by

$$B_{\alpha, \beta, \gamma, \delta, h} = \prod_{p|h} \left(\frac{B^{(0)} - p^{-1}B^{(1)} + p^{-2}B^{(2)}}{(p^{-\gamma} - p^{-\delta})(1 - p^{-2-\alpha-\beta-\gamma-\delta})} \right),$$

where

$$\begin{aligned} B^{(0)} &= p^{-\gamma(1+h_p)} - p^{-\delta(1+h_p)}, \\ B^{(1)} &= (p^{-\alpha} + p^{-\beta})p^{-\gamma-\delta}(p^{-\gamma h_p} - p^{-\delta h_p}), \\ B^{(2)} &= p^{-\alpha-\beta-\gamma-\delta}(p^{-\delta-\gamma h_p} - p^{-\gamma-\delta h_p}), \end{aligned}$$

and $h = \prod_p p^{h_p}$ is the prime factorization of h .

Using Lemma 2.6.1 and the definition (2.6.3), we can evaluate each of the sixteen quantities $f(p^{\ell_1}, p^{\ell_2}, p^{\ell_3}, p^{\ell_4})$ with $0 \leq \ell_1, \ell_2, \ell_3, \ell_4 \leq 1$. For example, if $\alpha, \beta, \gamma, \delta$ are small enough, then a straightforward calculation shows that

$$\begin{aligned} f(p, 1, 1, 1) &= \left\{ \frac{-1}{p^{1+s_1}} \right\} \left\{ \frac{p^{-2\gamma} - p^{-2\delta} - p^{-1}(p^{-\alpha} + p^{-\beta})p^{-\gamma-\delta}(p^{-\gamma} - p^{-\delta})}{(p^{-\gamma} - p^{-\delta})(1 - p^{-2-\alpha-\beta-\gamma-\delta})} \right\} \\ &= -\frac{1}{p^{1+s_1+\gamma}} - \frac{1}{p^{1+s_1+\delta}} + O\left(\frac{1}{p^{3/2}}\right) \end{aligned}$$

for $\sigma_1 > -1/8$, say, with absolute implied constant. Similarly, we can evaluate each of the other fifteen quantities in a straightforward way. The details are left to the reader. From the results of these calculations, we deduce that if $\alpha, \beta, \gamma, \delta$ are small enough, then

$$\begin{aligned} & \sum_{0 \leq \ell_1, \ell_2, \ell_3, \ell_4 < \infty} f(p^{\ell_1}, p^{\ell_2}, p^{\ell_3}, p^{\ell_4}) \\ &= 1 - \frac{1}{p^{1+s_1+\gamma}} - \frac{1}{p^{1+s_1+\delta}} - \frac{1}{p^{1+s_2+\gamma}} - \frac{1}{p^{1+s_2+\delta}} - \frac{1}{p^{1+s_3+\alpha}} - \frac{1}{p^{1+s_3+\beta}} \\ & \quad - \frac{1}{p^{1+s_4+\alpha}} - \frac{1}{p^{1+s_4+\beta}} + \frac{1}{p^{1+s_1+s_3}} + \frac{1}{p^{1+s_1+s_4}} + \frac{1}{p^{1+s_2+s_3}} + \frac{1}{p^{1+s_2+s_4}} + O\left(\frac{1}{p^{3/2}}\right) \end{aligned}$$

for prime p and complex s_1, s_2, s_3, s_4 having real parts $\sigma_j > -1/8$. Therefore, if $\sigma_j > \max\{|\alpha|, |\beta|, |\gamma|, |\delta|\}$ for $j = 1, 2, 3, 4$, then

$$\begin{aligned} & \prod_p \left(\sum_{0 \leq \ell_1, \ell_2, \ell_3, \ell_4 < \infty} f(p^{\ell_1}, p^{\ell_2}, p^{\ell_3}, p^{\ell_4}) \right) \\ &= \zeta^{-1}(1+s_1+\gamma) \zeta^{-1}(1+s_1+\delta) \zeta^{-1}(1+s_2+\gamma) \zeta^{-1}(1+s_2+\delta) \\ & \quad \times \zeta^{-1}(1+s_3+\alpha) \zeta^{-1}(1+s_3+\beta) \zeta^{-1}(1+s_4+\alpha) \zeta^{-1}(1+s_4+\beta) \\ & \quad \times \zeta(1+s_1+s_3) \zeta(1+s_1+s_4) \zeta(1+s_2+s_3) \zeta(1+s_2+s_4) F(s_1, s_2, s_3, s_4), \end{aligned} \tag{2.6.5}$$

where $F(s_1, s_2, s_3, s_4) = F(s_1, s_2, s_3, s_4; \alpha, \beta, \gamma, \delta)$ is an Euler product satisfying

$$F(s_1, s_2, s_3, s_4) = \prod_p \left(1 + O\left(\frac{1}{p^{3/2}}\right) \right) \tag{2.6.6}$$

uniformly for all small enough $\alpha, \beta, \gamma, \delta$ and all complex s_1, s_2, s_3, s_4 that have real parts $\sigma_j > -1/8$.

In view of the assumption (2.6.2) and the definition (2.6.3), we now deduce from (2.6.1), (2.6.4), and (2.6.5) that

$$Q(\alpha, \beta, \gamma, \delta) = \frac{1}{(2\pi i)^4} \int_{(\eta_1)} \int_{(\eta_2)} \int_{(\eta_3)} \int_{(\eta_4)} \mathcal{F}(s_1, s_2, s_3, s_4) ds_4 ds_3 ds_2 ds_1, \tag{2.6.7}$$

where

$$\begin{aligned}
\mathcal{F}(s_1, s_2, s_3, s_4) &= \mathcal{F}(s_1, s_2, s_3, s_4; \alpha, \beta, \gamma, \delta, y) = \frac{y^{s_1+s_2+s_3+s_4}}{s_1^3 s_2^3 s_3^3 s_4^3} \times \\
&\times \zeta^{-1}(1+s_1+\gamma) \zeta^{-1}(1+s_1+\delta) \zeta^{-1}(1+s_2+\gamma) \zeta^{-1}(1+s_2+\delta) \\
&\times \zeta^{-1}(1+s_3+\alpha) \zeta^{-1}(1+s_3+\beta) \zeta^{-1}(1+s_4+\alpha) \zeta^{-1}(1+s_4+\beta) \\
&\times \zeta(1+s_1+s_3) \zeta(1+s_1+s_4) \zeta(1+s_2+s_3) \zeta(1+s_2+s_4) F(s_1, s_2, s_3, s_4).
\end{aligned} \tag{2.6.8}$$

Recall that we have chosen $\eta_j = c_j/L$ for $j = 1, 2, 3, 4$, where the constants c_j satisfy the assumption (2.6.2).

2.7 Initial shifting of contours

To continue with our evaluation of $Q(\alpha, \beta, \gamma, \delta)$, the idea is to move the lines of integration in (2.6.7) and write $Q(\alpha, \beta, \gamma, \delta)$ in terms of the residues of $\mathcal{F}(s_1, s_2, s_3, s_4)$ at its poles. This turns out to be a rather involved process, and will occupy us for the next few sections. Recall that we use the notation $s_j = \sigma_j + it_j$, that $L = \log T$, and that A denotes a positive absolute constant that is not necessarily the same for each instance. Recall also that we are assuming (2.6.2). The more trivial estimations will be left to the reader.

We first truncate the lines of integration in (2.6.7), as follows. For $m = 1, 2, 3, 4$, let E_m be the line segment from $\eta_m - imL^{17}$ to $\eta_m + imL^{17}$. The exponent 17 in L^{17} is arbitrary, and the discussions that follow remain valid if we replace 17 by any larger real number. Since $\eta_m = c_m/L$ and $y = T^\theta$, it follows that $y^{s_j} \ll 1$ when $\sigma_j = \eta_j$. Using this, Lemma 2.2.2, and (2.6.6), we can deduce from (2.6.7) and (2.6.8) that

$$Q = \frac{1}{(2\pi i)^4} \int_{E_1} \int_{E_2} \int_{E_3} \int_{E_4} \mathcal{F}(s_1, s_2, s_3, s_4) ds_4 ds_3 ds_2 ds_1 + O(L^{-A}). \tag{2.7.1}$$

The next step is to move E_4 to the left and write Q in terms of an integral of residues of $\mathcal{F}$. Let $c > 0$ be an absolute constant that is so small that we

can apply Lemma 2.2.2 to the zeta-functions in the definition (2.6.8) of $\mathcal{F}$ when $-c(\log L)^{-1} \leq \sigma_4 \leq 1$ and $s_j \in E_j$ for $j = 1, 2, 3$. We can take $c = c_0/99$, say, where c_0 is as in Lemma 2.2.2. Let D_4 be the line segment from $-c(\log L)^{-1} - i4L^{17}$ to $-c(\log L)^{-1} + i4L^{17}$. We move E_4 to D_4 and proceed in a way similar to that in the above discussion containing equations (2.4.13) through (2.4.16). That is, we integrate $\mathcal{F}$ with respect to s_4 along the rectangle determined by E_4 and D_4 . We use the residue theorem to write that the integral equals a sum of residues of $\mathcal{F}$. Then we integrate both sides of the resulting equation with respect to s_3 , s_2 , and s_1 along E_3 , E_2 , and E_1 , respectively. The contribution of the integrals along the horizontal sides of the rectangle is small because of Lemma 2.2.2, (2.6.6), the bound $y^{s_1+s_2+s_3+s_4} \ll 1$, and the small size of the factor s_4^{-3} of $\mathcal{F}$. Similarly, the contribution of the integral along D_4 is small because $y^{s_4} \ll y^{-c/\log L}$ for $s_4 \in D_4$. The result we arrive at from (2.7.1) and this process of moving E_4 to D_4 is

$$Q = \frac{1}{(2\pi i)^3} \int_{E_1} \int_{E_2} \int_{E_3} \left\{ \text{Res}_{s_4=0} \mathcal{F} + \text{Res}_{s_4=-s_1} \mathcal{F} + \text{Res}_{s_4=-s_2} \mathcal{F} \right\} ds_3 ds_2 ds_1 + O(L^{-A}). \quad (2.7.2)$$

In a similar way, we move E_3 to the line segment D_3 that runs from $-c(\log L)^{-1} - i3L^{17}$ to $-c(\log L)^{-1} + i3L^{17}$ to deduce from (2.7.2) that

$$\begin{aligned} Q = & \frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} \left\{ \text{Res}_{s_3=0} \text{Res}_{s_4=0} \mathcal{F} + \text{Res}_{s_3=0} \text{Res}_{s_4=-s_1} \mathcal{F} + \text{Res}_{s_3=0} \text{Res}_{s_4=-s_2} \mathcal{F} \right. \\ & + \text{Res}_{s_3=-s_1} \text{Res}_{s_4=0} \mathcal{F} + \text{Res}_{s_3=-s_1} \text{Res}_{s_4=-s_1} \mathcal{F} + \text{Res}_{s_3=-s_1} \text{Res}_{s_4=-s_2} \mathcal{F} \\ & \left. + \text{Res}_{s_3=-s_2} \text{Res}_{s_4=0} \mathcal{F} + \text{Res}_{s_3=-s_2} \text{Res}_{s_4=-s_1} \mathcal{F} + \text{Res}_{s_3=-s_2} \text{Res}_{s_4=-s_2} \mathcal{F} \right\} ds_2 ds_1 + O(L^{-A}). \end{aligned}$$

Write this as

$$\begin{aligned} Q = & \frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} \left\{ \begin{aligned} & R_{00} + R_{01} + R_{02} \\ & + R_{10} + R_{11} + R_{12} \\ & + R_{20} + R_{21} + R_{22} \end{aligned} \right\} ds_2 ds_1 + O(L^{-A}). \end{aligned} \quad (2.7.3)$$

We treat each of the nine integrals $\iint R_{\ell\nu}$ in (2.7.3) one by one in the following sections. As mentioned at the beginning of this section, the basic idea is to move the paths of integration in (2.7.3) and express Q as a sum of residues of $\mathcal{F}$. When we “move” one path of integration to another, we always do so by connecting the two paths with horizontal line segments and then applying the residue theorem to write one path integral in terms of the other. The contributions of the integrals along the horizontal line segments are always bounded by $\ll L^{-A}$. These bounds follow from using $y^{s_j} \ll 1$, Lemma 2.2.2, (2.6.6), and the fact that $|s_m| \geq |t_m| \asymp L^{17}$ for some $m = 1, 2, 3, 4$. The calculations involved in bounding the integrals of these horizontal segments are straightforward, and are thus left to the reader. From now on, we will move paths of integration without further mention of the horizontal line segments.

Another thing to keep in mind in the calculations below is that the factors $\zeta^{-1}(1 \pm s_j + \xi)$ with $\xi = \alpha, \beta, \gamma$, or δ are holomorphic functions of s_j in the domains we are concerned with. This is because we will only work within the known zero-free region of $\zeta(s)$ given by Lemma 2.2.2. Also, the Euler product F defined in (2.6.5) is holomorphic by (2.6.6). Thus we can determine the poles of each $R_{\ell\nu}$ by looking only at the factors s_j^{-1} and $\zeta(1 + s_j \pm s_k)$ in our expression for $R_{\ell\nu}$.

2.8 Evaluation of $\iint R_{12}$ and $\iint R_{21}$

If $f(s)$ is a function with a simple pole at $s = \omega$, then

$$\operatorname{Res}_{s=\omega} f = \lim_{s \rightarrow \omega} (s - \omega) f(s). \quad (2.8.1)$$

From this, the definition (2.6.8), and the equation $\lim_{s \rightarrow 0} s\zeta(1+s) = 1$, it follows that

$$\begin{aligned} R_{12} &= \frac{1}{s_1^6 s_2^6} \zeta^{-1}(1+s_1+\gamma) \zeta^{-1}(1+s_1+\delta) \zeta^{-1}(1+s_2+\gamma) \zeta^{-1}(1+s_2+\delta) \\ &\quad \times \zeta^{-1}(1-s_1+\alpha) \zeta^{-1}(1-s_1+\beta) \zeta^{-1}(1-s_2+\alpha) \zeta^{-1}(1-s_2+\beta) \quad (2.8.2) \\ &\quad \times \zeta(1+s_1-s_2) \zeta(1+s_2-s_1) F(s_1, s_2, -s_1, -s_2) \end{aligned}$$

when $s_1 \in E_1$ and $s_2 \in E_2$. For a fixed $s_1 \in E_1$, let Γ_0 be the path defined by

$$\Gamma_0(t_2) = \frac{c}{\log(|t_1| + |t_2| + 3)} + it_2, \quad -2L^{17} \leq t_2 \leq 2L^{17}.$$

Note that Γ_0 lies to the right of E_2 for large enough L . Thus it also lies to the right of E_1 by our assumption (2.6.2). Hence, if $s_1 \in E_1$, then R_{12} has no poles as a function of s_2 on the region between E_2 and Γ_0 . Thus, the result of moving E_2 to Γ_0 is

$$\int_{E_1} \int_{E_2} R_{12} ds_2 ds_1 = \int_{E_1} \int_{\Gamma_0} R_{12} ds_2 ds_1 + O(L^{-A}). \quad (2.8.3)$$

To bound the integral on the right-hand side, we estimate the factors in (2.8.2) as follows. We take $s_1 \in E_1$ and $s_2 \in \Gamma_0$ in (2.8.2), and we recall our assumption (2.6.2). First, by Lemma 2.2.2, we see that

$$\zeta(1 \pm s_1 \mp s_2) \ll \max \left\{ \frac{1}{\sigma_2 - \sigma_1}, \log(|t_1| + |t_2| + 2) \right\} \ll \log(|t_1| + |t_2| + 3).$$

Second, if $|t_1| \leq 1$ then we can use the power series of $\zeta^{-1}(s)$ near $s = 1$ to write

$$\frac{1}{s_1 \zeta(1 \pm s_1 + \alpha)} \ll 1.$$

Otherwise, we use Lemma 2.2.2. Thus, if $-\infty < t_1 < \infty$ then

$$\frac{1}{s_1 \zeta(1 \pm s_1 + \alpha)} \ll \min \left\{ 1, \frac{\log(|t_1| + 3)}{|t_1|} \right\}.$$

The same estimate holds when s_1 is replaced by s_2 or α is replaced by β , γ , or δ .

Therefore

$$\begin{aligned} & \frac{1}{s_1^4 s_2^4} \zeta^{-1}(1 + s_1 + \gamma) \zeta^{-1}(1 + s_1 + \delta) \zeta^{-1}(1 + s_2 + \gamma) \zeta^{-1}(1 + s_2 + \delta) \\ & \quad \times \zeta^{-1}(1 - s_1 + \alpha) \zeta^{-1}(1 - s_1 + \beta) \zeta^{-1}(1 - s_2 + \alpha) \zeta^{-1}(1 - s_2 + \beta) \quad (2.8.4) \\ & \ll \min \left\{ 1, \frac{\log(|t_1| + 3)}{|t_1|} \right\}^4 \min \left\{ 1, \frac{\log(|t_2| + 3)}{|t_2|} \right\}^4. \end{aligned}$$

Third, it is true that

$$\frac{1}{s_1^2 s_2^2} \ll \frac{1}{\sigma_1^2 \sigma_2^2} \ll L^2 \log^2(|t_1| + |t_2| + 3).$$

Fourth, it follows from (2.6.6) that $F(s_1, s_2, -s_1, -s_2) \ll 1$.

From all these bounds, we arrive at

$$\int_{E_1} \int_{\Gamma_0} R_{12} \, ds_2 \, ds_1 \ll L^2.$$

It follows from this and (2.8.3) that

$$\int_{E_1} \int_{E_2} R_{12} \, ds_2 \, ds_1 \ll L^2. \quad (2.8.5)$$

A similar argument gives

$$\int_{E_1} \int_{E_2} R_{21} \, ds_2 \, ds_1 \ll L^2. \quad (2.8.6)$$

2.9 Evaluation of $\iint R_{22}$

From (2.8.1), the definition (2.6.8), and the equation $\lim_{s \rightarrow 0} s \zeta(1 + s) = 1$, it follows that

$$\begin{aligned} R_{22} &= \frac{y^{s_1 - s_2}}{s_1^3 s_2^9} \zeta^{-1}(1 + s_1 + \gamma) \zeta^{-1}(1 + s_1 + \delta) \zeta^{-1}(1 + s_2 + \gamma) \zeta^{-1}(1 + s_2 + \delta) \\ & \quad \times \zeta^{-1}(1 - s_2 + \alpha) \zeta^{-1}(1 - s_2 + \beta) \zeta^{-1}(1 - s_2 + \alpha) \zeta^{-1}(1 - s_2 + \beta) \\ & \quad \times \zeta(1 + s_1 - s_2) \zeta(1 + s_1 - s_2) F(s_1, s_2, -s_2, -s_2) \end{aligned} \quad (2.9.1)$$

when $s_1 \in E_1$ and $s_2 \in E_2$. Let Υ_2 be the line segment from $c(\log L)^{-1} - i2L^{17}$ to $c(\log L)^{-1} + i2L^{17}$. Note that Υ_2 lies to the right of E_2 for large enough L . Thus it also lies to the right of E_1 by (2.6.2). Hence, if $s_1 \in E_1$, then R_{22} has no poles as a function of s_2 inside the rectangle determined by E_2 and Υ_2 . Therefore the result when we move E_2 to Υ_2 is

$$\int_{E_1} \int_{E_2} R_{22} ds_2 ds_1 = \int_{E_1} \int_{\Upsilon_2} R_{22} ds_2 ds_1 + O(L^{-A}). \quad (2.9.2)$$

Using (2.9.1), Lemma 2.2.2, (2.6.6), and the bound $y^{-s_2} \ll y^{-c/\log L}$, we can show that

$$\int_{E_1} \int_{\Upsilon_2} R_{22} ds_2 ds_1 \ll L^A y^{-c/\log L} \ll L^{-A}. \quad (2.9.3)$$

It follows from this and (2.9.2) that

$$\int_{E_1} \int_{E_2} R_{22} ds_2 ds_1 \ll L^{-A}. \quad (2.9.4)$$

2.10 Evaluation of $\iint R_{11}$

We use (2.8.1) to deduce from (2.6.8) that if $s_1 \in E_1$ and $s_2 \in E_2$, then

$$\begin{aligned} R_{11} &= \frac{y^{s_2-s_1}}{s_1^9 s_2^3} \zeta^{-1}(1+s_1+\gamma) \zeta^{-1}(1+s_1+\delta) \zeta^{-1}(1+s_2+\gamma) \zeta^{-1}(1+s_2+\delta) \\ &\quad \times \zeta^{-1}(1-s_1+\alpha) \zeta^{-1}(1-s_1+\beta) \zeta^{-1}(1-s_1+\alpha) \zeta^{-1}(1-s_1+\beta) \\ &\quad \times \zeta(1+s_2-s_1) \zeta(1+s_2-s_1) F(s_1, s_2, -s_1, -s_1). \end{aligned} \quad (2.10.1)$$

Let D_2 be the line segment from $-c(\log L)^{-1} - i2L^{17}$ to $-c(\log L)^{-1} + i2L^{17}$. We move E_2 to D_2 and arrive at

$$\begin{aligned} \frac{1}{2\pi i} \int_{E_1} \int_{E_2} R_{11} ds_2 ds_1 &= \frac{1}{2\pi i} \int_{E_1} \int_{D_2} R_{11} ds_2 ds_1 \\ &\quad + \int_{E_1} \operatorname{Res}_{s_2=0} R_{11} ds_1 + \int_{E_1} \operatorname{Res}_{s_2=s_1} R_{11} ds_1 + O(L^{-A}) \\ &= J_{11} + I_{110} + I_{111} + O(L^{-A}), \end{aligned} \quad (2.10.2)$$

say. We can bound J_{11} in the same way we showed (2.9.3). That is, we use (2.10.1) and $y^{s_2} \ll y^{-c/\log L}$ to write

$$J_{11} \ll L^{-A}. \quad (2.10.3)$$

To estimate the integral I_{110} in (2.10.2), let Υ_1 be the line segment from $c(\log L)^{-1} - iL^{17}$ to $c(\log L)^{-1} + iL^{17}$. Observe that Υ_1 lies to the right of E_1 when L is large enough. Thus, we see from (2.10.1) that $\text{Res}_{s_2=0} R_{11}$ has no poles between E_1 and Υ_1 . Hence we deduce from moving E_1 to Υ_1 that

$$I_{110} = \int_{\Upsilon_1} \text{Res}_{s_2=0} R_{11} ds_1 + O(L^{-A}).$$

The integral on the right-hand side is $\ll L^{-A}$ as in (2.9.3), because R_{11} has the factor y^{-s_1} in (2.10.1). Therefore

$$I_{110} \ll L^{-A}. \quad (2.10.4)$$

We next estimate the integral I_{111} in (2.10.2). If $f(s)$ is a function with a double pole at $s = \omega$, then

$$\text{Res} f = \lim_{s \rightarrow \omega} \frac{d}{ds} (s - \omega)^2 f(s).$$

It follows from this, (2.10.1), and straightforward differentiation that

$$\begin{aligned} \text{Res}_{s_2=s_1} R_{11} &= \frac{\log y}{s_1^{12}} \zeta^{-2}(1 + s_1 + \gamma) \zeta^{-2}(1 + s_1 + \delta) \\ &\quad \times \zeta^{-2}(1 - s_1 + \alpha) \zeta^{-2}(1 - s_1 + \beta) F(s_1, s_1, -s_1, -s_1) + X_0(s_1), \end{aligned} \quad (2.10.5)$$

where $X_0(s_1) = X_0(s_1; \alpha, \beta, \gamma, \delta)$ is an expression involving ζ^{-1} , ζ'/ζ , F , and $\partial F/\partial s_2$. The function $X_0(s_1)$ is independent of y . Now let Γ_1 be the path defined by

$$\Gamma_1(t_1) = \frac{c}{\log(|t_1| + 3)} + it_1, \quad -L^{17} \leq t_1 \leq L^{17}. \quad (2.10.6)$$

Note that Γ_1 lies to the right of E_1 when L is large. From (2.10.5), we can deduce that $\text{Res}_{s_2=s_1} R_{11}$ has no poles between E_1 and Γ_1 . Thus, the result of moving E_1 to Γ_1 is

$$I_{111} = \int_{\Gamma_1} \text{Res}_{s_2=s_1} R_{11} ds_1 + O(L^{-A}). \quad (2.10.7)$$

We use Lemma 2.2.2 and (2.6.6) to bound each term on the right-hand side of (2.10.5) with $s_1 \in \Gamma_1$ to see that

$$\int_{\Gamma_1} \operatorname{Res}_{s_2=s_1} R_{11} ds_1 = O(\log y) + O(1) \ll L.$$

This and (2.10.7) imply

$$I_{111} \ll L.$$

From this, (2.10.4), (2.10.3), and (2.10.2), we arrive at

$$\int_{E_1} \int_{E_2} R_{11} ds_2 ds_1 \ll L. \quad (2.10.8)$$

2.11 Evaluation of $\iint R_{02}$ and $\iint R_{20}$

Applying (2.8.1) and the residue theorem, we see for $s_1 \in E_1$ and $s_2 \in E_2$ that

$$\begin{aligned} R_{02} &= -\frac{1}{2\pi i} \oint_{\mathcal{C}_3} \frac{y^{s_1+s_3}}{s_1^3 s_2^6 s_3^3} \times \\ &\quad \times \zeta^{-1}(1+s_1+\gamma) \zeta^{-1}(1+s_1+\delta) \zeta^{-1}(1+s_2+\gamma) \zeta^{-1}(1+s_2+\delta) \\ &\quad \times \zeta^{-1}(1+s_3+\alpha) \zeta^{-1}(1+s_3+\beta) \zeta^{-1}(1-s_2+\alpha) \zeta^{-1}(1-s_2+\beta) \\ &\quad \times \zeta(1+s_1+s_3) \zeta(1+s_1-s_2) \zeta(1+s_2+s_3) F(s_1, s_2, s_3, -s_2) ds_3, \end{aligned} \quad (2.11.1)$$

where $\mathcal{C}_3$ is the circle $|s_3| = c_1/2L$, taken once in the positive direction. Let D_1 be the line segment from $-c(\log L)^{-1} - iL^{17}$ to $-c(\log L)^{-1} + iL^{17}$. We interchange the order of integration and then move E_1 to D_1 to deduce that

$$\begin{aligned} \frac{1}{2\pi i} \int_{E_1} \int_{E_2} R_{02} ds_2 ds_1 &= \frac{1}{2\pi i} \int_{E_2} \int_{E_1} R_{02} ds_1 ds_2 \\ &= \frac{1}{2\pi i} \int_{E_2} \int_{D_1} R_{02} ds_1 ds_2 + \int_{E_2} \operatorname{Res}_{s_1=0} R_{02} ds_2 + O(L^{-A}) \\ &= J_{02} + I_{02} + O(L^{-A}), \end{aligned} \quad (2.11.2)$$

say. Note that the pole of R_{02} at $s_1 = s_2$ does not leave a residue because it is not between E_1 and D_1 when $s_2 \in E_2$, by (2.6.2). Using (2.11.1), we can estimate the

integral J_{02} in (2.11.2) in the same way we arrived at (2.9.3). In other words, we can write

$$J_{02} \ll L^A y^{-c/\log L} \ll L^{-A} \quad (2.11.3)$$

because of the presence of the factor y^{s_1} in (2.11.1).

To evaluate the expression I_{02} in (2.11.2), we use the residue theorem to write $\text{Res}_{s_1=0} R_{02}$ as an integral along a circle. The result is

$$I_{02} = \frac{1}{2\pi i} \int_{E_2} \oint_{\mathcal{C}_1} R_{02} \, ds_1 \, ds_2, \quad (2.11.4)$$

where $\mathcal{C}_1$ is the circle $|s_1| = c_1/L$, taken once in the positive direction. Define the path Γ_2 by the equation

$$\Gamma_2(t_2) = \frac{c}{\log(|t_2| + 3)} + it_2, \quad -2L^{17} \leq t_2 \leq 2L^{17}.$$

This path lies to the right of E_2 when L is large. If $s_1 \in \mathcal{C}_1$ and $s_3 \in \mathcal{C}_3$, then the integrand in (2.11.1) is holomorphic as a function of s_2 between E_2 and Γ_2 , by (2.6.2). Therefore we deduce from (2.11.4) and moving E_2 to Γ_2 that

$$I_{02} = \frac{1}{2\pi i} \int_{\Gamma_2} \oint_{\mathcal{C}_1} R_{02} \, ds_1 \, ds_2 + O(L^{-A}). \quad (2.11.5)$$

We need to estimate the integral on the right-hand side. To do this, we bound the factors in (2.11.1) as follows. Let $s_2 \in \Gamma_2$, $s_1 \in \mathcal{C}_1$, and $s_3 \in \mathcal{C}_3$, so that $|s_1| = c_1/L$ and $|s_3| = c_1/2L$. Also, recall the assumption (2.6.2). First, we see by the Laurent series of $\zeta(s)$ about $s = 1$ that

$$\zeta(1 + s_1 + s_3) \ll \frac{1}{|s_1 + s_3|} \ll L.$$

Second, we see from Lemma 2.2.2 that

$$\zeta(1 + s_1 - s_2) \ll \max \left\{ \frac{1}{\sigma_2 - \sigma_1}, \log(|t_1| + |t_2| + 2) \right\} \ll \log(|t_2| + 3),$$

and similarly $\zeta(1 + s_2 + s_3) \ll \log(|t_2| + 3)$. Third, we can prove the inequality

$$\begin{aligned} & \frac{1}{s_1^2 s_2^4 s_3^2} \zeta^{-1}(1 + s_1 + \gamma) \zeta^{-1}(1 + s_1 + \delta) \zeta^{-1}(1 + s_2 + \gamma) \zeta^{-1}(1 + s_2 + \delta) \\ & \quad \times \zeta^{-1}(1 + s_3 + \alpha) \zeta^{-1}(1 + s_3 + \beta) \zeta^{-1}(1 - s_2 + \alpha) \zeta^{-1}(1 - s_2 + \beta) \\ & \ll \min \left\{ 1, \frac{\log(|t_2| + 3)}{|t_2|} \right\}^4 \end{aligned}$$

in the same way we showed (2.8.4). Fourth, it is true that

$$\frac{1}{s_1 s_2^2 s_3} \ll \frac{1}{|s_1|(\sigma_2)^2 |s_3|} \asymp L^2 \log^2(|t_2| + 3).$$

Fifth, the estimate $y^{s_1+s_3} \ll 1$ follows from the fact that $s_1, s_3 \ll 1/L$. And finally, sixth, it follows from (2.6.6) that $F(s_1, s_2, s_3, -s_2) \ll 1$.

Using all these bounds, we deduce from (2.11.1) that

$$\frac{1}{2\pi i} \int_{\Gamma_2} \oint_{\mathcal{C}_1} R_{02} \, ds_1 \, ds_2 \ll L. \quad (2.11.6)$$

From this, (2.11.5), (2.11.3), and (2.11.2), it follows that

$$\frac{1}{2\pi i} \int_{E_1} \int_{E_2} R_{02} \, ds_2 \, ds_1 \ll L. \quad (2.11.7)$$

A similar argument gives

$$\frac{1}{2\pi i} \int_{E_1} \int_{E_2} R_{20} \, ds_2 \, ds_1 \ll L. \quad (2.11.8)$$

2.12 Evaluation of $\iint R_{01}$ and $\iint R_{10}$

We use (2.8.1), (2.6.8), and the residue theorem to write

$$\begin{aligned} R_{01} &= -\frac{1}{2\pi i} \oint_{\mathcal{C}_3} \frac{y^{s_2+s_3}}{s_1^6 s_2^3 s_3^3} \times \\ & \quad \times \zeta^{-1}(1 + s_1 + \gamma) \zeta^{-1}(1 + s_1 + \delta) \zeta^{-1}(1 + s_2 + \gamma) \zeta^{-1}(1 + s_2 + \delta) \\ & \quad \times \zeta^{-1}(1 + s_3 + \alpha) \zeta^{-1}(1 + s_3 + \beta) \zeta^{-1}(1 - s_1 + \alpha) \zeta^{-1}(1 - s_1 + \beta) \\ & \quad \times \zeta(1 + s_1 + s_3) \zeta(1 + s_2 + s_3) \zeta(1 + s_2 - s_1) F(s_1, s_2, s_3, -s_1) \, ds_3 \end{aligned} \quad (2.12.1)$$

for $s_1 \in E_1$ and $s_2 \in E_2$, where $\mathcal{C}_3$ is the circle $|s_3| = c_1/2L$, taken once in the positive direction. Let D_2 be the line segment from $-c(\log L)^{-1} - i2L^{17}$ to $-c(\log L)^{-1} + i2L^{17}$. Move E_2 to D_2 to see that

$$\begin{aligned} \frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} R_{01} ds_2 ds_1 &= \frac{1}{(2\pi i)^2} \int_{E_1} \int_{D_2} R_{01} ds_2 ds_1 \\ &+ \frac{1}{2\pi i} \int_{E_1} \operatorname{Res}_{s_2=0} R_{01} ds_1 + \frac{1}{2\pi i} \int_{E_1} \operatorname{Res}_{s_2=s_1} R_{01} ds_1 + O(L^{-A}) \quad (2.12.2) \\ &= J_{01} + I_{010} + I_{011} + O(L^{-A}), \end{aligned}$$

say. By the presence of the factor y^{s_2} in (2.12.1), it follows that

$$J_{01} \ll L^A y^{-c/\log L} \ll L^{-A} \quad (2.12.3)$$

in the same way that (2.9.3) holds.

To bound the integral I_{010} in (2.12.2), apply the residue theorem to write it as

$$I_{010} = \frac{1}{(2\pi i)^2} \int_{E_1} \oint_{\mathcal{C}_2} R_{01} ds_2 ds_1, \quad (2.12.4)$$

where $\mathcal{C}_2$ is the circle $|s_2| = 3c_1/4L$, taken once in the positive direction. Let Γ_1 be the path defined by (2.10.6). Since there are no poles of $\operatorname{Res}_{s_2=0} R_{01}$ between E_1 and Γ_1 , we deduce from (2.12.4) and moving E_1 to Γ_1 that

$$I_{010} = \frac{1}{(2\pi i)^2} \int_{\Gamma_1} \oint_{\mathcal{C}_2} R_{01} ds_2 ds_1 + O(L^{-A}). \quad (2.12.5)$$

We can use (2.12.1) and arguments similar to those leading up to (2.11.6) to show that

$$\frac{1}{(2\pi i)^2} \int_{\Gamma_1} \oint_{\mathcal{C}_2} R_{01} ds_2 ds_1 \ll L.$$

From this and (2.12.5), we arrive at

$$I_{010} \ll L. \quad (2.12.6)$$

The next step is to estimate the integral I_{011} in (2.12.2). We use (2.8.1) with (2.12.1) and pass the limit under the integral sign to write

$$\begin{aligned} \operatorname{Res}_{s_2=s_1} R_{01} &= -\frac{1}{2\pi i} \oint_{\mathcal{C}_3} \frac{y^{s_1+s_3}}{s_1^9 s_3^3} \times \\ &\quad \times \zeta^{-2}(1+s_1+\gamma) \zeta^{-2}(1+s_1+\delta) \zeta^{-1}(1+s_3+\alpha) \zeta^{-1}(1+s_3+\beta) \\ &\quad \times \zeta^{-1}(1-s_1+\alpha) \zeta^{-1}(1-s_1+\beta) \zeta^2(1+s_1+s_3) F(s_1, s_1, s_3, -s_1) ds_3. \end{aligned} \quad (2.12.7)$$

Let D_1 be the line segment from $-c(\log L)^{-1} - iL^{17}$ to $-c(\log L)^{-1} + iL^{17}$. Move E_1 to D_1 to deduce that

$$I_{011} = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} + \frac{1}{2\pi i} \int_{D_1} \operatorname{Res}_{s_2=s_1} R_{01} ds_1 + O(L^{-A}).$$

The integral on the right-hand side is $\ll L^{-A}$ as in (2.9.3), since the factor y^{s_1} is present in (2.12.7). Therefore

$$I_{011} = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} + O(L^{-A}).$$

It now follows from this, (2.12.6), (2.12.3), and (2.12.2) that

$$\frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} R_{01} ds_2 ds_1 = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} + O(L). \quad (2.12.8)$$

By a similar argument, we can show that

$$\frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} R_{10} ds_2 ds_1 = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{10} + O(L). \quad (2.12.9)$$

2.13 Evaluation of $\iint R_{00}$

By the definition (2.6.8) of $\mathcal{F}$ and the residue theorem, we can write

$$R_{00} = \frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_3} \oint_{\mathcal{C}_4} \mathcal{F} ds_4 ds_3 \quad (2.13.1)$$

for $s_1 \in E_1$ and $s_2 \in E_2$, where $\mathcal{C}_3$ is the circle $|s_3| = c_1/2L$ and $\mathcal{C}_4$ is the circle $|s_4| = c_1/2L$, both taken once along the positive direction. Let D_2 be the line

segment from $-c(\log L)^{-1} - i2L^{17}$ to $-c(\log L)^{-1} + i2L^{17}$. Move E_2 to D_2 to see that

$$\begin{aligned} & \frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} R_{00} \, ds_2 \, ds_1 \\ &= \frac{1}{(2\pi i)^2} \int_{E_1} \int_{D_2} R_{00} \, ds_2 \, ds_1 + \frac{1}{2\pi i} \int_{E_1} \operatorname{Res}_{s_2=0} R_{00} \, ds_1 + O(L^{-A}) \quad (2.13.2) \\ &= J_{00} + I_{00} + O(L^{-A}), \end{aligned}$$

say. Because of the presence of the factor y^{s_2} of $\mathcal{F}$ in (2.13.1), we can show that

$$J_{00} \ll L^A y^{-c/\log L} \ll L^{-A}. \quad (2.13.3)$$

To evaluate the integral I_{00} in (2.13.2), let D_1 be the line segment from $-c(\log L)^{-1} - iL^{17}$ to $-c(\log L)^{-1} + iL^{17}$. We move E_1 to D_1 and write

$$I_{00} = \frac{1}{2\pi i} \int_{D_1} \operatorname{Res}_{s_2=0} R_{00} \, ds_1 + \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} + O(L^{-A}).$$

The integral on the right-hand side is $\ll L^{-A}$ since $\mathcal{F}$ has the factor y^{s_1} in (2.13.1).

Hence

$$I_{00} = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} + O(L^{-A}).$$

From this, (2.13.3), and (2.13.2), it follows that

$$\frac{1}{(2\pi i)^2} \int_{E_1} \int_{E_2} R_{00} \, ds_2 \, ds_1 = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} + O(L^{-A}). \quad (2.13.4)$$

2.14 Putting together the estimates

We now collect together the nine estimates (2.8.5), (2.8.6), (2.9.4), (2.10.8), (2.11.7), (2.11.8), (2.12.8), (2.12.9), and (2.13.4), and then insert them into (2.7.3) to deduce that

$$Q(\alpha, \beta, \gamma, \delta) = \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} + \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} + \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{10} + O(L^2). \quad (2.14.1)$$

In this section, we simplify this further and prove an asymptotic formula for $Q(\alpha, \beta, \gamma, \delta)$ that has a polynomial main term. We continue to assume that $\alpha, \beta, \gamma, \delta \ll 1/L$.

We first write the main terms in (2.14.1) as path integrals. By (2.13.1) and the residue theorem, we can write

$$\operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} = \frac{1}{(2\pi i)^4} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \oint_{\mathcal{C}_3} \oint_{\mathcal{C}_4} \mathcal{F} \, ds_4 \, ds_3 \, ds_2 \, ds_1, \quad (2.14.2)$$

where $\mathcal{C}_1$ is the circle $|s_1| = c_1/L$ and $\mathcal{C}_2$ is the circle $|s_2| = c_1/L$, both taken once in the positive direction. Similarly, from the residue theorem and (2.12.7), it follows that

$$\begin{aligned} \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} &= -\frac{1}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_3} \frac{y^{s_1+s_3}}{s_1^9 s_3^3} \times \\ &\quad \times \zeta^{-2}(1+s_1+\gamma) \zeta^{-2}(1+s_1+\delta) \zeta^{-1}(1+s_3+\alpha) \zeta^{-1}(1+s_3+\beta) \\ &\quad \times \zeta^{-1}(1-s_1+\alpha) \zeta^{-1}(1-s_1+\beta) \zeta^2(1+s_1+s_3) F(s_1, s_1, s_3, -s_1) \, ds_3 \, ds_1. \end{aligned} \quad (2.14.3)$$

Similarly, or by symmetry, the same equation holds with R_{10} in place of R_{01} .

We put these equations into simpler forms by using Laurent series expansions, as follows. By repeated applications of (2.4.24), we see that

$$F(s_1, s_2, s_3, s_4; \alpha, \beta, \gamma, \delta) = F(0, 0, 0, 0; 0, 0, 0, 0) + O(L^{-1})$$

when each variable is $\ll 1/L$. We can show that $F(0, \dots, 0) = \zeta(2)$ by a straightforward computation using the definition (2.6.5) of F . Thus

$$F(s_1, s_2, s_3, s_4; \alpha, \beta, \gamma, \delta) = \zeta(2) + O(L^{-1})$$

when each variable is $\ll 1/L$. We use this, the definition (2.6.8) of $\mathcal{F}$, and the

Laurent series of $\zeta(s)$ and $\zeta^{-1}(s)$ near $s = 1$ to simplify (2.14.2) to

$$\begin{aligned} \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=0} R_{00} &= \frac{\zeta(2)}{(2\pi i)^4} \oint_{c_1} \oint_{c_2} \oint_{c_3} \oint_{c_4} \frac{y^{s_1+s_2+s_3+s_4}}{s_1^3 s_2^3 s_3^3 s_4^3} \times \\ &\quad \times (s_1 + \gamma)(s_1 + \delta)(s_2 + \gamma)(s_2 + \delta)(s_3 + \alpha)(s_3 + \beta)(s_4 + \alpha)(s_4 + \beta) \\ &\quad \times (s_1 + s_3)^{-1}(s_1 + s_4)^{-1}(s_2 + s_3)^{-1}(s_2 + s_4)^{-1} ds_4 ds_3 ds_2 ds_1 + O(L^3). \end{aligned} \quad (2.14.4)$$

Similarly, (2.14.3) simplifies to

$$\begin{aligned} \operatorname{Res}_{s_1=0} \operatorname{Res}_{s_2=s_1} R_{01} &= -\frac{\zeta(2)}{(2\pi i)^2} \oint_{c_1} \oint_{c_3} \frac{y^{s_1+s_3}}{s_1^9 s_3^3} (s_1 + \gamma)^2 (s_1 + \delta)^2 (s_3 + \alpha)(s_3 + \beta) \\ &\quad \times (-s_1 + \alpha)(-s_1 + \beta)(s_1 + s_3)^{-2} ds_3 ds_1 + O(L^3). \end{aligned} \quad (2.14.5)$$

The same formula holds with R_{10} in place of R_{01} .

We may express the factor $y^{s_1+s_2+s_3+s_4}$ in (2.14.4) as an infinite sum using the series expansion of $\exp(z)$. The same is true for $y^{s_1+s_3}$ in (2.14.5). Also, we may write each factor $(s_j + s_k)^{-1}$ in (2.14.4) and (2.14.5) as the infinite sum

$$\frac{1}{s_j(1 + s_k/s_j)} = \frac{1}{s_j} - \frac{s_k}{s_j^2} + \frac{s_k^2}{s_j^3} + \cdots.$$

By Cauchy's theorem, all but finitely many terms from these sums contribute zero to the values of the integrals. With these observations, we now deduce from (2.14.1), (2.14.4), and (2.14.5) that

$$Q(\alpha, \beta, \gamma, \delta) = \zeta(2)P(\alpha, \beta, \gamma, \delta) + O(L^3), \quad (2.14.6)$$

where $P(\alpha, \beta, \gamma, \delta)$ is defined by

$$\begin{aligned}
P(\alpha, \beta, \gamma, \delta) = & \frac{1}{(2\pi i)^4} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \oint_{\mathcal{C}_3} \oint_{\mathcal{C}_4} \left\{ \sum_{n=0}^{12} \frac{(\log y)^n (s_1 + s_2 + s_3 + s_4)^n}{n! s_1^5 s_2^5 s_3^3 s_4^3} \right\} \\
& \times (s_1 + \gamma)(s_1 + \delta)(s_2 + \gamma)(s_2 + \delta)(s_3 + \alpha)(s_3 + \beta)(s_4 + \alpha)(s_4 + \beta) \\
& \times \left(1 - \frac{s_3}{s_1} - \frac{s_3}{s_2} + \frac{s_3^2}{s_1^2} + \frac{s_3^2}{s_1 s_2} + \frac{s_3^2}{s_2^2} \right) \left(1 - \frac{s_4}{s_1} - \frac{s_4}{s_2} + \frac{s_4^2}{s_1^2} + \frac{s_4^2}{s_1 s_2} + \frac{s_4^2}{s_2^2} \right) \\
& \times ds_4 ds_3 ds_2 ds_1 \\
& - \frac{2}{(2\pi i)^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_3} \left\{ \sum_{n=0}^{12} \frac{(\log y)^n (s_1 + s_3)^n}{n! s_1^{11} s_3^3} \right\} \\
& \times (s_1 + \gamma)^2 (s_1 + \delta)^2 (s_3 + \alpha)(s_3 + \beta)(-s_1 + \alpha)(-s_1 + \beta) \\
& \times \left(1 - 2\frac{s_3}{s_1} + 3\frac{s_3^2}{s_1^2} \right) ds_3 ds_1.
\end{aligned} \tag{2.14.7}$$

Note that it immediately follows from the residue theorem that $P(\alpha, \beta, \gamma, \delta)$ is a polynomial in $\alpha, \beta, \gamma, \delta$, and $\log y$.

2.15 Proof of Theorem 1.1.3

Having evaluated $Q(\alpha, \beta, \gamma, \delta)$, we can now complete the proof of Theorem 1.1.3. We first simplify (2.5.10). Observe that $|s_j| \asymp 1/L$ for each variable of integration s_j in (2.14.4) and (2.14.5). Therefore, it follows from (2.14.1), (2.14.4), and (2.14.5) that

$$Q(\alpha, \beta, \gamma, \delta) \ll L^4$$

for $\alpha, \beta, \gamma, \delta \ll 1/L$. Hence, if $|z_j| = 1/L$ for $j = 1, 2, 3, 4$, then

$$Q(z_1 + i\rho, z_2 + i\rho, -z_3 - i\rho, -z_4 - i\rho) \ll L^4 \tag{2.15.1}$$

because $\rho \asymp 1/L$ by (2.5.5). Moreover, we see from the definition (2.5.1) and the Laurent series of $\zeta(s)$ near the points $s = 1$ and $s = 2$ that

$$A_{z_1, z_2, -z_3, -z_4}(0) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell)^2 = \frac{1}{\zeta(2)} (z_1 - z_2)(z_3 - z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell) + O\left(\frac{1}{L^9}\right)$$

when $z_j \ll 1/L$ for $j = 1, 2, 3, 4$. We insert this into (2.5.10). We bound the contribution of the error term $O(L^{-9})$ by applying (2.15.1) and the fact that $|z_j| = 1/L$ and $t^{\pm z_j} \ll 1$ for each j . The result is

$$\begin{aligned} \mathcal{J} = & \int_{-\infty}^{\infty} \frac{w(t)}{(2\pi i)^4} \oint \oint \oint \oint \left(\frac{t}{2\pi}\right)^{\frac{z_1 + z_2 - z_3 - z_4}{2}} (z_1 - z_2)(z_3 - z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell) \\ & \times \left(\frac{4}{\zeta(2)(\log y)^8}\right) Q(z_1 + i\rho, z_2 + i\rho, -z_3 - i\rho, -z_4 - i\rho) \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \\ & + O\left(\frac{W}{L}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned}$$

where W is defined by (1.1.5).

We now insert into this formula the expression (2.14.6) with the appropriate values for $\alpha, \beta, \gamma, \delta$. We can easily bound the contribution of the error term $O(L^3)$ in (2.14.6) using the fact that $|z_j| = 1/L$ and $t^{\pm z_j} \ll 1$ for each j . Furthermore, we can write the factor $(t/2\pi)^{(z_1 + z_2 - z_3 - z_4)/2}$ as an infinite sum via the series expansion of $\exp(z)$. Only the first few terms from this infinite sum actually contribute to the value of the integral, because of Cauchy's theorem and the fact that P is a polynomial. Hence the result we arrive at is

$$\begin{aligned} \mathcal{J} = & \int_{-\infty}^{\infty} \frac{w(t)}{(2\pi i)^4} \oint \oint \oint \oint \left\{ \sum_{m=0}^4 \frac{(\log \frac{t}{2\pi})^m}{m! 2^m} (z_1 + z_2 - z_3 - z_4)^m \right\} \\ & \times (z_1 - z_2)(z_3 - z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell) \\ & \times \left(\frac{4}{(\log y)^8}\right) P(z_1 + i\rho, z_2 + i\rho, -z_3 - i\rho, -z_4 - i\rho) \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \\ & + O\left(\frac{W}{L}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}). \end{aligned}$$

We next simplify this equation by evaluating the path integrals. Write the equation as

$$\mathcal{J} = \frac{4}{(\log y)^8} \int_{-\infty}^{\infty} w(t) \mathcal{R}(t; y, \rho) dt + O\left(\frac{W}{L}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}). \quad (2.15.2)$$

By the residue theorem and the definition (2.14.7) of P , it follows that $\mathcal{R}(t; y, \rho)$ equals the sum of the coefficients of certain monomials of the type

$$s_1^h s_2^k s_3^q s_4^r z_1^3 z_2^3 z_3^3 z_4^3, \quad h + k + q + r = 12, \quad (2.15.3)$$

in the polynomial $\mathcal{R}_0 \mathcal{R}_1 - 2\mathcal{R}_0 \mathcal{R}_2$, where

$$\begin{aligned} \mathcal{R}_0 &= \left\{ \sum_{m=0}^4 \frac{(\log \frac{t}{2\pi})^m}{m! 2^m} (z_1 + z_2 - z_3 - z_4)^m \right\} (z_1 - z_2)(z_3 - z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell), \\ \mathcal{R}_1 &= \left\{ \sum_{n=0}^{12} \frac{1}{n!} (\log y)^n (s_1 + s_2 + s_3 + s_4)^n \right\} (s_1 - z_3 - i\rho)(s_1 - z_4 - i\rho) \\ &\quad \times (s_2 - z_3 - i\rho)(s_2 - z_4 - i\rho)(s_3 + z_1 + i\rho)(s_3 + z_2 + i\rho) \\ &\quad \times (s_4 + z_1 + i\rho)(s_4 + z_2 + i\rho), \end{aligned}$$

and

$$\begin{aligned} \mathcal{R}_2 &= \left\{ \sum_{n=0}^{12} \frac{1}{n!} (\log y)^n (s_1 + s_3)^n \right\} (s_1 - z_3 - i\rho)^2 (s_1 - z_4 - i\rho)^2 \\ &\quad \times (s_3 + z_1 + i\rho)(s_3 + z_2 + i\rho)(-s_1 + z_1 + i\rho)(-s_1 + z_2 + i\rho). \end{aligned}$$

Observe that the exponents of the z_j in (2.15.3) add up to 12, and the same can be said of the s_k . Therefore $\mathcal{R}(t; y, \rho)$ is the sum of terms of the form

$$\Lambda_{m,n} \left(\log \frac{t}{2\pi} \right)^m (\log y)^n (i\rho)^{m+n-8},$$

where $0 \leq m \leq 4$, $0 \leq n \leq 12$, and each $\Lambda_{m,n}$ is an absolute constant that is computable. We write $\mathcal{R}(t; y, \rho)$ as this sum to express (2.15.2) as

$$\begin{aligned} \mathcal{J} &= 4 \sum_{m,n} \Lambda_{m,n} (\log y)^{n-8} (i\rho)^{m+n-8} \int_{-\infty}^{\infty} w(t) \left(\log \frac{t}{2\pi} \right)^m dt \\ &\quad + O\left(\frac{W}{L}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned} \quad (2.15.4)$$

where the sum runs through all pairs m, n of integers satisfying $0 \leq m \leq 4$ and $0 \leq n \leq 12$.

We can further simplify (2.15.4) by using the properties of $w(t)$ and the definitions of y and ρ . Indeed, recall that $L = \log T$ and W is defined by (1.1.5). Since $w(t)$ is supported on $[T/2, 4T]$, it follows that

$$\int_{-\infty}^{\infty} w(t) \left(\log \frac{t}{2\pi} \right)^m dt = WL^m \left(1 + O\left(\frac{1}{L}\right) \right).$$

Also, the definition (2.5.5) of ρ implies that

$$\rho^{m+n-8} = \left(\frac{2a}{\log \frac{T}{2\pi}} \right)^{m+n-8} = \left(\frac{2a}{L} \right)^{m+n-8} \left(1 + O\left(\frac{1}{L}\right) \right).$$

Moreover, it follows from the definition $y = T^\theta$ that $\log y = \theta L$. We insert these into (2.15.4) and arrive at

$$\mathcal{J} = \sum_{m,n} \lambda_{m,n} \theta^{n-8} a^{m+n-8} W + O\left(\frac{W}{L}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}),$$

where $\lambda_{m,n} = 4\Lambda_{m,n}(2i)^{m+n-8}$ and the sum runs through all pairs m, n of integers with $0 \leq m \leq 4$ and $0 \leq n \leq 12$. This completes the proof of Theorem 1.1.3, with

$$C(a, \theta) = \sum_{m,n} \lambda_{m,n} \theta^{n-8} a^{m+n-8}. \quad (2.15.5)$$

2.16 Proofs of the corollaries

In this section, we choose a specific $w(t)$ to prove Corollaries 1.1.1 and 1.1.2. There may be many ways of doing this, and we do not claim that the method here is the best in any sense.

Let $h(x)$ be the smooth function

$$h(x) = \begin{cases} e^{-1/x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

For each positive $\eta > 0$ and each interval $[u, \tau]$, define the function $\psi(x) = \psi(x; u, \tau, \eta)$ by

$$\psi(x) = \left(\frac{h(\tau - x + \eta)}{h(\tau - x + \eta) + h(x - \tau)} \right) \left(\frac{h(x - u + \eta)}{h(x - u + \eta) + h(u - x)} \right). \quad (2.16.1)$$

The properties of $\psi(x)$ are summarized in the following proposition.

Proposition 2.16.1. *The function $\psi(x)$ is smooth and nonnegative on $(-\infty, \infty)$. It takes the value 1 on $[u, \tau]$ and is supported on $[u - \eta, \tau + \eta]$. Moreover,*

$$\psi^{(j)}(x) \ll_j \exp\left(\frac{6j}{\eta}\right) \quad (2.16.2)$$

for all $j = 0, 1, 2, \dots$

Proof. Since $h(x)$ is smooth and nonnegative, the first sentence of the proposition follows from the fact that the denominator in the definition (2.16.1) of $\psi(x)$ is never zero. The second sentence follows from the fact that $h(x) = 0$ for $x \leq 0$ and $h(x) > 0$ for $x > 0$. Thus it is left to prove (2.16.2). Observe that

$$h^{(j)}(x) \ll_j 1 \quad (2.16.3)$$

for all $j \geq 0$. Since $h(x)$ is nonnegative, we can show that

$$h(\tau - x + \eta) + h(x - \tau) \geq \exp\left(-\frac{2}{\eta}\right) \quad (2.16.4)$$

for all real x by considering the two cases $x \geq \tau + \frac{\eta}{2}$ and $x \leq \tau + \frac{\eta}{2}$ separately. Now apply Lemma 2.2.3 with $f(t) = 1/t$ and $g(t) = h(\tau - t + \eta) + h(t - \tau)$ to deduce from (2.16.3) and (2.16.4) that

$$\frac{d^j}{dt^j} \left(\frac{1}{h(\tau - t + \eta) + h(t - \tau)} \right) \ll_j \exp\left(\frac{2(j+1)}{\eta}\right)$$

for all $j \geq 0$. Similarly, we can prove that

$$\frac{d^j}{dt^j} \left(\frac{1}{h(t - u + \eta) + h(u - t)} \right) \ll_j \exp\left(\frac{2(j+1)}{\eta}\right)$$

for $j \geq 0$. With these bounds and (2.16.3), we can prove (2.16.2) for $j \geq 1$ by a straightforward computation using the product rule. The case $j = 0$ of (2.16.2) trivially follows from the fact that $\psi(x) \leq 1$. $\square$

We now start our proof of Corollary 1.1.1. Let $b \leq 1$ be a real number with $\theta < 2b - \frac{3}{2}$. For large T , we take $\eta = L^{-1/2}$, $u = T/T^b$, and $\tau = (T + T^b)/T^b$ in the definition (2.16.1) of $\psi(x)$ and define

$$w(t) = \psi\left(\frac{t}{T^b}\right). \quad (2.16.5)$$

It immediately follows from Proposition 2.16.1 that $w(t)$ is smooth and nonnegative, and that

$$w(t) = 1 \quad \text{for } t \in [T, T + T^b] \quad (2.16.6)$$

and

$$w(t) = 0 \quad \text{for } t \notin \left[T - \frac{T^b}{\sqrt{\log T}}, T + T^b + \frac{T^b}{\sqrt{\log T}}\right]. \quad (2.16.7)$$

Hence, since $b \leq 1$, it follows that $w(t)$ is supported on $[T/2, 4T]$ for large T . Differentiating (2.16.5) j times and applying (2.16.2), we see that

$$w^{(j)}(t) \ll_j T_0^{-j} \quad (2.16.8)$$

for all $j \geq 0$, where

$$T_0 = T^b e^{-6\sqrt{\log T}}. \quad (2.16.9)$$

Note that $T^{1/2+\varepsilon} \ll T_0 \ll T$ since the inequalities $b \leq 1$ and $0 < \theta < 2b - \frac{3}{2}$ imply that $\frac{3}{4} < b \leq 1$. Hence $w(t)$ satisfies the conditions in the hypothesis of Theorem 1.1.2. Thus we can apply Theorem 1.1.2 with $w(t)$ defined by (2.16.5).

We need to simplify the right-hand side of (1.1.6) for our choice of $w(t)$. It follows from (2.16.6), (2.16.7), and the definition (1.1.5) of W that

$$W = (1 + o(1))T^b \quad (2.16.10)$$

as $T \rightarrow \infty$. This simplifies the main terms and shows that the first error term on the right-hand side of (1.1.6) is $o(T^b)$ for our specific $w(t)$. To simplify the second error term, observe that it follows from (2.16.9) that

$$T_0 \gg_\varepsilon T^{b-\varepsilon} \quad (2.16.11)$$

for arbitrarily small $\varepsilon > 0$. Thus the second error term in (1.1.6) is bounded by

$$\ll \frac{T^{\theta\ell+\theta}}{T^{(b-\varepsilon)\ell}} T^{\frac{3}{2}+\frac{\ell}{2}+\varepsilon} = T^{\ell(\theta-b+\varepsilon+\frac{1}{2})+\theta+\frac{3}{2}+\varepsilon}.$$

Since $\theta < 2b - \frac{3}{2}$ and $b \leq 1$, it follows that $\theta - b + \frac{1}{2} < 0$. Thus, if $A > 0$ is any constant, then we can choose ε small enough and ℓ large enough such that

$$\ell \left(\theta - b + \varepsilon + \frac{1}{2} \right) + \theta + \frac{3}{2} + \varepsilon < -A.$$

Hence the second error term in (1.1.6) is $O(T^{-A})$ for a suitably chosen ℓ . Next, to simplify the third error term, observe that our assumption $\theta < 2b - \frac{3}{2}$ implies that $\frac{\theta}{2} + \frac{3}{4} < b$ and so

$$y^{1/2} T^{3/4} \log^3 T = T^{\frac{\theta}{2}+\frac{3}{4}} \log^3 T = o(T^b)$$

as $T \rightarrow \infty$. This ends the proof of Corollary 1.1.1.

Now we prove Corollary 1.1.2. Let $b \leq 1$ be a real number with $\theta < \frac{13}{22}b - \frac{6}{11}$. We continue to use the function $w(t)$ defined by (2.16.5). As we have proved above, $w(t)$ satisfies the conditions in the hypothesis of Theorem 1.1.3, and so we can apply Theorem 1.1.3. We can simplify the main term and the first error term in (1.1.8) using (2.16.10). To bound the second error term, observe that our assumption $\theta < \frac{13}{22}b - \frac{6}{11}$ implies

$$\frac{11}{2}\theta + 3 - \frac{9}{4}b < b.$$

It follows from this and (2.16.11) that the second error term in (1.1.8) is

$$\ll y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4} \ll T^{\frac{11}{2}\theta+3-\frac{9}{4}b+\varepsilon} = o(T^b) \quad (2.16.12)$$

for small enough ε . This proves Corollary 1.1.2.

2.17 The fourth moment of $Z(t')M(t)$

We are almost ready to complete the proof of Theorem 1.1.1. Before we can do so, we need to prove an asymptotic formula for each of the integrals on the right-hand

side of (2.1.1). We do this for one of them in this section, and prove the following theorem.

Theorem 2.17.1. *Let $0 < \theta < \frac{1}{22}$ and $a \in \mathbb{R}$ be fixed. Suppose that $w(t)$ is defined by (2.16.5). If $\frac{12}{13} + \frac{22}{13}\theta < b < 1 - 2\theta$, then*

$$\int_{-\infty}^{\infty} |Z(t')|^4 |M(t)|^4 w(t) dt = C(a, \theta) T^b + o(T^b) \quad (2.17.1)$$

as $T \rightarrow \infty$, where $C(a, \theta)$ is the same as that in Theorem 1.1.3.

Proof. For brevity, let $\mathcal{J}_0$ denote the integral on the left-hand side of (2.17.1). Recall that $|Z(t)| = |\zeta(\frac{1}{2} + it)|$. We use the definition (1.1.4) of $M(t)$ to write

$$\mathcal{J}_0 = \sum_{n_1, n_2, n_3, n_4 \leq y} \left\{ \prod_{j=1}^4 \frac{\mu(n_j)}{n_j^{1/2}} \left(\frac{\log(y/n_j)}{\log y} \right)^2 \right\} \int_{-\infty}^{\infty} \left(\frac{(n_3 n_4)'}{(n_1 n_2)'} \right)^{-it} |\zeta(\frac{1}{2} + it')|^4 w(t) dt, \quad (2.17.2)$$

where $(n_1 n_2)' = n_1 n_2 / (n_1 n_2, n_3 n_4)$ and $(n_3 n_4)' = n_3 n_4 / (n_1 n_2, n_3 n_4)$.

We first estimate the inner integral on the right-hand side. Recall that $w(t)$ satisfies the property (2.16.7). Thus, if $w(t) \neq 0$, then it follows from (2.2.5) that

$$-t = -t' + \frac{2a}{\log \frac{T}{2\pi}} + O\left(\frac{T^{b-1}}{\log^2 T}\right) \quad (2.17.3)$$

since $b < 1$. This formula implies that

$$\left(\frac{(n_3 n_4)'}{(n_1 n_2)'} \right)^{-it} = \left(\frac{(n_3 n_4)'}{(n_1 n_2)'} \right)^{-it' + i\rho} \left(1 + O\left(\frac{T^{b-1}}{\log T}\right) \right)$$

for $n_1, n_2, n_3, n_4 \leq y$, where ρ is defined by (2.5.5). We insert this into (2.17.2) and apply the definition (1.1.4) of $M(t)$ to deduce that

$$\mathcal{J}_0 = \int_{-\infty}^{\infty} |\zeta(\frac{1}{2} + it')|^4 |M(t' - \rho)|^4 w(t) dt + \mathcal{E}, \quad (2.17.4)$$

where

$$\mathcal{E} \ll \frac{y^2 T^{b-1}}{\log T} \int_{-\infty}^{\infty} |\zeta(\frac{1}{2} + it')|^4 w(t) dt. \quad (2.17.5)$$

We need to bound the integral on the right-hand side of (2.17.5). To do this, we define $g(t)$ to be the inverse function of $t \mapsto t'$. Recall that t' is defined by (1.1.3). Hence $g(t)$ is defined by

$$\vartheta(g(t)) - \vartheta(t) = -a \quad (2.17.6)$$

for large enough t . Note that $g(t)' = g(t') = t$. Since a can be any fixed real number in the hypothesis of Lemma 2.2.4, $g(t)$ can take the role of t' in its statement. That is, we may replace a by $-a$ in the statement of Lemma 2.2.4 to deduce that

$$\frac{d}{dt}g(t) = 1 + O\left(\frac{1}{t \log^2 t}\right), \quad (2.17.7)$$

and

$$\frac{d^n}{dt^n}g(t) \ll_n \frac{1}{t^n \log t}, \quad n \geq 2, \quad (2.17.8)$$

for large enough t .

Using the properties of $g(t)$, we can bound the integral on the right-hand side of (2.17.5) via Proposition 2.5.1. Indeed, make a change of variable and use the estimate (2.17.7) to deduce that

$$\begin{aligned} \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it')|^4 w(t) dt &= \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it)|^4 w(g(t)) \frac{dg}{dt}(t) dt \\ &\ll \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it)|^4 w(g(t)) dt. \end{aligned} \quad (2.17.9)$$

Note that $w(g(t))$ is smooth because both $w(t)$ and $g(t)$ are. Also, it follows from (2.16.7) and (2.2.5) that $w(g(t))$ is supported on $[T/2, 4T]$ for large T . To bound the derivatives of $w(g(t))$, we use Lemma 2.2.3 with $f(t) = w(t)$ and $g(t)$ defined by (2.17.6). We apply the estimates (2.16.8), (2.17.7), and (2.17.8) to the resulting equation for $(d^j/dt^j)w(g(t))$ to see that it is $\ll_j T_0^{-j}$ for all $j \geq 0$, where T_0 is defined by (2.16.9). Therefore the function $w(g(t))$ satisfies the conditions in the hypothesis of Theorem 1.1.3. Hence we can apply Proposition 2.5.1 with

$h = k = 1$ and $\rho = 0$ to arrive at

$$\begin{aligned} & \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it)|^4 w(g(t)) dt \\ &= \int_{-\infty}^{\infty} \frac{w(g(t))}{4(2\pi i)^4} \oint \oint \oint \oint K(z_1, z_2, z_3, z_4) \prod_{1 \leq j < \ell \leq 4} (z_j - z_\ell)^2 \prod_{j=1}^4 \frac{dz_j}{z_j^4} dt \quad (2.17.10) \\ & \quad + O(T^{3/4+\varepsilon}(T/T_0)^{9/4}), \end{aligned}$$

where

$$K(z_1, z_2, z_3, z_4) = \left(\frac{t}{2\pi} \right)^{\frac{z_1+z_2-z_3-z_4}{2}} A_{z_1, z_2, -z_3, -z_4}(0).$$

We choose each path of integration to be a circle with center 0 and radius $1/\log T$.

To estimate the right-hand side of (2.17.10), first observe that a change of variable and (2.2.6) give

$$\int_{-\infty}^{\infty} w(g(t)) dt = \int_{-\infty}^{\infty} w(t) \frac{dt'}{dt} dt \ll W,$$

where W is defined by (1.1.5). Using this, the definition (2.5.1), and the Laurent series of $\zeta(s)$ near $s = 1$, we can bound the integral on the right-hand side of (2.17.10) by

$$\ll W \log^4 T.$$

This bound is $\ll T^b \log^4 T$ by (2.16.10). By (2.16.9) and our assumption $b > 12/13$, the error term in (2.17.10) is $o(T^b)$. From these bounds and (2.17.10), it follows that

$$\int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it)|^4 w(g(t)) dt \ll T^b \log^4 T.$$

From this, (2.17.9), and (2.17.5), we arrive at

$$\mathcal{E} \ll y^2 T^{2b-1} \log^3 T = T^{2\theta+2b-1} \log^3 T.$$

Thus $\mathcal{E} = o(T^b)$ by our assumption $b < 1 - 2\theta$. This and (2.17.4) together give

$$\mathcal{J}_0 = \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it')|^4 |M(t' - \rho)|^4 w(t) dt + o(T^b) \quad (2.17.11)$$

as $T \rightarrow \infty$.

The next step is to estimate the integral on the right-hand side of (2.17.11).

By a change of variable, we see that

$$\begin{aligned} & \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it')|^4 |M(t' - \rho)|^4 w(t) dt \\ &= \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it + i\rho)|^4 |M(t)|^4 w(g(t + \rho)) \frac{dg}{dt}(t + \rho) dt. \end{aligned} \quad (2.17.12)$$

It follows from (2.16.7) and (2.2.5) that $w(g(t + \rho))$ is supported on $[T/2, 4T]$ for large enough T . Thus, by (2.17.7), if $w(g(t + \rho))$ is nonzero, then

$$\frac{dg}{dt}(t + \rho) = 1 + O\left(\frac{1}{T \log^2 T}\right).$$

Insert this into (2.17.12) to deduce that

$$\begin{aligned} & \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it')|^4 |M(t' - \rho)|^4 w(t) dt \\ &= \left(1 + O\left(\frac{1}{T \log^2 T}\right)\right) \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it + i\rho)|^4 |M(t)|^4 w(g(t + \rho)) dt. \end{aligned} \quad (2.17.13)$$

By arguments similar to those below (2.17.9), we can show that $w(g(t + \rho))$ satisfies the conditions in the statement of Theorem 1.1.3, with T_0 defined by (2.16.9).

Therefore we can apply Theorem 1.1.3 to write

$$\begin{aligned} \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it + i\rho)|^4 |M(t)|^4 w(g(t + \rho)) dt &= C(a, \theta) W_g \\ &+ O\left(\frac{W_g}{\log T}\right) + O(y^{11/2} T^{3/4+\varepsilon} (T/T_0)^{9/4}), \end{aligned} \quad (2.17.14)$$

where

$$W_g = \int_{-\infty}^{\infty} w(g(t + \rho)) dt.$$

By a change of variable and (2.2.6), we see that

$$W_g = \int_{-\infty}^{\infty} w(t) \frac{dt'}{dt} dt = (1 + o(1)) W,$$

where W is defined by (1.1.5). It follows from this and (2.16.10) that

$$W_g = (1 + o(1))T^b \quad (2.17.15)$$

as $T \rightarrow \infty$. Moreover, since $b > \frac{12}{13} + \frac{22}{13}\theta$, the bound (2.16.12) applies to the second error term in (2.17.14). By this, (2.17.15), and (2.17.14), we conclude that

$$\int_{-\infty}^{\infty} |\zeta(\frac{1}{2} + it + i\rho)|^4 |M(t)|^4 w(g(t + \rho)) dt = C(a, \theta) T^b + o(T^b)$$

as $T \rightarrow \infty$. This together with (2.17.13) and (2.17.11) complete the proof of Theorem 2.17.1. $\square$

2.18 Proof of Theorem 1.1.1

We now prove Theorem 1.1.1. To do this, we choose suitable values for the parameters θ , a , and b . Then we finish the arguments we started in Section 2.1.

Let $0 < \theta < 1/48$. This guarantees that we can choose a real number b with $\frac{12}{13} + \frac{22}{13}\theta < b < 1 - 2\theta$. Also, let $a > 0$ such that the right-hand side of (1.1.6) is negative for large T . We may take $a = \pi$, say.

Our choice of the parameters θ, a, b allows us to apply Corollary 1.1.1, which we proved using the function $w(t)$ defined by (2.16.5). Hence, from (1.1.7) and (2.1.2), we deduce that

$$-\left(\cos a + \frac{4 \sin a}{3a\theta} + o(1)\right) T^b \leq \int_{\mathcal{S}} |Z(t)Z(t')| |M(t)|^2 w(t) dt.$$

We may square both sides because both are positive by our choice of a . Thus

$$\left(\cos a + \frac{4 \sin a}{3a\theta} + o(1)\right)^2 T^{2b} \leq \left(\int_{\mathcal{S}} |Z(t)Z(t')| |M(t)|^2 w(t) dt\right)^2. \quad (2.18.1)$$

The restrictions on θ, a, b also allow us to apply Corollary 1.1.2 and Theorem 2.17.1. We use the special case of Corollary 1.1.2 with $a = 0$ to write

$$\left(\int_{-\infty}^{\infty} |Z(t)|^4 |M(t)|^4 w(t) dt\right)^{1/2} = (C(0, \theta) + o(1))^{1/2} T^{b/2}. \quad (2.18.2)$$

Also, the conclusion of Theorem 2.17.1 gives

$$\left(\int_{-\infty}^{\infty} |Z(t')|^4 |M(t)|^4 w(t) dt \right)^{1/2} = (C(a, \theta) + o(1))^{1/2} T^{b/2}. \quad (2.18.3)$$

We now see from (2.1.1), (2.18.1), (2.18.2), and (2.18.3) that

$$(\Delta + o(1))T^b \leq \mathcal{N}, \quad (2.18.4)$$

where $\mathcal{N} = \int_{-\infty}^{\infty} \mathbf{1}_{\mathcal{S}}(t) w(t) dt$ and

$$\Delta = \frac{(\cos a + \frac{4 \sin a}{3a\theta})^2}{C(0, \theta)^{1/2} C(a, \theta)^{1/2}}.$$

The properties (2.16.6) and (2.16.7) of $w(t)$ imply that

$$\begin{aligned} \mathcal{N} &= \int_{-\infty}^{\infty} \mathbf{1}_{\mathcal{S}}(t) w(t) dt = \int_T^{T+T^b} \mathbf{1}_{\mathcal{S}}(t) dt + O\left(\frac{T^b}{\sqrt{\log T}}\right) \\ &= \mathcal{M} + o(T^b), \end{aligned} \quad (2.18.5)$$

where $\mathcal{M}$ is the measure of the set $\mathcal{S} \cap [T, T + T^b]$. Hence (2.18.4) gives a lower bound for the value of $\mathcal{M}$.

From this estimate for $\mathcal{M}$, we can deduce a lower bound for the number of zeros of $\zeta(s)$ on the critical line, as follows. Recall that $\mathcal{S}$ is the set of $t \in [T/2, 4T]$ for which $Z(t)Z(t') < 0$. Let $t \in \mathcal{S} \cap [T, T + T^b]$, so that $Z(t)$ and $Z(t')$ are of opposite sign. This implies that there is a γ such that $t < \gamma < t'$ and $Z(\gamma) = 0$. Since $t \in [T, T + T^b]$, the inequality $t < \gamma < t'$ implies $T < \gamma < (T + T^b)'$, which gives $T < \gamma < T + T^b + 1$ for large T by (2.2.5). Now (2.17.3) holds since $t \in [T, T + T^b]$. Thus it follows from the inequality $t < \gamma < t'$ that

$$t < \gamma < t + \frac{2a}{\log \frac{T}{2\pi}} + O\left(\frac{T^{b-1}}{\log^2 T}\right).$$

We can also write this as

$$\gamma - \frac{2a}{\log T}(1 + o(1)) < t < \gamma.$$

Hence we have proved that the set $\mathcal{S} \cap [T, T + T^b]$ is contained in the union of all intervals of length $(1 + o(1))2a/\log T$ that have right endpoints at zeros γ of $Z(t)$ satisfying $T < \gamma < T + T^b + 1$. The number of such intervals is at most $N_0(T + T^b + 1) - N_0(T)$ because $Z(\gamma) = 0$ if and only if $\zeta(\frac{1}{2} + i\gamma) = 0$. We thus conclude that

$$\mathcal{M} \leq (1 + o(1)) \frac{2a}{\log T} \left(N_0(T + T^b + 1) - N_0(T) \right).$$

It follows from this, (2.18.5), and (2.18.4) that

$$(\Delta + o(1))T^b \leq \frac{2a}{\log T} \left(N_0(T + T^b + 1) - N_0(T) \right).$$

We rearrange this and arrive at

$$\left(\frac{\pi}{a} \Delta + o(1) \right) \frac{T^b}{2\pi} \log T \leq N_0(T + T^b + 1) - N_0(T).$$

By the well-known bound $N(T + 1) - N(T) \ll \log T$ (see Theorem 9.2 of Titchmarsh [64]) and the fact that $N_0(T) \leq N(T)$, it follows that

$$N_0(T + T^b + 1) = N_0(T + T^b) + O(\log T).$$

Hence

$$\left(\frac{\pi}{a} \Delta + o(1) \right) \frac{T^b}{2\pi} \log T \leq N_0(T + T^b) - N_0(T). \quad (2.18.6)$$

The following elementary argument now shows that Theorem 1.1.1 follows from (2.18.6). For large T , let $U_1, U_2, U_3, \dots$ be defined by $U_1 = T^b$ and the recursive relation

$$U_j = (T + U_1 + U_2 + \dots + U_{j-1})^b, \quad j = 2, 3, 4, \dots$$

Let r be the unique integer for which

$$U_1 + \dots + U_r \leq T < U_1 + \dots + U_r + U_{r+1}.$$

Note that the definition of r implies that

$$U_1 + \dots + U_r > T - U_{r+1} = T - (T + U_1 + U_2 + \dots + U_r)^b \geq T - (2T)^b.$$

Hence, since $T - (2T)^b = (1 + o(1))T$, it follows that

$$(1 + o(1))T < U_1 + \cdots + U_r \leq T. \quad (2.18.7)$$

By (2.18.6), we can write

$$\left(\frac{\pi}{a}\Delta + o(1)\right) \frac{U_j}{2\pi} \log T \leq N_0(T + U_1 + U_2 + \cdots + U_j) - N_0(T + U_1 + U_2 + \cdots + U_{j-1})$$

for $j = 1, 2, 3, \dots, r$. Add the corresponding sides of these r inequalities to deduce that

$$\left(\frac{\pi}{a}\Delta + o(1)\right) \frac{U_1 + \cdots + U_r}{2\pi} \log T \leq N_0(T + U_1 + \cdots + U_r) - N_0(T).$$

It follows from this and (2.18.7) that

$$\left(\frac{\pi}{a}\Delta + o(1)\right) \frac{T}{2\pi} \log T \leq N_0(2T) - N_0(T).$$

Now replace T in turn by $\frac{T}{2}, \frac{T}{4}, \frac{T}{8}, \dots$ and add the results to conclude that

$$\left(\frac{\pi}{a}\Delta + o(1)\right) \frac{T}{2\pi} \log T \leq N_0(T).$$

From this and the asymptotic formula (1.0.1), we deduce Theorem 1.1.1 with

$$\kappa \geq \frac{\pi}{a}\Delta = \frac{\pi \left(\cos a + \frac{4 \sin a}{3a\theta}\right)^2}{aC(0, \theta)^{1/2}C(a, \theta)^{1/2}} > 0. \quad (2.18.8)$$

This completes the proof of Theorem 1.1.1.

2.19 Numerical computations

We can use any standard computer package to compute the coefficients of the polynomial $\mathcal{R}(t; y, \rho)$ in (2.15.2). By such numerical computations we are able to determine the values of the coefficients $\lambda_{m,n}$ in (2.15.5) and see that

$$\begin{aligned} C(a, \theta) = & \frac{724}{31185}\theta^4 a^8 + \frac{14684}{42525}\theta^2 a^6 + \frac{2896}{10395}\theta^3 a^6 + \frac{64}{45}a^4 \\ & + \frac{7136}{2835}\theta a^4 + \frac{362}{315}\theta^2 a^4 + \frac{724}{3465}\theta^3 a^4 + \frac{52}{45}\theta^{-2} a^2 \\ & + \frac{272}{63}\theta^{-1} a^2 + \frac{88}{15}a^2 + \frac{416}{135}\theta a^2 + \frac{2}{9}\theta^{-4} \\ & + \frac{64}{45}\theta^{-3} + \frac{10}{3}\theta^{-2} + \frac{68}{21}\theta^{-1} + 1. \end{aligned}$$

Note that

$$C(0, \theta) = \frac{2}{9}\theta^{-4} + \frac{64}{45}\theta^{-3} + \frac{10}{3}\theta^{-2} + \frac{68}{21}\theta^{-1} + 1.$$

This agrees with the conjecture (6.29) of Conrey and Snaith [19].

With the above explicit form of $C(a, \theta)$, we can numerically optimize the value of the lower bound (2.18.8) for κ . For fixed θ close to $1/48$, we find that the value $a \approx 4.3120526$ is optimal, and we arrive at the lower bound

$$\kappa > 0.0001049.$$

If we assume the (yet unproved) statement that Theorems 1.1.2 and 1.1.3 hold for *any* $\theta > 0$, then we can optimize numerically by varying both a and θ . We find that the values $a \approx 3.0407338$ and $\theta \approx 0.9967280$ are optimal, and they give

$$\kappa > 0.0086729.$$

3 Zeros of the Riemann zeta-function near the critical line

In this chapter, we prove the results described in Section 1.2. As in the previous chapter, we follow common practice in analytic number theory and let ε denote an arbitrarily small positive constant that is not necessarily the same in each instance. We also use the standard notation $e(z)$ for $\exp(2\pi iz)$. We consider the variables $T > 1$ and $y > 1$ as large parameters going to infinity.

3.1 Lemmas

We first state and prove a number of technical lemmas we will need in our proofs. The following lemma contains some basic estimates for the function $W(\xi)$ defined by (1.2.7). Its proof is reminiscent of that of Lemma 2.1 of Soundararajan [61].

Lemma 3.1.1. *Let $y > 1$ and $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$, with $0 < \epsilon_0 < \frac{1}{4}$ small enough that*

$$\left| \frac{\Gamma(2-2\sigma)\zeta(2-2\sigma)}{\zeta(2\sigma)} \right| < \frac{e^2}{2}. \quad (3.1.1)$$

If $W(\xi)$ is defined by (1.2.7), then

$$(i) \quad W(y) = 1 + O(y^{1-2\sigma}),$$

(ii) $W(y) \gg 1$ provided that y is large enough,

(iii) $W(y/h) \ll \exp\left(-\frac{h}{2y}\right)$ for $h \geq 4y$, and

(iv) $W(y/h) \ll 1$ for all $h > 0$.

The implied constants may depend on ϵ_0 , but not on y , σ , or h .

Proof. We use the residue theorem and move the line of integration in (1.2.7) to the line $\operatorname{Re}(s) = -\frac{3}{4}$. We estimate the new integral using Stirling's formula and the fact that (see (5.1.1) of Titchmarsh [64])

$$\zeta(s) \ll |\operatorname{Im}(s)|^2 \quad (3.1.2)$$

uniformly for $\operatorname{Re}(s) \geq 0$ and $|\operatorname{Im}(s)| \geq 1$. The result is

$$W(\xi) = 1 + \frac{\Gamma(2-2\sigma)\zeta(2-2\sigma)}{\zeta(2\sigma)}\xi^{1-2\sigma} + O(\xi^{-3/4}) \quad (3.1.3)$$

for $\xi > 0$. Taking $\xi = y$ leads to (i) since $y^{-3/4} < y^{1-2\sigma}$ for $\sigma < \frac{3}{4}$. To prove (ii), observe that $y^{1-2\sigma} \leq e^{-2}$ for $\frac{1}{2} + \frac{1}{\log y} \leq \sigma$. It follows from this, (3.1.1), and (3.1.3) with $\xi = y$ that $W(y) \geq \frac{1}{2} + O(y^{-3/4})$. Statement (ii) then follows for sufficiently large y . Next, to show (iii), use (1.2.7) and the property $z\Gamma(z) = \Gamma(1+z)$ of the gamma function to write

$$W(\xi) = \frac{1}{2\pi i} \int_{(r)} \Gamma(s+3) \frac{\zeta(2\sigma+s)}{\zeta(2\sigma)} \zeta(1+s) \frac{\xi^s}{(s+1)(s+2)} ds \quad (3.1.4)$$

for $\xi > 0$. Note that Cauchy's theorem implies that changing the value of r does not change the value of $W(\xi)$, so long as we keep r positive. If $\operatorname{Re}(s) = r \geq 1$, then $|\zeta(2\sigma+s)| \leq \zeta(2\sigma)$, $\zeta(1+s) \ll 1$, and $|\Gamma(s+3)| \leq \Gamma(r+3)$. Thus it follows from (3.1.4) that

$$W(\xi) \ll \Gamma(r+3)\xi^r \int_{(r)} \frac{|ds|}{|s^2|} \ll \Gamma(r+3)\xi^r$$

for $r \geq 1$. From this and Stirling's formula, we arrive at

$$W(\xi) \ll \frac{(r+3)^{r+\frac{5}{2}}}{e^r} \xi^r.$$

Now take $r = \frac{1}{\xi} - 3$, which is ≥ 1 for $0 < \xi \leq \frac{1}{4}$, to deduce that

$$W(\xi) \ll \exp\left(-\frac{1}{2\xi}\right)$$

for $0 < \xi \leq \frac{1}{4}$. Statement (iii) follows upon taking $\xi = y/h$. Finally, to prove (iv), take $\xi = y/h$ in (3.1.3) to deduce that $W(y/h) \ll 1$ for $h \leq 4y$. From this fact and (iii), we arrive at (iv). $\square$

Recall that we use $e(z)$ to denote $e^{2\pi iz}$. If H, K are relatively prime integers with $K > 0$, then define

$$D(s, \alpha, \beta, H/K) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{m^{s+\alpha} n^{s+\beta}} e\left(\frac{mnH}{K}\right) \quad (3.1.5)$$

for complex s, α, β such that $\operatorname{Re}(s + \alpha) > 1$ and $\operatorname{Re}(s + \beta) > 1$. We define this function for all other complex s, α, β by analytic continuation. For the same H, K , define

$$S(w, \alpha, \beta, H/K) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{m^{\alpha} n^{\beta}} e\left(\frac{mnH}{K}\right) e(mnw) \quad (3.1.6)$$

for complex α, β, w with $\operatorname{Im}(w) > 0$.

Lemma 3.1.2. *Let H and K be relatively prime integers with $K > 0$. If $\frac{1}{2} < \sigma \leq \frac{3}{4}$, then*

$$|D(0, 2\sigma - 1, 0, H/K)| \leq 2K^2.$$

Proof. Let $\zeta(s, a)$ be the Hurwitz zeta-function defined by $\zeta(s, a) = \sum_{n=0}^{\infty} (n+a)^{-s}$ for $\operatorname{Re}(s) > 1$ and by analytic continuation for all other complex s . By grouping together terms in the definition (3.1.5) that have the same residues of m and n (mod K), we see that

$$D(s, \alpha, \beta, H/K) = \sum_{a=1}^K \sum_{b=1}^K e\left(\frac{abH}{K}\right) K^{-2s-\alpha-\beta} \zeta(s + \alpha, a/K) \zeta(s + \beta, b/K). \quad (3.1.7)$$

It follows from Euler's summation formula that (see, for example, (2.1.2)—(2.1.4) of Titchmarsh [64])

$$\zeta(s, a) = \frac{1}{2a^s} - \frac{a^{1-s}}{1-s} - s \int_0^\infty \frac{x - [x] - \frac{1}{2}}{(x+a)^{s+1}} dx$$

for $\operatorname{Re}(s) > -1$. This implies $|\zeta(2\sigma - 1, a/K)| \leq 2 + 2K^{2\sigma-1}$ and $|\zeta(0, a/K)| \leq \frac{1}{2}$ for $1 \leq a \leq K$. From these bounds and (3.1.7), we deduce that

$$|D(0, 2\sigma - 1, 0, H/K)| \leq \sum_{a=1}^K \sum_{b=1}^K K^{1-2\sigma} (K^{2\sigma-1} + 1) \leq 2K^2.$$

□

Lemma 3.1.3. *Let H and K be relatively prime integers with $K > 0$. Suppose that α, β, w are complex numbers with $-1 < \operatorname{Re}(\alpha) < 2$, $-1 < \operatorname{Re}(\beta) < 2$, and $\operatorname{Im}(w) > 0$. If*

$$\max\{1, \operatorname{Re}(\alpha), \operatorname{Re}(\beta), 1 - \operatorname{Re}(\alpha), 1 - \operatorname{Re}(\beta)\} < c < 2,$$

then

$$\begin{aligned} S(w, \alpha, \beta, H/K) &= K^{-1+\alpha-\beta} \zeta(1+\beta-\alpha) \Gamma(1-\alpha) (-2\pi i w)^{\alpha-1} \\ &\quad + K^{-1+\beta-\alpha} \zeta(1+\alpha-\beta) \Gamma(1-\beta) (-2\pi i w)^{\beta-1} \\ &\quad + D(0, \alpha, \beta, H/K) + \frac{1}{2\pi^2 i} \int_{(c)} (-2\pi i w)^{s-1} \Gamma(1-s) \left(\frac{K}{2\pi}\right)^{2s-1-\alpha-\beta} \times \\ &\quad \times \Gamma(s-\alpha) \Gamma(s-\beta) \left\{ D(s, -\alpha, -\beta, \overline{H}/K) \cos\left(\frac{\pi}{2}(\alpha-\beta)\right) \right. \\ &\quad \left. - D(s, -\alpha, -\beta, -\overline{H}/K) \cos\left(\frac{\pi}{2}(2-2s+\alpha+\beta)\right) \right\} ds, \end{aligned}$$

where the function $s \mapsto (-iw)^s$ is the branch with $|\arg(-iw)| < \frac{\pi}{2}$.

Proof. This is Lemma 5 in Conrey [15], so we merely sketch its proof. We insert the Mellin transform

$$\frac{1}{2\pi i} \int_{(c)} \Gamma(s) z^{-s} ds = e^{-z}, \quad |\arg z| < \frac{\pi}{2}, \quad (3.1.8)$$

with $z = -2\pi imnw$ into the definition (3.1.6) and interchange the order of summation. The result is

$$S(w, \alpha, \beta, H/K) = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) (-2\pi iw)^{-s} D(s, \alpha, \beta, H/K) ds$$

because of the definition (3.1.5). We then move the line of integration to the line $\operatorname{Re}(s) = 1 - c$ and use the residue theorem to write $S(w, \alpha, \beta, H/K)$ as the sum of the new integral and the residues of the integrand at $s = 0$, $1 - \alpha$, and $1 - \beta$. We then insert Estermann's functional equation

$$\begin{aligned} D(s, \alpha, \beta, H/K) &= \frac{1}{\pi} \left(\frac{K}{2\pi} \right)^{1-2s-\alpha-\beta} \Gamma(1-s-\alpha) \Gamma(1-s-\beta) \times \\ &\quad \times \left\{ D(1-s, -\alpha, -\beta, \overline{H}/K) \cos\left(\frac{\pi}{2}(\alpha-\beta)\right) \right. \\ &\quad \left. - D(1-s, -\alpha, -\beta, -\overline{H}/K) \cos\left(\frac{\pi}{2}(2s+\alpha+\beta)\right) \right\} \end{aligned}$$

into the new integral, and make the change of variable $s \mapsto 1 - s$. $\square$

For complex s , $\operatorname{Im}(z) \neq 0$, and $j = 1, 2$, define $G_j(z, s) = G_j(z, s; \sigma, u, \Delta)$ by

$$\begin{aligned} G_j(z, s) &= z^{-\sigma+iu} \exp\left(-\frac{1}{4}\Delta^2(\log z)^2\right) (-i\varrho(z-1))^{s-1} \\ &\quad \times (2\pi)^{2\sigma-s-1} \Gamma(s-2\sigma+1) f_j(s) / \sin \pi s, \end{aligned} \tag{3.1.9}$$

where $\varrho = \operatorname{sgn}(\operatorname{Im}(z))$, $f_1(s) = \cos\left(\frac{\pi}{2}(2\sigma-1)\right)$, and $f_2(s) = \cos\left(\frac{\pi}{2}(1-2s+2\sigma)\right)$. In this definition, the factor $z^{-\sigma+iu}$ and the exponent $\log z$ are defined as having $|\arg z| < \pi$, while the factor $(-i\varrho(z-1))^{s-1}$ is defined to have $|\arg(-i\varrho(z-1))| < \pi/2$.

Lemma 3.1.4. *Let $\frac{1}{2} \leq \sigma \leq \frac{3}{4}$, $0 < \eta < \frac{1}{4}$, $u > 0$, $0 < \lambda < \frac{\pi}{2}$, and $\Delta > e^{99}$. Suppose that $\lambda\Delta \ll 1$ and $\lambda u \ll 1$. Let $G_j(z, s)$ be defined by (3.1.9). If $c = 2\sigma + \eta$, then*

$$\int_0^{\infty e^{\pm i\lambda}} \int_{(c)} (1 + |s|) |G_j(z, s) ds dz| \ll_{\eta} \lambda^{-\eta-\frac{5}{2}} \left(\frac{\log \Delta}{\Delta} \right)^{\eta+\frac{7}{2}}$$

for $j = 1, 2$. On the other hand, if $c = 2\sigma + \eta - 1$, then

$$\int_0^{\infty e^{\pm i\lambda}} \int_{(c)} (1 + |s|) |G_j(z, s)| ds dz \ll_{\eta} \lambda^{2\sigma - \frac{7}{2}} \left(\frac{\log \Delta}{\Delta} \right)^{\eta + \frac{5}{2}}$$

for $j = 1, 2$.

Proof. The proof is similar to the proof of Lemma 5 of Balasubramanian, Conrey, and Heath-Brown [3]. We assume that the sign of $\pm\lambda$ is positive, as a similar argument applies to the other case. Write $z = xe^{i\lambda}$ and $s = c + it$, with $x > 0$ and $t \in \mathbb{R}$. We bound the factors of $G_j(z, s)$ as follows:

$$\begin{aligned} z^{-\sigma + iu} &\ll x^{-\sigma}, \\ \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) &\ll \exp\left(-\frac{\Delta^2}{4}\log^2 x\right), \\ \Gamma(s - 2\sigma + 1) &\ll (1 + |t|)^{c - 2\sigma + \frac{1}{2}} e^{-\frac{\pi}{2}|t|} \end{aligned}$$

by Stirling's formula,

$$\frac{f_j(s)}{\sin \pi s} \ll 1,$$

and

$$(-i\varrho(z - 1))^{s-1} \ll a(x, \lambda) e^{-b(x, \lambda)t} e^{\frac{\pi}{2}t},$$

where

$$a(x, \lambda) = |xe^{i\lambda} - 1|^{c-1} \tag{3.1.10}$$

and

$$b(x, \lambda) = \arg(xe^{i\lambda} - 1), \tag{3.1.11}$$

with the value of $\arg$ in the interval $(0, \pi)$. It follows from these bounds that

$$\begin{aligned} &\int_0^{\infty e^{i\lambda}} \int_{(c)} (1 + |s|) |G_j(z, s)| ds dz \\ &\ll \int_0^{\infty} \int_{-\infty}^{\infty} a(x, \lambda) x^{-\sigma} \exp\left(-\frac{\Delta^2}{4}\log^2 x\right) (1 + |t|)^{c - 2\sigma + \frac{3}{2}} e^{-\vartheta|t|} dt dx, \end{aligned} \tag{3.1.12}$$

where $\vartheta = b(x, \lambda)$ for $t \geq 0$, and $\vartheta = \pi - b(x, \lambda)$ for $t < 0$. Using the integral definition of the gamma function, we can show that

$$\int_0^\infty (1+t)^{c-2\sigma+\frac{3}{2}} e^{-\vartheta t} dt \ll \vartheta^{-c+2\sigma-\frac{5}{2}}. \quad (3.1.13)$$

We split the right-hand side of (3.1.12) into two parts, one with $t \geq 0$ and the other with $t < 0$, and use (3.1.13) to bound each part to arrive at

$$\int_0^{\infty e^{i\lambda}} \int_{(c)} (1+|s|) |G_j(z, s)| ds dz \ll I + J, \quad (3.1.14)$$

where

$$I = \int_0^\infty a(x, \lambda) x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) b(x, \lambda)^{-c+2\sigma-\frac{5}{2}} dx \quad (3.1.15)$$

and

$$J = \int_0^\infty a(x, \lambda) x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) (\pi - b(x, \lambda))^{-c+2\sigma-\frac{5}{2}} dx. \quad (3.1.16)$$

To bound these integrals, we need estimates for $a(x, \lambda)$ and $b(x, \lambda)$. We may write the definition (3.1.11) of $b(x, \lambda)$ in terms of arctan, and use the properties of arctan to prove that

$$b(x, \lambda) \gg \begin{cases} \lambda & \text{if } x > 0, \\ \frac{\lambda \Delta}{\log \Delta} & \text{if } |x-1| \leq \frac{\log \Delta}{\Delta}, \end{cases} \quad (3.1.17)$$

and

$$\pi - b(x, \lambda) \gg \begin{cases} x\lambda & \text{if } 0 < x \leq \frac{1}{4}, \\ \lambda & \text{if } x \geq \frac{1}{4}, \\ \frac{\lambda \Delta}{\log \Delta} & \text{if } |x-1| \leq \frac{\log \Delta}{\Delta}. \end{cases} \quad (3.1.18)$$

Moreover, since $|xe^{i\lambda} - 1|^2 = x^2 - 2x \cos \lambda + 1$ and

$$\lambda^2 \ll x^2 - 2x \cos \lambda + 1 \ll \max\{1, x^2\},$$

it follows from the definition (3.1.10) of $a(x, \lambda)$ that

$$a(x, \lambda) \ll \max\{1, x^{c-1}\} \quad (3.1.19)$$

for $c > 1$, while

$$a(x, \lambda) \ll \lambda^{c-1} \quad (3.1.20)$$

for $c < 1$. In view of these estimates, we bound the integrals I and J as follows. We split the integral I defined by (3.1.15) into three parts and write $I = I_1 + I_2 + I_3$, where I_1 is the part with $0 < x \leq 1 - \frac{\log \Delta}{\Delta}$, I_2 the part with $1 - \frac{\log \Delta}{\Delta} \leq x \leq 1 + \frac{\log \Delta}{\Delta}$, and I_3 the part with $x > 1 + \frac{\log \Delta}{\Delta}$. We use (3.1.17) to estimate $b(x, \lambda)$ and (3.1.19) or (3.1.20) to estimate $a(x, \lambda)$, depending on whether $c > 1$ or $c < 1$. The results when $c > 1$ are

$$\begin{aligned} I_1 &\ll \int_0^{1 - \frac{\log \Delta}{\Delta}} x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \lambda^{-c+2\sigma-\frac{5}{2}} dx \\ &\ll \exp\left(-\frac{1}{4} \log^2 \Delta\right) \lambda^{-c+2\sigma-\frac{5}{2}}, \end{aligned}$$

$$\begin{aligned} I_2 &\ll \int_{1 - \frac{\log \Delta}{\Delta}}^{1 + \frac{\log \Delta}{\Delta}} x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \left(\frac{\lambda \Delta}{\log \Delta}\right)^{-c+2\sigma-\frac{5}{2}} dx \\ &\ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{-c+2\sigma-\frac{5}{2}}, \end{aligned}$$

and

$$\begin{aligned} I_3 &\ll \int_{1 + \frac{\log \Delta}{\Delta}}^{\infty} x^{c-1-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \lambda^{-c+2\sigma-\frac{5}{2}} dx \\ &\ll \exp\left(-\frac{1}{8} \log^2 \Delta\right) \lambda^{-c+2\sigma-\frac{5}{2}}; \end{aligned}$$

therefore

$$I = I_1 + I_2 + I_3 \ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{-c+2\sigma-\frac{5}{2}} \quad (3.1.21)$$

for $c > 1$. When $c < 1$, the estimate (3.1.20) gives rise to an extra factor λ^{c-1} , and so the result is

$$I \ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{2\sigma-\frac{7}{2}} \quad (3.1.22)$$

for $c < 1$. Similarly, we split the integral J defined by (3.1.16) into four parts and write $J = J_1 + J_2 + J_3 + J_4$, where J_1 is the part with $0 < x < \frac{1}{4}$, J_2 the part with $\frac{1}{4} \leq x < 1 - \frac{\log \Delta}{\Delta}$, J_3 the part with $1 - \frac{\log \Delta}{\Delta} \leq x \leq 1 + \frac{\log \Delta}{\Delta}$, and J_4 the part with $x > 1 + \frac{\log \Delta}{\Delta}$. We use (3.1.18) to estimate $\pi - b(x, \lambda)$ and (3.1.19) or (3.1.20) to estimate $a(x, \lambda)$, depending on whether $c > 1$ or $c < 1$. The results when $c > 1$ are

$$\begin{aligned}
J_1 &\ll \int_0^{1/4} x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) (x\lambda)^{-c+2\sigma-\frac{5}{2}} dx \\
&\ll \exp\left(-\frac{1}{8} \log^2 \Delta\right) \lambda^{-c+2\sigma-\frac{5}{2}}, \\
J_2 &\ll \int_{1/4}^{1-\frac{\log \Delta}{\Delta}} x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \lambda^{-c+2\sigma-\frac{5}{2}} dx \\
&\ll \exp\left(-\frac{1}{4} \log^2 \Delta\right) \lambda^{-c+2\sigma-\frac{5}{2}}, \\
J_3 &\ll \int_{1-\frac{\log \Delta}{\Delta}}^{1+\frac{\log \Delta}{\Delta}} x^{-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \left(\frac{\lambda \Delta}{\log \Delta}\right)^{-c+2\sigma-\frac{5}{2}} dx \\
&\ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{-c+2\sigma-\frac{5}{2}},
\end{aligned}$$

and

$$\begin{aligned}
J_4 &\ll \int_{1+\frac{\log \Delta}{\Delta}}^{\infty} x^{c-1-\sigma} \exp\left(-\frac{\Delta^2}{4} \log^2 x\right) \lambda^{-c+2\sigma-\frac{5}{2}} dx \\
&\ll \exp\left(-\frac{1}{8} \log^2 \Delta\right) \lambda^{-c+2\sigma-\frac{5}{2}};
\end{aligned}$$

therefore

$$J = J_1 + J_2 + J_3 + J_4 \ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{-c+2\sigma-\frac{5}{2}} \quad (3.1.23)$$

for $c > 1$. When $c < 1$, the bound (3.1.20) gives an extra factor of λ^{c-1} , and so the estimate is

$$J \ll \left(\frac{\log \Delta}{\Delta}\right)^{c-2\sigma+\frac{7}{2}} \lambda^{2\sigma-\frac{7}{2}} \quad (3.1.24)$$

for $c < 1$. The lemma now follows from (3.1.14), (3.1.21), and (3.1.23) for $c = 2\sigma + \eta$, and from (3.1.14), (3.1.22), and (3.1.24) for $c = 2\sigma + \eta - 1$. $\square$

The following two lemmas are essentially from Conrey [15] and Deshouillers and Iwaniec [23].

Lemma 3.1.5. *Let v and q be relatively prime positive integers. For an integer b with $(b, v) = 1$, let $\bar{b}$ satisfy $b\bar{b} \equiv 1 \pmod{v}$. If n is any integer, then*

$$\sum_{\substack{r \leq b \leq w \\ (b, vq)=1}} e\left(\frac{n\bar{b}}{v}\right) \ll_{\varepsilon} (vq)^{\varepsilon} v^{1/2} (n, v)^{1/2} \left(q + \frac{w-r}{v}\right).$$

Proof. If c is a positive integer, then Weil's bound for Kloosterman sums implies (see, for example, Lemma 4 of Hooley [39])

$$\sum_{\substack{a \in I \\ (a, c)=1}} e\left(\frac{m\bar{a}}{c}\right) \ll_{\varepsilon} (m, c)^{1/2} c^{1/2+\varepsilon} \quad (3.1.25)$$

for any integer m and any interval I of length not more than c , where $\bar{a}$ is any integer that satisfies $a\bar{a} \equiv 1 \pmod{c}$. Now write

$$\sum_{\substack{r \leq b \leq w \\ (b, vq)=1}} e\left(\frac{n\bar{b}}{v}\right) = \sum_{\substack{r \leq b \leq w \\ (b, vq)=1}} e\left(\frac{nq\bar{b}}{vq}\right). \quad (3.1.26)$$

We choose the integers $\bar{b}$ on the left-hand side of (3.1.26) in a such a way that they satisfy $b\bar{b} \equiv 1 \pmod{vq}$ as well. This is possible since v and q are relatively prime. The interval $[r, w]$ can be split into at most $1 + (w-r)/(vq)$ intervals of length not more than vq . It follows from this and (3.1.25) with $c = vq$ and $m = nq$ that

$$\sum_{\substack{r \leq b \leq w \\ (b, vq)=1}} e\left(\frac{nq\bar{b}}{vq}\right) \ll_{\varepsilon} \left(1 + \frac{w-r}{vq}\right) (nq, vq)^{1/2} (vq)^{1/2+\varepsilon}.$$

The lemma follows from this, (3.1.26), and the fact that $(nq, vq) = (n, v)q$. $\square$

Lemma 3.1.6. *For integers a, b relatively prime to v , let $\bar{a}$ and $\bar{b}$ satisfy $a\bar{a} \equiv b\bar{b} \equiv 1 \pmod{v}$. If $A, B, N, V > 0$ and $C(a, \ell) \ll 1$ then*

$$\sum_{B < b \leq 2B} \sum_{\substack{V < v \leq 2V \\ (b, v) = 1}} \left| \sum_{\substack{A < a \leq 2A \\ (a, v) = 1}} \sum_{N < \ell \leq 2N} C(a, \ell) e\left(\ell \frac{\bar{a}\bar{b}}{v}\right) \right| \\ \ll_{\varepsilon} (ABNV)^{\varepsilon} \left\{ \sum_{(a, n, u, v) \in E} A^a N^n (AB)^u V^v \right\}^{1/4},$$

where E is the set containing $(-2, 2, 4, 4)$, $(1, 2, 3, 4)$, $(3, 2, 3, 3)$, $(-1, 4, 3, 4)$, $(1, 4, 3, 3)$, $(0, 4, 4, 2)$, and $(1, 3, 4, 2)$.

Proof. Lemma 1 of Deshouillers and Iwaniec [23] implies that

$$\sum_{B < b \leq 2B} \sum_{\substack{V < v \leq 2V \\ (b, v) = 1}} \left| \sum_{\substack{A < a \leq 2A \\ (a, v) = 1}} \sum_{N < \ell \leq 2N} C(a, \ell) e\left(\ell \frac{\bar{a}\bar{b}}{v}\right) \right| \\ \ll_{\varepsilon} (ABNV)^{\frac{1}{2} + \varepsilon} \left\{ (VB)^{\frac{1}{2}} + (N + A)^{\frac{1}{4}} \left[VB(N + A)(V + A^2) + NA^2B^2 \right]^{\frac{1}{4}} \right\}.$$

Let $U = AB$ and write this upper bound as

$$(UNV)^{\frac{1}{2} + \varepsilon} \left\{ (A^{-1}VU)^{\frac{1}{2}} + (N + A)^{\frac{1}{4}} \left[A^{-1}VU(N + A)(V + A^2) + NU^2 \right]^{\frac{1}{4}} \right\}.$$

We use $x^{1/4} + y^{1/4} \ll (x + y)^{1/4}$ and $(x + y)^2 \ll x^2 + y^2$ to bound this, and the result is the lemma. $\square$

Recall the functional equation

$$\zeta(s) = \chi(s)\zeta(1 - s), \quad (3.1.27)$$

where $\chi(s)$ is defined by

$$\chi(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi}{2}s\right) \Gamma(1 - s). \quad (3.1.28)$$

The following is Lemma 3 of Conrey [15]. At the center of its proof is the Plancherel Theorem from Fourier analysis.

Lemma 3.1.7. *Let $\Delta > 0$, and let s_0 , β , and ω be complex numbers with $\omega \neq 0$.*

If $\max\{1, \operatorname{Re}(\beta)\} < c < 2$, then

$$\begin{aligned} & \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s-s_0)^2}{\Delta^2}\right) \chi(1+\beta-s)\omega^{-s} ds \\ &= \omega^{-\beta} \int_0^\infty z^{s_0-\beta} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) (e(-\omega z) + e(\omega z)) \frac{dz}{z}. \end{aligned}$$

Proof. Since this is Lemma 3 in Conrey [15], we merely sketch its proof. It suffices to show that

$$\begin{aligned} & \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s-s_0)^2}{\Delta^2}\right) \chi(1-s)\omega^{-s} ds \\ &= \int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) (e(-\omega z) + e(\omega z)) \frac{dz}{z} \end{aligned} \tag{3.1.29}$$

for $c > 0$, as the lemma follows from this and a change of variable. Assume first that ω is real. Use the definition (3.1.28) of $\chi(1-s)$ to write the left-hand side of (3.1.29) as $J_1 + J_2$, where

$$J_1 = \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s-s_0)^2}{\Delta^2}\right) \Gamma(s)(2\pi)^{-s} e^{-i\frac{\pi}{2}s} \omega^{-s} ds$$

and

$$J_2 = \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s-s_0)^2}{\Delta^2}\right) \Gamma(s)(2\pi)^{-s} e^{i\frac{\pi}{2}s} \omega^{-s} ds.$$

To evaluate J_1 , we write it as

$$J_1 = \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} F(s)G(s)x^{-s} ds, \tag{3.1.30}$$

where

$$F(s) = \exp\left(\frac{(s-s_0)^2}{\Delta^2}\right), \tag{3.1.31}$$

$$G(s) = \Gamma(s)(2\pi)^{-s} e^{-i\frac{\pi}{2}s} e^{i\epsilon s} \tag{3.1.32}$$

and

$$x = \omega e^{i\varepsilon}. \quad (3.1.33)$$

We then apply the formula (see (2.1.17) of Titchmarsh [63])

$$\frac{1}{2\pi i} \int_{(c)} F(s)G(s)x^{-s} ds = \int_0^\infty f\left(\frac{1}{z}\right) g(xz) \frac{dz}{z}, \quad (3.1.34)$$

where

$$f(z) = \frac{1}{2\pi i} \int_{(c)} F(s)z^{-s} ds \quad (3.1.35)$$

and

$$g(z) = \frac{1}{2\pi i} \int_{(c)} G(s)z^{-s} ds. \quad (3.1.36)$$

The formula (3.1.34) is an analogue of the Plancherel Theorem for Mellin transforms. It follows from (3.1.31) and (3.1.35) that

$$f(z) = \frac{\Delta}{2\sqrt{\pi}} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) z^{-s_0}, \quad (3.1.37)$$

while (3.1.8), (3.1.32), and (3.1.36) imply

$$g(z) = e(-ze^{-i\varepsilon}). \quad (3.1.38)$$

From (3.1.33), (3.1.34), (3.1.37), and (3.1.38), we deduce that

$$\frac{1}{2\pi i} \int_{(c)} F(s)G(s)x^{-s} ds = \frac{\Delta}{2\sqrt{\pi}} \int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) e(-\omega z) \frac{dz}{z}.$$

This and (3.1.30) now give

$$J_1 = \int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) e(-\omega z) \frac{dz}{z}.$$

A similar argument applies to J_2 and leads to

$$J_2 = \int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) e(\omega z) \frac{dz}{z}.$$

The lemma now follows from adding these two expressions for J_1 and J_2 . $\square$

The following is Lemma 4 of Balasubramanian, Conrey, and Heath-Brown [3].

Lemma 3.1.8. *If s_0 is a complex number and $\Delta > 0$, then*

$$\int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \frac{dz}{z} = \frac{2\sqrt{\pi}}{\Delta} \exp\left(\frac{s_0^2}{\Delta^2}\right).$$

Proof. We make the change of variable $v = \frac{1}{2}\Delta \log z$ to write

$$\int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \frac{dz}{z} = \frac{2}{\Delta} \int_{-\infty}^\infty \exp\left(-v^2 + \frac{2vs_0}{\Delta}\right) dv.$$

We complete the square on the right-hand side to deduce that

$$\int_0^\infty z^{s_0} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \frac{dz}{z} = \frac{2}{\Delta} \exp\left(\frac{s_0^2}{\Delta^2}\right) \int_{-\infty}^\infty \exp\left(-\left(v - \frac{s_0}{\Delta}\right)^2\right) dv. \quad (3.1.39)$$

Cauchy's theorem implies

$$\int_{-\infty}^\infty \exp\left(-\left(v - \frac{s_0}{\Delta}\right)^2\right) dv = \int_{-\infty}^\infty \exp(-v^2) dv = \sqrt{\pi}.$$

This and (3.1.39) now prove the lemma. $\square$

Lemma 3.1.9. *Let $\frac{1}{2} < \sigma \leq \frac{3}{4}$, $0 < \delta < \frac{1}{3}$, $0 < \lambda < \frac{\pi}{2}$, and $T > 1$. Let the function $z \mapsto (z-1)^{2\sigma-2}$ be the branch with $|\arg(z-1)| < \pi$. If $\Delta = T^{1-\delta}$ and $\frac{T}{2} \leq u \leq 4T$, then*

$$\int_0^{\infty e^{\pm i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) (z-1)^{2\sigma-2} dz \ll T^{1-2\sigma} + T^{-\delta},$$

with absolute implied constant, provided that $T^{2\delta-1}$ is sufficiently small.

Proof. For brevity, let J denote the integral in the conclusion of the lemma. We use Cauchy's theorem to move the contour of integration to the positive real axis, with an indent around $z = 1$. Since $2\sigma - 2 > -1$, the value of the integral over the indent goes to zero as the length of the indent goes to zero. Therefore

$$J = \int_0^\infty z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) (z-1)^{2\sigma-2} dz.$$

We split this integral into two parts and write

$$J = J_1 + J_2, \quad (3.1.40)$$

where J_1 is the part with $0 < z < 1$, and J_2 the part with $1 < z < \infty$, and treat these parts separately. We first treat the latter part. A change of variable $z \mapsto e^x$ gives

$$J_2 = \int_0^\infty e^{x(1-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (e^x - 1)^{2\sigma-2} dx. \quad (3.1.41)$$

If $x \geq T^{2\delta-1}$, then $\exp\left(-\frac{1}{8}\Delta^2 x^2\right) \leq \exp\left(-\frac{1}{8}T^{2\delta}\right)$. From this and (3.1.41), it follows that

$$J_2 = \int_0^{T^{2\delta-1}} e^{x(1-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (e^x - 1)^{2\sigma-2} dx + O\left(\exp\left(-\frac{1}{8}T^{2\delta}\right)\right). \quad (3.1.42)$$

To estimate the integral on the right-hand side, write

$$\begin{aligned} (e^x - 1)^{2\sigma-2} &= x^{2\sigma-2} e^{-x/2} + e^{-x/2} \left\{ \frac{(e^x - 1)^{2\sigma-1}}{e^{x/2} - e^{-x/2}} - x^{2\sigma-2} \right\} \\ &= x^{2\sigma-2} e^{-x/2} + O(x^{2\sigma-1} e^{-x/2}) \end{aligned}$$

for $0 < x \leq T^{2\delta-1}$ when $T^{2\delta-1}$ is sufficiently small. It follows from this that

$$\begin{aligned} &\int_0^{T^{2\delta-1}} e^{x(1-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (e^x - 1)^{2\sigma-2} dx \\ &= \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx + O\left(\int_0^{T^{2\delta-1}} x^{2\sigma-1} dx\right). \end{aligned}$$

The error term is $\ll (T^{2\delta-1})^{2\sigma} \leq T^{-\delta}$ because $\sigma > \frac{1}{2}$ and $\delta < \frac{1}{3}$. This and (3.1.42) imply

$$J_2 = \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx + O(T^{-\delta}). \quad (3.1.43)$$

We next estimate the integral on the right-hand side by writing

$$\begin{aligned} \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx &= \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx \\ &\quad + \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} \left(\exp\left(-\frac{\Delta^2 x^2}{4}\right) - 1 \right) x^{2\sigma-2} dx. \end{aligned} \quad (3.1.44)$$

Make a change of variable $x \mapsto \frac{2}{\Delta}x$ to see that

$$\begin{aligned} \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} \left(\exp\left(-\frac{\Delta^2 x^2}{4}\right) - 1 \right) x^{2\sigma-2} dx \\ = \left(\frac{2}{\Delta} \right)^{2\sigma-1} \int_0^{T^\delta/2} e^{\frac{2x}{\Delta}(\frac{1}{2}-\sigma+iu)} H_1(x) dx, \end{aligned} \quad (3.1.45)$$

where

$$H_1(x) = x^{2\sigma-2} (\exp(-x^2) - 1).$$

Observe that $H_1(x) \ll \min\{x^{2\sigma}, x^{2\sigma-2}\}$ and $H_1'(x) \ll \min\{x^{2\sigma-1}, x^{2\sigma-3}\}$. Thus integration by parts gives

$$\int_0^{T^\delta/2} e^{\frac{2x}{\Delta}(\frac{1}{2}-\sigma+iu)} H_1(x) dx \ll \frac{\Delta}{u} \ll T^{-\delta}.$$

It follows from this, (3.1.45), (3.1.44), and (3.1.43) that

$$J_2 = \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx + O(T^{-\delta}).$$

We may extend the range of integration of the integral on the right-hand side to infinity because integration by parts implies

$$\int_{T^{2\delta-1}}^\infty e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx \ll \frac{1}{u} (T^{2\delta-1})^{2\sigma-2} \ll T^{-2\delta}.$$

Hence

$$J_2 = \int_0^\infty e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx + O(T^{-\delta}). \quad (3.1.46)$$

We let $\vartheta = \arg(\sigma - \frac{1}{2} - iu)$ be the value in the interval $(-\pi, \pi)$, and write

$$\int_0^\infty e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx = (\sigma - \frac{1}{2} - iu)^{1-2\sigma} \int_0^{\infty e^{i\vartheta}} e^{-z} z^{2\sigma-2} dz.$$

We then use Cauchy's theorem to move the contour of integration to the positive real axis to deduce that

$$\int_0^\infty e^{x(\frac{1}{2}-\sigma+iu)} x^{2\sigma-2} dx = \Gamma(2\sigma-1) (\sigma - \frac{1}{2} - iu)^{1-2\sigma}.$$

It follows from this and (3.1.46) that

$$J_2 = \Gamma(2\sigma - 1)(\sigma - \tfrac{1}{2} - iu)^{1-2\sigma} + O(T^{-\delta}). \quad (3.1.47)$$

Having estimated the integral J_2 in (3.1.40), we next treat the integral J_1 . Make a change of variable $z \mapsto e^{-x}$ to write

$$J_1 = \varrho \int_0^\infty e^{x(\sigma-1-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (1 - e^{-x})^{2\sigma-2} dx, \quad (3.1.48)$$

where

$$\varrho = \exp(i\pi(2\sigma - 2)\operatorname{sgn}(\pm\lambda)), \quad (3.1.49)$$

which depends on the sign of $\pm\lambda$ that we use for the integral J in the conclusion of the lemma. Since $\exp(-\frac{1}{8}\Delta^2 x^2) \leq \exp(-\frac{1}{8}T^{2\delta})$ for $x \geq T^{2\delta-1}$, it follows from (3.1.48) that

$$J_1 = \varrho \int_0^{T^{2\delta-1}} e^{x(\sigma-1-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (1 - e^{-x})^{2\sigma-2} dx + O\left(\exp\left(-\frac{1}{8}T^{2\delta}\right)\right). \quad (3.1.50)$$

To estimate the integral on the right-hand side, write

$$\begin{aligned} (1 - e^{-x})^{2\sigma-2} &= x^{2\sigma-2} e^{x/2} + e^{x/2} \left\{ \frac{(1 - e^{-x})^{2\sigma-1}}{e^{x/2} - e^{-x/2}} - x^{2\sigma-2} \right\} \\ &= x^{2\sigma-2} e^{x/2} + O(x^{2\sigma-1} e^{x/2}). \end{aligned}$$

We deduce from this that

$$\begin{aligned} &\int_0^{T^{2\delta-1}} e^{x(\sigma-1-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) (1 - e^{-x})^{2\sigma-2} dx \\ &= \int_0^{T^{2\delta-1}} e^{x(\sigma-\frac{1}{2}-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx + O\left(\int_0^{T^{2\delta-1}} x^{2\sigma-1} dx\right). \end{aligned}$$

The error term is $\ll (T^{2\delta-1})^{2\sigma} \leq T^{-\delta}$ because $\sigma > \frac{1}{2}$ and $\delta < \frac{1}{3}$. It follows from this and (3.1.50) that

$$J_1 = \varrho \int_0^{T^{2\delta-1}} e^{x(\sigma-\frac{1}{2}-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx + O(T^{-\delta}). \quad (3.1.51)$$

We estimate the integral on the right-hand side by writing

$$\begin{aligned} \int_0^{T^{2\delta-1}} e^{x(\sigma-\frac{1}{2}-iu)} \exp\left(-\frac{\Delta^2 x^2}{4}\right) x^{2\sigma-2} dx &= \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma-iu)} x^{2\sigma-2} dx \\ &+ \int_0^{T^{2\delta-1}} e^{x(\sigma-\frac{1}{2}-iu)} \left(\exp\left(-\frac{\Delta^2 x^2}{4}\right) - e^{x(1-2\sigma)} \right) x^{2\sigma-2} dx. \end{aligned} \quad (3.1.52)$$

Make a change of variable $x \mapsto \frac{2}{\Delta}x$ to see that

$$\begin{aligned} \int_0^{T^{2\delta-1}} e^{x(\sigma-\frac{1}{2}-iu)} \left(\exp\left(-\frac{\Delta^2 x^2}{4}\right) - e^{x(1-2\sigma)} \right) x^{2\sigma-2} dx \\ = \left(\frac{2}{\Delta} \right)^{2\sigma-1} \int_0^{T^\delta/2} e^{\frac{2x}{\Delta}(\frac{1}{2}-\sigma-iu)} H_2(x) dx, \end{aligned} \quad (3.1.53)$$

where

$$H_2(x) = x^{2\sigma-2} \left(\exp\left(-x^2 + \frac{4\sigma-2}{\Delta}x\right) - 1 \right).$$

Observe that $H_2(x) \ll \min\{x^{2\sigma-1}, x^{2\sigma-2}\}$ and

$$H_2'(x) \ll \begin{cases} (2\sigma-1)x^{2\sigma-2} + x^{2\sigma-1} & \text{if } 0 < x \leq 1, \\ x^{2\sigma-3} & \text{if } x \geq 1. \end{cases}$$

Therefore integration by parts gives

$$\int_0^{T^\delta/2} e^{\frac{2x}{\Delta}(\frac{1}{2}-\sigma-iu)} H_2(x) dx \ll \frac{\Delta}{u} \ll T^{-\delta}.$$

It follows from this, (3.1.53), (3.1.52), and (3.1.51) that

$$J_1 = \varrho \int_0^{T^{2\delta-1}} e^{x(\frac{1}{2}-\sigma-iu)} x^{2\sigma-2} dx + O(T^{-\delta}).$$

We may extend the range of integration to infinity and arrive at

$$J_1 = \varrho \int_0^\infty e^{x(\frac{1}{2}-\sigma-iu)} x^{2\sigma-2} dx + O(T^{-\delta}). \quad (3.1.54)$$

in the same way we deduced (3.1.46). Then we move the contour of integration to the positive real axis, as in our derivation of (3.1.47), to conclude that

$$J_1 = \varrho \Gamma(2\sigma-1)(\sigma-\frac{1}{2}+iu)^{1-2\sigma} + O(T^{-\delta}), \quad (3.1.55)$$

where $\arg(\sigma - \frac{1}{2} + iu)$ is the value in $(-\pi, \pi)$.

It now follows from (3.1.40), (3.1.47), and (3.1.55) that

$$J = \Gamma(2\sigma - 1) \left((\sigma - \frac{1}{2} - iu)^{1-2\sigma} + \varrho(\sigma - \frac{1}{2} + iu)^{1-2\sigma} \right) + O(T^{-\delta}). \quad (3.1.56)$$

Now the definition (3.1.49) of ϱ and the power series of $\exp$ imply that

$$\varrho = -1 + O(2\sigma - 1).$$

It follows from this and (3.1.56) that

$$J = \Gamma(2\sigma - 1) \left((\sigma - \frac{1}{2} - iu)^{1-2\sigma} - (\sigma - \frac{1}{2} + iu)^{1-2\sigma} \right) + O(u^{1-2\sigma}) + O(T^{-\delta}) \quad (3.1.57)$$

since $(2\sigma - 1)\Gamma(2\sigma - 1) = \Gamma(2\sigma) \ll 1$. Furthermore, the definition of sine gives

$$\begin{aligned} & (\sigma - \frac{1}{2} - iu)^{1-2\sigma} - (\sigma - \frac{1}{2} + iu)^{1-2\sigma} \\ &= 2i|\sigma - \frac{1}{2} + iu|^{1-2\sigma} \sin((2\sigma - 1) \arg(\sigma - \frac{1}{2} + iu)). \end{aligned}$$

Since $|\arg(\sigma - \frac{1}{2} + iu)| < \frac{\pi}{2}$ and $|\sin x| \leq |x|$ for real x , it follows that

$$(\sigma - \frac{1}{2} - iu)^{1-2\sigma} - (\sigma - \frac{1}{2} + iu)^{1-2\sigma} \ll (2\sigma - 1)u^{1-2\sigma}.$$

This and (3.1.57) imply

$$J \ll u^{1-2\sigma} + T^{-\delta}.$$

The lemma now follows from this and the assumption that $u \geq T/2$. $\square$

3.2 Relating the density of zeros to the second moment

In this section, we relate $N(\sigma, T)$ to the mollified second moment (1.2.4) via standard arguments using Littlewood's lemma. Here and in the rest of the chapter,

we assume that $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$, with $\epsilon_0 > 0$ small enough and $y > 1$ large enough that the conditions in Lemma 3.1.1 are satisfied. First, observe that (ii) and (iv) of Lemma 3.1.1 imply

$$\zeta(s)M(s) = 1 + O\left(\frac{1}{2^{\operatorname{Re}(s)}}\right) \quad (3.2.1)$$

as $\operatorname{Re}(s) \rightarrow \infty$ since the first coefficient of the Dirichlet series definition (1.2.6) equals 1. We choose $b_0 > 1$ such that the error term in (3.2.1) has modulus $\leq \frac{1}{2}$ for $\operatorname{Re}(s) \geq b_0$. Our choice implies that $\zeta(s)M(s)$ has no zeros with real part $\geq b_0$.

Let $\nu(\xi; T, 2T)$ denote the number of zeros $\beta_1 + i\gamma_1$ of $\zeta(s)M(s)$ that have $\beta_1 > \xi$ and $T < \gamma_1 \leq 2T$. Since a zero of $\zeta(s)$ is also a zero of $\zeta(s)M(s)$, it follows that

$$\int_{\sigma}^1 (N(\xi, 2T) - N(\xi, T)) d\xi \leq \int_{\sigma}^1 \nu(\xi; T, 2T) d\xi \leq \int_{\sigma}^{b_0} \nu(\xi; T, 2T) d\xi. \quad (3.2.2)$$

Let $b \geq b_0$. Littlewood's lemma (see, for example, (9.9.1) of Titchmarsh [64]) gives

$$\int_{\sigma}^b \nu(\xi; T, 2T) d\xi = -\frac{1}{2\pi i} \int_{\mathcal{R}} \log(\zeta(s)M(s)) ds, \quad (3.2.3)$$

where $\mathcal{R}$ is the rectangle with corners $b + iT$, $b + i2T$, $\sigma + i2T$, $\sigma + iT$ taken in the positive direction. Here, $\log(\zeta(s)M(s))$ is defined by continuous variation from the point b , provided that $\operatorname{Im}(s)$ is not the ordinate of a zero or pole of $\zeta(s)M(s)$. Otherwise, the definition is

$$\log(\zeta(s)M(s)) = \lim_{\epsilon \rightarrow 0^+} \log(\zeta(s + i\epsilon)M(s + i\epsilon)).$$

We take the real parts of both sides of (3.2.3) to deduce that

$$\begin{aligned} \int_{\sigma}^b \nu(\xi; T, 2T) d\xi &= \frac{1}{2\pi} \int_T^{2T} \log |\zeta(\sigma + it)M(\sigma + it)| dt \\ &+ \frac{1}{2\pi} \int_{\sigma}^b \arg(\zeta(\xi + i2T)M(\xi + i2T)) d\xi - \frac{1}{2\pi} \int_{\sigma}^b \arg(\zeta(\xi + iT)M(\xi + iT)) d\xi \\ &- \frac{1}{2\pi} \int_T^{2T} \log |\zeta(b + it)M(b + it)| dt, \end{aligned} \quad (3.2.4)$$

where $\arg(\zeta(s)M(s))$ is defined as the imaginary part of $\log(\zeta(s)M(s))$. The equation (3.2.1) implies

$$\lim_{b \rightarrow \infty} \int_T^{2T} \log |\zeta(b+it)M(b+it)| dt = 0.$$

Thus we may take $b \rightarrow \infty$ on both sides of (3.2.4) to deduce that

$$\begin{aligned} \int_{\sigma}^{\infty} \nu(\xi; T, 2T) d\xi &= \frac{1}{2\pi} \int_T^{2T} \log |\zeta(\sigma+it)M(\sigma+it)| dt \\ &+ \frac{1}{2\pi} \int_{\sigma}^{\infty} \arg(\zeta(\xi+i2T)M(\xi+i2T)) d\xi - \frac{1}{2\pi} \int_{\sigma}^{\infty} \arg(\zeta(\xi+iT)M(\xi+iT)) d\xi. \end{aligned} \quad (3.2.5)$$

Since $\zeta(s)M(s)$ has no zeros with real part $\geq b_0$, it is true that

$$\int_{\sigma}^{\infty} \nu(\xi; T, 2T) d\xi = \int_{\sigma}^{b_0} \nu(\xi; T, 2T) d\xi.$$

We deduce from this and (3.2.5) that

$$\begin{aligned} \int_{\sigma}^{b_0} \nu(\xi; T, 2T) d\xi &= \frac{1}{2\pi} \int_T^{2T} \log |\zeta(\sigma+it)M(\sigma+it)| dt \\ &+ \frac{1}{2\pi} \int_{\sigma}^{\infty} \arg(\zeta(\xi+i2T)M(\xi+i2T)) d\xi - \frac{1}{2\pi} \int_{\sigma}^{\infty} \arg(\zeta(\xi+iT)M(\xi+iT)) d\xi. \end{aligned} \quad (3.2.6)$$

Our choice for b_0 , described below (3.2.1), also implies

$$\int_{b_0}^{\infty} \arg(\zeta(\xi+iT)M(\xi+iT)) d\xi \ll \int_{b_0}^{\infty} 2^{-\xi} d\xi \ll 1, \quad (3.2.7)$$

and similarly when T is replaced by $2T$. Moreover, (ii) and (iv) of Lemma 3.1.1 and the definition (1.2.6) of $M(s)$ imply that

$$M(s) \ll y^{1+\varepsilon} \quad (3.2.8)$$

for $\operatorname{Re}(s) \geq 0$, and so a standard argument using Jensen's theorem gives (see §9.4 of Titchmarsh [64])

$$\int_{\sigma}^{b_0} \arg(\zeta(\xi+iT)M(\xi+iT)) d\xi \ll \log T + \log y,$$

and similarly when T is changed to $2T$. This and (3.2.7) give

$$\int_{\sigma}^{\infty} \arg(\zeta(\xi + iT)M(\xi + iT)) d\xi \ll \log T + \log y,$$

and the same bound holds when T is replaced by $2T$. It follows from this, (3.2.6), and (3.2.2) that

$$\int_{\sigma}^1 (N(\xi, 2T) - N(\xi, T)) d\xi \leq \frac{1}{2\pi} \int_T^{2T} \log |\zeta(\sigma + it)M(\sigma + it)| dt + O(\log(Ty)). \quad (3.2.9)$$

Since $\log$ is a concave function, it follows from Jensen's inequality that

$$\begin{aligned} \int_T^{2T} \log |\zeta(\sigma + it)M(\sigma + it)| dt &= \frac{1}{2} \int_T^{2T} \log |\zeta(\sigma + it)M(\sigma + it)|^2 dt \\ &\leq \frac{T}{2} \log \left\{ \frac{1}{T} \int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \right\}. \end{aligned}$$

This and (3.2.9) now give the following lemma, which summarizes the results of this section.

Lemma 3.2.1. *Let $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$, with $\epsilon_0 > 0$ small enough and $y > 1$ large enough that the conditions in Lemma 3.1.1 are satisfied. Then*

$$\begin{aligned} \int_{\sigma}^1 (N(\xi, 2T) - N(\xi, T)) d\xi \\ \leq \frac{T}{4\pi} \log \left\{ \frac{1}{T} \int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \right\} + O(\log(Ty)) \end{aligned}$$

for sufficiently large T .

3.3 Relating the second moment to the Kloosterman sum

In this section, we begin the analysis of the mollified second moment (1.2.4), which also appears in the conclusion of Lemma 3.2.1. The first step is to approximate

the second moment by a smoothed version that enables us to use Fourier analysis, in the form of Lemma 3.1.7, to study the moment. Define $w(t)$ by

$$w(t) = \frac{1}{\Delta\sqrt{\pi}} \int_{T-\Delta\log T}^{2T+\Delta\log T} \exp\left(-\frac{(t-u)^2}{\Delta^2}\right) du \quad (3.3.1)$$

for $\Delta > 0$ and $T > 1$. Observe that the change of variable $(t-u)/\Delta \mapsto \xi$ gives

$$\frac{1}{\Delta\sqrt{\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{(t-u)^2}{\Delta^2}\right) du = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp(-\xi^2) d\xi = 1. \quad (3.3.2)$$

Moreover, if $|t-u| \geq \Delta\log T$, then

$$\exp\left(-\frac{(t-u)^2}{2\Delta^2}\right) \leq \exp\left(-\frac{1}{2}\log^2 T\right). \quad (3.3.3)$$

It follows from this, (3.3.2), and (3.3.1) that

$$w(t) = 1 + O\left(\exp\left(-\frac{1}{2}\log^2 T\right)\right)$$

for $T \leq t \leq 2T$. From this, (3.2.8), and the well-known bound for the second moment of zeta (see Theorem 7.2(A) of Titchmarsh [64]), we arrive at

$$\begin{aligned} \int_T^{2T} |\zeta(\sigma+it)M(\sigma+it)|^2 dt &= \int_T^{2T} |\zeta(\sigma+it)M(\sigma+it)|^2 w(t) dt \\ &\quad + O\left(y^{2+\varepsilon} \exp\left(-\frac{1}{4}\log^2 T\right)\right). \end{aligned}$$

We insert into this the definition (3.3.1) of $w(t)$ and extend the interval of integration for the t -integral to $(-\infty, \infty)$ to deduce that

$$\int_T^{2T} |\zeta(\sigma+it)M(\sigma+it)|^2 dt \leq \int_{T-\Delta\log T}^{2T+\Delta\log T} g(u) du + O\left(y^{2+\varepsilon} \exp\left(-\frac{1}{4}\log^2 T\right)\right), \quad (3.3.4)$$

where $g(u) = g(u; \Delta, \sigma, y, \varepsilon)$ is defined by

$$g(u) = \frac{1}{\Delta\sqrt{\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{(t-u)^2}{\Delta^2}\right) |\zeta(\sigma+it)M(\sigma+it)|^2 dt. \quad (3.3.5)$$

We see from (3.3.4) that an estimate for $g(u)$ leads to a bound for the mollified second moment. Most of the rest of the chapter is about proving an asymptotic

formula for $g(u)$. We start by writing $g(u)$ as a contour integral. The definition (1.2.7) of $W(\xi)$ implies that $W(\xi)$ is real-valued for $\xi > 0$. It follows from this and the definition (1.2.6) of $M(s)$ that $\overline{M(\bar{s})} = M(s)$, and so

$$|\zeta(\sigma + it)M(\sigma + it)|^2 = \zeta(\sigma + it)\zeta(\sigma - it)M(\sigma + it)M(\sigma - it).$$

Thus we may write (3.3.5) as

$$g(u) = \frac{1}{i\Delta\sqrt{\pi}} \int_{(\sigma)} \exp\left(\frac{(s - \sigma - iu)^2}{\Delta^2}\right) \zeta(s)\zeta(2\sigma - s)M(s)M(2\sigma - s) ds.$$

We apply the residue theorem and move the line of integration to the line $\operatorname{Re}(s) = c$, where $2\sigma < c < 2$. The residue at the pole $s = 1$ is

$$\frac{1}{i\Delta\sqrt{\pi}} \exp\left(\frac{(1 - \sigma - iu)^2}{\Delta^2}\right) \zeta(2\sigma - 1)M(1)M(2\sigma - 1),$$

which is $\ll y^{2+\varepsilon} \exp(-\log^2 T)$ for $\Delta \geq 1$ and $u \geq \Delta \log T$, by (3.2.8). Hence the result of applying the residue theorem is

$$\begin{aligned} g(u) &= \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s - \sigma - iu)^2}{\Delta^2}\right) \zeta(s)\zeta(2\sigma - s)M(s)M(2\sigma - s) ds \\ &\quad + O(y^{2+\varepsilon} \exp(-\log^2 T)) \end{aligned} \tag{3.3.6}$$

for $\Delta \geq 1$ and $u \geq \Delta \log T$. The functional equation (3.1.27) implies that

$$\zeta(2\sigma - s) = \chi(2\sigma - s)\zeta(1 - 2\sigma + s).$$

We insert this into (3.3.6) and expand the zeta-functions and the factors $M(s)$ and $M(2\sigma - s)$ into Dirichlet series, using the definition (1.2.6), to deduce that

$$\begin{aligned} g(u) &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{n^{1-2\sigma}} \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{k^{2\sigma}} \left(\frac{W(y/h)}{W(y)}\right) \left(\frac{W(y/k)}{W(y)}\right) \times \\ &\quad \times \frac{1}{i\Delta\sqrt{\pi}} \int_{(c)} \exp\left(\frac{(s - \sigma - iu)^2}{\Delta^2}\right) \chi(2\sigma - s) \left(\frac{mnh}{k}\right)^{-s} ds \\ &\quad + O(y^{2+\varepsilon} \exp(-\log^2 T)). \end{aligned}$$

The interchange of order of summation is justified by absolute convergence. It follows from this and Lemma 3.1.7 with $\omega = mn h/k$ and $\beta = 2\sigma - 1$ that

$$\begin{aligned} g(u) = & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{n^{1-2\sigma}} \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{k^{2\sigma}} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) (J_1 + J_2) \\ & + O(y^{2+\varepsilon} \exp(-\log^2 T)), \end{aligned} \quad (3.3.7)$$

where

$$J_1 = \left(\frac{mn h}{k} \right)^{1-2\sigma} \int_0^{\infty} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) e\left(\frac{mn h}{k}z\right) dz$$

and J_2 is the same, except with $e(mn h z/k)$ replaced by $e(-mn h z/k)$. We apply Cauchy's theorem and move the path of integration of J_1 to the half-line $\{re^{i\lambda} : r > 0\}$ and the path of integration of J_2 to the half-line $\{re^{-i\lambda} : r > 0\}$, where $0 < \lambda < \frac{\pi}{2}$. We then write

$$e\left(\frac{mn h}{k}z\right) = e\left(\frac{mn h}{k}\right) e\left(\frac{mn h}{k}(z-1)\right)$$

and interchange the order of summation in (3.3.7) to deduce that

$$\begin{aligned} g(u) = & \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma-1}k} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) (I_1 + I_2) \\ & + O(y^{2+\varepsilon} \exp(-\log^2 T)), \end{aligned}$$

where

$$I_1 = \int_0^{\infty e^{i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) S\left(\frac{H}{K}(z-1), 2\sigma-1, 0, H/K\right) dz$$

and

$$I_2 = \int_0^{\infty e^{-i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) S\left(-\frac{H}{K}(z-1), 2\sigma-1, 0, -H/K\right) dz,$$

where the function S is defined by (3.1.6), $H = h/(h, k)$, and $K = k/(h, k)$. From this and Lemma 3.1.3 with $\alpha = 2\sigma - 1$ and $\beta = 0$, we arrive at

$$g(u) = \mathfrak{M}_1 + \mathfrak{M}_2 + \mathfrak{D}_1 + \mathfrak{D}_2 + \mathfrak{E}, \quad (3.3.8)$$

where $\mathfrak{M}_1$, $\mathfrak{M}_2$, $\mathfrak{D}_1$, $\mathfrak{D}_2$, and $\mathfrak{E}$ are defined by

$$\mathfrak{M}_1 = \zeta(2\sigma) \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma}k^{2\sigma}} (h, k)^{2\sigma} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \mathcal{I} \quad (3.3.9)$$

with

$$\begin{aligned} \mathcal{I} = & -\frac{1}{2\pi i} \int_0^{\infty e^{i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \frac{dz}{z-1} \\ & + \frac{1}{2\pi i} \int_0^{\infty e^{-i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \frac{dz}{z-1}, \end{aligned} \quad (3.3.10)$$

$$\begin{aligned} \mathfrak{M}_2 = & \Gamma(2-2\sigma)\zeta(2-2\sigma) \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{hk} (h, k)^{2-2\sigma} \times \\ & \times \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \mathcal{J} \end{aligned} \quad (3.3.11)$$

with

$$\begin{aligned} \mathcal{J} = & \int_0^{\infty e^{i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \left(-2\pi i(z-1)\right)^{2\sigma-2} dz \\ & + \int_0^{\infty e^{-i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) \left(2\pi i(z-1)\right)^{2\sigma-2} dz, \end{aligned} \quad (3.3.12)$$

$$\begin{aligned} \mathfrak{D}_1 = & \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma-1}k} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \times \\ & \times D(0, 2\sigma-1, 0, H/K) \int_0^{\infty e^{i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) dz, \end{aligned} \quad (3.3.13)$$

$$\begin{aligned} \mathfrak{D}_2 = & \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma-1}k} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \times \\ & \times D(0, 2\sigma-1, 0, -H/K) \int_0^{\infty e^{-i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) dz, \end{aligned} \quad (3.3.14)$$

and, since $\Gamma(1-s)\Gamma(s) = \pi/\sin(\pi s)$,

$$\begin{aligned} \mathfrak{E} = & \frac{1}{2\pi i} \int_0^{\infty e^{i\lambda}} \int_{(c)} \left(G_1(z, s) \mathcal{B}(s) + G_2(z, s) \overline{\mathcal{B}(\overline{s})} \right) ds dz \\ & + \frac{1}{2\pi i} \int_0^{\infty e^{-i\lambda}} \int_{(c)} \left(G_1(z, s) \overline{\mathcal{B}(\overline{s})} + G_2(z, s) \mathcal{B}(s) \right) ds dz, \end{aligned} \quad (3.3.15)$$

with $G_j(z, s)$ defined by (3.1.9) and $\mathcal{B}(s)$ defined by

$$\begin{aligned} \mathcal{B}(s) = & \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma-1}k} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \times \\ & \times \left(\frac{H}{K} \right)^{s-1} K^{2s-2\sigma} D(s, 1-2\sigma, 0, \overline{H}/K). \end{aligned} \quad (3.3.16)$$

We end this section with the following lemma that summarizes the results of the section.

Lemma 3.3.1. *Let $T > 1$, $\Delta > 1$, and $u \geq \Delta \log T$. If $g(u)$ is defined by (3.3.5), then (3.3.8) holds for $0 < \lambda < \frac{\pi}{2}$ and $1 < c < 2$, where the function D is defined by (3.1.5), H denotes $h/(h, k)$, K denotes $k/(h, k)$, and $\overline{H}$ denotes the inverse of $H \pmod{K}$.*

3.4 The error terms

In this section, we estimate the quantity $\mathfrak{E}$ defined by (3.3.15). To do this, we break the sum $\mathcal{B}(s)$ defined by (3.3.16) into parts, and bound the contribution of each part. To facilitate the estimations, we take

$$\Delta = T^{1-\delta}, \quad \lambda = \frac{1}{T}, \quad \text{and} \quad c = 2\sigma + \eta \quad (3.4.1)$$

in Lemma 3.3.1, where $\delta, \eta > 0$ are small positive numbers to be chosen later. Since $c > 2\sigma$, we can write the function D in (3.3.16) as a series through its definition (3.1.5). Thus

$$\begin{aligned} \mathcal{B}(s) = & \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma-1}k} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \times \\ & \times \left(\frac{H}{K} \right)^{s-1} K^{2s-2\sigma} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{m^{s+1-2\sigma} n^s} e\left(\frac{mn\overline{H}}{K} \right). \end{aligned} \quad (3.4.2)$$

We group together terms with the same (h, k) and with the same mn , and let $q = (h, k)$ and $\ell = mn$ to write (3.4.2) as

$$\begin{aligned} \mathcal{B}(s) &= \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_{\substack{H, K \leq y^{1+\varepsilon}/q \\ (HK, q) = (H, K) = 1}} \frac{\mu(H)\mu(K)}{H^{2\sigma-s}K^{2\sigma-s}} \left(\frac{W(y/Hq)}{W(y)} \right) \times \\ &\quad \times \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{\ell=1}^{\infty} \frac{\rho(\ell)}{\ell^s} e\left(\frac{\ell \bar{H}}{K}\right), \end{aligned}$$

where $\rho(\ell)$ is defined by

$$\rho(\ell) = \sum_{m|\ell} m^{2\sigma-1}. \quad (3.4.3)$$

By (iii) of Lemma 3.1.1, we may extend the range of summation over H to infinity, with small error, provided $y^\varepsilon > 4$. That is,

$$\begin{aligned} \mathcal{B}(s) &= \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_{\substack{H \geq 1 \\ (H, q) = 1}} \sum_{\substack{K \leq y^{1+\varepsilon}/q \\ (K, q) = (H, K) = 1}} \frac{\mu(H)\mu(K)}{H^{2\sigma-s}K^{2\sigma-s}} \left(\frac{W(y/Hq)}{W(y)} \right) \times \\ &\quad \times \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{\ell=1}^{\infty} \frac{\rho(\ell)}{\ell^s} e\left(\frac{\ell \bar{H}}{K}\right) + O(\exp(-y^\varepsilon/4)), \end{aligned}$$

where, in bounding the error term, we used (iii) of Lemma 3.1.1 to bound $W(y/Hq)$, (iv) to bound $W(y/Kq)$, and (ii) to bound $W(y)^{-2}$. To facilitate the succeeding estimations, we partition the ranges of summation over K and ℓ into dyadic intervals and write

$$\begin{aligned} \mathcal{B}(s) &= \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N \sum_{\substack{H \geq 1 \\ (H, q) = 1}} \sum_{\substack{V < K \leq 2V \\ (K, Hq) = 1}} \frac{\mu(H)\mu(K)}{H^{2\sigma-s}K^{2\sigma-s}} \times \\ &\quad \times \left(\frac{W(y/Hq)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e\left(\frac{\ell \bar{H}}{K}\right) + O(\exp(-y^\varepsilon/4)), \end{aligned} \quad (3.4.4)$$

where V runs through real numbers $\geq 1/2$ that are of the form $2^{-n}y^{1+\varepsilon}/q$ with n a positive integer, and N runs through integral powers of 2 that are $\geq 1/2$.

The next step is to apply Vaughan's identity (see Conrey [15]) that states that

$$\mu(H) = c_1(H) + c_2(H) + c_3(H), \quad (3.4.5)$$

where

$$c_1(H) = \sum_{\substack{\alpha\beta\gamma=H \\ \alpha>y^{1/4} \\ \beta>y^{1/4}}} \mu_1(\alpha)\mu_1(\beta)\mu(\gamma) \quad (3.4.6)$$

with

$$\mu_1(n) = - \sum_{\substack{d|n \\ d\leq y^{1/4}}} \mu(d), \quad (3.4.7)$$

$$c_2(H) = \begin{cases} 2\mu(H) & \text{if } H \leq y^{1/4} \\ 0 & \text{else,} \end{cases} \quad (3.4.8)$$

and

$$c_3(H) = - \sum_{\substack{\alpha\beta\gamma=H \\ \alpha\leq y^{1/4} \\ \beta\leq y^{1/4}}} \mu(\alpha)\mu(\beta). \quad (3.4.9)$$

Vaughan's identity (3.4.5) can be proved by equating the Dirichlet series coefficients on both sides of the equation

$$\frac{1}{\zeta} = \frac{1}{\zeta}(1 - \zeta M_0)^2 + 2M_0 - \zeta M_0^2,$$

where $\zeta = \zeta(s)$ is the Riemann zeta-function and

$$M_0 = M_0(s) = \sum_{n\leq y^{1/4}} \frac{\mu(n)}{n^s}.$$

Inserting Vaughan's identity (3.4.5) for $\mu(H)$ into (3.4.4) gives

$$\mathcal{B}(s) = \sum_{q\leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N (S_1 + S_2 + S_3) + O(\exp(-y^\varepsilon/4)), \quad (3.4.10)$$

where

$$\begin{aligned} S_1 = & \sum_{\substack{\alpha>y^{1/4} \\ (\alpha,q)=1}} \sum_{\substack{\beta>y^{1/4} \\ (\beta,q)=1}} \sum_{\substack{\gamma\geq 1 \\ (\gamma,q)=1}} \sum_{\substack{V<K\leq 2V \\ (K,\alpha\beta\gamma q)=1}} \frac{\mu_1(\alpha)\mu_1(\beta)\mu(\gamma)\mu(K)}{(\alpha\beta\gamma K)^{2\sigma-s}} \times \\ & \times \left(\frac{W(y/\alpha\beta\gamma q)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N<\ell\leq 2N} \frac{\rho(\ell)}{\ell^s} e\left(\ell \frac{\overline{\alpha\beta\gamma}}{K}\right), \end{aligned} \quad (3.4.11)$$

$$\begin{aligned}
S_2 &= 2 \sum_{\substack{H \leq y^{1/4} \\ (H,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,Hq)=1}} \frac{\mu(H)\mu(K)}{H^{2\sigma-s}K^{2\sigma-s}} \left(\frac{W(y/Hq)}{W(y)} \right) \times \\
&\quad \times \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e \left(\frac{\ell \bar{H}}{K} \right),
\end{aligned} \tag{3.4.12}$$

and

$$\begin{aligned}
S_3 &= - \sum_{\substack{\alpha \leq y^{1/4} \\ (\alpha,q)=1}} \sum_{\substack{\beta \leq y^{1/4} \\ (\beta,q)=1}} \sum_{\substack{\gamma \geq 1 \\ (\gamma,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,\alpha\beta\gamma q)=1}} \frac{\mu(\alpha)\mu(\beta)\mu(K)}{(\alpha\beta\gamma K)^{2\sigma-s}} \times \\
&\quad \times \left(\frac{W(y/\alpha\beta\gamma q)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e \left(\ell \frac{\overline{\alpha\beta\gamma}}{K} \right).
\end{aligned} \tag{3.4.13}$$

To estimate S_1 , we first partition the ranges of summation over α , β , and γ into dyadic intervals and write (3.4.11) as

$$S_1 = \sum_{\alpha_0 \geq y^{1/4}} \sum_{\beta_0 \geq y^{1/4}} \sum_{\gamma_0} L(\alpha_0, \beta_0, \gamma_0; s), \tag{3.4.14}$$

where α_0 and β_0 both run through real numbers of the form $2^n y^{1/4}$ with n a nonnegative integer, γ_0 runs through integral powers of 2 that are $\geq 1/2$, and $L(\alpha_0, \beta_0, \gamma_0; s)$ is defined by

$$\begin{aligned}
L(\alpha_0, \beta_0, \gamma_0; s) &= \sum_{\substack{\alpha_0 < \alpha \leq 2\alpha_0 \\ (\alpha,q)=1}} \sum_{\substack{\beta_0 < \beta \leq 2\beta_0 \\ (\beta,q)=1}} \sum_{\substack{\gamma_0 < \gamma \leq 2\gamma_0 \\ (\gamma,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,\alpha\beta\gamma q)=1}} \frac{\mu_1(\alpha)\mu_1(\beta)\mu(\gamma)\mu(K)}{(\alpha\beta\gamma K)^{2\sigma-s}} \times \\
&\quad \times \left(\frac{W(y/\alpha\beta\gamma q)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e \left(\ell \frac{\overline{\alpha\beta\gamma}}{K} \right).
\end{aligned} \tag{3.4.15}$$

By (iii) of Lemma 3.1.1, we may truncate the range of summation over $\alpha_0, \beta_0, \gamma_0$ in (3.4.14) to $\{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}\}$, with negligible error, provided $y^\varepsilon > 4$. Indeed, the definition (3.4.7) of $\mu_1(n)$ implies that

$$|\mu_1(n)| \leq d(n) \ll n^\varepsilon, \tag{3.4.16}$$

while the definition (3.4.3) of $\rho(\ell)$ gives

$$|\rho(\ell)| \leq d(\ell)\ell^{2\sigma-1} \ll \ell^{2\sigma-1+\varepsilon}. \quad (3.4.17)$$

We use these together with Lemma 3.1.1 (iii), (iv), and (ii) to bound $W(y/\alpha\beta\gamma q)$, $W(y/Kq)$, and $W(y)^{-2}$, respectively, when $\alpha_0\beta_0\gamma_0 > y^{1+\varepsilon}$, to deduce from (3.4.14) that

$$S_1 = \sum_{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}} L(\alpha_0, \beta_0, \gamma_0; s) + O(N^{2\sigma-c+\varepsilon} \exp(-y^\varepsilon/4)), \quad (3.4.18)$$

where $\alpha_0, \beta_0, \gamma_0$ have the same meanings as in (3.4.14). The next task is to estimate $L(\alpha_0, \beta_0, \gamma_0; s)$ for $\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}$.

Lemma 3.4.1. *Let $L(\alpha_0, \beta_0, \gamma_0; s)$ be defined by (3.4.15), and let $\alpha_0 \geq y^{1/4}$, $\beta_0 \geq y^{1/4}$, and $\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}$. If $y \leq T^{8/13}$ and $\operatorname{Re}(s) = x$ with $0 < x < 2$, then*

$$\begin{aligned} L(\alpha_0, \beta_0, \gamma_0; s) &\ll (\alpha_0\beta_0\gamma_0 V)^{x-2\sigma} N^{2\sigma-x-1+\varepsilon} \times \\ &\times \max\{TN, \alpha_0\beta_0\gamma_0 V\} T^{-\frac{1}{2}} y^{\frac{7}{8}+\varepsilon}. \end{aligned}$$

Proof. We see from the definition (3.4.15) that $L(\alpha_0, \beta_0, \gamma_0; s)$ is symmetric in the variables α_0 and β_0 . Thus we may assume without loss of generality that $\alpha_0 \leq \beta_0$. We group together the terms in (3.4.15) that have the same $\beta\gamma$ and let

$$A = \alpha_0 \quad \text{and} \quad B = \beta_0\gamma_0 \quad (3.4.19)$$

to write (3.4.15) as

$$\begin{aligned} L(\alpha_0, \beta_0, \gamma_0; s) &= \sum_{\substack{A < a \leq 2A \\ (a,q)=1}} \sum_{\substack{B < b \leq 4B \\ (b,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,abq)=1}} \frac{\mu_1(a)\mu_2(b)\mu(K)}{(abK)^{2\sigma-s}} \times \\ &\times \left(\frac{W(y/abq)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e\left(\ell \frac{\overline{ab}}{K}\right), \end{aligned} \quad (3.4.20)$$

where $\mu_2(b)$ is defined by

$$\mu_2(b) = \sum_{\substack{\beta\gamma=b \\ \beta_0 < \beta \leq 2\beta_0 \\ \gamma_0 < \gamma \leq 2\gamma_0}} \mu_1(\beta)\mu(\gamma). \quad (3.4.21)$$

We insert the definition (1.2.7) of $W(y/abq)$ into (3.4.20) and arrive at

$$\begin{aligned}
L(\alpha_0, \beta_0, \gamma_0; s) &= \frac{1}{2\pi i} \int_{(r)} \Gamma(z+1) \frac{\zeta(2\sigma+z)}{\zeta(2\sigma)} \zeta(1+z) \left(\frac{y}{q}\right)^z \times \\
&\times \sum_{\substack{A < a \leq 2A \\ (a,q)=1}} \sum_{\substack{B < b \leq 4B \\ (b,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,abq)=1}} \frac{\mu_1(a)\mu_2(b)\mu(K)}{(ab)^{2\sigma-s+z} K^{2\sigma-s}} \left(\frac{W(y/Kq)}{W(y)^2}\right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e\left(\ell \frac{\overline{ab}}{K}\right) dz.
\end{aligned} \tag{3.4.22}$$

Now define $C(a, \ell)$ by

$$C(a, \ell) = \frac{\mu_1(a)\rho(\ell)}{a^{2\sigma-s+z}\ell^s} A^{2\sigma-x+r-\varepsilon} N^{-2\sigma+x+1-\varepsilon} \tag{3.4.23}$$

and rearrange (3.4.22) to write it as

$$\begin{aligned}
L(\alpha_0, \beta_0, \gamma_0; s) &= A^{-2\sigma+x-r+\varepsilon} N^{2\sigma-x-1+\varepsilon} \left(\frac{1}{2\pi i}\right) \int_{(r)} \Gamma(z+1) \frac{\zeta(2\sigma+z)}{\zeta(2\sigma)} \times \\
&\times \zeta(1+z) \left(\frac{y}{q}\right)^z \sum_{\substack{B < b \leq 4B \\ (b,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,bq)=1}} \frac{\mu_2(b)\mu(K)}{b^{2\sigma-s+z} K^{2\sigma-s}} \left(\frac{W(y/Kq)}{W(y)^2}\right) \times \\
&\times \sum_{\substack{A < a \leq 2A \\ (a,Kq)=1}} \sum_{N < \ell \leq 2N} C(a, \ell) e\left(\ell \frac{\overline{ab}}{K}\right) dz.
\end{aligned}$$

We choose $r = (\log y)^{-1}$ and use the following estimates to bound the right-hand side: $A^{-r} \ll 1$,

$$\frac{|\zeta(2\sigma+z)|}{\zeta(2\sigma)} \leq 1, \tag{3.4.24}$$

$$\zeta(1+z) \ll \frac{1}{r} = \log y, \tag{3.4.25}$$

$$\left(\frac{y}{q}\right)^z \ll 1, \tag{3.4.26}$$

$$\left| \frac{\mu_2(b)\mu(K)}{b^{2\sigma-s+z} K^{2\sigma-s}} \right| \leq d_3(b) b^{-2\sigma+x-r} K^{-2\sigma+x} \ll B^{-2\sigma+x+\varepsilon} V^{-2\sigma+x}$$

by (3.4.16) and (3.4.21),

$$\frac{W(y/Kq)}{W(y)^2} \ll 1 \tag{3.4.27}$$

by (ii) and (iv) of Lemma 3.1.1, and

$$\int_{(r)} |\Gamma(z+1)| |dz| \ll 1 \quad (3.4.28)$$

by Stirling's formula. The result is

$$\begin{aligned} L(\alpha_0, \beta_0, \gamma_0; s) &\ll (\log y)(ABV)^{-2\sigma+x+\varepsilon} N^{2\sigma-x-1+\varepsilon} \times \\ &\times \sum_{\substack{B < b \leq 4B \\ (b,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,bq)=1}} \left| \sum_{\substack{A < a \leq 2A \\ (a,Kq)=1}} \sum_{N < \ell \leq 2N} C(a, \ell) e\left(\ell \frac{\overline{ab}}{K}\right) \right|. \end{aligned} \quad (3.4.29)$$

The definition (3.4.23) of $C(a, \ell)$ and the bounds (3.4.16) and (3.4.17) imply that

$$C(a, \ell) \ll 1,$$

and so we may apply Lemma 3.1.6 to the sum on the right-hand side of (3.4.29).

Hence

$$\begin{aligned} L(\alpha_0, \beta_0, \gamma_0; s) &\ll (\log y)(ABV)^{-2\sigma+x+\varepsilon} N^{2\sigma-x-1+\varepsilon} \times \\ &\times \left\{ \sum_{(a,n,u,v) \in E} A^a N^n (AB)^u V^v \right\}^{1/4}, \end{aligned} \quad (3.4.30)$$

where E is the set defined in Lemma 3.1.6. Now for $(a, n, u, v) \in E$, it is true that

$$\begin{aligned} A^a N^n (AB)^u V^v &= A^a (TN)^n (ABV)^{4-n} T^{-n} (AB)^{u+n-4} V^{v+n-4} \\ &\leq A^a (TN)^n (ABV)^{4-n} T^{-n} y^{2n+u+v-8+\varepsilon} \end{aligned}$$

because

$$AB = \alpha_0 \beta_0 \gamma_0 \leq y^{1+\varepsilon}, \quad \text{and} \quad V \leq y^{1+\varepsilon} \quad (3.4.31)$$

by hypothesis (see (3.4.4), (3.4.18), and (3.4.19)). This implies that

$$A^a N^n (AB)^u V^v \leq A^a \max\{TN, ABV\}^4 T^{-n} y^{2n+u+v-8+\varepsilon} \quad (3.4.32)$$

since $2 \leq n \leq 4$ for all $(a, n, u, v) \in E$. Recall that we have assumed $\alpha_0 \leq \beta_0$ at the beginning of this proof. Since $\gamma_0 \geq \frac{1}{2}$, it follows from (3.4.19) and the

assumption $\alpha_0 \leq \beta_0$ that $A^2 \leq 2AB$, and so $A^2 \leq 2y^{1+\varepsilon}$ by (3.4.31). This fact and the hypothesis $\alpha_0 \geq y^{1/4}$ give

$$y^{1/4} \leq A \ll y^{\frac{1}{2}+\varepsilon}.$$

It follows from this and (3.4.32) that

$$A^a N^n (AB)^u V^v \ll \max\{TN, ABV\}^4 T^{-n} y^{\frac{a}{2}+2n+u+v-8+\varepsilon} \quad (3.4.33)$$

for $(a, n, u, v) \in E$ with positive or zero a , and

$$A^a N^n (AB)^u V^v \ll \max\{TN, ABV\}^4 T^{-n} y^{\frac{a}{4}+2n+u+v-8+\varepsilon} \quad (3.4.34)$$

for $(a, n, u, v) \in E$ with negative a . By evaluating the right-hand sides of (3.4.33) and (3.4.34) for each $(a, n, u, v) \in E$, we see that the largest bound implied by (3.4.33) and (3.4.34) is

$$A^a N^n (AB)^u V^v \ll \max\{TN, ABV\}^4 T^{-2} y^{\frac{7}{2}+\varepsilon}$$

because of the hypothesis $y \leq T^{8/13}$. This and (3.4.30) imply that

$$\begin{aligned} L(\alpha_0, \beta_0, \gamma_0; s) &\ll (ABV)^{-2\sigma+x+\varepsilon} N^{2\sigma-x-1+\varepsilon} \times \\ &\times \max\{TN, ABV\} T^{-\frac{1}{2}} y^{\frac{7}{8}+\varepsilon}. \end{aligned}$$

The lemma now follows from this, (3.4.19), and (3.4.31). $\square$

We now have the necessary estimates to bound the contribution of S_1 to $\mathfrak{E}$. We consider only the integral of $G_1(z, s)\mathcal{B}(s)$ in (3.3.15), as similar arguments allow us to bound the other integrals in (3.3.15). Using (3.4.10), we may write the said integral as

$$\int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \mathcal{B}(s) ds dz = \mathfrak{E}_1 + \mathfrak{E}_2 + \mathfrak{E}_3 + O(\mathfrak{F}), \quad (3.4.35)$$

where

$$\mathfrak{E}_j = \int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N S_j ds dz \quad (3.4.36)$$

for $j = 1, 2, 3$, and

$$\mathfrak{F} = \exp(-y^\varepsilon/4) \int_0^{\infty e^{i\lambda}} \int_{(c)} |G_1(z, s) ds dz|. \quad (3.4.37)$$

It follows from (3.4.36) with $j = 1$, (3.4.18), and the definition (3.4.37) that

$$\begin{aligned} \mathfrak{E}_1 &= \int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N \sum_{\alpha_0 \beta_0 \gamma_0 \leq y^{1+\varepsilon}} L(\alpha_0, \beta_0, \gamma_0; s) ds dz \\ &\quad + O \left\{ \mathfrak{F} \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N N^{2\sigma-c+\varepsilon} \right\}. \end{aligned} \quad (3.4.38)$$

So that the sum over N in the error term converges, we choose η in (3.4.1) to be

$$\eta = 2\varepsilon. \quad (3.4.39)$$

With this choice of η , we see that the error term in (3.4.38) is $O(y^\varepsilon \mathfrak{F})$ because

$$\sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V 1 \ll (\log y)^2.$$

Hence

$$\begin{aligned} \mathfrak{E}_1 &= \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N \sum_{\alpha_0 \beta_0 \gamma_0 \leq y^{1+\varepsilon}} \int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \times \\ &\quad \times L(\alpha_0, \beta_0, \gamma_0; s) ds dz + O(y^\varepsilon \mathfrak{F}). \end{aligned} \quad (3.4.40)$$

Write this as

$$\mathfrak{E}_1 = \mathfrak{E}_{11} + \mathfrak{E}_{12} + O(y^\varepsilon \mathfrak{F}), \quad (3.4.41)$$

where $\mathfrak{E}_{11}$ is the sum of the terms in (3.4.40) that have $TN > \alpha_0 \beta_0 \gamma_0 V$ and $\mathfrak{E}_{12}$ is the sum of the rest of the terms. To estimate $\mathfrak{E}_{11}$, assume $y \leq T^{8/13}$ and apply Lemma 3.4.1 to deduce that

$$\begin{aligned} \mathfrak{E}_{11} &\ll \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_{N > \alpha_0 \beta_0 \gamma_0 V/T} \sum_{\alpha_0 \beta_0 \gamma_0 \leq y^{1+\varepsilon}} (\alpha_0 \beta_0 \gamma_0 V)^{c-2\sigma} N^{2\sigma-c+\varepsilon} \times \\ &\quad \times T^{\frac{1}{2}} y^{\frac{7}{8}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c)} |G_1(z, s) ds dz|. \end{aligned} \quad (3.4.42)$$

Our choice for the value of c given by (3.4.1) and (3.4.39) implies that the sum over N in (3.4.42) converges. It also implies that $(\alpha_0\beta_0\gamma_0V)^{c-2\sigma} \ll y^\varepsilon$ by (3.4.31). Hence it follows from (3.4.42) that

$$\mathfrak{E}_{11} \ll T^{\frac{1}{2}} y^{\frac{7}{8}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c)} |G_1(z, s) ds dz|.$$

From this and the definition (3.4.37), we arrive at

$$\mathfrak{E}_{11} + y^\varepsilon \mathfrak{F} \ll T^{\frac{1}{2}} y^{\frac{7}{8}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c)} |G_1(z, s) ds dz|.$$

To this we apply the first assertion of Lemma 3.1.4 with λ and Δ given by (3.4.1). The result is

$$\mathfrak{E}_{11} + y^\varepsilon \mathfrak{F} \ll T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} \quad (3.4.43)$$

for $y \leq T^{8/13}$. Next, recall from (3.4.41) that $\mathfrak{E}_{12}$ is the sum of the terms in (3.4.40) that have $TN \leq \alpha_0\beta_0\gamma_0V$. To estimate $\mathfrak{E}_{12}$, we let $c' = 2\sigma + \eta - 1$ and move the line of integration of the s -integral to $\operatorname{Re}(s) = c'$, leaving a residue from the pole at $s = 1$. We may bound the contribution of the residue using the definition (3.1.9) of $G_1(z, s)$, Lemma 3.1.8, and Lemma 3.4.1 with $s = 1$. We thus arrive at

$$\begin{aligned} \mathfrak{E}_{12} = & \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_{N \leq \alpha_0\beta_0\gamma_0V/T} \sum_{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}} \int_0^{\infty e^{i\lambda}} \int_{(c')} G_1(z, s) \times \\ & \times L(\alpha_0, \beta_0, \gamma_0; s) ds dz + O(\exp(-\tfrac{1}{2} \log^2 T)) \end{aligned}$$

for $u \geq \Delta \log T$ and $y \leq T^{8/13}$. From this and Lemma 3.4.1, we deduce that

$$\begin{aligned} \mathfrak{E}_{12} \ll & \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_{N \leq \alpha_0\beta_0\gamma_0V/T} \sum_{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}} (\alpha_0\beta_0\gamma_0V)^{c'-2\sigma+1} N^{2\sigma-c'-1+\varepsilon} \times \\ & \times T^{-\frac{1}{2}} y^{\frac{7}{8}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c')} |G_1(z, s) ds dz| + O(\exp(-\tfrac{1}{2} \log^2 T)). \end{aligned}$$

It follows from this, (3.4.19), and our choice $c' = 2\sigma + \eta - 1$ with (3.4.39) that

$$\mathfrak{E}_{12} \ll T^{-\frac{1}{2}} y^{\frac{7}{8}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c')} |G_1(z, s) ds dz| + O(\exp(-\tfrac{1}{2} \log^2 T)).$$

From this and the second assertion of Lemma 3.1.4 with λ and Δ given by (3.4.1), we arrive at

$$\mathfrak{E}_{12} \ll T^{\frac{1}{2}-2\sigma+\frac{5}{2}\delta+\varepsilon} y^{\frac{7}{8}} \leq T^{-\frac{1}{2}+\frac{5}{2}\delta+\varepsilon} y^{\frac{7}{8}}.$$

It now follows from this, (3.4.43), and (3.4.41) that

$$\mathfrak{E}_1 \ll T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} \quad (3.4.44)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$.

Having bounded the contribution of S_1 to $\mathfrak{E}$, we next estimate the contribution $\mathfrak{E}_2$ of S_2 , which is defined by (3.4.36) with (3.4.12). For this, we do not need exponential sum estimates to deduce a bound of the same size as (3.4.44). We simply take the absolute value of each term in (3.4.12) to arrive at

$$S_2 \ll \sum_{\substack{H \leq y^{1/4} \\ (H,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,Hq)=1}} (HK)^{c-2\sigma} \sum_{N < \ell \leq 2N} \ell^{2\sigma-1-c+\varepsilon}$$

by (3.4.17) and (ii) and (iv) of Lemma 3.1.1. Thus

$$S_2 \ll (y^{1/4}V)^{c-2\sigma+1} N^{2\sigma-c+\varepsilon} \ll y^{\frac{1}{4}+\varepsilon} V N^{-\varepsilon}$$

by our choice for c given by (3.4.1) and (3.4.39). It follows from this and (3.4.36) with $j = 2$ that

$$\mathfrak{E}_2 \ll y^{\frac{5}{4}+\varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c)} |G_1(z, s) ds dz|.$$

To this we apply the first assertion of Lemma 3.1.4 with λ and Δ given by (3.4.1), and the result is

$$\mathfrak{E}_2 \ll T^{-1+\frac{7}{2}\delta+\varepsilon} y^{\frac{5}{4}} \quad (3.4.45)$$

for $y \ll T$.

It is left to bound the contribution $\mathfrak{E}_3$ of S_3 , which is defined by (3.4.36) and (3.4.13). As in our treatment of S_1 , we first partition the ranges of summation over α , β , and γ into dyadic intervals and write (3.4.13) as

$$S_3 = \sum_{\alpha_0 \leq \frac{1}{2}y^{1/4}} \sum_{\beta_0 \leq \frac{1}{2}y^{1/4}} \sum_{\gamma_0} J(\alpha_0, \beta_0, \gamma_0; s), \quad (3.4.46)$$

where α_0 and β_0 both run through real numbers $\geq \frac{1}{2}$ that are of the form $2^{-n}y^{1/4}$, with n a positive integer, γ_0 runs through powers of 2 that are $\geq \frac{1}{2}$, and the function $J(\alpha_0, \beta_0, \gamma_0; s)$ is defined by

$$J(\alpha_0, \beta_0, \gamma_0; s) = - \sum_{\substack{\alpha_0 < \alpha \leq 2\alpha_0 \\ (\alpha, q)=1}} \sum_{\substack{\beta_0 < \beta \leq 2\beta_0 \\ (\beta, q)=1}} \sum_{\substack{\gamma_0 < \gamma \leq 2\gamma_0 \\ (\gamma, q)=1}} \sum_{\substack{V < K \leq 2V \\ (K, \alpha\beta\gamma q)=1}} \frac{\mu(\alpha)\mu(\beta)\mu(K)}{(\alpha\beta\gamma K)^{2\sigma-s}} \times \\ \times \left(\frac{W(y/\alpha\beta\gamma q)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e \left(\ell \frac{\overline{\alpha\beta\gamma}}{K} \right). \quad (3.4.47)$$

As in our estimation of S_1 (see (3.4.18)), we may use (3.4.17) and Lemma 3.1.1 to truncate the range of summation in (3.4.46) and deduce that

$$S_3 = \sum_{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}} J(\alpha_0, \beta_0, \gamma_0; s) + O(N^{2\sigma-c+\varepsilon} \exp(-y^\varepsilon/4)), \quad (3.4.48)$$

where $\alpha_0, \beta_0, \gamma_0$ have the same meanings described below (3.4.46). We may bound $J(\alpha_0, \beta_0, \gamma_0; s)$ in the same way we deduced Lemma 3.4.1, provided we assume in addition that $\alpha_0\beta_0 \geq y^{1/4}$.

Lemma 3.4.2. *Let $J(\alpha_0, \beta_0, \gamma_0; s)$ be defined by (3.4.47), and let $\alpha_0\beta_0 \geq y^{1/4}$ and $\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}$. If $y \leq T^{8/13}$ and $\text{Re}(s) = x$ with $0 < x < 2$, then*

$$J(\alpha_0, \beta_0, \gamma_0; s) \ll (\alpha_0\beta_0\gamma_0 V)^{x-2\sigma} N^{2\sigma-x-1+\varepsilon} \times \\ \times \max\{TN, \alpha_0\beta_0\gamma_0 V\} T^{-\frac{1}{2}} y^{\frac{7}{8}+\varepsilon}.$$

Proof. We group together the terms in (3.4.47) that have the same $\alpha\beta$, and let

$$A = \alpha_0\beta_0 \quad \text{and} \quad B = \gamma_0 \quad (3.4.49)$$

to write (3.4.47) as

$$J(\alpha_0, \beta_0, \gamma_0; s) = - \sum_{\substack{A < a \leq 4A \\ (a, q)=1}} \sum_{\substack{B < b \leq 2B \\ (b, q)=1}} \sum_{\substack{V < K \leq 2V \\ (K, abq)=1}} \frac{\mu_3(a)\mu(K)}{(abK)^{2\sigma-s}} \times \\ \times \left(\frac{W(y/abq)}{W(y)} \right) \left(\frac{W(y/Kq)}{W(y)} \right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} e \left(\ell \frac{\overline{ab}}{K} \right), \quad (3.4.50)$$

where $\mu_3(a)$ is defined by

$$\mu_3(a) = \sum_{\substack{\alpha\beta=a \\ \alpha_0 < \beta \leq 2\alpha_0 \\ \beta_0 < \beta \leq 2\beta_0}} \mu(\alpha)\mu(\beta). \quad (3.4.51)$$

The expression (3.4.50) for $J(\alpha_0, \beta_0, \gamma_0; s)$ is similar to (3.4.20). Moreover, since $\alpha_0\beta_0 \geq y^{1/4}$ by hypothesis and $\alpha_0 \leq y^{1/4}$ and $\beta_0 \leq y^{1/4}$ by their definitions in (3.4.46), it follows from (3.4.49) that

$$y^{1/4} \leq A \leq y^{1/2}.$$

Therefore the same arguments in the proof of Lemma 3.4.1 apply to the right-hand side of (3.4.50), and from them we arrive at the desired bound for $J(\alpha_0, \beta_0, \gamma_0; s)$. $\square$

Lemma 3.4.2 gives an estimate for $J(\alpha_0, \beta_0, \gamma_0; s)$ when $\alpha_0\beta_0 \geq y^{1/4}$. For the opposite case $\alpha_0\beta_0 < y^{1/4}$, we have the following lemma.

Lemma 3.4.3. *Let $J(\alpha_0, \beta_0, \gamma_0; s)$ be defined by (3.4.47), and suppose that $\alpha_0\beta_0 \leq y^{1/4}$ and $\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}$. If $\operatorname{Re}(s) = c$ with c given by (3.4.1) and (3.4.39), then*

$$J(\alpha_0, \beta_0, \gamma_0; s) \ll (1 + |s|)N^{-\varepsilon}y^{\frac{7}{4}+\varepsilon}.$$

Proof. We insert the definition (1.2.7) of $W(y/abq)$ into (3.4.50) and rearrange it to deduce that

$$\begin{aligned} J(\alpha_0, \beta_0, \gamma_0; s) &= -\frac{1}{2\pi i} \int_{(r)} \Gamma(z+1) \frac{\zeta(2\sigma+z)}{\zeta(2\sigma)} \zeta(1+z) \left(\frac{y}{q}\right)^z \sum_{\substack{A < a \leq 4A \\ (a,q)=1}} \sum_{\substack{V < K \leq 2V \\ (K,aq)=1}} \times \\ &\times \frac{\mu_3(a)\mu(K)}{a^{2\sigma-s+z}K^{2\sigma-s}} \left(\frac{W(y/Kq)}{W(y)^2}\right) \sum_{N < \ell \leq 2N} \frac{\rho(\ell)}{\ell^s} \sum_{\substack{B < b \leq 2B \\ (b,Kq)=1}} b^{-2\sigma+s-z} e\left(\ell \frac{\overline{ab}}{K}\right) dz, \end{aligned} \quad (3.4.52)$$

where we choose $r = (\log y)^{-1}$. Partial summation and Lemma 3.1.5 gives for s, z with $\operatorname{Re}(s) = c$ and $\operatorname{Re}(z) = r$ that

$$\sum_{\substack{B < b \leq 2B \\ (b, Kq) = 1}} b^{-2\sigma + s - z} e\left(\ell \frac{\overline{ab}}{K}\right) \ll (1 + |s| + |z|) B^{-2\sigma + c} (Kq)^\varepsilon K^{1/2} (\ell \overline{a}, K) \left(q + \frac{B}{K}\right),$$

with $(\ell \overline{a}, K) = (\ell, K)$ since a is relatively prime to K . We use this and the following estimates to bound the right-hand side of (3.4.52): the estimate (3.4.17), the bounds (3.4.24) through (3.4.28),

$$\int_{(r)} |z \Gamma(z + 1)| |dz| \ll 1$$

by Stirling's formula, and

$$\frac{\mu_3(a) \mu(K)}{a^{2\sigma - s + z} K^{2\sigma - s}} \ll d_2(a) a^{-2\sigma + c - r} K^{-2\sigma + c} \ll A^\varepsilon (AV)^{c - 2\sigma}$$

by (3.4.51). The result is

$$\begin{aligned} J(\alpha_0, \beta_0, \gamma_0; s) &\ll (1 + |s|) y^\varepsilon A^{c - 2\sigma + 1 + \varepsilon} V^{c - 2\sigma + \frac{1}{2} + \varepsilon} N^{2\sigma - 1 + \varepsilon - c} \times \\ &\times B^{c - 2\sigma} q^\varepsilon \left(q + \frac{B}{V}\right) \sum_{V < K \leq 2V} \sum_{N < \ell \leq 2N} (\ell, K). \end{aligned} \quad (3.4.53)$$

Since $(\ell, K) = \sum_{d|(\ell, K)} \varphi(d)$, where φ is the Euler totient function, it is true that

$$\begin{aligned} \sum_{V < K \leq 2V} \sum_{N < \ell \leq 2N} (\ell, K) &= \sum_{d \leq 4NV} \varphi(d) \sum_{\substack{V < K \leq 2V \\ d|K}} \sum_{\substack{N < \ell \leq 2N \\ d|\ell}} 1 \\ &\leq 4NV \sum_{d \leq 4NV} \frac{\varphi(d)}{d^2} \ll (NV)^{1 + \varepsilon}. \end{aligned}$$

It follows from this and (3.4.53) that

$$J(\alpha_0, \beta_0, \gamma_0; s) \ll (1 + |s|) y^\varepsilon A^{c - 2\sigma + 1 + \varepsilon} V^{c - 2\sigma + \frac{3}{2} + \varepsilon} N^{2\sigma + \varepsilon - c} B^{c - 2\sigma} q^\varepsilon \left(q + \frac{B}{V}\right). \quad (3.4.54)$$

By our choice of c given by (3.4.1) and (3.4.39), we may simplify (3.4.54) to

$$J(\alpha_0, \beta_0, \gamma_0; s) \ll (1 + |s|) (yABVq)^\varepsilon N^{-\varepsilon} \left(qAV^{\frac{3}{2}} + ABV^{\frac{1}{2}}\right). \quad (3.4.55)$$

The hypotheses $\alpha_0\beta_0 \leq y^{1/4}$ and $\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}$ and the definitions (3.4.49) of A and B imply that $A \leq y^{1/4}$ and $AB \leq y^{1+\varepsilon}$. Also, recall that $q \leq y^{1+\varepsilon}$ and $V \leq y^{1+\varepsilon}/q$ (see (3.4.4) and the description of V below it). From these bounds and (3.4.55), we arrive at

$$J(\alpha_0, \beta_0, \gamma_0; s) \ll (1 + |s|)y^\varepsilon N^{-\varepsilon} \left(y^{\frac{7}{4}} + y^{\frac{3}{2}} \right).$$

The lemma now follows. $\square$

Having bounded $J(\alpha_0, \beta_0, \gamma_0; s)$, we now continue with our estimation of $\mathfrak{E}_3$. We insert (3.4.48) into the definition (3.4.36) of $\mathfrak{E}_3$ to deduce that

$$\begin{aligned} \mathfrak{E}_3 = & \sum_{q \leq y^{1+\varepsilon}} \frac{\mu^2(q)}{q^{2\sigma}} \sum_V \sum_N \sum_{\alpha_0\beta_0\gamma_0 \leq y^{1+\varepsilon}} \int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \times \\ & \times J(\alpha_0, \beta_0, \gamma_0; s) ds dz + O(y^\varepsilon \mathfrak{F}) \end{aligned} \quad (3.4.56)$$

in the same way we proved (3.4.40), where $\mathfrak{F}$ is defined by (3.4.37). In view of the conditions of Lemmas 3.4.2 and 3.4.3, we split the sum to write (3.4.56) as

$$\mathfrak{E}_3 = \mathfrak{E}_{31} + \mathfrak{E}_{32} + O(y^\varepsilon \mathfrak{F}), \quad (3.4.57)$$

where $\mathfrak{E}_{31}$ is the sum of the terms in (3.4.56) that have $\alpha_0\beta_0 \geq y^{1/4}$, while $\mathfrak{E}_{32}$ is the sum of the rest. We may apply Lemma 3.4.2 to $\mathfrak{E}_{31}$, and so arguments similar to our proof of (3.4.44) lead to

$$\mathfrak{E}_{31} \ll T^{-\frac{1}{2} + \frac{7}{2}\delta + \varepsilon} y^{\frac{7}{8}} \quad (3.4.58)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$. On the other hand, Lemma 3.4.3 implies

$$\mathfrak{E}_{32} \ll y^{\frac{7}{4} + \varepsilon} \int_0^{\infty e^{i\lambda}} \int_{(c)} (1 + |s|) |G_1(z, s) ds dz|.$$

It follows from this, (3.4.1), (3.4.39), and the first assertion of Lemma 3.1.4 that

$$\mathfrak{E}_{32} \ll T^{-1 + \frac{7}{2}\delta + \varepsilon} y^{\frac{7}{4}}. \quad (3.4.59)$$

It now follows from (3.4.57), (3.4.58), (3.4.59), and the bound for $y^\varepsilon \mathfrak{F}$ given by (3.4.43) that

$$\mathfrak{E}_3 \ll T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} + T^{-1+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{4}} \quad (3.4.60)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$.

Putting together the estimates, we deduce from (3.4.35), (3.4.44), (3.4.45), (3.4.60), and the bound for $\mathfrak{F}$ given by (3.4.43) that

$$\int_0^{\infty e^{i\lambda}} \int_{(c)} G_1(z, s) \mathcal{B}(s) ds dz \ll T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} + T^{-1+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{4}}$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$. Similar arguments apply to the other integrals in (3.3.15), and we conclude that

$$\mathfrak{E} \ll T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} + T^{-1+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{4}} \quad (3.4.61)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$, provided we choose Δ, λ, c to be given by (3.4.1) and (3.4.39).

3.5 The main terms

Having bounded the term $\mathfrak{E}$ in (3.3.8), we now proceed to the estimation of the terms $\mathfrak{M}_1$ and $\mathfrak{M}_2$. Recall that we have chosen Δ, λ in Lemma 3.3.1 to be given by (3.4.1), and so we assume the same throughout this section. We first estimate $\mathfrak{M}_1$ defined by (3.3.9) and (3.3.10). Observe that the residue theorem and (3.3.10) imply that

$$\mathcal{I} = 1$$

because the only pole of the integrand is at $z = 1$, which has residue 1. It follows from this and (3.3.9) that

$$\mathfrak{M}_1 = \zeta(2\sigma) \sum_{h \leq y^{1+\varepsilon}} \sum_{k \leq y^{1+\varepsilon}} \frac{\mu(h)\mu(k)}{h^{2\sigma} k^{2\sigma}} (h, k)^{2\sigma} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right).$$

By (ii) and (iii) of Lemma 3.1.1, we may extend the range of summation to infinity, with small error, provided $y^\varepsilon \geq 4$. That is,

$$\begin{aligned} \mathfrak{M}_1 &= \zeta(2\sigma) \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{h^{2\sigma}k^{2\sigma}} (h, k)^{2\sigma} \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \\ &\quad + O(\exp(-y^\varepsilon/4)). \end{aligned}$$

We insert into this the definitions of $W(y/h)$ and $W(y/k)$ given by (1.2.7), and interchange the order of summation to arrive at

$$\begin{aligned} \mathfrak{M}_1 &= \frac{1}{(2\pi i)^2 W(y)^2 \zeta(2\sigma)} \int_{(r)} \int_{(r)} y^{s_1+s_2} \Gamma(s_1+1) \Gamma(s_2+1) \zeta(2\sigma+s_1) \zeta(2\sigma+s_2) \times \\ &\quad \times \zeta(1+s_1) \zeta(1+s_2) \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{h^{2\sigma+s_1} k^{2\sigma+s_2}} (h, k)^{2\sigma} ds_2 ds_1 + O(\exp(-y^\varepsilon/4)), \end{aligned} \quad (3.5.1)$$

where we choose r to equal $(\log y)^{-1}$. The interchange of order of summation is justified by absolute convergence. We next express the inner sum as an Euler product using Lemma 2.2.1 and write

$$\begin{aligned} \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{h^{2\sigma+s_1} k^{2\sigma+s_2}} (h, k)^{2\sigma} &= \prod_p \left(1 - \frac{1}{p^{2\sigma+s_1}} - \frac{1}{p^{2\sigma+s_2}} + \frac{1}{p^{2\sigma+s_1+s_2}} \right) \\ &= \frac{\zeta(2\sigma+s_1+s_2)}{\zeta(2\sigma+s_1)\zeta(2\sigma+s_2)} F_1(s_1, s_2; 2\sigma), \end{aligned} \quad (3.5.2)$$

where $F_1(s_1, s_2; 2\sigma)$ is the Euler product

$$F_1(s_1, s_2; 2\sigma) = \prod_p \frac{\left(1 - \frac{1}{p^{2\sigma+s_1}} - \frac{1}{p^{2\sigma+s_2}} + \frac{1}{p^{2\sigma+s_1+s_2}} \right) \left(1 - \frac{1}{p^{2\sigma+s_1+s_2}} \right)}{\left(1 - \frac{1}{p^{2\sigma+s_1}} \right) \left(1 - \frac{1}{p^{2\sigma+s_2}} \right)}. \quad (3.5.3)$$

This Euler product satisfies

$$F_1(s_1, s_2; 2\sigma) = \prod_p \left(1 + O\left(\frac{1}{p^{9/8}} \right) \right) \ll 1 \quad (3.5.4)$$

for s_1, s_2, σ such that $2\sigma + \operatorname{Re}(s_1) > \frac{9}{16}$, $2\sigma + \operatorname{Re}(s_2) > \frac{9}{16}$, and $2\sigma + \operatorname{Re}(s_1) + \operatorname{Re}(s_2) > \frac{9}{16}$. We insert (3.5.2) into (3.5.1) and cancel the factors $\zeta(2\sigma+s_1)$ and $\zeta(2\sigma+s_2)$

to deduce that

$$\begin{aligned} \mathfrak{M}_1 &= \frac{1}{(2\pi i)^2 W(y)^2 \zeta(2\sigma)} \int_{(r)} \int_{(r)} y^{s_1+s_2} \Gamma(s_1+1) \Gamma(s_2+1) \zeta(1+s_1) \zeta(1+s_2) \times \\ &\quad \times \zeta(2\sigma+s_1+s_2) F_1(s_1, s_2; 2\sigma) ds_2 ds_1 + O(\exp(-y^\varepsilon/4)). \end{aligned} \quad (3.5.5)$$

Now we apply the residue theorem and move the line of integration of the s_2 -integral to the line $\operatorname{Re}(s_2) = -2\sigma + \frac{3}{4}$, leaving residues at the poles $s_2 = 0$ and $s_2 = 1 - 2\sigma - s_1$. We estimate the new double integral using (ii) of Lemma 3.1.1, Stirling's formula, (3.1.2), (3.5.4), and the fact that $\zeta(1+s_1) \ll \log y$ and

$$y^{s_1+s_2} \ll y^{\frac{3}{4}-2\sigma}$$

when $\operatorname{Re}(s_1) = r = (\log y)^{-1}$ and $\operatorname{Re}(s_2) = -2\sigma + \frac{3}{4}$. Hence the result of applying the residue theorem to (3.5.5) is

$$\mathfrak{M}_1 = \mathfrak{M}_{11} + \mathfrak{M}_{12} + O\left(y^{\frac{3}{4}-2\sigma+\varepsilon}\right), \quad (3.5.6)$$

where

$$\mathfrak{M}_{11} = \frac{1}{(2\pi i) W(y)^2 \zeta(2\sigma)} \int_{(r)} y^{s_1} \Gamma(s_1+1) \zeta(1+s_1) \zeta(2\sigma+s_1) F_1(s_1, 0; 2\sigma) ds_1 \quad (3.5.7)$$

and

$$\begin{aligned} \mathfrak{M}_{12} &= \frac{1}{(2\pi i) W(y)^2 \zeta(2\sigma)} \int_{(r)} y^{1-2\sigma} \Gamma(s_1+1) \Gamma(2-2\sigma-s_1) \times \\ &\quad \times \zeta(1+s_1) \zeta(2-2\sigma-s_1) F_1(s_1, 1-2\sigma-s_1; 2\sigma) ds_1. \end{aligned} \quad (3.5.8)$$

To estimate $\mathfrak{M}_{11}$, move the line of integration in (3.5.7) to the line $\operatorname{Re}(s_1) = -2\sigma + \frac{3}{4}$, leaving residues at the poles $s_1 = 0$ and $s_1 = 1 - 2\sigma$. We bound the new integral in the same way we estimated the error term in (3.5.6), and arrive at

$$\mathfrak{M}_{11} = \frac{F_1(0, 0; 2\sigma)}{W(y)^2} + \frac{\Gamma(2-2\sigma) \zeta(2-2\sigma) F_1(1-2\sigma, 0; 2\sigma)}{W(y)^2 \zeta(2\sigma)} y^{1-2\sigma} + O\left(y^{\frac{3}{4}-2\sigma+\varepsilon}\right). \quad (3.5.9)$$

A straightforward calculation using the definition (3.5.3) of $F_1(s_1, s_2; 2\sigma)$ shows that $F_1(0, 0; 2\sigma) = 1$ and $F_1(1 - 2\sigma, 0; 2\sigma) = 1$. We insert these into (3.5.9), and then apply (ii) of Lemma 3.1.1 and our assumption (3.1.1) to deduce that

$$\mathfrak{M}_{11} = \frac{1}{W(y)^2} + O(y^{1-2\sigma}).$$

From this and (i) of Lemma 3.1.1, it follows that

$$\mathfrak{M}_{11} = 1 + O(y^{1-2\sigma}). \quad (3.5.10)$$

Next, to estimate $\mathfrak{M}_{12}$, we use Cauchy's theorem to move the line of integration in (3.5.8) to the line $\operatorname{Re}(s_1) = \frac{1}{8}$. Note that there are no residues because the integrand is holomorphic between $\operatorname{Re}(s_1) = r$ and $\operatorname{Re}(s_1) = \frac{1}{8}$. We then bound the new integral using (ii) of Lemma 3.1.1, Stirling's formula, (3.1.2), (3.5.4), and the fact that $\zeta(1 + s_1) \ll 1$ for $\operatorname{Re}(s_1) = \frac{1}{8}$ to deduce that

$$\mathfrak{M}_{12} \ll y^{1-2\sigma}. \quad (3.5.11)$$

From (3.5.6), (3.5.10), and (3.5.11), we conclude that

$$\mathfrak{M}_1 = 1 + O(y^{1-2\sigma}). \quad (3.5.12)$$

Having evaluated the term $\mathfrak{M}_1$ in (3.3.8), our next task is to estimate the term $\mathfrak{M}_2$, which is defined by (3.3.11) and (3.3.12). By (ii) and (iii) of Lemma 3.1.1, we may extend the range of summation in (3.3.11) to infinity, with small error, and deduce that

$$\begin{aligned} \mathfrak{M}_2 = & \Gamma(2 - 2\sigma)\zeta(2 - 2\sigma) \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{hk} (h, k)^{2-2\sigma} \times \\ & \times \left(\frac{W(y/h)}{W(y)} \right) \left(\frac{W(y/k)}{W(y)} \right) \mathcal{J} + O(|\mathcal{J}| \exp(-y^\varepsilon/4)). \end{aligned}$$

We insert into this the definitions of $W(y/h)$ and $W(y/k)$ given by (1.2.7) and interchange the order of summation to arrive at

$$\begin{aligned} \mathfrak{M}_2 = & \mathcal{J} \frac{\Gamma(2-2\sigma)\zeta(2-2\sigma)}{(2\pi i)^2 W(y)^2 \zeta^2(2\sigma)} \int_{(r)} \int_{(r)} y^{s_1+s_2} \Gamma(s_1+1) \Gamma(s_2+1) \zeta(2\sigma+s_1) \times \\ & \times \zeta(2\sigma+s_2) \zeta(1+s_1) \zeta(1+s_2) \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{h^{1+s_1} k^{1+s_2}} (h, k)^{2-2\sigma} ds_2 ds_1 \\ & + O(|\mathcal{J}| \exp(-y^\varepsilon/4)), \end{aligned} \quad (3.5.13)$$

where we again choose r to equal $(\log y)^{-1}$. The interchange of order of summation is justified by absolute convergence. We express the inner sum as an Euler product through Lemma 2.2.1 and write

$$\begin{aligned} \sum_{h=1}^{\infty} \sum_{k=1}^{\infty} \frac{\mu(h)\mu(k)}{h^{1+s_1} k^{1+s_2}} (h, k)^{2-2\sigma} &= \prod_p \left(1 - \frac{1}{p^{1+s_1}} - \frac{1}{p^{1+s_2}} + \frac{1}{p^{2\sigma+s_1+s_2}} \right) \\ &= \frac{\zeta(2\sigma+s_1+s_2)}{\zeta(1+s_1)\zeta(1+s_2)} F_2(s_1, s_2; 2\sigma), \end{aligned} \quad (3.5.14)$$

where $F_2(s_1, s_2; 2\sigma)$ is the Euler product

$$F_2(s_1, s_2; 2\sigma) = \prod_p \frac{\left(1 - \frac{1}{p^{1+s_1}} - \frac{1}{p^{1+s_2}} + \frac{1}{p^{2\sigma+s_1+s_2}} \right) \left(1 - \frac{1}{p^{2\sigma+s_1+s_2}} \right)}{\left(1 - \frac{1}{p^{1+s_1}} \right) \left(1 - \frac{1}{p^{1+s_2}} \right)}. \quad (3.5.15)$$

This Euler product satisfies

$$F_2(s_1, s_2; 2\sigma) = \prod_p \left(1 + O\left(\frac{1}{p^{9/8}} \right) \right) \ll 1 \quad (3.5.16)$$

for s_1, s_2, σ such that $1 + \operatorname{Re}(s_1) > \frac{9}{16}$, $1 + \operatorname{Re}(s_2) > \frac{9}{16}$, and $2\sigma + \operatorname{Re}(s_1) + \operatorname{Re}(s_2) > \frac{9}{16}$. We insert (3.5.14) into (3.5.13) and cancel the factors $\zeta(1+s_1)$ and $\zeta(1+s_2)$ to deduce that

$$\begin{aligned} \mathfrak{M}_2 = & \mathcal{J} \frac{\Gamma(2-2\sigma)\zeta(2-2\sigma)}{(2\pi i)^2 W(y)^2 \zeta^2(2\sigma)} \int_{(r)} \int_{(r)} y^{s_1+s_2} \Gamma(s_1+1) \Gamma(s_2+1) \zeta(2\sigma+s_1) \times \\ & \times \zeta(2\sigma+s_2) \zeta(2\sigma+s_1+s_2) F_2(s_1, s_2; 2\sigma) ds_2 ds_1 + O(|\mathcal{J}| \exp(-y^\varepsilon/4)). \end{aligned} \quad (3.5.17)$$

We next apply the residue theorem and move the line of integration of the s_2 -integral to the line $\operatorname{Re}(s_2) = -2\sigma + \frac{3}{4}$, leaving residues at the poles $s_2 = 1 - 2\sigma$ and $s_2 = 1 - 2\sigma - s_1$. We estimate the new double integral using (ii) of Lemma 3.1.1, our assumption (3.1.1), Stirling's formula, (3.1.2), (3.5.16), and the fact that

$$|\zeta(2\sigma + s_1)| \leq \zeta(1 + r) \ll \frac{1}{r} = \log y$$

and $y^{s_1+s_2} \ll y^{\frac{3}{4}-2\sigma}$ when $\operatorname{Re}(s_1) = r = (\log y)^{-1}$ and $\operatorname{Re}(s_2) = -2\sigma + \frac{3}{4}$. Thus the result of applying the residue theorem to (3.5.17) is

$$\mathfrak{M}_2 = \mathfrak{M}_{21} + \mathfrak{M}_{22} + O\left(|\mathcal{J}|y^{\frac{3}{4}-2\sigma+\varepsilon}\right), \quad (3.5.18)$$

where

$$\begin{aligned} \mathfrak{M}_{21} = & \mathcal{J} \frac{\Gamma^2(2-2\sigma)\zeta(2-2\sigma)}{(2\pi i)W(y)^2\zeta^2(2\sigma)} \int_{(r)} y^{s_1+1-2\sigma} \Gamma(s_1+1) \times \\ & \times \zeta(2\sigma+s_1)\zeta(1+s_1)F_2(s_1, 1-2\sigma; 2\sigma) ds_1 \end{aligned} \quad (3.5.19)$$

and

$$\begin{aligned} \mathfrak{M}_{22} = & \mathcal{J} \frac{\Gamma(2-2\sigma)\zeta(2-2\sigma)}{(2\pi i)W(y)^2\zeta^2(2\sigma)} \int_{(r)} y^{1-2\sigma} \Gamma(s_1+1)\Gamma(2-2\sigma-s_1) \times \\ & \times \zeta(2\sigma+s_1)\zeta(1-s_1)F_2(s_1, 1-2\sigma-s_1; 2\sigma) ds_1. \end{aligned} \quad (3.5.20)$$

To estimate $\mathfrak{M}_{21}$, move the line of integration in (3.5.19) to the line $\operatorname{Re}(s_1) = -2\sigma + \frac{3}{4}$, leaving residues at the poles $s_1 = 0$ and $s_1 = 1 - 2\sigma$. We estimate the new integral in the same way we bounded the error term in (3.5.18), and arrive at

$$\begin{aligned} \mathfrak{M}_{21} = & \mathcal{J} \frac{\Gamma^2(2-2\sigma)\zeta(2-2\sigma)F_2(0, 1-2\sigma; 2\sigma)}{W(y)^2\zeta(2\sigma)} y^{1-2\sigma} \\ & + \mathcal{J} \frac{\Gamma^3(2-2\sigma)\zeta^2(2-2\sigma)F_2(1-2\sigma, 1-2\sigma; 2\sigma)}{W(y)^2\zeta^2(2\sigma)} y^{2-4\sigma} + O\left(|\mathcal{J}|y^{\frac{3}{4}-2\sigma+\varepsilon}\right). \end{aligned}$$

It follows from this, our assumption (3.1.1), (ii) of Lemma 3.1.1, and the bound (3.5.16) that

$$\mathfrak{M}_{21} \ll |\mathcal{J}|y^{1-2\sigma}. \quad (3.5.21)$$

Next, to estimate $\mathfrak{M}_{22}$, we use Cauchy's theorem to move the line of integration in (3.5.20) to the line $\operatorname{Re}(s_1) = \frac{1}{8}$. Note that there are no residues because the integrand is holomorphic between $\operatorname{Re}(s_1) = r$ and $\operatorname{Re}(s_1) = \frac{1}{8}$. We bound the new integral using our assumption (3.1.1), (ii) of Lemma 3.1.1, Stirling's formula, (3.1.2), (3.5.16), and the fact that $\zeta(2\sigma + s_1) \ll 1$ for $\operatorname{Re}(s_1) = \frac{1}{8}$ to deduce that

$$\mathfrak{M}_{22} \ll |\mathcal{J}|y^{1-2\sigma}. \quad (3.5.22)$$

It now follows from (3.5.18), (3.5.21), and (3.5.22) that

$$\mathfrak{M}_2 \ll |\mathcal{J}|y^{1-2\sigma}.$$

From this, the definition (3.3.12) of $\mathcal{J}$, and Lemma 3.1.9, we arrive at

$$\mathfrak{M}_2 \ll y^{1-2\sigma}. \quad (3.5.23)$$

3.6 Proof of Theorem 1.2.1

In this section, we finish our estimation of $g(u)$ and complete the proof of Theorem 1.2.1. From (3.3.8), (3.4.61), (3.5.12), and (3.5.23), we arrive at

$$g(u) = 1 + O(y^{1-2\sigma}) + O\left(T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon}y^{\frac{7}{8}} + T^{-1+\frac{7}{2}\delta+\varepsilon}y^{\frac{7}{4}}\right) + \mathfrak{D}_1 + \mathfrak{D}_2 \quad (3.6.1)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$, with Δ and λ given by (3.4.1). Recall the definition (3.3.13) of $\mathfrak{D}_1$. By Cauchy's theorem, we may move the path of integration of the z -integral in (3.3.13) to the positive real axis. We may then apply Lemma 3.1.8 to deduce that

$$\begin{aligned} \int_0^{\infty e^{i\lambda}} z^{-\sigma+iu} \exp\left(-\frac{\Delta^2}{4}(\log z)^2\right) dz &= \frac{2\sqrt{\pi}}{\Delta} \exp\left(\frac{(1-\sigma+iu)^2}{\Delta^2}\right) \\ &\ll \exp(-\log^2 T) \end{aligned}$$

for $u \geq \Delta \log T$. We use this, Lemma 3.1.2, and (ii) and (iv) of Lemma 3.1.1 to bound the right-hand side of the definition (3.3.13) of $\mathfrak{D}_1$. The result is

$$\mathfrak{D}_1 \ll \exp(-\tfrac{1}{2}\log^2 T)$$

for $y \leq T$. Similarly, $\mathfrak{D}_2 \ll \exp(-\frac{1}{2} \log^2 T)$. It follows from these and (3.6.1) that

$$g(u) = 1 + O(y^{1-2\sigma}) + O\left(T^{-\frac{1}{2}+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{8}} + T^{-1+\frac{7}{2}\delta+\varepsilon} y^{\frac{7}{4}}\right) \quad (3.6.2)$$

for $y \leq T^{8/13}$ and $u \geq \Delta \log T$.

Now we set $y = T^\theta$ with $0 < \theta < \frac{4}{7}$, as assumed in Theorem 1.2.1, and choose values for the parameters δ and ϵ_0 introduced in (3.4.1) and Lemma 3.1.1. We choose $0 < \delta < \frac{1}{3}$ small enough, depending on θ , such that

$$7\delta + \frac{7}{4}\theta < 1. \quad (3.6.3)$$

Then, having chosen δ , we choose $0 < \epsilon_0 < \frac{1}{4}$ small enough, depending on θ and δ , such that

$$4\theta\epsilon_0 < \delta, \quad (3.6.4)$$

$$7\delta + \left(\frac{7}{4} + 4\epsilon_0\right)\theta < 1, \quad (3.6.5)$$

and such that our assumption (3.1.1) holds for $\frac{1}{2} \leq \sigma \leq \frac{1}{2} + \epsilon_0$. Note that it is possible to choose such an ϵ_0 because of (3.6.3) and the fact that $\delta > 0$. It follows from (3.6.5) that

$$-\frac{1}{2} + \frac{7}{2}\delta + \frac{7}{8}\theta + \varepsilon < -2\epsilon_0\theta \leq (1 - 2\sigma)\theta \quad (3.6.6)$$

and

$$-1 + \frac{7}{2}\delta + \frac{7}{4}\theta + \varepsilon < -2\epsilon_0\theta \leq (1 - 2\sigma)\theta \quad (3.6.7)$$

for $\sigma \leq \frac{1}{2} + \epsilon_0$ and small enough $\varepsilon > 0$. From (3.6.2), (3.6.6), and (3.6.7), we conclude that

$$g(u) = 1 + O(y^{1-2\sigma}) \quad (3.6.8)$$

for $y = T^\theta$, $u \geq \Delta \log T$, and $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$.

Having estimated $g(u)$, we now return to the problem of bounding $N(\sigma, T)$. We deduce from (3.3.4) and (3.6.8) that

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \leq T + O(\Delta \log T) + O(Ty^{1-2\sigma}). \quad (3.6.9)$$

The definition of Δ given by (3.4.1) and the condition (3.6.4) in our choice of ϵ_0 imply that

$$\Delta \log T \ll T^{1-\frac{\delta}{2}} \leq T^{1-2\epsilon_0\theta} \leq T^{1+\theta(1-2\sigma)} \quad (3.6.10)$$

for $\sigma \leq \frac{1}{2} + \epsilon_0$. From (3.6.9), (3.6.10), and the fact that $y = T^\theta$, we arrive at

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \leq T + O(T^{1+\theta(1-2\sigma)}). \quad (3.6.11)$$

It follows from this, Lemma 3.2.1, and the inequality $\log(1+x) \leq x$ that

$$\int_\sigma^1 (N(\xi, 2T) - N(\xi, T)) d\xi \ll T^{1+\theta(1-2\sigma)} \quad (3.6.12)$$

$0 < \theta < \frac{4}{7}$ and $\frac{1}{2} + \frac{1}{\log y} \leq \sigma \leq \frac{1}{2} + \epsilon_0$. In fact, (3.6.12) holds for $0 < \theta < \frac{4}{7}$ and $\frac{1}{2} \leq \sigma \leq \frac{1}{2} + \epsilon_0$. Indeed, if $N(T)$ is the total number of nontrivial zeros of $\zeta(s)$ that have imaginary part between 0 and T , then (1.0.1) implies

$$N(T) \ll T \log T. \quad (3.6.13)$$

Thus

$$\int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{\log y}} (N(\xi, 2T) - N(\xi, T)) d\xi \leq \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{\log y}} N(T) d\xi \ll T,$$

and it follows from this and (3.6.12) with $\sigma = \frac{1}{2} + \frac{1}{\log y}$ that

$$\int_{\frac{1}{2}}^1 (N(\xi, 2T) - N(\xi, T)) d\xi \ll T.$$

Hence (3.6.12) holds for $0 < \theta < \frac{4}{7}$ and $\frac{1}{2} \leq \sigma \leq \frac{1}{2} + \epsilon_0$.

We now complete the proof of Theorem 1.2.1. If $\sigma \geq \frac{1}{2} + \epsilon_0$, then the conclusion of Theorem 1.2.1 follows immediately from Ingham's theorem (1.2.2). If $\frac{1}{2} \leq \sigma \leq$

$\frac{1}{2} + \frac{1}{\log T}$, then the conclusion of Theorem 1.2.1 follows immediately from (3.6.13).

Now suppose that $\frac{1}{2} + \frac{1}{\log T} \leq \sigma \leq \frac{1}{2} + \epsilon_0$. As in Selberg [59], we write

$$\begin{aligned} N(\sigma, 2T) - N(\sigma, T) &\leq \frac{\int_{\sigma - \frac{1}{\log T}}^{\sigma} N(\xi, 2T) - N(\xi, T) d\xi}{\int_{\sigma - \frac{1}{\log T}}^{\sigma} 1 d\xi} \\ &= \log T \int_{\sigma - \frac{1}{\log T}}^{\sigma} N(\xi, 2T) - N(\xi, T) d\xi. \end{aligned}$$

It follows from this and (3.6.12) that

$$N(\sigma, 2T) - N(\sigma, T) \ll T^{1+\theta(1-2\sigma)} \log T.$$

We replace T in turn by $\frac{T}{2}, \frac{T}{4}, \frac{T}{8}, \dots$, and add the results to conclude that

$$N(\sigma, T) \ll T^{1+\theta(1-2\sigma)} \log T.$$

This completes the proof of Theorem 1.2.1.

3.7 Proof of Theorem 1.2.2

We now prove Theorem 1.2.2. We choose ϵ_0 as in the previous section, so that (3.6.8) holds for $u \geq \Delta \log T$. In view of (3.6.11), we only need to prove that the reverse inequality holds. To this end, define the function $v(t)$ by

$$v(t) = \frac{1}{\Delta\sqrt{\pi}} \int_{T+\Delta \log T}^{2T-\Delta \log T} \exp\left(-\frac{(t-u)^2}{\Delta^2}\right) du. \quad (3.7.1)$$

It follows from this definition and (3.3.2) that $0 \leq v(t) \leq 1$ for all real t . Hence

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \geq \int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 v(t) dt. \quad (3.7.2)$$

We insert the definition (3.7.1) of $v(t)$ into the right-hand side of (3.7.2) to deduce that

$$\begin{aligned} &\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \\ &\geq \frac{1}{\Delta\sqrt{\pi}} \int_{T+\Delta \log T}^{2T-\Delta \log T} \int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 \exp\left(-\frac{(t-u)^2}{\Delta^2}\right) dt du. \end{aligned} \quad (3.7.3)$$

If $|t - u| \geq \Delta \log T$, then (3.3.3) holds. Thus it follows from (3.1.2) and (3.2.8) that

$$\left(\int_{-\infty}^T + \int_{2T}^{\infty} \right) |\zeta(\sigma + it)M(\sigma + it)|^2 \exp \left(-\frac{(t - u)^2}{\Delta^2} \right) dt \ll \exp \left(-\frac{1}{4} \log^2 T \right)$$

for $T + \Delta \log T \leq u \leq 2T - \Delta \log T$. From this and (3.7.3), we arrive at

$$\begin{aligned} \int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt &\geq \frac{1}{\Delta \sqrt{\pi}} \int_{T+\Delta \log T}^{2T-\Delta \log T} \int_{-\infty}^{\infty} |\zeta(\sigma + it)M(\sigma + it)|^2 \times \\ &\quad \times \exp \left(-\frac{(t - u)^2}{\Delta^2} \right) dt du + O \left(\exp \left(-\frac{1}{8} \log^2 T \right) \right). \end{aligned}$$

This is the same as

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \geq \int_{T+\Delta \log T}^{2T-\Delta \log T} g(u) du + O \left(\exp \left(-\frac{1}{8} \log^2 T \right) \right), \quad (3.7.4)$$

by the definition (3.3.5) of $g(u)$. We insert (3.6.8) into (3.7.4) to deduce that

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt \geq T + O(\Delta \log T) + O(Ty^{1-2\sigma}).$$

From this, (3.6.10), (3.6.11), and the definition $y = T^\theta$, we arrive at

$$\int_T^{2T} |\zeta(\sigma + it)M(\sigma + it)|^2 dt = T + O(Ty^{1-2\sigma}).$$

This completes the proof of Theorem 1.2.2.

4 The pair correlation conjecture and the alternative hypothesis

In this chapter, we prove the results described in Section 1.3. As usual, we consider the variable $T > 1$ as a large parameter going to infinity.

4.1 Lemmas

In each of the following lemmas we assume there is a number T_0 such that the function $f(\alpha, T)$ is defined for $\alpha \geq 0$ and $T \geq T_0$, and $f(\cdot, T)$ is Lebesgue measurable for each such T .

Lemma 4.1.1. *Let $f(\alpha) = f(\alpha, T)$ be nonnegative and let μ be a positive Borel measure on $[0, \infty)$ such that $\int_0^\infty e^{-bx} d\mu(x) < \infty$ for all $b \geq 1$. If*

$$\lim_{T \rightarrow \infty} \int_0^\infty f(\alpha, T) e^{-b\alpha} d\alpha = \int_0^\infty e^{-b\alpha} d\mu(\alpha) \quad (4.1.1)$$

for all fixed positive integers b , then

$$\mu[0, d] + o(1) \leq \int_0^d f(\alpha, T) d\alpha \leq \mu[0, d] + o(1) \quad (4.1.2)$$

as $T \rightarrow \infty$, for all fixed $d > 0$.

Proof. The proof uses Karamata's method (see for example §7.53 of [62]). Let $0 < \delta < e^{-d}$, and set $\eta = -\log(e^{-d} - \delta)$. Define $k(u)$ by

$$k(u) = \begin{cases} 1/u & \text{if } e^{-d} \leq u \leq 1, \\ (u - e^{-\eta}) \frac{e^d}{\delta} & \text{if } e^{-\eta} \leq u \leq e^{-d}, \\ 0 & \text{otherwise.} \end{cases}$$

By the Weierstrass approximation theorem, given any $\varepsilon > 0$, there is a polynomial $P(u) = \sum_{n=0}^N a_n u^n$ such that

$$k(u) < P(u) < k(u) + \varepsilon \quad (4.1.3)$$

for $u \in [0, 1]$. Now $uk(u) = 1$ for $e^{-d} \leq u \leq 1$. Thus, by (4.1.3), $1 < uP(u)$ for $e^{-d} \leq u \leq 1$. Also, since $k(u) \geq 0$, we have $0 < uP(u)$ for $0 < u \leq e^{-d}$. Hence $1 < e^{-\alpha}P(e^{-\alpha})$ for $0 \leq \alpha \leq d$ and $0 < e^{-\alpha}P(e^{-\alpha})$ for $\alpha \geq d$. Since f is nonnegative,

$$\int_0^d f(\alpha) d\alpha \leq \int_0^d f(\alpha) e^{-\alpha} P(e^{-\alpha}) d\alpha \leq \int_0^\infty f(\alpha) e^{-\alpha} P(e^{-\alpha}) d\alpha. \quad (4.1.4)$$

By (4.1.1), we have

$$\begin{aligned} \int_0^\infty f(\alpha) e^{-\alpha} P(e^{-\alpha}) d\alpha &= \sum_{n=0}^N a_n \int_0^\infty f(\alpha) e^{-\alpha(n+1)} d\alpha \\ &\sim \sum_{n=0}^N a_n \left(\int_0^\infty e^{-(n+1)\alpha} d\mu(\alpha) \right) = \int_0^\infty e^{-\alpha} P(e^{-\alpha}) d\mu(\alpha). \end{aligned} \quad (4.1.5)$$

It follows from (4.1.4) and (4.1.5) that

$$\int_0^d f(\alpha) d\alpha \leq \int_0^\infty e^{-\alpha} P(e^{-\alpha}) d\mu(\alpha) + o(1) \quad (4.1.6)$$

as $T \rightarrow \infty$. By (4.1.3) we have $e^{-\alpha}P(e^{-\alpha}) < e^{-\alpha}k(e^{-\alpha}) + \varepsilon e^{-\alpha}$ for $\alpha \geq 0$. From this and (4.1.6) we see that

$$\int_0^d f(\alpha) d\alpha \leq \int_0^\infty e^{-\alpha} k(e^{-\alpha}) d\mu(\alpha) + \varepsilon \int_0^\infty e^{-\alpha} d\mu(\alpha) + o(1). \quad (4.1.7)$$

If $e^{-\eta} \leq u \leq e^{-d}$, then by the definitions of $k(u)$ and η , we have

$$uk(u) = u(u - e^{-\eta}) \frac{e^d}{\delta} \leq e^{-d}(e^{-d} - e^{-\eta}) \frac{e^d}{\delta} = e^{-d}(\delta) \frac{e^d}{\delta} = 1.$$

Thus $uk(u) \leq 1$ for $e^{-\eta} \leq u \leq 1$. Hence $e^{-\alpha}k(e^{-\alpha}) \leq 1$ for $0 \leq \alpha \leq \eta$. Also, since $k(u) = 0$ for $u \leq e^{-\eta}$, we have $e^{-\alpha}k(e^{-\alpha}) = 0$ for $\alpha \geq \eta$. Thus

$$\begin{aligned} \int_0^\infty e^{-\alpha}k(e^{-\alpha}) d\mu(\alpha) &= \int_0^\eta e^{-\alpha}k(e^{-\alpha}) d\mu(\alpha) + \int_\eta^\infty e^{-\alpha}k(e^{-\alpha}) d\mu(\alpha) \\ &\leq \int_0^\eta d\mu(\alpha) + 0 \\ &= \mu[0, \eta). \end{aligned}$$

We find from this and (4.1.7) that

$$\int_0^d f(\alpha) d\alpha \leq \mu[0, \eta) + \varepsilon \int_0^\infty e^{-\alpha} d\mu(\alpha) + o(1). \quad (4.1.8)$$

Now $\mu[0, \eta) \rightarrow \mu[0, d]$ as $\eta \rightarrow d^+$. Since $\eta = -\log(e^{-d} - \delta)$, we have that $\eta \rightarrow d^+$ as $\delta \rightarrow 0^+$. Thus, making δ and ε small enough in (4.1.8), we obtain the second inequality in (4.1.2).

To prove the first inequality in (4.1.2), define $\tau = -\log(e^{-d} + \delta)$ and define $\ell(u)$ by

$$\ell(u) = \begin{cases} 1/u & \text{if } e^{-\tau} \leq u \leq 1, \\ (u - e^{-d}) \frac{e^\tau}{\delta} & \text{if } e^{-d} \leq u \leq e^{-\tau}, \\ 0 & \text{otherwise.} \end{cases}$$

By the Weierstrass approximation theorem, given any $\varepsilon > 0$, there is a polynomial $Q(u) = \sum_{n=0}^M b_n u^n$ such that

$$\ell(u) - \varepsilon < Q(u) < \ell(u) \quad (4.1.9)$$

for $u \in [0, 1]$. Now $u\ell(u) = 1$ for $e^{-\tau} \leq u \leq 1$. Also, if $e^{-d} \leq u \leq e^{-\tau}$, then by the definition of τ ,

$$u\ell(u) = u(u - e^{-d}) \frac{e^\tau}{\delta} \leq e^{-\tau}(e^{-\tau} - e^{-d}) \frac{e^\tau}{\delta} = e^{-\tau}(\delta) \frac{e^\tau}{\delta} = 1.$$

Hence $u\ell(u) \leq 1$ for $e^{-d} \leq u \leq 1$. Therefore, by (4.1.9), $1 > uQ(u)$ for $e^{-d} \leq u \leq 1$. Also, $\ell(u) = 0$ for $0 \leq u \leq e^{-d}$, so by (4.1.9) we have $0 > uQ(u)$ for $0 < u \leq e^{-d}$. Hence, by a change of variable, $1 > e^{-\alpha}Q(e^{-\alpha})$ for $0 \leq \alpha \leq d$, and $0 > e^{-\alpha}Q(e^{-\alpha})$ for $\alpha \geq d$. Since f is nonnegative,

$$\int_0^d f(\alpha) d\alpha \geq \int_0^d f(\alpha) e^{-\alpha} Q(e^{-\alpha}) d\alpha \geq \int_0^\infty f(\alpha) e^{-\alpha} Q(e^{-\alpha}) d\alpha. \quad (4.1.10)$$

By (4.1.1), we have

$$\begin{aligned} \int_0^\infty f(\alpha) e^{-\alpha} Q(e^{-\alpha}) d\alpha &= \sum_{n=0}^M b_n \int_0^\infty f(\alpha) e^{-\alpha(n+1)} d\alpha \\ &\sim \sum_{n=0}^M b_n \left(\int_0^\infty e^{-(n+1)\alpha} d\mu(\alpha) \right) = \int_0^\infty e^{-\alpha} Q(e^{-\alpha}) d\mu(\alpha). \end{aligned} \quad (4.1.11)$$

It follows from (4.1.10) and (4.1.11) that

$$\int_0^d f(\alpha) d\alpha \geq \int_0^\infty e^{-\alpha} Q(e^{-\alpha}) d\mu(\alpha) + o(1) \quad (4.1.12)$$

as $T \rightarrow \infty$. By (4.1.9) we have $e^{-\alpha}Q(e^{-\alpha}) > e^{-\alpha}\ell(e^{-\alpha}) - \varepsilon e^{-\alpha}$ for $\alpha \geq 0$. It follows from this and (4.1.12) that

$$\int_0^d f(\alpha) d\alpha \geq \int_0^\infty e^{-\alpha}\ell(e^{-\alpha}) d\mu(\alpha) - \varepsilon \int_0^\infty e^{-\alpha} d\mu(\alpha) + o(1). \quad (4.1.13)$$

By the definition of $\ell(u)$, we have $u\ell(u) = 1$ for $e^{-\tau} \leq u \leq 1$. Also, $\ell(u) \geq 0$ for $0 \leq u \leq e^{-\tau}$. Thus $e^{-\alpha}\ell(e^{-\alpha}) = 1$ for $0 \leq \alpha \leq \tau$, and $e^{-\alpha}\ell(e^{-\alpha}) \geq 0$ for $\alpha \geq \tau$.

We now see that

$$\begin{aligned} \int_0^\infty e^{-\alpha}\ell(e^{-\alpha}) d\mu(\alpha) &= \int_0^\tau e^{-\alpha}\ell(e^{-\alpha}) d\mu(\alpha) + \int_\tau^\infty e^{-\alpha}\ell(e^{-\alpha}) d\mu(\alpha) \\ &\geq \int_0^\tau d\mu(\alpha) + 0 \\ &= \mu[0, \tau]. \end{aligned}$$

It follows from this and (4.1.13) that

$$\int_0^d f(\alpha) d\alpha \geq \mu[0, \tau] - \varepsilon \int_0^\infty e^{-\alpha} d\mu(\alpha) + o(1). \quad (4.1.14)$$

Now $\mu[0, \tau) \rightarrow \mu[0, d)$ as $\tau \rightarrow d^-$. Furthermore, $\tau = -\log(e^{-d} + \delta)$, so $\tau \rightarrow d^-$ as $\delta \rightarrow 0^+$. Thus, making δ and ε small enough in (4.1.14), we obtain the first inequality in (4.1.2). $\square$

Lemma 4.1.2. *Suppose that*

$$\int_0^x |f(\alpha, T)| d\alpha \ll x \quad (4.1.15)$$

uniformly for all large x and T , that μ is a positive Borel measure on $[0, \infty)$ such that $\mu[0, d] < \infty$ for $d > 0$, and that r is a real continuous function such that $\int_0^\infty |r(\alpha)| d\mu(\alpha) < \infty$. Assume that there exists a function $r_1(x)$ and an $x_0 > 0$ such that

- (i) $|r(x)| \leq r_1(x)$ for $x \geq x_0$;
- (ii) $r'_1(x)$ exists for $x \geq x_0$, and r'_1 is Riemann integrable over closed subintervals of $[x_0, \infty)$;
- (iii) $xr_1(x) \rightarrow 0$ as $x \rightarrow \infty$; and
- (iv) $\int_{x_0}^\infty x|r'_1(x)| dx < \infty$.

If for each fixed $d > 0$, we have

$$\mu[0, d) + o(1) \leq \int_0^d f(\alpha, T) d\alpha \leq \mu[0, d] + o(1) \quad (4.1.16)$$

as $T \rightarrow \infty$, then

$$\lim_{T \rightarrow \infty} \int_0^\infty f(\alpha, T) r(\alpha) d\alpha = \int_0^\infty r(\alpha) d\mu(\alpha). \quad (4.1.17)$$

Proof. Let x_0 be as in our hypotheses and let $B' > B > x_0$. By condition (ii)

$$\begin{aligned} \int_B^{B'} |f(\alpha, T)| r_1(\alpha) d\alpha &= \left(r_1(\alpha) \int_0^\alpha |f(u, T)| du \right) \Big|_{\alpha=B}^{\alpha=B'} \\ &\quad - \int_B^{B'} \left(\int_0^\alpha |f(u, T)| du \right) r'_1(\alpha) d\alpha. \end{aligned} \quad (4.1.18)$$

The assumptions (4.1.15), (iii), and (iv) allow us to let $B' \rightarrow \infty$ in (4.1.18) and obtain

$$\begin{aligned} \int_B^\infty |f(\alpha, T)| r_1(\alpha) d\alpha &= -r_1(B) \int_0^B |f(u, T)| du \\ &\quad - \int_B^\infty \left(\int_0^\alpha |f(u, T)| du \right) r_1'(\alpha) d\alpha. \end{aligned} \quad (4.1.19)$$

By the same assumptions, if $\varepsilon > 0$, then the right-hand side of (4.1.19) is less than ε for B and T large enough. Thus by (i),

$$\left| \int_B^\infty f(\alpha, T) r(\alpha) d\alpha \right| \leq \int_B^\infty |f(\alpha, T)| r_1(\alpha) d\alpha < \varepsilon \quad (4.1.20)$$

for B and T large enough. Furthermore, since $\int_0^\infty |r(\alpha)| d\mu(\alpha) < \infty$, we see that

$$\left| \int_B^\infty r(\alpha) d\mu(\alpha) \right| < \varepsilon \quad (4.1.21)$$

for B large enough. Since $\mu[a, b] < \infty$ for $b > a \geq 0$, not all points of $[0, \infty)$ can have positive measure. Thus, we may choose a B such that (4.1.20) and (4.1.21) hold and such that $\mu\{B\} = 0$. By (4.1.15) we can also choose an $\eta > 0$ so small that

$$\eta \mu[0, B] < \varepsilon \quad \text{and} \quad \eta \int_0^B |f(\alpha, T)| d\alpha < \varepsilon \quad (4.1.22)$$

for all large T . Let $x_1, x_2, \dots$ be the distinct points in $[0, B]$ with positive μ -measure. Since $\mu[0, B]$ is finite, there can only be countably many such points. For the same reason, a positive integer ν can be chosen such that

$$2 \left(\max_{0 \leq u \leq B} |r(u)| \right) \sum_{k=\nu+1}^\infty \mu\{x_k\} < \varepsilon. \quad (4.1.23)$$

Since r is continuous, we can partition $[0, B]$ into small subintervals $[c, d]$ such that $|r(x) - r(y)| < \eta$ for all x, y in $[c, d]$. We may also choose the endpoints of each subinterval $[c, d]$ so that none, except possibly $c = 0$, is equal to $x_1, x_2, \dots, x_\nu$.

Now, for each subinterval $[c, d]$ we have

$$\begin{aligned}
& \left| \int_c^d f(\alpha) r(\alpha) d\alpha - \int_c^d r(\alpha) d\mu(\alpha) \right| \\
&= \left| \int_c^d f(\alpha)(r(\alpha) - r(d)) d\alpha + \int_c^d f(\alpha)r(d) d\alpha \right. \\
&\quad \left. - \int_c^d r(d) d\mu(\alpha) - \int_c^d (r(\alpha) - r(d)) d\mu(\alpha) \right| \\
&\leq \int_c^d |f(\alpha)| |r(\alpha) - r(d)| d\alpha + |r(d)| \left| \int_c^d f(\alpha) d\alpha - \int_c^d d\mu(\alpha) \right| \\
&\quad + \int_c^d |r(\alpha) - r(d)| d\mu(\alpha).
\end{aligned}$$

Thus, since $|r(x) - r(y)| < \eta$ for x, y in $[c, d]$, we have

$$\begin{aligned}
& \left| \int_c^d f(\alpha) r(\alpha) d\alpha - \int_c^d r(\alpha) d\mu(\alpha) \right| \\
&\leq \eta \int_c^d |f(\alpha)| d\alpha + |r(d)| \left| \int_c^d f(\alpha) d\alpha - \int_c^d d\mu(\alpha) \right| + \eta \int_c^d d\mu(\alpha).
\end{aligned} \tag{4.1.24}$$

Now replace d by c in (4.1.16) and subtract the resulting formula from (4.1.16) to obtain

$$\mu(c, d) + o(1) \leq \int_c^d f(\alpha, T) d\alpha \leq \mu[c, d] + o(1).$$

Thus

$$-\mu\{c\} + o(1) \leq \int_c^d f(\alpha, T) d\alpha - \mu[c, d] \leq \mu\{d\} + o(1),$$

and we find that

$$\left| \int_c^d f(\alpha, T) d\alpha - \mu[c, d] \right| \leq \mu\{c\} + \mu\{d\} + o(1).$$

Hence

$$\left| \int_c^d f(\alpha) d\alpha - \int_c^d d\mu(\alpha) \right| = \left| \int_c^d f(\alpha) d\alpha - \mu[c, d] \right| \leq \mu\{c\} + \mu\{d\} + o(1).$$

It follows from this and (4.1.24) that

$$\begin{aligned}
& \left| \int_c^d f(\alpha) r(\alpha) d\alpha - \int_c^d r(\alpha) d\mu(\alpha) \right| \\
&\leq \eta \int_c^d |f(\alpha)| d\alpha + |r(d)| (\mu\{c\} + \mu\{d\} + o(1)) + \eta \int_c^d d\mu(\alpha).
\end{aligned} \tag{4.1.25}$$

We will use this estimate for subintervals $[c, d]$ with $c > 0$, but a different estimate when $c = 0$. When $c = 0$ we proceed as follows. By (4.1.16) we have

$$o(1) \leq \int_0^d f(\alpha, T) d\alpha - \mu[0, d] \leq \mu\{d\} + o(1).$$

Thus

$$\left| \int_0^d f(\alpha, T) d\alpha - \mu[0, d] \right| \leq \mu\{d\} + o(1).$$

That is,

$$\left| \int_0^d f(\alpha) d\alpha - \int_0^d d\mu(\alpha) \right| = \left| \int_0^d f(\alpha) d\alpha - \mu[0, d] \right| \leq \mu\{d\} + o(1).$$

It follows from this and (4.1.24) that

$$\begin{aligned} & \left| \int_0^d f(\alpha) r(\alpha) d\alpha - \int_0^d r(\alpha) d\mu(\alpha) \right| \\ & \leq \eta \int_0^d |f(\alpha)| d\alpha + |r(d)|(\mu\{d\} + o(1)) + \eta \int_0^d d\mu(\alpha). \end{aligned} \quad (4.1.26)$$

Next we sum our estimates over all the subintervals $[c, d]$. Let $\sum_{[c, d]}$ denote a sum over all the subintervals $[c, d]$ of our partition of $[0, B]$, and let $\sum'_{[c, d]}$ denote a sum that excludes the term $\mu\{c\}$ if $c = 0$. Then by (4.1.25) and (4.1.26) we see that

$$\begin{aligned} & \left| \int_0^B f(\alpha) r(\alpha) d\alpha - \int_0^B r(\alpha) d\mu(\alpha) \right| = \left| \sum_{[c, d]} \left(\int_c^d f(\alpha) r(\alpha) d\alpha - \int_c^d r(\alpha) d\mu(\alpha) \right) \right| \\ & \leq \sum_{[c, d]} \left| \int_c^d f(\alpha) r(\alpha) d\alpha - \int_c^d r(\alpha) d\mu(\alpha) \right| \\ & \leq \eta \sum_{[c, d]} \int_c^d |f(\alpha)| d\alpha + \sum'_{[c, d]} |r(d)|(\mu\{c\} + \mu\{d\} + o(1)) + \eta \sum_{[c, d]} \int_c^d d\mu(\alpha). \end{aligned}$$

Hence

$$\begin{aligned} & \left| \int_0^B f(\alpha) r(\alpha) d\alpha - \int_0^B r(\alpha) d\mu(\alpha) \right| \\ & \leq \eta \int_0^B |f(\alpha)| d\alpha + \sum'_{[c, d]} |r(d)|(\mu\{c\} + \mu\{d\} + o(1)) + \eta \int_0^B d\mu(\alpha). \end{aligned} \quad (4.1.27)$$

To estimate the sum on the right-hand side, we first note that

$$\begin{aligned}
\sum_{[c,d]}' |r(d)|(\mu\{c\} + \mu\{d\} + o(1)) &\leq \left(\max_{0 \leq u \leq B} |r(u)| \right) \sum_{[c,d]}' (\mu\{c\} + \mu\{d\} + o(1)) \\
&= \left(\max_{0 \leq u \leq B} |r(u)| \right) \left(\sum_{[c,d]}' \mu\{c\} + \sum_{[c,d]} \mu\{d\} \right) + o(1).
\end{aligned} \tag{4.1.28}$$

Since none of the nonzero endpoints of our partition are among the points $x_1, \dots, x_\nu$, we see that

$$\sum_{[c,d]}' \mu\{c\} \leq \sum_{k=\nu+1}^{\infty} \mu\{x_k\} \quad \text{and} \quad \sum_{[c,d]} \mu\{d\} \leq \sum_{k=\nu+1}^{\infty} \mu\{x_k\}.$$

Using these estimates in (4.1.28), we obtain

$$\sum_{[c,d]}' |r(d)|(\mu\{c\} + \mu\{d\} + o(1)) \leq 2 \left(\max_{0 \leq u \leq B} |r(u)| \right) \sum_{k=\nu+1}^{\infty} \mu\{x_k\} + o(1).$$

From this and (4.1.23) we now find that

$$\sum_{[c,d]}' |r(d)|(\mu\{c\} + \mu\{d\} + o(1)) < \varepsilon + o(1).$$

Inserting this in (4.1.27), we obtain

$$\left| \int_0^B f(\alpha) r(\alpha) d\alpha - \int_0^B r(\alpha) d\mu(\alpha) \right| \leq \eta \int_0^B |f(\alpha)| d\alpha + \eta \int_0^B d\mu(\alpha) + \varepsilon + o(1). \tag{4.1.29}$$

Since $\int_0^B d\mu(\alpha) = \mu[0, B)$, it follows from (4.1.22) and (4.1.29) that

$$\left| \int_0^B f(\alpha) r(\alpha) d\alpha - \int_0^B r(\alpha) d\mu(\alpha) \right| \leq 3\varepsilon + o(1). \tag{4.1.30}$$

Finally, by (4.1.20), (4.1.21), and (4.1.30), we see that

$$\begin{aligned}
&\left| \int_0^\infty f(\alpha) r(\alpha) d\alpha - \int_0^\infty r(\alpha) d\mu(\alpha) \right| \\
&\leq \left| \int_B^\infty f(\alpha) r(\alpha) d\alpha \right| + \left| \int_0^B f(\alpha) r(\alpha) d\alpha - \int_0^B r(\alpha) d\mu(\alpha) \right| + \left| \int_B^\infty r(\alpha) d\mu(\alpha) \right| \\
&< \varepsilon + 3\varepsilon + \varepsilon + o(1) = 5\varepsilon + o(1).
\end{aligned}$$

□

Lemma 4.1.3. *Assume that $f(\alpha, T) \geq 0$ and that*

$$\int_0^x f(\alpha, T) d\alpha \ll x \quad (4.1.31)$$

uniformly for $x \geq 1$ and large enough T . Let μ be a positive Borel measure on $[0, \infty)$ such that

$$\int_0^\infty \min\{1, \alpha^{-2}\} d\mu(\alpha) < \infty.$$

If

$$\lim_{T \rightarrow \infty} \int_0^\infty \left(\frac{\sin \beta \alpha}{\alpha} \right)^2 f(\alpha, T) d\alpha = \int_0^\infty \left(\frac{\sin \beta \alpha}{\alpha} \right)^2 d\mu(\alpha) \quad (4.1.32)$$

for each fixed $\beta > 0$, then

$$\mu[0, d) + o(1) \leq \int_0^d f(\alpha, T) d\alpha \leq \mu[0, d] + o(1) \quad (4.1.33)$$

as $T \rightarrow \infty$ for all fixed $d > 0$.

Proof. Our approach is similar to that used to prove Lemma 4 of [33]. Let $\eta > 0$, and define

$$K(x) = K_\eta(x) = \frac{\sin 2\pi x + \sin 2\pi(1 + \eta)x}{2\pi x(1 - 4\eta^2 x^2)}.$$

Then

$$\hat{K}_\eta(t) \stackrel{\text{def}}{=} \int_{-\infty}^\infty K_\eta(x) e^{-2\pi i x t} dx = \begin{cases} 1 & \text{if } |t| \leq 1, \\ \cos^2(\pi(|t| - 1)/2\eta) & \text{if } 1 \leq |t| \leq 1 + \eta, \\ 0 & \text{if } |t| \geq 1 + \eta. \end{cases} \quad (4.1.34)$$

Note that $K(x)$ is an even function and

$$K^{(j)}(x) \ll_\eta \min\{1, x^{-3}\} \quad (4.1.35)$$

for $x \geq 0$ and $j = 0, 1, 2$. Integrating by parts twice, we have

$$\hat{K}(t) = \int_0^\infty K''(x) \left(\frac{\sin \pi t x}{\pi t} \right)^2 dx \quad (4.1.36)$$

for all real t . To prove the lemma, we first show that

$$\int_0^\infty f(\alpha, T) \hat{K}\left(\frac{\alpha}{d}\right) d\alpha = \int_0^\infty \hat{K}\left(\frac{\alpha}{d}\right) d\mu(\alpha) + o(1) \quad (4.1.37)$$

as $T \rightarrow \infty$, for all fixed $d > 0$. We substitute (4.1.36) for $\hat{K}$ and interchange the order of integration to obtain

$$\int_0^\infty f(\alpha, T) \hat{K}\left(\frac{\alpha}{d}\right) d\alpha - \int_0^\infty \hat{K}\left(\frac{\alpha}{d}\right) d\mu(\alpha) = \left(\frac{d}{\pi}\right)^2 \int_0^\infty K''(x) R\left(\frac{\pi x}{d}, T\right) dx, \quad (4.1.38)$$

where

$$R(\kappa, T) = \int_0^\infty f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha}\right)^2 d\alpha - \int_0^\infty \left(\frac{\sin \kappa \alpha}{\alpha}\right)^2 d\mu(\alpha). \quad (4.1.39)$$

The validity of the interchange in the order of integration will be justified below.

From (4.1.38) we see that (4.1.37) will follow from

$$\lim_{T \rightarrow \infty} \int_0^\infty K''(x) R\left(\frac{\pi x}{d}, T\right) dx = 0. \quad (4.1.40)$$

We write

$$\int_0^\infty K''(x) R\left(\frac{\pi x}{d}, T\right) dx = \int_0^D + \int_D^\infty \quad (4.1.41)$$

and estimate the two integrals on the right-hand side separately.

In order to estimate the integral over $[0, D]$, we first show that

$$R(\kappa, T) \ll 1 + \kappa^2 \quad (4.1.42)$$

uniformly for $\kappa \geq 0$ and large enough T . By integration by parts and (4.1.31),

$$\int_1^\infty \frac{f(\alpha, T)}{\alpha^2} d\alpha = \frac{1}{\alpha^2} \int_0^\alpha f(\xi, T) d\xi \Big|_{\alpha=1}^\infty + 2 \int_1^\infty \frac{1}{\alpha^3} \int_0^\alpha f(\xi, T) d\xi d\alpha \ll 1 \quad (4.1.43)$$

for T sufficiently large. It follows that

$$\int_1^\infty f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha}\right)^2 d\alpha \leq \int_1^\infty \frac{f(\alpha, T)}{\alpha^2} d\alpha \ll 1$$

for such T . Also, by (4.1.31) we have

$$\begin{aligned} \int_0^1 f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha} \right)^2 d\alpha &= \kappa^2 \int_0^1 f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\kappa \alpha} \right)^2 d\alpha \\ &\leq \kappa^2 \int_0^1 f(\alpha, T) d\alpha \\ &\ll \kappa^2 \end{aligned}$$

for all sufficiently large T . Hence

$$\begin{aligned} \int_0^\infty f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha} \right)^2 d\alpha &= \int_0^1 f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha} \right)^2 d\alpha + \int_1^\infty f(\alpha, T) \left(\frac{\sin \kappa \alpha}{\alpha} \right)^2 d\alpha \\ &\ll \kappa^2 + 1 \end{aligned} \tag{4.1.44}$$

for T large enough. A similar computation shows that

$$\int_0^\infty \left(\frac{\sin \kappa \alpha}{\alpha} \right)^2 d\mu(\alpha) \ll \kappa^2 + 1. \tag{4.1.45}$$

The bound (4.1.42) now follows from (4.1.39), (4.1.44), and (4.1.45). Now by (4.1.32), we have

$$\lim_{T \rightarrow \infty} R(\kappa, T) = 0$$

for fixed $\kappa \geq 0$. By this, (4.1.42), and Lebesgue's dominated convergence theorem, we now see that

$$\lim_{T \rightarrow \infty} \int_0^D K''(x) R\left(\frac{\pi x}{d}, T\right) dx = 0 \tag{4.1.46}$$

for fixed $D \geq 0$.

Next we show that the integral over $[D, \infty)$ in (4.1.41) is small. That is, we show that if $d > 0$ is fixed, then

$$\int_D^\infty K''(x) R\left(\frac{\pi x}{d}, T\right) dx \ll \frac{1}{D} \tag{4.1.47}$$

uniformly for $D \geq 1$ and all T sufficiently large. For $D \geq 1$ write

$$\begin{aligned} \int_D^\infty \int_0^\infty f(\alpha, T) K''(x) \left(\frac{\sin \pi \frac{\alpha}{d} x}{\alpha} \right)^2 d\alpha dx \\ = \int_D^\infty \int_1^\infty \dots d\alpha dx + \int_D^\infty \int_0^1 \dots d\alpha dx = J_1 + J_2, \end{aligned}$$

say. By (4.1.35),

$$J_1 \ll \int_D^\infty \int_1^\infty \frac{f(\alpha, T)}{x^3 \alpha^2} d\alpha dx = \frac{1}{2D^2} \int_1^\infty \frac{f(\alpha, T)}{\alpha^2} d\alpha.$$

It follows from this and (4.1.43) that

$$J_1 \ll \frac{1}{D^2} \quad (4.1.48)$$

uniformly for large T . To estimate J_2 , let $h(\alpha, x) = \left(\sin \pi \frac{\alpha}{d} x / \pi \frac{\alpha}{d}\right)^2$, and observe that

$$h(\alpha, x) \ll x^2 \quad \text{and} \quad h_x(\alpha, x) \stackrel{\text{def}}{=} \frac{\partial h}{\partial x}(\alpha, x) \ll x \quad (4.1.49)$$

uniformly for $\alpha \geq 0$. Using this and (4.1.31), we see that

$$\begin{aligned} \int_0^1 f(\alpha, T) h(\alpha, x) d\alpha &\ll x^2 \int_0^1 f(\alpha, T) d\alpha \ll x^2 \quad \text{and} \\ \int_0^1 f(\alpha, T) h_x(\alpha, x) d\alpha &\ll x \int_0^1 f(\alpha, T) d\alpha \ll x \end{aligned}$$

uniformly for $x \geq 0$ and all sufficiently large T . It follows from these estimates, (4.1.35), and integration by parts that

$$\begin{aligned} \left(\frac{d}{\pi}\right)^2 J_2 &= \int_D^\infty \int_0^1 f(\alpha, T) K''(x) h(\alpha, x) d\alpha dx \\ &= K'(x) \int_0^1 f(\alpha, T) h(\alpha, x) d\alpha \Big|_{x=D}^\infty - \int_D^\infty K'(x) \int_0^1 f(\alpha, T) h_x(\alpha, x) d\alpha dx \\ &\ll \frac{1}{D}. \end{aligned}$$

Combining this and (4.1.48), we now find that

$$\int_D^\infty \int_0^\infty f(\alpha, T) K''(x) \left(\frac{\sin \pi \frac{\alpha}{d} x}{\alpha}\right)^2 d\alpha dx \ll \frac{1}{D}$$

for all sufficiently large T . A similar computation gives

$$\int_D^\infty \int_0^\infty K''(x) \left(\frac{\sin \pi \frac{\alpha}{d} x}{\alpha}\right)^2 d\mu(\alpha) dx \ll \frac{1}{D}.$$

Using these and the definition of $R(\kappa, T)$ from (4.1.39), we obtain (4.1.47).

By (4.1.46), (4.1.47), and (4.1.41), we obtain (4.1.40) upon taking D large enough. The formula (4.1.37) now follows from (4.1.38) and (4.1.40).

Having established (4.1.37), we now complete the proof of the lemma. If $\chi_{[a,b]}$ is the characteristic function of the interval $[a, b]$, then by (4.1.34) we have

$$\chi_{[0,1]}(t) \leq \hat{K}_\eta(t) \leq \chi_{[0,1+\eta]}(t) \quad (4.1.50)$$

for $t \geq 0$. Thus

$$\begin{aligned} \int_0^\infty f(\alpha, T) \hat{K}_\eta \left(\frac{(1+\eta)\alpha}{d} \right) d\alpha \\ \leq \int_0^d f(\alpha, T) d\alpha \leq \int_0^\infty f(\alpha, T) \hat{K}_\eta \left(\frac{\alpha}{d} \right) d\alpha. \end{aligned} \quad (4.1.51)$$

Using (4.1.37) to replace the first and third integrals in (4.1.51), we obtain

$$\begin{aligned} \int_0^\infty \hat{K}_\eta \left(\frac{(1+\eta)\alpha}{d} \right) d\mu(\alpha) + o(1) \\ \leq \int_0^d f(\alpha, T) d\alpha \leq \int_0^\infty \hat{K}_\eta \left(\frac{\alpha}{d} \right) d\mu(\alpha) + o(1). \end{aligned} \quad (4.1.52)$$

By (4.1.50), we have

$$\int_0^\infty \hat{K}_\eta \left(\frac{(1+\eta)\alpha}{d} \right) d\mu(\alpha) \geq \int_0^\infty \chi_{[0,1]} \left(\frac{(1+\eta)\alpha}{d} \right) d\mu(\alpha) = \mu[0, d/(1+\eta)]$$

and

$$\int_0^\infty \hat{K}_\eta \left(\frac{\alpha}{d} \right) d\mu(\alpha) \leq \int_0^\infty \chi_{[0,1+\eta]} \left(\frac{\alpha}{d} \right) d\mu(\alpha) = \mu[0, d(1+\eta)].$$

It follows from this and (4.1.52) that

$$\mu[0, d/(1+\eta)] + o(1) \leq \int_0^d f(\alpha, T) d\alpha \leq \mu[0, d(1+\eta)] + o(1)$$

as $T \rightarrow \infty$, for all fixed $d > 0$ and $\eta > 0$. Since $\mu[0, \delta] \rightarrow \mu[0, d]$ as $\delta \rightarrow d^-$ and $\mu[0, \delta] \rightarrow \mu[0, d]$ as $\delta \rightarrow d^+$, making η small gives (4.1.33).

It remains to show that the interchange in the order of integration in (4.1.38) is valid. Write

$$\begin{aligned}
\int_0^\infty \hat{K}\left(\frac{\alpha}{d}\right) d\mu(\alpha) &= \int_0^\infty \int_0^\infty K''(x) \left(\frac{\sin \pi \frac{\alpha}{d} x}{\pi \frac{\alpha}{d}}\right)^2 dx d\mu(\alpha) \\
&= \int_1^\infty \int_0^\infty \dots dx d\mu(\alpha) + \int_0^1 \int_0^\infty \dots dx d\mu(\alpha) \\
&= I_1 + I_2, \quad \text{say.}
\end{aligned}$$

By (4.1.35) and the fact that $|\sin x| \leq 1$ for real x , the double integral I_1 converges absolutely. Thus the order of integration can be interchanged in I_1 by Fubini's theorem. For I_2 we use (4.1.35), (4.1.49), integration by parts (twice), and Fubini's theorem (twice) to see that

$$\begin{aligned}
I_2 &= \int_0^1 \int_0^\infty K''(x) h(\alpha, x) dx d\mu(\alpha) = - \int_0^1 \int_0^\infty K'(x) h_x(\alpha, x) dx d\mu(\alpha) \\
&= - \int_0^\infty \int_0^1 K'(x) h_x(\alpha, x) d\mu(\alpha) dx = \int_0^\infty K''(x) \int_0^x \int_0^1 h_x(\alpha, \xi) d\mu(\alpha) d\xi dx \\
&= \int_0^\infty K''(x) \int_0^1 \int_0^x h_x(\alpha, \xi) d\xi d\mu(\alpha) dx = \int_0^\infty K''(x) \int_0^1 h(\alpha, x) d\mu(\alpha) dx.
\end{aligned}$$

A similar computation using (4.1.43) justifies the interchange in the order of integration on the right-hand side of

$$\int_0^\infty f(\alpha, T) \hat{K}\left(\frac{\alpha}{d}\right) d\alpha = \int_0^\infty \int_0^\infty f(\alpha, T) K''(x) \left(\frac{\sin \pi \frac{\alpha}{d} x}{\pi \frac{\alpha}{d}}\right)^2 dx d\alpha.$$

This proves (4.1.38). □

4.2 Proof of Theorem 1.3.2

By a result of Goldston [30, (2.6)], on RH there exists a T_0 such that

$$\int_c^d F(\alpha, T) d\alpha \ll \max\{1, d - c\} \tag{4.2.1}$$

uniformly for real $c \leq d$ and $T \geq T_0$. Thus (4.1.15) holds for $f(\alpha, T) = F(\alpha, T)$ (we can dispense with the absolute value sign because $F(\alpha, T)$ is nonnegative). We will use this fact repeatedly below without mention.

To prove Theorem 1.3.2 we show that (A) $\Rightarrow$ (B), (B) $\Rightarrow$ (A), (A) $\Rightarrow$ (C), (C) $\Rightarrow$ (A), (A) $\Rightarrow$ (D), and (D) $\Rightarrow$ (A).

First we show that (A) $\Rightarrow$ (B). Suppose that (A) holds. Take $r(x) = r_1(x) = e^{-2bx}$ in Lemma 4.1.2 to get

$$\lim_{T \rightarrow \infty} \int_0^\infty F(\alpha, T) e^{-2b\alpha} d\alpha = \int_0^\infty e^{-2b\alpha} d\mu(\alpha). \quad (4.2.2)$$

By Theorem 1 of Goldston, Gonek, and Montgomery [32], stated in a slightly different form,

$$I(b, T) = \left(\int_0^\infty F(\alpha, T) e^{-2b\alpha} d\alpha - \frac{1}{2} \right) (1 + o(1)) T \log^2 T \quad (4.2.3)$$

as $T \rightarrow \infty$, uniformly for $T^{-1} \log^3 T \leq b \ll 1$. Inserting (4.2.2) in (4.2.3), we obtain

$$I(b, T) = \left(\int_0^\infty e^{-2b\alpha} d\mu(\alpha) + o(1) - \frac{1}{2} \right) (1 + o(1)) T \log^2 T$$

as $T \rightarrow \infty$ for fixed $b > 0$, which is (B).

To prove (B) $\Rightarrow$ (A), suppose that (B) holds. Then combining (B) and (4.2.3), we see that (4.2.2) holds for all $b > 0$. Now apply Lemma 4.1.1 to obtain (A).

To show (A) $\Rightarrow$ (C), we make the change of variable $v = \frac{\log x}{\log T}$ in the definition of $J(\beta, T)$ and see that

$$\begin{aligned} J(\beta, T) \frac{T}{\log^2 T} &= \int_0^\beta (\psi(T^v + T^{v-1}) - \psi(T^v) - T^{v-1})^2 \frac{dv}{T^{v-1} \log T} \\ &= \int_0^\beta W_1(v, T) dv, \quad \text{say.} \end{aligned} \quad (4.2.4)$$

A special case of a result of Goldston [30, Theorem 2] is¹

$$\begin{aligned} \int_1^\infty \left(\psi \left(x + \frac{x}{T} \right) - \psi(x) - \frac{x}{T} \right)^2 e^{-b \frac{\log x}{\log T}} \frac{dx}{x^2} \\ \sim \left(\int_0^\infty F(\alpha, T) e^{-b\alpha} d\alpha - \frac{1}{2} \right) \frac{\log^2 T}{T} \end{aligned}$$

¹This can also be proved using (4.2.3) and an explicit formula due to Selberg, as in [32].

as $T \rightarrow \infty$, for fixed $b > 0$. From this and our previous change of variable we find that

$$\int_0^\infty W_1(v, T) e^{-bv} dv \sim \int_0^\infty F(\alpha, T) e^{-b\alpha} d\alpha - \frac{1}{2}. \quad (4.2.5)$$

Let $\omega(v, T)$ be a nonnegative function such that

$$\int_0^\beta \omega(v, T) e^{-bv} dv \sim \frac{1}{2} \quad \text{as } T \rightarrow \infty \quad (4.2.6)$$

for fixed $b \geq 0$, $\beta > 0$ (for instance we can take $\omega(v, T) = T^{-2v} \log T$). Let $W(v, T) = \omega(v, T) + W_1(v, T)$. Then combining (4.2.5) and (4.2.6), we obtain

$$\int_0^\infty W(v, T) e^{-bv} dv \sim \int_0^\infty F(\alpha, T) e^{-b\alpha} d\alpha. \quad (4.2.7)$$

Now suppose that (A) holds. Take $f(\alpha, T) = F(\alpha, T)$ and $r(\alpha) = r_1(\alpha) = e^{-b\alpha}$ in Lemma 4.1.2 to see that

$$\lim_{T \rightarrow \infty} \int_0^\infty F(\alpha, T) e^{-b\alpha} d\alpha = \int_0^\infty e^{-b\alpha} d\mu(\alpha). \quad (4.2.8)$$

It follows from this and (4.2.7) that

$$\lim_{T \rightarrow \infty} \int_0^\infty W(v, T) e^{-bv} dv = \int_0^\infty e^{-b\alpha} d\mu(\alpha) \quad (4.2.9)$$

for all $b > 0$. Thus, by Lemma 4.1.1 with $f(\alpha, T) = W(\alpha, T)$, we have

$$\mu[0, \beta) + o(1) \leq \int_0^\beta W(v, T) dv \leq \mu[0, \beta] + o(1) \quad (4.2.10)$$

as $T \rightarrow \infty$ for any fixed $\beta > 0$. Since $W(v, T) = W_1(v, T) + \omega(v, T)$, it follows from (4.2.10) and (4.2.6) that

$$\mu[0, \beta) - \frac{1}{2} + o(1) \leq \int_0^\beta W_1(v, T) dv \leq \mu[0, \beta] - \frac{1}{2} + o(1)$$

as $T \rightarrow \infty$ for any fixed $\beta > 0$. From this and (4.2.4) we obtain (C). This proves (A) $\Rightarrow$ (C).

Next we prove that (C) $\Rightarrow$ (A). Let $W(v, T)$, $W_1(v, T)$, and $\omega(v, T)$ be as in the previous paragraph. By Theorem 1 of Goldston and Montgomery [33], if RH is true, then

$$\int_1^X \left(\psi \left(x + \frac{x}{T} \right) - \psi(x) - \frac{x}{T} \right)^2 \frac{dx}{x^2} \ll \frac{(\log X)(\log 2T)}{T}$$

for $T \geq 1$ and $X \geq 2$. Thus, letting $\beta \geq 1$ and $X = T^\beta$, we see that

$$J(\beta, T) \ll \frac{\beta(\log T)(\log 2T)}{T}$$

for $T \geq 2$. It follows from this and (4.2.4) that

$$\int_0^\beta W_1(v, T) dv \ll \beta$$

uniformly for $\beta \geq 1$ and $T \geq 2$. Since $W(v, T) = W_1(v, T) + \omega(v, T)$, by (4.2.6) we have

$$\int_0^\beta W(v, T) dv \ll \beta \quad \text{uniformly for } \beta \geq 1 \text{ and } T \geq 2. \quad (4.2.11)$$

Thus (4.1.15) holds with $f(\alpha, T) = W(\alpha, T)$ (note that $W(\alpha, T)$ is nonnegative). Now assume that (C) is true. Since $W(v, T) = W_1(v, T) + \omega(v, T)$, we see from (4.2.4), (4.2.6), and (C) that (4.2.10) holds as $T \rightarrow \infty$ for any fixed $\beta > 0$. By (4.2.10) and (4.2.11) the hypotheses of Lemma 4.1.2 are satisfied with $f(\alpha, T) = W(\alpha, T)$ and $r(\alpha) = r_1(\alpha) = e^{-b\alpha}$. Hence we have (4.2.9). It now follows from (4.2.9) and (4.2.7) that (4.2.8) holds for all $b > 0$. Applying Lemma 4.1.1, we finally obtain (A).

To prove (A) $\Rightarrow$ (D), we use the formula (6.14) of [31] (see also (2.3) and (2.5) of [30]), namely

$$\int_{-\infty}^{\infty} F(\alpha, T) \left(\frac{\sin(\pi\beta\alpha)}{\pi\beta\alpha} \right)^2 d\alpha = \frac{1}{\beta} N^*(T) + \frac{2}{\beta^2} \int_0^\beta N(T, u) du + O\left(\frac{\beta(1+\beta)}{\log^2 T} \right).$$

Since $F(\alpha, T) = F(-\alpha, T)$, we can also write this as

$$\frac{\beta}{2} N^*(T) + \int_0^\beta N(T, u) du = \int_0^\infty F(\alpha, T) \left(\frac{\sin(\pi\beta\alpha)}{\pi\alpha} \right)^2 d\alpha + O\left(\frac{\beta^3(1+\beta)}{\log^2 T} \right). \quad (4.2.12)$$

Suppose that (A) holds. By (4.2.1) and (A) we may use Lemma 4.1.2 with $f(\alpha) = F(\alpha)$, $r(\alpha) = (\sin(\pi\beta\alpha)/\pi\alpha)^2$ and $r_1(\alpha) = (\pi\alpha)^{-2}$ to obtain

$$\lim_{T \rightarrow \infty} \int_0^\infty F(\alpha, T) \left(\frac{\sin(\pi\beta\alpha)}{\pi\alpha} \right)^2 d\alpha = \int_0^\infty \left(\frac{\sin(\pi\beta\alpha)}{\pi\alpha} \right)^2 d\mu(\alpha). \quad (4.2.13)$$

From this and (4.2.12) we obtain (D). This proves (A) $\Rightarrow$ (D).

To prove that (D) $\Rightarrow$ (A), suppose that (D) holds. Then by (4.2.12) we have (4.2.13). We may therefore apply Lemma 4.1.3 to obtain (A). This completes the proof of Theorem 1.3.2.

4.3 Proof of Theorem 1.3.3

Recall that $g(\alpha) = |\alpha|$ for $|\alpha| \leq 1$, and is defined for all $\mathbb{R}$ by periodicity. Let ν be the measure on $[0, \infty)$ defined by $\nu(0) = \frac{1}{2}$, $\nu(2n) = 1$ for all integers $n \geq 1$, and $d\nu(\alpha) = g(\alpha) d\alpha$ for $2n < \alpha < 2n+2$, $n \geq 0$. When $\mu = \nu$, we can easily evaluate the expressions in (A), (B), and (C) involving μ and see that the statements (A), (B), and (C) are the same as (A2), (B2), and (C2), respectively. Thus (A2), (B2), and (C2) are equivalent statements by Theorem 1.3.2. However, showing that they are equivalent to (D2) is not as straightforward. To do this, we will define a new statement (D') and show that, when $\mu = \nu$, the statement (D) is equivalent to (D'), which in turn is equivalent to (D2).

Lemma 4.3.1. *Assume RH. Let μ be a positive Borel measure on $[0, \infty)$ for which the function $\alpha \mapsto \min\{1, \alpha^{-2}\}$ is integrable over $[0, \infty)$, and define*

$$\varphi(\beta) = \varphi(\beta, \mu) = \int_0^\infty \left(\frac{\sin \pi \beta \alpha}{\pi \alpha} \right)^2 d\mu(\alpha).$$

If $\varphi(\beta)$ is absolutely continuous on each closed (finite) subinterval of $[0, \infty)$, then (D) is equivalent to

$$(D') \quad \frac{1}{2}N^*(T) + N(T, \beta) = \varphi'(\beta) + o(1)$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$ for which the derivative exists.

Proof. Suppose that $\varphi(\beta)$ is absolutely continuous on any closed subinterval of $[0, \infty)$. Then $\varphi'(\beta)$ exists almost everywhere. Also, $\varphi(\beta) = \int_0^\beta \varphi'(x) dx$ for all

$\beta \geq 0$. To show that (D) $\Rightarrow$ (D'), assume that (D) holds and suppose that $\beta > 0$ is a real number such that $\varphi'(\beta)$ exists. Since $N(T, u)$ is an increasing function of u for fixed T , we have

$$\frac{1}{h} \int_{\beta-h}^{\beta} N(T, u) du \leq N(T, \beta) \leq \frac{1}{h} \int_{\beta}^{\beta+h} N(T, u) du$$

for all small enough $h > 0$. It follows from this and (D) that

$$\frac{\varphi(\beta) - \varphi(\beta - h)}{h} + o(1) \leq \frac{1}{2}N^*(T) + N(T, \beta) \leq \frac{\varphi(\beta + h) - \varphi(\beta)}{h} + o(1).$$

Making h small enough, we obtain (D').

Now suppose that (D') holds. To show that (D') $\Rightarrow$ (D), we use the estimate

$$N(T, \beta) \ll 1 + \beta \quad \text{uniformly for } \beta > 0 \text{ and large } T, \quad (4.3.1)$$

which follows from Lemma 9 of [33]. By (4.3.1) and (D'), if $\beta > 0$ is such that $\varphi'(\beta)$ exists, then

$$N^*(T) \ll 1 + \beta + |\varphi'(\beta)|.$$

Hence, since the left-hand side does not depend on β , we have

$$N^*(T) \ll 1.$$

This and (4.3.1) gives

$$\frac{N^*(T)}{2} + N(T, u) \ll 1 + u \quad \text{uniformly for } u > 0 \text{ and large } T.$$

Thus, by Lebesgue's dominated convergence theorem and (D') we have

$$\lim_{T \rightarrow \infty} \int_0^{\beta} \left(\frac{N^*(T)}{2} + N(T, u) \right) du = \int_0^{\beta} \varphi'(u) du = \varphi(\beta),$$

which is (D). This proves that (D') $\Rightarrow$ (D), thereby completing the proof of the lemma. $\square$

To show that Lemma 4.3.1 is applicable when $\mu = \nu$, we need to show that $\varphi(\beta, \nu)$ is absolutely continuous. We do this by explicitly calculating $\varphi(\beta, \nu)$ and computing its derivative. By the definition of ν (see the beginning of this section), we have

$$\begin{aligned}\varphi(\beta, \nu) &= \int_0^\infty \left(\frac{\sin \pi \beta \alpha}{\pi \alpha} \right)^2 d\nu(\alpha) \\ &= \frac{\beta^2}{2} + \sum_{n=1}^\infty \left(\frac{\sin 2\pi \beta n}{2\pi n} \right)^2 + \int_0^\infty g(\alpha) \left(\frac{\sin \pi \beta \alpha}{\pi \alpha} \right)^2 d\alpha.\end{aligned}\quad (4.3.2)$$

To evaluate the sum on the right-hand side of (4.3.2) we integrate the Fourier series expansion of $x - [x] - \frac{1}{2}$ term-by-term to see that

$$\frac{(y - [y]) - (y - [y])^2}{2} = \sum_{n=1}^\infty \left(\frac{\sin \pi n y}{\pi n} \right)^2 \quad (4.3.3)$$

for all real y . (This can also be obtained by directly computing the Fourier series expansion of the left-hand side of (4.3.3).) It follows from (4.3.3) that the sum on the right-hand side of (4.3.2) equals

$$\sum_{n=1}^\infty \left(\frac{\sin 2\pi \beta n}{2\pi n} \right)^2 = \frac{(2\beta - [2\beta]) - (2\beta - [2\beta])^2}{8}. \quad (4.3.4)$$

To evaluate the integral on the right-hand side of (4.3.2), we expand g as a Fourier series:

$$g(\alpha) = \sum_{n=-\infty}^\infty \frac{1}{2} \left(\frac{1 - e^{-\pi i n}}{\pi i n} \right)^2 e^{\pi i n \alpha}. \quad (4.3.5)$$

Use this and the Fourier transform pair

$$r(\alpha) = \left(\frac{\sin(\pi \beta \alpha)}{\pi \beta \alpha} \right)^2 \quad \text{and} \quad \hat{r}(u) = \frac{1}{\beta} \max \left\{ 0, 1 - \frac{|u|}{\beta} \right\} \quad (4.3.6)$$

to write

$$\begin{aligned}\int_{-\infty}^\infty g(\alpha) \left(\frac{\sin(\pi \beta \alpha)}{\pi \beta \alpha} \right)^2 d\alpha &= \sum_{n=-\infty}^\infty \frac{1}{2} \left(\frac{1 - e^{-\pi i n}}{\pi i n} \right)^2 \int_{-\infty}^\infty \left(\frac{\sin(\pi \beta \alpha)}{\pi \beta \alpha} \right)^2 e^{\pi i n \alpha} d\alpha \\ &= \frac{1}{2\beta} - \frac{1}{\pi^2 \beta} \sum_{1 \leq n \leq 2\beta} \left(\frac{1 - (-1)^n}{n} \right)^2 \left(1 - \frac{n}{2\beta} \right).\end{aligned}\quad (4.3.7)$$

The interchange in order of summation is justified by absolute convergence. Inserting (4.3.4) and (4.3.7) in (4.3.2), we obtain

$$\begin{aligned} \varphi(\beta, \nu) = & \frac{\beta}{4} + \frac{\beta^2}{2} + \frac{(2\beta - [2\beta]) - (2\beta - [2\beta])^2}{8} \\ & - \frac{1}{2\pi^2} \sum_{0 < n < 2\beta} \left(\frac{1 - (-1)^n}{n} \right)^2 \left(\beta - \frac{n}{2} \right). \end{aligned} \quad (4.3.8)$$

We see from this that $\varphi(\beta, \nu)$ is differentiable at each $\beta > 0$ that is not a half-integer. Thus $\varphi(\beta, \nu)$ is absolutely continuous on every closed subinterval of $[0, \infty)$.

It now follows that we may apply Lemma 4.3.1 with $\mu = \nu$. Using (4.3.8), we obtain

Lemma 4.3.2. *Assume RH. Denote $[\beta]$ by m . When $\mu = \nu$, the statement (D) is equivalent to*

$$\frac{1}{2}N^*(T) + N(T, \beta) \sim \begin{cases} m + \frac{1}{2} - \frac{1}{2\pi^2} \sum_{n=0}^{m-1} \frac{4}{(2n+1)^2} & \text{if } m < \beta < m + \frac{1}{2}, \\ m + 1 - \frac{1}{2\pi^2} \sum_{n=0}^m \frac{4}{(2n+1)^2} & \text{if } m + \frac{1}{2} < \beta < m + 1 \end{cases}$$

as $T \rightarrow \infty$, for all fixed $\beta > 0$ not a half-integer.

(4.3.9)

To complete the proof of Theorem 1.3.3, we need to show that (4.3.9) is equivalent to (D2). Observe that if both $N^*(T) \sim 1$ and (4.3.9) are true, then (D2) holds. Similarly, if both $N^*(T) \sim 1$ and (D2) hold, then (4.3.9) is true. Thus, to prove that (4.3.9) is equivalent to (D2), it suffices to show that each of them implies $N^*(T) \sim 1$.

First we prove that (4.3.9) implies $N^*(T) \sim 1$. Recall that $N(T)$ is the number of zeros of $\zeta(s)$ with $0 < \gamma \leq T$ (counting multiplicity). If $\lambda > 0$, then by the

definition of $N^*(T)$, the definition of $F(\alpha, T)$, and the Fourier transform pair (4.3.6), we have

$$\begin{aligned}
 \left(\frac{T}{2\pi} \log T \right)^{-1} N(T) &\leq N^*(T) \\
 &\leq \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{0 < \gamma, \gamma' \leq T} \left(\frac{\sin \frac{\lambda}{2}(\gamma - \gamma') \log T}{\frac{\lambda}{2}(\gamma - \gamma') \log T} \right)^2 w(\gamma - \gamma') \\
 &= \int_0^\infty F(\alpha, T) \cdot \frac{2}{\lambda} \max \left\{ 0, 1 - \frac{\alpha}{\lambda} \right\} d\alpha.
 \end{aligned} \tag{4.3.10}$$

Now suppose that (4.3.9) is true. Then by Lemma 4.3.2, the statement (D) holds with $\mu = \nu$. Hence, by Theorem 1.3.2, the statement (A) is true with $\mu = \nu$. It follows from this and Lemma 4.1.2 with $f(\alpha, T) = F(\alpha, T)$, $\mu = \nu$, $r(\alpha) = \frac{2}{\lambda} \max \left\{ 0, 1 - \frac{\alpha}{\lambda} \right\}$, and $r_1(\alpha) = 0$, that

$$\lim_{T \rightarrow \infty} \int_0^\infty F(\alpha, T) \cdot \frac{2}{\lambda} \max \left\{ 0, 1 - \frac{\alpha}{\lambda} \right\} d\alpha = \int_0^\infty \frac{2}{\lambda} \max \left\{ 0, 1 - \frac{\alpha}{\lambda} \right\} d\nu(\alpha). \tag{4.3.11}$$

By a straightforward calculation using the definition of ν , we may write the right-hand side of (4.3.11) as

$$\begin{aligned}
 &\int_0^\infty \frac{2}{\lambda} \max \left\{ 0, 1 - \frac{\alpha}{\lambda} \right\} d\nu(\alpha) = \\
 &= \begin{cases} \frac{\vartheta}{\lambda^2} + \frac{\vartheta^3}{3\lambda^2} + \sum_{n=0}^{[\lambda/2]-1} \left(\frac{4}{\lambda} - \frac{8n+4}{\lambda^2} \right) & \text{if } 0 \leq \vartheta \leq 1, \\ -\frac{\vartheta}{\lambda^2} + \frac{2\vartheta^2}{\lambda^2} - \frac{\vartheta^3}{3\lambda^2} + \frac{2}{3\lambda^2} + \sum_{n=0}^{[\lambda/2]-1} \left(\frac{4}{\lambda} - \frac{8n+4}{\lambda^2} \right) & \text{if } 1 \leq \vartheta < 2, \end{cases}
 \end{aligned} \tag{4.3.12}$$

where $\vartheta = 2 \left(\frac{\lambda}{2} - \left[\frac{\lambda}{2} \right] \right)$. Using the fact that the sum of the first m positive integers is $m(m+1)/2$ and the fact that ϑ is bounded, we easily see that the limit of (4.3.12) as $\lambda \rightarrow \infty$ is equal to 1. It follows from this and (4.3.11) that making λ large

enough in (4.3.10) gives

$$\left(\frac{T}{2\pi} \log T\right)^{-1} N(T) \leq N^*(T) \leq 1 + o(1)$$

as $T \rightarrow \infty$. From this and the fact that

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T) \quad (4.3.13)$$

(this is also (1.0.1)), we obtain $N^*(T) \sim 1$ as $T \rightarrow \infty$. This proves that (4.3.9) implies $N^*(T) \sim 1$.

Next we show that (D2) implies $N^*(T) \sim 1$. First split the integral

$$\int_0^\infty F(\alpha, T) \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha$$

into two integrals, one over $[0, 1]$, and the other over $[1, \infty]$. For the integral over $[1, \infty]$ we use (4.2.1) and integration by parts, as in (4.1.43), to see that

$$\int_1^\infty F(\alpha, T) \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha \ll \int_1^\infty F(\alpha, T) \frac{d\alpha}{\alpha^2} \ll 1.$$

For the integral over $[0, 1]$ we use (1.3.1) to see that

$$\int_0^1 F(\alpha, T) \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha \sim \frac{\beta^2}{2} + \int_0^1 \alpha \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha.$$

If $\beta \geq 1$, then we may write the latter integral as

$$\begin{aligned} \int_0^1 \alpha \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha &= \int_0^{1/\beta} + \int_{1/\beta}^1 = \int_0^1 u \left(\frac{\sin \pi u}{\pi u}\right)^2 du + \int_{1/\beta}^1 \\ &\leq O(1) + \frac{1}{\pi^2} \int_{1/\beta}^1 \frac{d\alpha}{\alpha} = \frac{\log \beta}{\pi^2} + O(1). \end{aligned}$$

Hence for a fixed $\beta \geq 2$, say, we have

$$\int_0^\infty F(\alpha, T) \left(\frac{\sin \pi \beta \alpha}{\pi \alpha}\right)^2 d\alpha = \frac{1}{2}\beta^2 + O(\log \beta) + o(1). \quad (4.3.14)$$

as $T \rightarrow \infty$. The implied constant in the $O(\log \beta)$ term is absolute. We insert (4.3.14) into (4.2.12) and multiply through by $2/\beta$ to obtain

$$N^*(T) + \frac{2}{\beta} \int_0^\beta N(T, u) du = \beta + O\left(\frac{\log \beta}{\beta}\right) + o(1) \quad (4.3.15)$$

as $T \rightarrow \infty$ for all fixed $\beta \geq 2$. Now suppose that (D2) holds. By (4.3.8), the right-hand side of (D2) is equal to the derivative of $\varphi(\beta, \nu) - \beta/2$. From this fact, (4.3.1), (D2), and Lebesgue's dominated convergence theorem we have

$$\begin{aligned} \int_0^\beta N(T, u) du &\sim -\frac{\beta}{4} + \frac{\beta^2}{2} + \frac{(2\beta - [2\beta]) - (2\beta - [2\beta])^2}{8} \\ &\quad - \frac{1}{2\pi^2} \sum_{0 < n < 2\beta} \left(\frac{1 - (-1)^n}{n} \right)^2 \left(\beta - \frac{n}{2} \right). \end{aligned}$$

Inserting this into (4.3.15), we obtain

$$\begin{aligned} N^*(T) &\sim \frac{1}{2} - \frac{(2\beta - [2\beta]) - (2\beta - [2\beta])^2}{4\beta} \\ &\quad + \frac{1}{\pi^2} \sum_{0 < n < 2\beta} \left(\frac{1 - (-1)^n}{n} \right)^2 \left(1 - \frac{n}{2\beta} \right) + O\left(\frac{\log \beta}{\beta}\right) \end{aligned} \quad (4.3.16)$$

as $T \rightarrow \infty$ for all fixed $\beta \geq 2$. By the monotone convergence theorem and the special case $\alpha = 0$ of (4.3.5), we have

$$\begin{aligned} \lim_{\beta \rightarrow \infty} \frac{1}{\pi^2} \sum_{0 < n < 2\beta} \left(\frac{1 - (-1)^n}{n} \right)^2 \left(1 - \frac{n}{2\beta} \right) \\ = \frac{1}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{1 - (-1)^n}{n} \right)^2 = \frac{1}{2} - g(0) = \frac{1}{2}. \end{aligned}$$

We now see that $N^*(T) \sim 1$ on taking β large enough in (4.3.16). This proves that (D2) implies $N^*(T) \sim 1$. The proof of Theorem 1.3.3 is now complete.

Remark 1: As we mentioned in Section 1.3, Theorem 1.3.2 may be used to prove Theorem 1.3.1 by taking μ to be the measure defined by $\mu(0) = 1/2$, $d\mu(\alpha) = \alpha d\alpha$ for $0 < \alpha \leq 1$, and $d\mu(\alpha) = d\alpha$ for $\alpha > 1$. The process in doing this is similar to the one in this section.

Remark 2: A stronger conclusion holds in Lemma 4.1.2 in the special case when $r(\alpha) = (\sin \beta \alpha / \alpha)^2$ and $r_1(\alpha) = \alpha^{-2}$. Namely, the conclusion (4.1.17) of the

lemma holds uniformly for all β in any fixed closed interval. It follows from this and our proof of Theorem 1.3.2 that if (A) holds, then (D) holds uniformly for all β in any fixed, closed interval. Therefore, by our proof of Lemma 4.3.1, if (A) holds, then (D') holds uniformly for all β in any fixed, closed subinterval of an open interval on which $\varphi'(\beta)$ exists and is continuous. Hence, by our proof of Theorem 1.3.3, if (D2) holds, then it holds uniformly for all β in any fixed closed subinterval of $[0, \infty)$ not containing half-integers. Similarly, if (D1) holds, then it holds uniformly on any fixed closed interval. This uniformity in (D1) was first observed by Gallagher and Mueller [29].

4.4 The Alternative Hypothesis

In this section we prove Proposition 4.4.1, which states that the Alternative Hypothesis and the assumption that $N^*(T) \sim 1$ imply (A2). We also prove Theorem 1.3.4. First, we need the following lemma.

Lemma 4.4.1. *Suppose the Alternative Hypothesis is true. There is a function $\Psi(T)$ such that $\Psi(T) \rightarrow \infty$ as $T \rightarrow \infty$, $\Psi(T) = o(\log T)$, and such that the zeros of $\zeta(s)$ have the following property: if γ and γ' are ordinates of zeros for which*

$$\frac{T}{\log^2 T} < \gamma, \gamma' \leq T \quad \text{and} \quad \left| \frac{\gamma - \gamma'}{2\pi} \log T \right| \leq M, \quad (4.4.1)$$

then there is an integer k such that

$$(\gamma - \gamma') \log T = \pi k + O \left(|\gamma - \gamma'| \left(\Psi(T) + \frac{M \log T}{T} + \log \log T \right) \right).$$

Proof. Recall the definition of ψ from the statement of the Alternative Hypothesis. Let $\Psi(T) = \max\{\psi(\gamma) : T(\log T)^{-2} < \gamma \leq T\}$. The facts that $\Psi(T) \rightarrow \infty$ as $T \rightarrow \infty$ and $\Psi(T) = o(\log T)$ follow from the same properties of ψ . Let γ and γ' be as in the hypothesis of the Lemma. Suppose, without loss of generality, that

$\gamma' = \gamma_n$ and $\gamma = \gamma_m$ with $m \geq n$. From (1.3.4) and the definition of $\Psi(T)$, it follows that

$$\begin{aligned}\tilde{\gamma} - \tilde{\gamma}' &= \sum_{\ell=n}^{m-1} (\tilde{\gamma}_{\ell+1} - \tilde{\gamma}_\ell) \\ &= \sum_{\ell=n}^{m-1} \left(\frac{k_\ell}{2} + O((\gamma_{\ell+1} - \gamma_\ell)\psi(\gamma_\ell)) \right) \\ &= \frac{k}{2} + O(|\gamma - \gamma'| \Psi(T)),\end{aligned}\tag{4.4.2}$$

where $k = \sum_{\ell=n}^{m-1} k_\ell$. Now by (4.4.1), we have

$$\gamma \log \frac{\gamma}{\gamma'} \leq \gamma \left(\frac{\gamma - \gamma'}{\gamma'} \right) = \left(1 + \frac{\gamma - \gamma'}{\gamma'} \right) (\gamma - \gamma') \leq \left(1 + \frac{2\pi M \log T}{T} \right) (\gamma - \gamma').$$

Hence

$$\begin{aligned}\tilde{\gamma} - \tilde{\gamma}' &= \frac{1}{2\pi} (\gamma - \gamma') \log \gamma' + \frac{\gamma}{2\pi} \log \frac{\gamma}{\gamma'} \\ &= \frac{1}{2\pi} (\gamma - \gamma') (\log T + O(\log \log T)) + O \left(\left(1 + \frac{2\pi M \log T}{T} \right) (\gamma - \gamma') \right).\end{aligned}$$

This and (4.4.2) complete the proof of the Lemma. $\square$

We will prove Proposition 4.4.1 by first showing that if ν is the measure defined at the beginning of Section 4.3, then

$$\int_0^d \int_0^t F(\alpha, T) d\alpha dt = \int_0^d \nu[0, t] dt + o(1)\tag{4.4.3}$$

as $T \rightarrow \infty$ for any fixed $d > 0$. To prove this, we use the assumption that $N^*(T) \sim 1$ to handle the “diagonal terms” that have $\gamma = \gamma'$ in the definition of $F(\alpha, T)$. For the “off-diagonal terms” with $\gamma \neq \gamma'$, we use the Alternative Hypothesis, as follows.

Lemma 4.4.2. *Assume RH. Define*

$$G(\alpha, T) = \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{0 < \gamma' < \gamma \leq T} T^{i\alpha(\gamma - \gamma')} w(\gamma - \gamma').$$

If the Alternative Hypothesis is true, then

$$\int_{c+2}^{d+2} \int_0^t G(\alpha, T) d\alpha dt = \int_c^d \int_0^t G(\alpha, T) d\alpha dt + o(1)$$

as $T \rightarrow \infty$, for all fixed $c \leq d$.

Proof. By a change of variable, the conclusion of the Lemma is equivalent to

$$\int_c^d \int_t^{t+2} G(\alpha, T) d\alpha dt = o(1)$$

as $T \rightarrow \infty$, for all fixed $c \leq d$. To prove this formula, we integrate the definition of $G(\alpha, T)$ term-by-term to see that

$$\begin{aligned} \int_c^d \int_t^{t+2} G(\alpha, T) d\alpha dt &= \left(\frac{T}{2\pi} \log T \right)^{-1} \times \\ &\times \sum_{0 < \gamma' < \gamma \leq T} \frac{T^{i(d+2)(\gamma-\gamma')} - T^{i(c+2)(\gamma-\gamma')} - T^{id(\gamma-\gamma')} + T^{ic(\gamma-\gamma')}}{(i(\gamma - \gamma') \log T)^2} w(\gamma - \gamma'). \end{aligned}$$

Let $M > 1$ and write the above equation as

$$\int_c^d \int_t^{t+2} G(\alpha, T) d\alpha dt = Z_0 + Z_1 + Z_2, \quad (4.4.4)$$

where Z_0 is the sum of the terms with $\gamma' \leq T(\log T)^{-2}$, Z_1 is the sum of those that satisfy the conditions (4.4.1), and Z_2 is the sum of the remaining terms.

To bound Z_0 , observe that

$$\frac{T^{i(d+2)(\gamma-\gamma')} - T^{i(c+2)(\gamma-\gamma')} - T^{id(\gamma-\gamma')} + T^{ic(\gamma-\gamma')}}{(i(\gamma - \gamma') \log T)^2} = \int_c^d \int_t^{t+2} T^{i\alpha(\gamma-\gamma')} d\alpha dt,$$

which has absolute value at most $2(d - c)$ since $|T^{i\alpha(\gamma-\gamma')}| = 1$. Thus

$$Z_0 \ll \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\gamma' \leq \frac{T}{\log^2 T}} \sum_{0 < \gamma \leq T} w(\gamma - \gamma'). \quad (4.4.5)$$

Using

$$N(T + 1) - N(T) \ll \log T \quad (4.4.6)$$

(see [64], Theorem 9.2), we see that

$$\sum_{0 < \gamma \leq T} w(\gamma - t) \ll \log T$$

uniformly for all real t . From this, (4.3.13), and (4.4.5), it follows that

$$Z_0 \ll \frac{1}{\log T}. \quad (4.4.7)$$

To estimate Z_1 , let γ and γ' satisfy the conditions (4.4.1) and let α be a real number. By Lemma 4.4.1, we can write

$$\begin{aligned} T^{i(\alpha+2)(\gamma-\gamma')} - T^{i\alpha(\gamma-\gamma')} &= T^{i\alpha(\gamma-\gamma')} \left(\exp(i2(\gamma-\gamma') \log T) - 1 \right) \\ &= T^{i\alpha(\gamma-\gamma')} \left(\exp(i2\pi k + i\mathcal{E}) - 1 \right) \\ &= T^{i\alpha(\gamma-\gamma')} \left(\exp(i\mathcal{E}) - 1 \right), \end{aligned}$$

where $\mathcal{E} = \mathcal{E}(\gamma, \gamma', T)$ is a real number such that

$$\mathcal{E} = O \left(|\gamma - \gamma'| \left(\Psi(T) + \frac{M \log T}{T} + \log \log T \right) \right).$$

From the identity

$$\exp(i\mathcal{E}) - 1 = i \int_0^{\mathcal{E}} e^{i\theta} d\theta$$

and the fact that $|e^{i\theta}| = 1$, we see that

$$\exp(i\mathcal{E}) - 1 \ll |\gamma - \gamma'| \left(\Psi(T) + \frac{M \log T}{T} + \log \log T \right).$$

Therefore

$$T^{i(\alpha+2)(\gamma-\gamma')} - T^{i\alpha(\gamma-\gamma')} \ll |\gamma - \gamma'| \left(\Psi(T) + \frac{M \log T}{T} + \log \log T \right).$$

We now express each term in Z_1 as an integral, and then insert the above estimate, as follows. Write

$$\begin{aligned} Z_1 &= \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\substack{\frac{T}{\log^2 T} < \gamma' < \gamma \leq T \\ 0 < \left| \frac{\gamma - \gamma'}{2\pi} \log T \right| \leq M}} \int_c^d \frac{T^{i(t+2)(\gamma-\gamma')} - T^{it(\gamma-\gamma')}}{i(\gamma - \gamma') \log T} dt w(\gamma - \gamma') \\ &\ll \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\substack{\frac{T}{\log^2 T} < \gamma, \gamma' \leq T \\ 0 < \left| \frac{\gamma - \gamma'}{2\pi} \log T \right| \leq M}} \left(\frac{\Psi(T)}{\log T} + \frac{M}{T} + \frac{\log \log T}{\log T} \right) w(\gamma - \gamma'). \end{aligned}$$

Thus, since $w(\gamma - \gamma') \leq 1$, we see that

$$Z_1 \ll N(T, M) \left(\frac{\Psi(T)}{\log T} + \frac{M}{T} + \frac{\log \log T}{\log T} \right).$$

From this, (4.3.1), and the fact that $M > 1$, it follows that

$$Z_1 \ll \frac{M\Psi(T)}{\log T} + \frac{M^2}{T} + \frac{M \log \log T}{\log T}. \quad (4.4.8)$$

To bound Z_2 , we use the trivial estimate

$$|T^{i(d+2)(\gamma-\gamma')} - T^{i(c+2)(\gamma-\gamma')} - T^{id(\gamma-\gamma')} + T^{ic(\gamma-\gamma')}| \leq 4$$

to write

$$\begin{aligned} Z_2 &\ll \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\substack{\frac{T}{\log^2 T} < \gamma, \gamma' \leq T \\ M < \left| \frac{\gamma-\gamma'}{2\pi} \log T \right|}} \frac{1}{((\gamma - \gamma') \log T)^2} \\ &= \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\ell=0}^{\infty} \sum_{\substack{\frac{T}{\log^2 T} < \gamma, \gamma' \leq T \\ 2^\ell M < \left| \frac{\gamma-\gamma'}{2\pi} \log T \right| \leq 2^{\ell+1} M}} \frac{1}{((\gamma - \gamma') \log T)^2} \\ &\ll \left(\frac{T}{2\pi} \log T \right)^{-1} \sum_{\ell=0}^{\infty} \sum_{\substack{0 < \gamma, \gamma' \leq T \\ 0 < \left| \frac{\gamma-\gamma'}{2\pi} \log T \right| \leq 2^{\ell+1} M}} \frac{1}{(2^\ell M)^2} \\ &= \sum_{\ell=0}^{\infty} \frac{2N(T, 2^{\ell+1} M)}{(2^\ell M)^2}. \end{aligned}$$

Therefore, by (4.3.1), we obtain

$$Z_2 \ll \sum_{\ell=0}^{\infty} \frac{2^{\ell+1} M}{(2^\ell M)^2} \ll \frac{1}{M}. \quad (4.4.9)$$

It now follows from (4.4.4), (4.4.7), (4.4.8), and (4.4.9) that

$$\int_c^d \int_t^{t+2} G(\alpha, T) d\alpha dt \ll \frac{M\Psi(T)}{\log T} + \frac{M^2}{T} + \frac{M \log \log T}{\log T} + \frac{1}{M}.$$

We end the proof of the Lemma upon making M large and taking $T \rightarrow \infty$. $\square$

Now we can prove the claim we made at the beginning of this section.

Proposition 4.4.1. *Assume RH and suppose that $N^*(T) \sim 1$. If the Alternative Hypothesis is true, then (A2) holds.*

Proof. We first prove (4.4.3). If $0 \leq d \leq 1$, then we integrate (1.3.1) twice to see that

$$\int_0^d \int_0^t F(\alpha, T) d\alpha dt = \frac{d}{2} + \frac{d^3}{6} + o(1). \quad (4.4.10)$$

From the definition of $F(\alpha, T)$ in Section 1.3 and the definition of $G(\alpha, T)$ in Lemma 4.4.2, we have

$$F(\alpha, T) = N^*(T) + G(\alpha, T) + \overline{G(\alpha, T)}, \quad (4.4.11)$$

where $\bar{z}$ denotes the complex conjugate of z . Inserting (4.4.11) in (4.4.10) and using our assumption that $N^*(T) \sim 1$, we obtain

$$\int_0^d \int_0^t (G + \bar{G})(\alpha, T) d\alpha dt = \frac{d}{2} + \frac{d^3}{6} - \frac{d^2}{2} + o(1) \quad (4.4.12)$$

as $T \rightarrow \infty$ for $0 \leq d \leq 1$. Now let $d > 1$ and set $m = [(d-1)/2]$. Write

$$\int_0^d \left(\int_0^t G + \bar{G} d\alpha \right) dt = \int_0^1 + \int_1^3 + \int_3^5 + \cdots + \int_{2m+1}^d. \quad (4.4.13)$$

We use Lemma 4.4.2 repeatedly on the right-hand side of (4.4.13) to obtain

$$\int_0^d \left(\int_0^t G + \bar{G} d\alpha \right) dt = \int_0^1 + \int_{-1}^1 + \int_{-1}^1 + \cdots + \int_{-1}^{d-2m-2} + o(1).$$

Since $G + \bar{G}$ is an even function of α , we have $\int_{-1}^1 = 0$ and

$$\int_0^1 + \int_{-1}^{d-2m-2} = \int_0^1 + \int_{-1}^0 + \int_0^{d-2m-2} = \int_0^{|d-2m-2|}.$$

Thus

$$\int_0^d \left(\int_0^t G + \bar{G} d\alpha \right) dt = \int_0^{|d-2m-2|} \left(\int_0^t G + \bar{G} d\alpha \right) dt + o(1).$$

Since $0 \leq |d - 2m - 2| \leq 1$, it follows from this and (4.4.12) that

$$\int_0^d \left(\int_0^t G + \overline{G} d\alpha \right) dt = \frac{|d - 2m - 2|}{2} + \frac{|d - 2m - 2|^3}{6} - \frac{|d - 2m - 2|^2}{2} + o(1).$$

By this, (4.4.11), and our assumption that $N^*(T) \sim 1$, we have

$$\begin{aligned} \int_0^d \int_0^t F(\alpha, T) d\alpha dt \\ = \frac{d^2}{2} + \frac{|d - 2m - 2|}{2} + \frac{|d - 2m - 2|^3}{6} - \frac{|d - 2m - 2|^2}{2} + o(1) \end{aligned} \quad (4.4.14)$$

as $T \rightarrow \infty$, for all fixed $d > 1$. Notice that (4.4.14) reduces to (4.4.10) when $0 \leq d \leq 1$. Thus (4.4.14) is true for all fixed $d \geq 0$. By a straightforward computation using the definition of ν , we see that the right-hand side of (4.4.14) is equal to $\int_0^d \nu[0, t] dt + o(1)$. This proves (4.4.3).

Now we complete the proof of the proposition. Since $F(\alpha, T)$ is nonnegative, we may write

$$\frac{1}{h} \int_{d-h}^d \left(\int_0^t F d\alpha \right) dt \leq \int_0^d F(\alpha, T) d\alpha \leq \frac{1}{h} \int_d^{d+h} \left(\int_0^t F d\alpha \right) dt$$

for $d > 0$ and small enough $h > 0$. Thus by (4.4.3) we have

$$\frac{1}{h} \int_{d-h}^d \nu[0, t] dt + o(1) \leq \int_0^d F(\alpha, T) d\alpha \leq \frac{1}{h} \int_d^{d+h} \nu[0, t] dt + o(1)$$

as $T \rightarrow \infty$ for fixed $d > 0$ and fixed small $h > 0$. By making h small enough, we obtain (A) with $\mu = \nu$, that is, we obtain (A2). $\square$

Before we prove Theorem 1.3.4, we need to make a few definitions. Recall from Section 1.3 that we used $0 < \gamma_1 \leq \gamma_2 \leq \dots$ to denote the sequence of ordinates of all the zeros above the real line, counting multiplicity. We also defined $B_{k/2}(T)$ as the set of $\gamma_n \leq T$ such that $k/2$ is closest among all half-integers to $\tilde{\gamma}_{n+1} - \tilde{\gamma}_n$, and we wrote

$$p_{k/2} = p_{k/2}(T) \stackrel{\text{def}}{=} \left(\frac{T}{2\pi} \log T \right)^{-1} |B_{k/2}(T)|.$$

It is possible that γ_n belongs to two of the sets $B_{k/2}(T)$; this happens when $\tilde{\gamma}_{n+1} - \tilde{\gamma}_n$ equals the midpoint between two consecutive half-integers. Thus, for convenience, we will deal instead with the sets

$$B'_{k/2}(T) \stackrel{\text{def}}{=} \{\gamma_n \leq T : \frac{k}{2} - \frac{1}{4} \leq \tilde{\gamma}_{n+1} - \tilde{\gamma}_n < \frac{k}{2} + \frac{1}{4}\}, \quad k = 0, 1, 2, \dots,$$

which are pairwise disjoint. If γ_n is in $B'_{k/2}(T)$, then, without loss of generality, we may take $k_n = k$ in (1.3.4). With this convention, we can write

$$B'_{k/2}(T) = \{\gamma_n \leq T : k_n = k\}. \quad (4.4.15)$$

Note that $B'_{k/2}(T)$ is a subset of $B_{k/2}(T)$, and an ordinate γ_n that is in $B_{k/2}(T)$ is not in $B'_{k/2}(T)$ if and only if $\tilde{\gamma}_{n+1} - \tilde{\gamma}_n = \frac{k}{2} + \frac{1}{4}$. Thus if γ_n is in $B_{k/2}(T)$ and not in $B'_{k/2}(T)$, then $k_n = k + 1$ and so by (1.3.4) we have

$$\frac{1}{4} = |\tilde{\gamma}_{n+1} - \tilde{\gamma}_n - \frac{k_n}{2}| \ll (\gamma_{n+1} - \gamma_n)\psi(\gamma_n). \quad (4.4.16)$$

We will use this and the following Lemma to show that $B'_{k/2}(T)$ has essentially the same size as $B_{k/2}(T)$.

Lemma 4.4.3. *Assume RH and suppose the Alternative Hypothesis is true. If $\delta > 0$ is fixed, then*

$$\#\{0 < \gamma_n \leq T : (\gamma_{n+1} - \gamma_n)\psi(\gamma_n) \geq \delta\} = o(T \log T).$$

Proof. Recall the properties of the function $\psi(\gamma)$ from the statement of the Alternative Hypothesis. Define

$$\Psi_0(T) = \max_{\gamma_n \leq T} \psi(\gamma_n). \quad (4.4.17)$$

It is immediate from the properties of $\psi(\gamma)$ that $\Psi_0(T) \rightarrow \infty$ and $\Psi_0(T) = o(\log T)$ as $T \rightarrow \infty$. It follows from (4.4.17) that

$$\#\{0 < \gamma_n \leq T : (\gamma_{n+1} - \gamma_n)\psi(\gamma_n) \geq \delta\} \leq \#\{0 < \gamma_n \leq T : \gamma_{n+1} - \gamma_n \geq \lambda / \log T\}, \quad (4.4.18)$$

where $\lambda = \delta \log T / \Psi_0(T)$. By a result of Fujii (see §9.25 of [64]), we have

$$\#\{0 < \gamma_n \leq T : \gamma_{n+1} - \gamma_n \geq \lambda / \log T\} \ll N(T) \exp(-A\lambda^{1/2}(\log \lambda)^{-1/4}). \quad (4.4.19)$$

The Lemma now follows from (4.4.18), (4.4.19), (4.3.13), and the fact that $\lambda \rightarrow \infty$ as $T \rightarrow \infty$. $\square$

By (4.4.16) and Lemma 4.4.3, we see that there are at most $o(T \log T)$ ordinates γ_n that are in $B_{k/2}(T)$ and not in $B'_{k/2}(T)$. Hence

$$p_{k/2} = p'_{k/2} + o(1) \quad (4.4.20)$$

as $T \rightarrow \infty$.

We now prove Theorem 1.3.4. By (1.3.4), if $k_n = 0$ then

$$(\gamma_{n+1} - \gamma_n) \frac{\log \gamma_n}{2\pi} \leq \tilde{\gamma}_{n+1} - \tilde{\gamma}_n \ll (\gamma_{n+1} - \gamma_n) \psi(\gamma_n).$$

This implies that $\log \gamma_n \ll \psi(\gamma_n)$ since $\gamma_{n+1} - \gamma_n > 0$ by our hypothesis that all the zeros are simple. Since $\psi(\gamma) = o(\log \gamma)$, it follows that $k_n = 0$ for at most finitely many n . Thus

$$|B'_0(T)| = O(1), \quad (4.4.21)$$

and so $p_0 = o(1)$ as $T \rightarrow \infty$ by (4.4.20).

We next compute upper bounds for $p_{k/2}$ when $k \geq 1$. By (4.3.13) there exists a positive constant Q for which

$$N(T+Q) - N(T) > 0 \quad (4.4.22)$$

for all $T > 0$. Let $\delta > 0$ be small enough so that $k/2$ is the only half-integer in the interval $[k/2 - \delta, k/2 + \delta]$. Define $C_{k/2}(T)$ to be the subset of $B'_{k/2}(T)$ for which $T(\log T)^{-2} < \gamma_n \leq T - Q$ and $|\tilde{\gamma}_{n+1} - \tilde{\gamma}_n - k_n/2| < \delta/2$. Note that by (4.3.13), (4.4.6), and Lemma 4.4.3 we have

$$|B'_{k/2}(T)| = |C_{k/2}(T)| + o(T \log T). \quad (4.4.23)$$

Furthermore, define

$$E_{k/2}(T) = \left\{ (\gamma, \gamma') : 0 < \gamma, \gamma' \leq T, \frac{2\pi}{\log T} \left(\frac{k}{2} - \delta \right) < \gamma - \gamma' \leq \frac{2\pi}{\log T} \left(\frac{k}{2} + \delta \right) \right\}. \quad (4.4.24)$$

We will show for large T that if γ_n is in $C_{k/2}(T)$ then (γ_{n+1}, γ_n) is in $E_{k/2}(T)$. The mean value theorem of differential calculus gives that there is a real number x between γ_n and γ_{n+1} such that

$$\gamma_{n+1} \log \gamma_{n+1} - \gamma_n \log \gamma_n = (\gamma_{n+1} - \gamma_n)(\log x + 1).$$

Therefore

$$(\gamma_{n+1} - \gamma_n) \frac{\log T}{2\pi} = (\tilde{\gamma}_{n+1} - \tilde{\gamma}_n) \left(\frac{\log T}{\log x + 1} \right). \quad (4.4.25)$$

If γ_n is in $C_{k/2}(T)$, then $|\tilde{\gamma}_{n+1} - \tilde{\gamma}_n - k/2| < \delta/2$ by the definition of $C_{k/2}(T)$ and (4.4.15); thus

$$\left(\frac{k}{2} - \frac{\delta}{2} \right) (1 - o(1)) < (\tilde{\gamma}_{n+1} - \tilde{\gamma}_n) \left(\frac{\log T}{\log x + 1} \right) < \left(\frac{k}{2} + \frac{\delta}{2} \right) (1 + o(1))$$

as $T \rightarrow \infty$. From this and (4.4.25), it follows that

$$\frac{k}{2} - \delta < (\gamma_{n+1} - \gamma_n) \frac{\log T}{2\pi} < \frac{k}{2} + \delta$$

for large T . Moreover, from the definition of $C_{k/2}(T)$, we have $\gamma_n \leq T - Q$ and therefore $\gamma_{n+1} \leq T$ by (4.4.22). Hence, if γ_n is in $C_{k/2}(T)$ and T is large enough, then (γ_{n+1}, γ_n) is in $E_{k/2}(T)$. Thus $|C_{k/2}(T)| \leq |E_{k/2}(T)|$ for large enough T . It follows now from (4.4.24) and (D2), which holds by Corollary 1.3.1, that

$$|C_{k/2}(T)| \leq |E_{k/2}(T)| \sim \left(\frac{T}{2\pi} \log T \right) \cdot \begin{cases} \frac{1}{2} & \text{if } k \text{ is even,} \\ \frac{1}{2} - \frac{2}{\pi^2 k^2} & \text{if } k \text{ is odd.} \end{cases} \quad (4.4.26)$$

From this, (4.4.23), and (4.4.20), we now deduce the upper bound for p_1 in (1.3.7) and that the right-hand side of (1.3.6) is an upper bound for $p_{1/2}$.

Next we prove the lower bound for $p_{1/2}$. To do this, we first consider the set $E_{1/2}(T)$ defined by (4.4.24) with $k = 1$. Write the set as a disjoint union

$$E_{1/2}(T) = D_{1/2}(T) \cup V_{1/2}(T), \quad (4.4.27)$$

where $D_{1/2}(T)$ contains all the pairs (γ, γ') in $E_{1/2}(T)$ that have $\gamma' \leq T(\log T)^{-2}$ and $V_{1/2}(T)$ contains the rest. We claim that if (γ, γ') is in $V_{1/2}(T)$ then γ' is in $B'_{1/2}(T)$, provided T is large enough. To prove this, let (γ_m, γ_n) be in $V_{1/2}(T)$ and note that $m > n$ since $\gamma_m > \gamma_n$ by the definition of $E_{1/2}(T)$. By Lemma 4.4.1 and the fact that $\gamma_m - \gamma_n \ll 1/\log T$, we have

$$(\gamma_m - \gamma_n) \frac{\log T}{2\pi} = \frac{j}{2} + o(1)$$

for some integer j . Since (γ_m, γ_n) is in $E_{1/2}(T)$, we must have $j = 1$ for T large enough. From the proof of Lemma 4.4.1, we see that $j = \sum_{\ell=n}^{m-1} k_\ell$. By (4.4.21), (4.4.15), and the fact that $\gamma_n > T(\log T)^{-2}$, we see that if T is large enough then $k_\ell > 0$ for all $\ell \geq n$. Hence, since $j = 1$, the only possible values for m and the k_ℓ are $m = n + 1$ and $k_n = 1$. Therefore γ_n is in $B'_{1/2}(T)$ by (4.4.15), and we have proved our claim.

We have shown that the map $(\gamma, \gamma') \mapsto \gamma'$ from $V_{1/2}(T)$ is into $B'_{1/2}(T)$. This map is injective because of our assumption that all zeros are simple. Hence

$$|V_{1/2}(T)| \leq |B'_{1/2}(T)|. \quad (4.4.28)$$

To estimate the size of the set $D_{1/2}(T)$ in (4.4.27), write

$$\begin{aligned} |D_{1/2}(T)| &\leq \sum_{0 < \gamma' < \frac{T}{\log^2 T}} \sum_{\left(\frac{1}{2}-\delta\right) \frac{2\pi}{\log T} < (\gamma - \gamma') \leq \left(\frac{1}{2}+\delta\right) \frac{2\pi}{\log T}} 1 \\ &\leq \sum_{0 < \gamma' < \frac{T}{\log^2 T}} (N(\gamma' + 1) - N(\gamma')) \\ &\ll T \end{aligned}$$

for large enough T , by (4.3.13) and (4.4.6). From this, (4.4.28), and (4.4.27), it follows that

$$|E_{1/2}(T)| \leq O(T) + |B'_{1/2}(T)|.$$

Hence, by (4.4.24) and (D2), we have

$$\left(\frac{T}{2\pi} \log T\right) \left(\frac{1}{2} - \frac{2}{\pi^2}\right) \sim |E_{1/2}(T)| \leq O(T) + |B'_{1/2}(T)|.$$

From this and (4.4.20), it follows that

$$\frac{1}{2} - \frac{2}{\pi^2} + o(1) \leq p_{1/2}.$$

Combining this with the upper bound for $p_{1/2}$, we obtain (1.3.6).

To prove the lower bound for p_1 , we need the formulas

$$\sum_{k=0}^{\infty} p'_{k/2} = 1 + o(1) \tag{4.4.29}$$

$$\sum_{k=0}^{\infty} \left(\frac{k}{2}\right) p'_{k/2} = 1 + o(1). \tag{4.4.30}$$

The first formula follows from (4.3.13) and the disjoint union

$$\bigcup_{k=0}^{\infty} B'_{k/2}(T) = \{\gamma_n : \gamma_n \leq T\}.$$

To prove the second, observe that by (4.4.15) we have

$$\sum_{k=0}^{\infty} \left(\frac{k}{2}\right) |B'_{k/2}(T)| = \sum_{\gamma_n \leq T} \frac{k_n}{2}. \tag{4.4.31}$$

Let $\Psi_0(T)$ be as defined in (4.4.17). By (1.3.4), we have

$$\begin{aligned} \sum_{\gamma_n \leq T} \frac{k_n}{2} &= \sum_{\gamma_n \leq T} \left\{ \tilde{\gamma}_{n+1} - \tilde{\gamma}_n - O((\gamma_{n+1} - \gamma_n)\psi(\gamma_n)) \right\} \\ &= \tilde{\gamma}_{m+1} - \tilde{\gamma}_1 + O((\gamma_{m+1} - \gamma_1)\Psi_0(T)), \end{aligned}$$

where m is the largest integer for which $\gamma_m \leq T$. It follows from (4.4.22) that $T < \gamma_{m+1} \leq T + Q$ and so

$$\tilde{\gamma}_{m+1} \sim \frac{T}{2\pi} \log T \quad \text{and} \quad \gamma_{m+1} \sim T$$

as $T \rightarrow \infty$. Hence

$$\sum_{\gamma_n \leq T} \frac{k_n}{2} = (1 + o(1)) \frac{T}{2\pi} \log T + O(T\Psi_0(T)).$$

Since $\Psi_0(T) = o(\log T)$, it follows that

$$\sum_{\gamma_n \leq T} \frac{k_n}{2} = (1 + o(1)) \frac{T}{2\pi} \log T$$

as $T \rightarrow \infty$. From this and (4.4.31), we deduce (4.4.30).

Now we can prove the lower bound for p_1 . The following elegant trick is due to Yoonbok Lee. Multiplying both sides of (4.4.29) by $3/2$ and subtracting the respective sides of (4.4.30) from the result, we obtain

$$\left(\frac{3}{2}\right) p'_0 + p'_{1/2} + \left(\frac{1}{2}\right) p'_1 - \left(\frac{1}{2}\right) p'_2 - p'_{5/2} - \cdots = \frac{1}{2} + o(1).$$

Since $p'_{k/2} \geq 0$ for all k , it follows that

$$\left(\frac{3}{2}\right) p'_0 + p'_{1/2} + \left(\frac{1}{2}\right) p'_1 \geq \frac{1}{2} + o(1).$$

Therefore, by (4.4.20), we have

$$\left(\frac{3}{2}\right) p_0 + p_{1/2} + \left(\frac{1}{2}\right) p_1 \geq \frac{1}{2} + o(1).$$

The lower bound for p_1 in (1.3.7) now follows by inserting $p_0 = o(1)$ and our estimate (1.3.6) for $p_{1/2}$ into the inequality and rearranging.

A similar argument proves the lower bound for $p_{3/2}$ in (1.3.8), as follows. We multiply both sides of (4.4.29) by 2 and subtract the respective sides of (4.4.30) from the result to deduce that

$$2p'_0 + \left(\frac{3}{2}\right) p'_{1/2} + p'_1 + \left(\frac{1}{2}\right) p'_{3/2} - \left(\frac{1}{2}\right) p'_{5/2} - p'_3 - \cdots = 1 + o(1).$$

Since $p'_{k/2} \geq 0$ for all k , it follows that

$$2p'_0 + \left(\frac{3}{2}\right) p'_{1/2} + p'_1 + \left(\frac{1}{2}\right) p'_{3/2} \geq 1 + o(1).$$

Therefore, by (4.4.20), we have

$$2p_0 + \left(\frac{3}{2}\right) p_{1/2} + p_1 + \left(\frac{1}{2}\right) p_{3/2} \geq 1 + o(1).$$

It follows from this, $p_0 = o(1)$, (1.3.6), and the upper bound in (1.3.7) that

$$\left(\frac{3}{2}\right) \left(\frac{1}{2} - \frac{2}{\pi^2}\right) + \frac{1}{2} + \left(\frac{1}{2}\right) p_{3/2} \geq 1 + o(1).$$

This gives the lower bound in (1.3.8) after a rearrangement.

To prove the upper bound in (1.3.8), observe that (4.4.30) implies that

$$\left(\frac{1}{2}\right) p'_{1/2} + p'_1 + \left(\frac{3}{2}\right) p'_{3/2} \leq 1 + o(1)$$

because $p'_{k/2} \geq 0$ for all k . Thus, by (4.4.20),

$$\left(\frac{1}{2}\right) p_{1/2} + p_1 + \left(\frac{3}{2}\right) p_{3/2} \leq 1 + o(1).$$

From this, (1.3.6), and the lower bound in (1.3.7), we deduce that

$$\left(\frac{1}{2}\right) \left(\frac{1}{2} - \frac{2}{\pi^2}\right) + \frac{4}{\pi^2} + \left(\frac{3}{2}\right) p_{3/2} \leq 1 + o(1).$$

Rearranging this leads to the upper bound in (1.3.8).

Similarly, to prove (1.3.9), observe that (4.4.30) implies for $k \geq 4$ that

$$\left(\frac{1}{2}\right) p'_{1/2} + p'_1 + \left(\frac{3}{2}\right) p'_{3/2} + \left(\frac{k}{2}\right) p'_{k/2} \leq 1 + o(1),$$

and so (4.4.20) gives

$$\left(\frac{1}{2}\right) p_{1/2} + p_1 + \left(\frac{3}{2}\right) p_{3/2} + \left(\frac{k}{2}\right) p_{k/2} \leq 1 + o(1).$$

From this, (1.3.6), and the lower bounds in (1.3.7) and (1.3.8), we conclude that

$$\left(\frac{1}{2}\right) \left(\frac{1}{2} - \frac{2}{\pi^2}\right) + \frac{4}{\pi^2} + \left(\frac{3}{2}\right) \left(\frac{6}{\pi^2} - \frac{1}{2}\right) + \left(\frac{k}{2}\right) p_{k/2} \leq 1 + o(1).$$

This implies (1.3.9) and completes the proof of Theorem 1.3.4.

Remark 3: The upper bounds in (1.3.6), (1.3.7), and (4.4.26) can be proved using the assumption that $N^*(T) \sim 1$ instead of the stronger assumption that all the zeros are simple. To do this, we only need to define the sets $E_{k/2}(T)$ so that they count multiplicity. However, we did use the hypothesis that all the zeros are simple to prove the lower bound for $p_{1/2}$. This is because we required $E_{k/2}(T)$ to count multiplicity to be able to use (D2), and at the same time we needed $E_{k/2}(T)$ to not count multiplicity in proving (4.4.28).

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